Section IV

Introductory Virology

Chapter 48

Nature, structure and taxonomy of viruses

The term 'virus' (Latin, poison) refers to members of a unique class of infectious agents which are extremely small, contain only one type of nucleic acid and have an absolute dependence on living cells for replication (Box 48.1). The genomes of the viruses which infect animals are smaller than those of prokaryotic cells, ranging from about 2 kilobase pairs (kbp) to 200 kbp. In most viruses, the nucleic acid is present as a single molecule; in some RNA viruses the nucleic acid occurs in separate segments. Although the nucleic acid of viral genomes is usually linear, it is circular in some viruses. Genomes of DNA viruses can be single-stranded or double-stranded. Particular characteristics of DNA and RNA viruses are presented in Tables 48.1 and 48.2 and in Figs. 48.1 and 48.2.

There are two unique types of infectious agents, viroids and prions, which are structurally less complex than viruses. Viroids are composed of naked RNA and prions are proteinaceous infectious particles which are devoid of demonstrable nucleic acid.

Viruses were first recognized as unique infectious entities in the late nineteenth century. Knowledge of the nature of bacteria and fungi was well established by this time. Dmitri Ivanovsky, a Russian scientist, reported in 1892 that it was possible to transmit tobacco mosaic disease from diseased to healthy plants using filtered leaf extract as inoculum. The filters used by Ivanovsky were Chamberland filters, porcelain filters designed to remove bacteria from drinking water. In 1898, Martinus Beijerinck, unaware of the work of Ivanovsky, also demonstrated the filterability of the agent of tobacco mosaic disease. Moreover, he realised that the disease could not be due to a toxin as the filtered sap from infected plants could be used for serial transmission of the disease without loss of potency. In the same year, Loeffler and Frosch identified the first filterable agent from animals, the virus of foot-and-mouth disease. Yellow fever virus, a filterable agent pathogenic for humans, was described by

Box 48.1 Characteristics of viruses which can infect animals.

- Small infectious agents ranging in size from 20 to 300 nm
- Composed of nucleic acid surrounded by a protein coat; in addition, some have envelopes
- Contain only one type of nucleic acid, either DNA or RNA
- Unlike bacteria and fungi, viruses cannot replicate on inert media; viable host cells are required for replication
- Some viruses have an affinity for particular cell types

Table 48.1 Characteristics of the families of DNA viruses of veterinary importance.

Family	Virion size (nm)	Capsid symmetry	Envelope	Type of genome
Adenoviridae	70–90	Icosahedral	Absent	Linear, double-stranded DNA
Asfarviridae	175–215	Icosahedral	Present	Linear, double-stranded DNA
Circoviridae	17–22	Icosahedral	Absent	Circular molecule of positive-sense or ambisense, single-stranded DNA
Herpesviridae	120–200	Icosahedral	Present	Single molecule of linear, double-stranded DNA
Papillomaviridae	55	Icosahedral	Absent	Single molecule of circular, double-stranded DNA
Parvoviridae	18–26	Icosahedral	Absent	Single molecule of linear, positive-sense or negative-sense, single-stranded DNA
Poxviridae	300 x 200	Complex	Present	Single molecule of linear, double-stranded DNA

Table 48.2 Characteristics of the families of RNA viruses of veterinary importance.

Family	Virion size (nm)	Capsid symmetry	Envelope	Type of genome
Arteriviridae	40–60	Icosahedral	Present	Linear, single molecule of positive-sense, single-stranded RNA
Astroviridae	28–30	Icosahedral	Absent	Linear, single molecule of positive-sense, single-stranded RNA
Birnaviridae	60	Icosahedral	Absent	Two segments of linear, double-stranded RNA
Bornaviridae	90	Icosahedral	Present	Linear, single molecule of negative-sense, single-stranded RNA
Bunyaviridae	80–120	Helical	Present	Three segments of linear, negative-sense or ambisense single-stranded RNA
Caliciviridae	27–40	Icosahedral	Absent	Linear, single molecule of positive-sense, single-stranded RNA
Coronaviridae	120–160	Helical	Present	Linear, single molecule of positive-sense, single-stranded RNA
Flaviviridae	40–60	Icosahedral	Present	Linear, single molecule of positive-sense, single-stranded RNA
Orthomyxoviridae	80–120	Helical	Present	Six to eight segments of linear, negative-sense, single-stranded RNA
Paramyxoviridae	150–300	Helical	Present	Linear, single molecule of negative-sense, single-stranded RNA
Picornaviridae	30	Icosahedral	Absent	Linear, single molecule of positive-sense, single-stranded RNA
Reoviridae	60–80	Icosahedral	Absent	Ten to twelve segments of linear, double-stranded RNA
Retroviridae	80–100	Icosahedral	Present	Diploid, linear, positive-sense, single-stranded RNA
Rhabdoviridae	180 x 75	Helical	Present	Linear, single molecule of negative-sense, single-stranded RNA
Togaviridae	70	Icosahedral	Present	Linear, single molecule of positive-sense, single-stranded RNA

Walter Reed and his team in 1901. Ellerman and Bang, in 1908, demonstrated the oncogenic potential of a filterable agent, the cause of avian leukosis. In 1915, Frederick Twort observed that bacteria were susceptible to a filterable agent and, two years later, Felix D'Herelle, made a similar observation. D'Herelle named these viruses 'bacteriophages' and developed a technique for establishing their concentration in active preparations. Bacteriophages have proved to be particularly useful in studies on viral replication and bacterial genetics.

Initially, the only way to recover large quantities of virus was through infecting susceptible animals. In 1913, Steinhardt and his colleagues succeeded in growing vaccinia virus in explants of guinea-pig cornea embedded in clotted plasma. Some twenty years later, Furth and Sturmia used mice as a host species for propagating viruses, while Woodruff and Goodpasture were successful in propagating fowlpox virus on the chorioallantoic membrane of embryonated eggs. A major advance was made in the early 1950s with the development of single cell cultures. Factors which were critical in this development included the availability of antibiotics to control bacterial contamination and the use of trypsin to obtain cell suspensions from embryonic or adult tissue. The separated cells could then be grown as monolayers on glass surfaces. Continuous cell lines, capable of multiplying indefinitely, provided a reliable source of cells for virus cultivation.

In 1887, Buist observed vaccinia virus using a light microscope. However, because of the limited resolving power of this type of microscopy, the structure of the virus was not discernible. In 1939, Kausche and his co-workers employed the newly developed electron microscope and a metal shadowing technique to identify tobacco mosaic virus in purified preparations. Ultrastructural studies of viruses were greatly expanded and enhanced in the 1950s by the development of negative staining and methods for cutting ultrathin sections. X-ray diffraction methods have been applied to viruses since the 1930s, when it was discovered that simple viruses could be crystallized. The first complete high resolution structure of a crystalline virus, tomato bushy stunt virus, was obtained by Harrison and his co-workers in 1978. Computer analysis of the diffraction patterns obtained by such studies has contributed to knowledge of the molecular structure of viruses.

In recent years, remarkable advances have been made in our understanding of the epidemiology, pathogenesis and

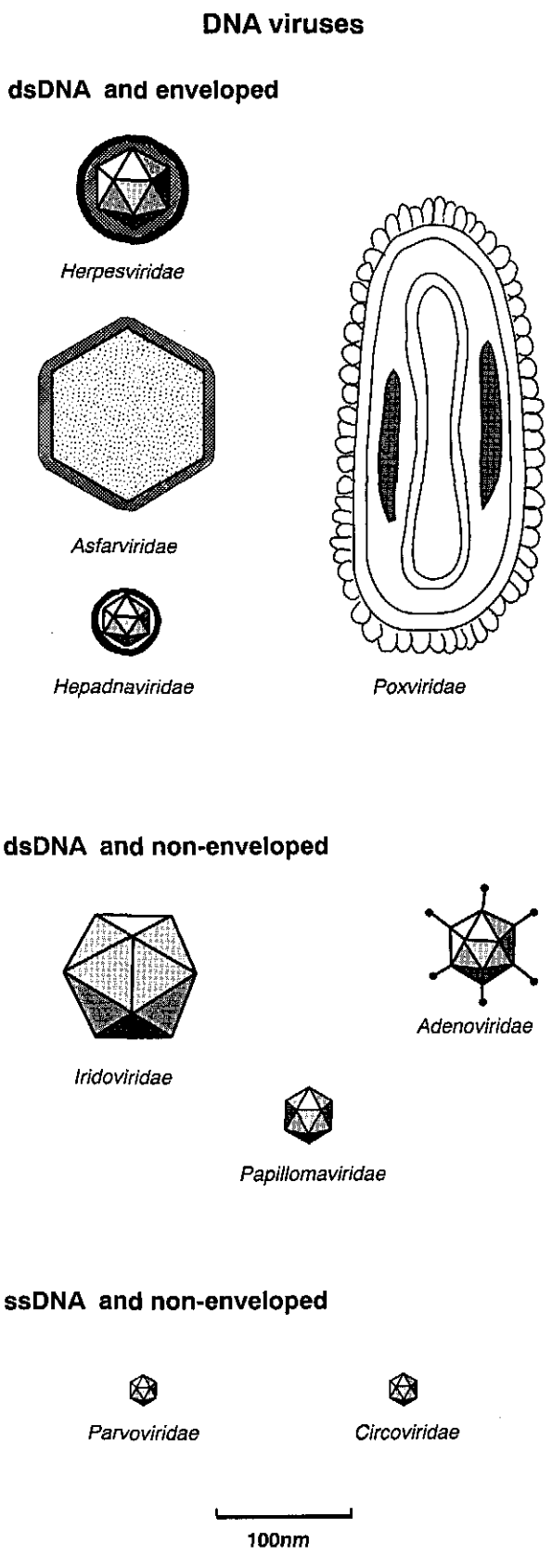


Figure 48.1 Diagrammatic representations of virions in families of DNA viruses of vertebrates. The genomes of these viruses are composed of either double-stranded (ds) or single-stranded (ss) DNA.

control of viral diseases. However, as new viral diseases are recognized in the animal and human populations, fresh challenges confront those engaged in virological research.

Origin of viruses

Viruses have evolved to a stage where they are considered to be among the most efficient and economic forms of microbial life. They can be categorized in three main groups on the basis of their nucleic acid composition: DNA viruses, RNA viruses and viruses which utilize both DNA and RNA for replication. Differences between these groups are significant and may be indicative of independent origins for each group. Although the origin of viruses is uncertain, three theories have been proposed to explain their evolution. They may have originated and evolved in parallel with primordial life forms. They may have arisen from segments of cellular nucleic acid which acquired the ability to replicate at the expense of the host cell. The third theory, the regressive theory of the origin of viruses, postulates that they arose from free-living organisms which gradually lost genetic information until they became totally dependent on the biosynthetic pathways of their host cells.

Structure of viruses

A fully assembled infective virus is termed a virion. The fundamental component of the virion is a nucleoprotein core with the ability to infect host cells and replicate in them, thus ensuring continued survival. The genome of vertebrate viruses is enclosed within a shell of proteins, called a capsid (Fig. 48.3). It is haploid except in retroviruses, in which it is diploid. The term nucleocapsid is used to describe the packaged form of the genome in the capsid. Each subunit of the capsid is composed of a folded polypeptide chain. Collections of these subunits constitute structural units or protomers which, in turn, comprise assembly units. The term capsomer or morphological unit is used to describe features such as protrusions seen on the surface of virus particles in electron micrographs. These often correspond to groups of protein subunits arranged about a local axis of symmetry. Capsids are, therefore, composed of multiples of one or more types of protein subunits. The orderly arrangement of similar protein-protein interfaces results in a symmetrical structure. Icosahedral and helical symmetries are the two types of capsid symmetry described in viruses (Fig. 48.4).

Closed-shell virions, isometric viruses, have structures based on icosahedral symmetry, a structural form which offers the maximum capacity and greatest strength for a given surface area. The icosahedron, one of the five platonic solids, has 20 equilateral triangles forming its faces, 30 edges and 12 vertices. At its simplest, a viral icosahedron has 60 identical structural units, three in each triangular face. Larger numbers of smaller units are

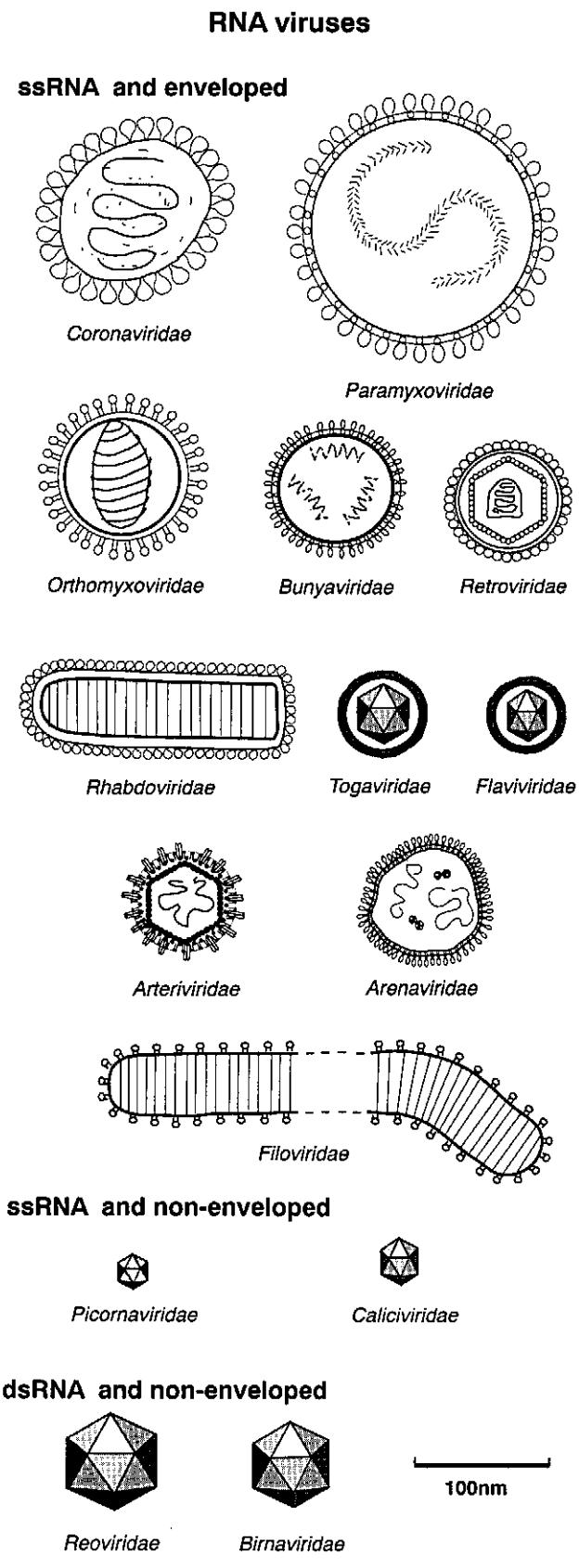


Figure 48.2 Diagrammatic representations of virions in families of RNA viruses of vertebrates. The genomes of these viruses are composed of either double-stranded (ds) or single-stranded (ss) RNA.

accommodated on the triangular faces only in specific multiples of 60, represented by the formula $T = h^2 + hk + k^2$ where h and k are integers having no common factors and T is the triangulation number. For caliciviruses $T = 3$ and for herpesviruses $T = 16$. The structural units at each vertex form groups of five termed pentons, while those on the faces form groups of six called hexons. There are two-fold, three-fold and five-fold axes of rotational symmetry which pass through edges, faces and vertices respectively. Viruses with icosahedral symmetry are often not seen as icosahedrons; they may appear in electron micrographs as spheres or hexagons. Icosahedral capsids are generally assembled in the host cell prior to incorporation of the viral nucleic acid. Some viral preparations may contain capsids devoid of nucleic acid. The nucleic acid of double-stranded DNA viruses is condensed into a form suitable for incorporation into capsids by the action of cellular histones and basic virus-encoded molecules. The protective capsid of many RNA viruses is formed by the insertion of protein units between each turn of the nucleic acid helix. The capsid protein helix, therefore, coincides with that of the nucleic acid and the length of the helix is determined by the length of the RNA molecule. Viral particles devoid of nucleic acid cannot be formed. In RNA viruses, each capsomere consists of a single polypeptide molecule.

In many types of viruses the nucleocapsid is covered by an envelope composed of a lipid bilayer and associated glycoproteins. The envelope is acquired when the nucleocapsid buds through a cellular membrane, usually the plasma membrane. In some viral infections, the envelope is acquired from the endoplasmic reticulum, the Golgi apparatus or the nuclear membrane. Proteins encoded by viral nucleic acid and integrated as glycoprotein into the appropriate membrane by the compartmentalization mechanisms of the host cell are an integral part of the viral envelope. These glycoproteins are associated with binding to receptors on host cells, membrane fusion, uncoating of the virion and destruction of receptors on host cells. A single envelope glycoprotein may have multiple functions. In most enveloped viruses, the envelope must be intact to maintain infectivity and, treatment with lipid solvents such as ether or chloroform, renders them non-infectious. Epitopes on envelope glycoproteins are often important for inducing protective immune responses in infected animals. Peplomers or spikes are knob-like projections from the envelope in certain viruses including coronaviruses, retroviruses, orthomyxoviruses, rhabdoviruses and paramyxoviruses. These structures are formed from oligomers of surface glycoproteins. They often bind to cell receptors and, in addition, they may have enzymatic activity. A layer of protein, termed matrix protein, is present between the nucleocapsid and the envelope in some enveloped viruses. This layer provides additional rigidity to the virion. Helical RNA viruses of animals are enveloped.

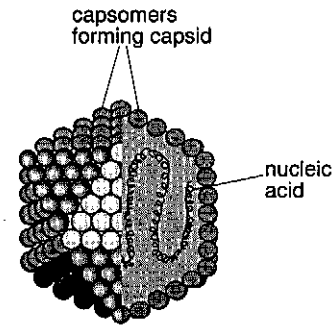


Figure 48.3 Diagrammatic representation of capsomers forming protective icosahedral capsid outside viral nucleic acid.

Taxonomy of viruses

The sole experimental indication of the minute size of viruses at the end of the nineteenth century was their ability to pass through filters which retained bacteria. Information relating to viruses derived largely from studies on the diseases which they caused. Early classification systems therefore, were based on their pathogenic effects and transmission patterns. In the 1930s, details of the structure and composition of viruses began to emerge. Subsequently, it was possible to group viruses on the basis of shared features of virions. During the 1950s and 1960s, several classification schemes were adopted. The International Committee on Nomenclature of Viruses (ICNV) was established in 1966 to develop a single universal taxonomic scheme. The successor to the ICNV, the International Committee on Taxonomy of Viruses (ICTV) established in 1973, developed and expanded the universal scheme in which virion characteristics are used to assign viruses to five main hierarchical levels namely order, family, subfamily, genus and species (Tables 48.3 to 48.8). Virus orders are designated by the suffix *-virales*. In orders, phylogenetically-related families are grouped together. Only two orders containing viruses of animals have been defined thus far. These are the *Mononegavirales* comprising the families *Paramyxoviridae, Rhabdoviridae, Bornaviridae* and *Filoviridae* and the *Nidovirales* comprising the families *Coronaviridae* and *Arteriviridae*. Families are, therefore, designated by the suffix *-viridae*. Of more than 50 families currently recognized, about 22 contain viruses of veterinary importance. The suffix *-virinae* denotes a subfamily. Viral genera are designated by the suffix *-virus*. More than 230 genera are recognized. The criteria for defining a genus differ from family to family. The species taxon is regarded as the most important level in the classification of viruses. However, its definition and application has always been difficult and controversial. In 1991, the ICTV accepted the definition of a virus species proposed by van Regenmortel (1990) which states that 'a virus species is defined as a polythetic class of viruses that constitutes a replicating lineage and occupies a particular ecological niche'. This implies that a virus species is defined by a combination of multiple properties and characteristics; no single or unique property is essential for species definition. This comprehensive type of taxonomy is the Adansonian system of classification. In the present scheme of virus taxonomy, the primary delineating criteria are the type and nature of the genome, the mode and site of viral replication and the structure of the virion. Currently, more than 1,500 virus species are recognized by the ICTV, with the

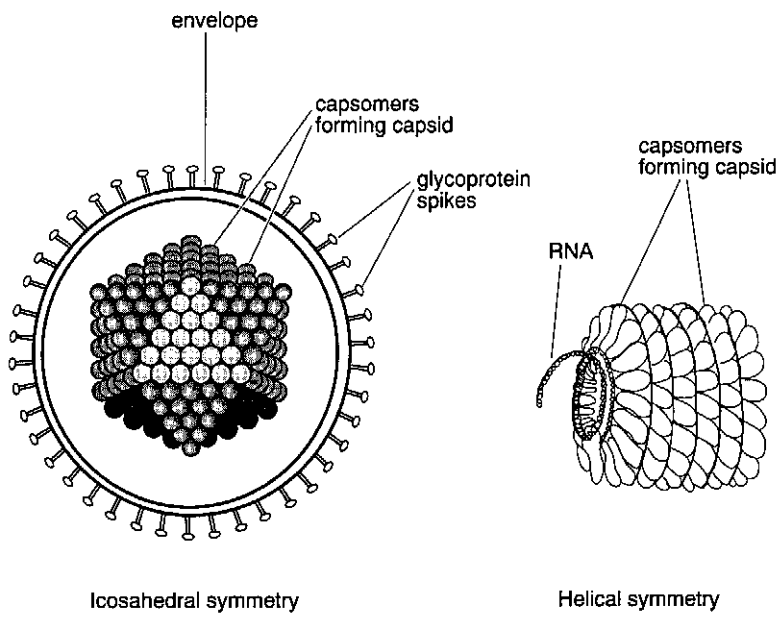


Figure 48.4 Diagrammatic representation of icosahedral and helical symmetry.

Table 48.3 The families containing the double-stranded DNA viruses of vertebrates.

Family	Subfamily	Genus	Type species
Adenoviridae		*Mastadenovirus*	Human adenovirus C
		Aviadenovirus	Fowl adenovirus A
Asfarviridae		*Asfivirus*	African swine fever virus
Herpesviridae	*Alphaherpesvirinae*	*Simplexvirus*	Human herpesvirus 1
		Varicellovirus	Human herpesvirus 3
		Marek's disease-like viruses	Gallid herpesvirus 2
		Infectious laryngotracheitis-like viruses	Gallid herpesvirus 1
	Betaherpesvirinae	*Cytomegalovirus*	Human herpesvirus 5
		Muromegalovirus	Murid herpesvirus 1
		Roseolovirus	Human herpesvirus 6
	Gammaherpesvirinae	*Lymphocryptovirus*	Human herpesvirus 4
		Rhadinovirus	Saimiriine herpesvirus 2
	Unnamed	Ictalurid herpes-like viruses	Ictalurid herpesvirus 1
Papillomaviridae		*Papillomavirus*	Cottontail rabbit papillomavirus
Poxviridae	*Chordopoxvirinae*	*Orthopoxvirus*	Vaccinia virus
		Parapoxvirus	Orf virus
		Avipoxvirus	Fowlpox virus
		Capripoxvirus	Sheeppox virus
		Leporipoxvirus	Myxoma virus
		Suipoxvirus	Swinepox virus
		Molluscipoxvirus	Molluscum contagiosum virus
		Yatapoxvirus	Yaba monkey tumour virus

Table 48.4 The families containing the single-stranded DNA viruses of vertebrates.

Family	Subfamily	Genus	Type species
Circoviridae		*Circovirus*	Porcine circovirus
		Gyrovirus	Chicken anaemia virus
Parvoviridae	*Parvovirinae*	*Parvovirus*	Mice minute virus
		Erythrovirus	B19 virus
		Dependovirus	Adeno-associated virus 2

Table 48.5 Retroviruses[a] of vertebrates

Family	Genus	Type species
Retroviridae	*Alpharetrovirus*	Avian leukosis virus
	Betaretrovirus	Mouse mammary tumour virus
	Gammaretrovirus	Murine leukaemia virus
	Deltaretrovirus	Bovine leukaemia virus
	Epsilonretrovirus	Walleye dermal sarcoma virus
	Lentivirus	Human imunodeficiency virus 1
	Spumavirus	Chimpanzee foamy virus 1

a RNA viruses with reverse transcriptase activity

Table 48.6 The families containing the double-stranded RNA viruses of vertebrates.

Family	Genus	Type species
Birnaviridae	*Avibirnavirus*	Infectious bursal disease virus
	Aquabirnavirus	Infectious pancreatic necrosis virus
Reoviridae	*Orthoreovirus*	Mammalian orthoreovirus
	Orbivirus	Bluetongue virus 1
	Rotavirus	Rotavirus A
	Coltivirus	Colarado tick fever virus
	Aquareovirus	Aquareovirus A

Table 48.7 Families containing negative-sense, single-stranded RNA viruses of vertebrates.

Family	Subfamily	Genus	Type species
Bornaviridae		*Bornavirus*	Borna disease virus
Bunyaviridae		*Bunyavirus*	Bunyamwera virus
		Hantavirus	Hantaan virus
		Nairovirus	Dugbe virus
		Phlebovirus	Rift valley fever virus
Orthomyxoviridae		*Influenzavirus A*	Influenza A virus
		Influenzavirus B	Influenza B virus
		Influenzavirus C	Influenza C virus
		Thogotovirus	Thogoto virus
Paramyxoviridae	*Paramyxovirinae*	*Respirovirus*	Sendai virus
		Morbillivirus	Measles virus
		Rubulavirus	Mumps virus
	Pneumovirinae	*Pneumovirus*	Human respiratory syncytial virus
		Metapneumovirus	Turkey rhinotracheitis virus
Rhabdoviridae		*Lyssavirus*	Rabies virus
		Vesiculovirus	Vesicular stomatitis Indiana virus
		Ephemerovirus	Bovine ephemeral fever virus
		Novirhabdovirus	Infectious haematopoietic necrosis virus

Table 48.8 Families containing the positive-sense, single-stranded RNA viruses of vertebrates.

Family	Genus	Type species
Arteriviridae	*Arterivirus*	Equine arteritis virus
Astroviridae	*Astrovirus*	Human astrovirus 1
Caliciviridae	*Vesivirus*	Swine vesicular exanthema virus
	Lagovirus	Rabbit haemorraghic disease virus
	Norwalk-like viruses	Norwalk virus
	Sapporo-like viruses	Sapporo virus
	Hepatitis E-like viruses	Hepatitis E virus
Coronaviridae	*Coronavirus*	Infectious bronchitis virus
	Torovirus	Equine torovirus
Flaviviridae	*Flavivirus*	Yellow fever virus
	Hepacivirus	Hepatitis C virus
	Pestivirus	Bovine viral diarrhoea virus 1
Picornaviridae	*Enterovirus*	Poliovirus
	Rhinovirus	Human rhinovirus A
	Hepatovirus	Hepatitis A virus
	Cardiovirus	Encephalomyocarditis virus
	Aphthovirus	Foot-and-mouth disease virus
	Parechovirus	Human parechovirus
	Erbovirus	Equine rhinitis B virus
	Kobuvirus	Aichi virus
	Teschovirus	Porcine teschovirus
Togaviridae	*Alphavirus*	Sindbis virus
	Rubivirus	Rubella virus

periodic addition of new species. In addition, international specialist groups monitor large numbers of strains and subtypes. These latter categories have become accepted for practical reasons such as vaccine development and the diagnosis of disease. No universal definitions or formal nomenclature are recognized for strains and subtypes of virus species.

The ease with which base sequencing can be carried out on viral nucleic acids has revolutionized approaches to virus taxonomy and phylogeny. Reference genome sequences are available for all viral taxa in databases such as GenBank (http://www.ncbi.nlm.nih.gov/GenBank) which permit substantial shortcuts to specific taxonomic placements. In addition, the use of statistical methods for comparing sequence similarities has contributed to studies on the evolution of viruses. The ICTV intends to reserve the hierarchical level or order solely for the recognition of phylogenetic relationships. Recently, interest has focussed on the study of virus evolution in real time at the subspecific level. This has resulted in the development of the concept of viral quasispecies (Eigen, 1993). This concept envisages that each virus species exists as a genetically diverse, rapidly evolving population of virions with non-identical but closely related mutant and recombinant viral genomes based on a consensus sequence. The population as a whole, the quasispecies, acts as a genetic pool which is subjected to a continuous process of variation, competition and selection. The population is in dynamic equilibrium, with the expansive force of mutation balanced by the constraining force of selection. The survival of the quasispecies depends on the stability of the consensus sequence, the complexity of the

uncoating may be completed at the nuclear pores.

The synthesis of viral proteins by host cells, which is the central event in replication of viruses, requires the production of viral mRNA. Those DNA viruses, which replicate in the nucleus, can avail of host cell transcriptases to synthesize viral mRNA. Other viruses utilize their own enzymes to generate mRNA. Viruses have evolved strategies which facilitate interference with the activity of cellular mRNA. Viruses direct the synthesis of either a separate mRNA for each gene or mRNA encompassing several genes. Eukaryotic cell protein-synthesizing mechanisms, however, translate only monocistronic messages. If a large precursor protein molecule is produced, cleavage into individual proteins is required and each family of viruses employs a unique strategy for this purpose.

Based on the nature of the genome and the pathways of mRNA synthesis, viruses of veterinary importance can be grouped in six classes (Baltimore, 1971). Central to this scheme is the designation of the genome of single-stranded RNA viruses as positive-sense or negative-sense nucleic acid (Fig. 49.1). In this context, the word 'sense' refers to nucleic acid polarity. The nucleic acid of positive-sense single-stranded RNA viruses is mRNA in sense and can be translated directly into virus protein.

In this chapter, specific viruses have been selected to illustrate the replicative mechanism of DNA and RNA viruses. However, individual viruses within these two groups frequently exhibit unique replicative methods. The mechanisms involved in the replication of DNA and RNA viruses, although having many similarities, are complex intracellular events which require separate discussion.

Replication of DNA viruses

Double-stranded DNA viruses, such as herpesviruses, papovaviruses and adenoviruses, which replicate in the nucleus of the cell, have a relatively direct replication strategy. The viral DNA is transcribed by cellular DNA-dependent RNA polymerase (transcriptase) to form mRNA. In contrast, the single-stranded DNA viruses, parvoviruses and circoviruses, which also replicate in cell nuclei, utilize cellular DNA polymerase to synthesize double-stranded DNA. This is then transcribed to mRNA by cellular transcriptases. Because of this transcription requirement, the replication of parvoviruses is largely confined to rapidly-dividing cells. Stages in the replication of a herpes virus, an enveloped double-stranded DNA virus are illustrated in Fig. 49.2.

A defined temporal sequence of events occurs during transcription and replication of DNA viruses. Specified genes encode for early proteins, which include the enzymes and other proteins necessary for virus replication and for suppression of the synthesis of host cell proteins. Subsequently, replication of viral nucleic acid and

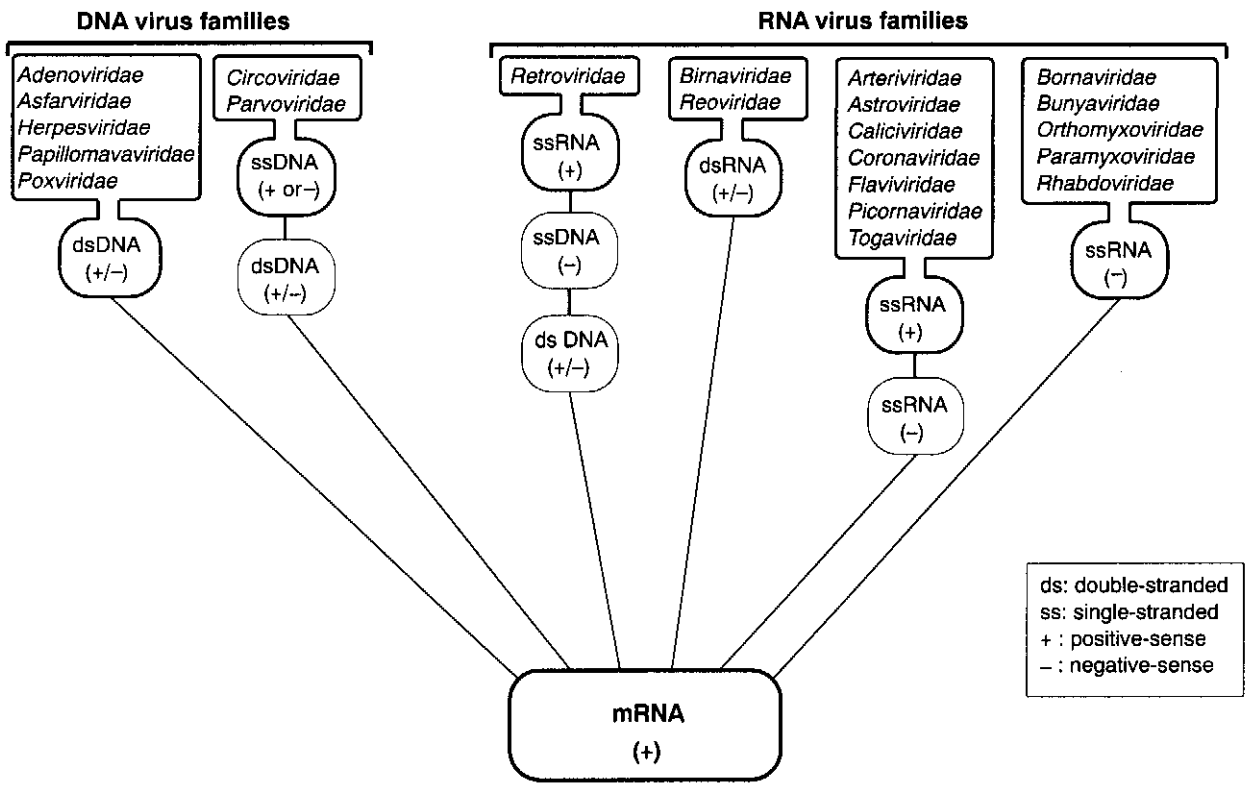


Figure 49.1 Families of DNA and RNA viruses of veterinary importance grouped according to genome types and pathways for messenger RNA synthesis (modified from Baltimore, 1971).

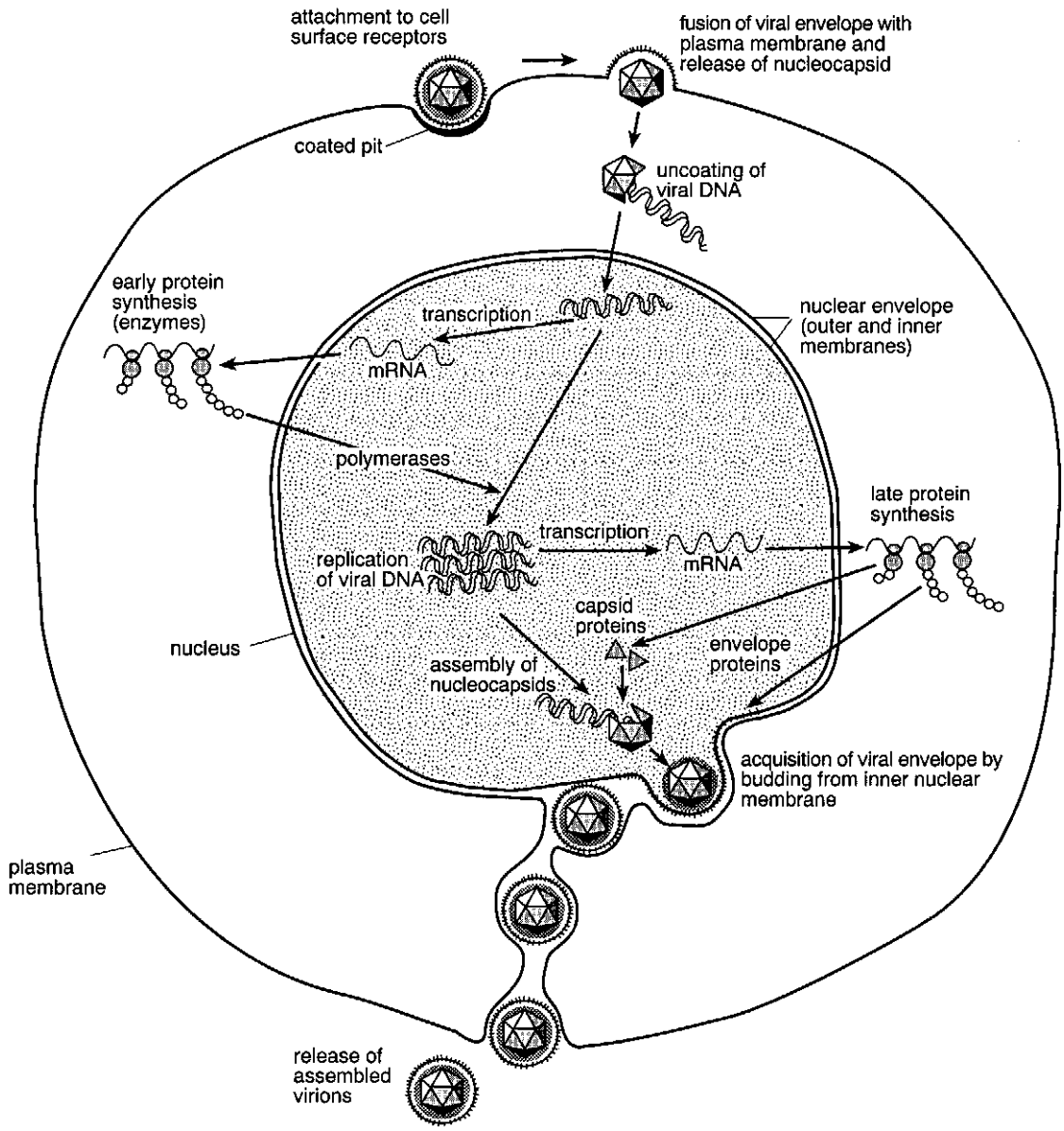


Figure 49.2 Stages in the replication of a herpes virus, an enveloped double-stranded DNA virus.

transcription of the genes which encode the late proteins occur. These late proteins, which are also often transcribed from newly-formed viral nucleic acid, are structural components synthesized late in the infection cycle. This temporal sequence is not clearly demonstrable in the replicative cycles of RNA viruses, in which most of the genetic information is expressed contemporaneously.

Replication of RNA viruses

Reoviruses and birnaviruses, double-stranded RNA viruses, have segmented genomes. Transcription occurs in the cytoplasm under the direction of a viral transcriptase. The negative-sense strand of each segment is transcribed to produce individual mRNA molecules. In contrast, the genomes of positive-sense, single-stranded RNA viruses can act directly as mRNA after infection (Fig. 49.3). The enzymes necessary for genome replication in these viruses are produced after infection by direct translation of virion RNA. This RNA can bind directly to ribosomes and is translated to yield a single polyprotein which is then cleaved to yield both functional and structural proteins. Because direct translation can occur, naked RNA extracted from such viruses is infectious. The positive-sense, single-stranded RNA viruses utilize a number of different synthetic pathways during replication. In togaviruses, only

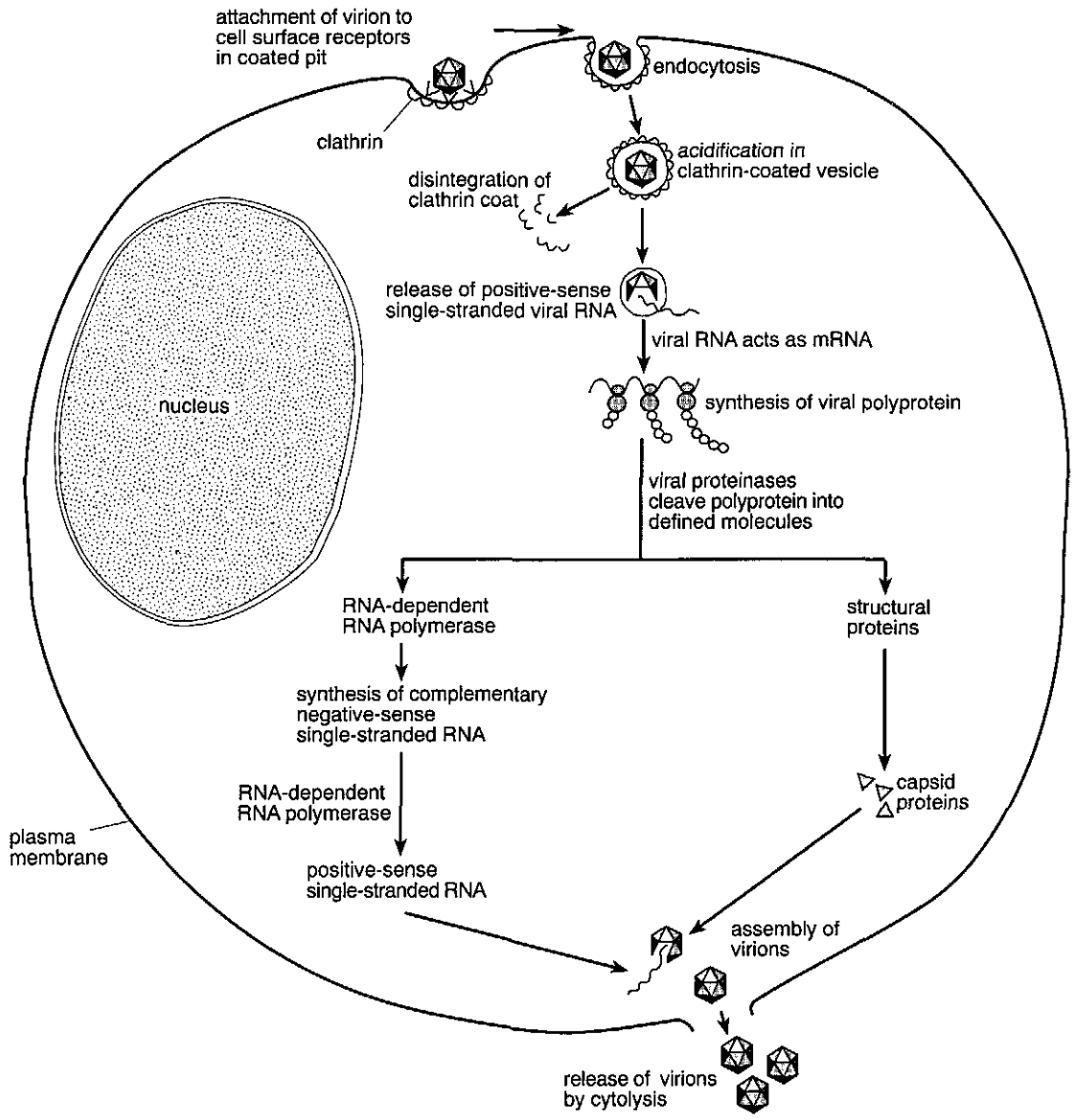


Figure 49.3 Stages in the replication of a picornavirus, a non-enveloped, positive-sense, single-stranded RNA virus.

about two-thirds of the viral RNA is directly translated during the first round of protein synthesis. Subsequently, full-length negative-sense RNA is synthesized and, from this, a full-length positive-sense RNA destined for encapsidation and a one-third length positive-sense RNA strand are formed. The genomes of caliciviruses, coronaviruses and arteriviruses also encode for mRNA which can be full length or shorter.

Negative-sense single-stranded RNA viruses possess an RNA-dependent RNA polymerase. The naked RNA of these viruses, unlike that of the positive-sense single-stranded RNA viruses, cannot initiate infection. After infection by the virion, the genomic RNA functions as a template for transcription of positive-sense mRNA and also for virus replication, utilizing the same polymerase. The positive-sense RNA subsequently serves as the template for synthesis of negative-sense genomic RNA. Most single-stranded, negative-sense RNA viruses replicate in the cytoplasm of the cell. Notable exceptions are orthomyxoviruses and Borna disease virus which replicate in the nucleus. Part of the segmented genome of some members of the *Bunyaviridae* is ambisense, utilizing a mixed replication strategy with features characteristic of both positive-sense and negative-sense single-stranded RNA viruses. Stages in the replication of a rhabdovirus, an enveloped, negative-sense, single-stranded RNA virus are illustrated in Fig. 49.4.

The genome of retroviruses consists of positive-sense, single-stranded RNA which does not function as messenger RNA. Instead, a single-stranded DNA copy is

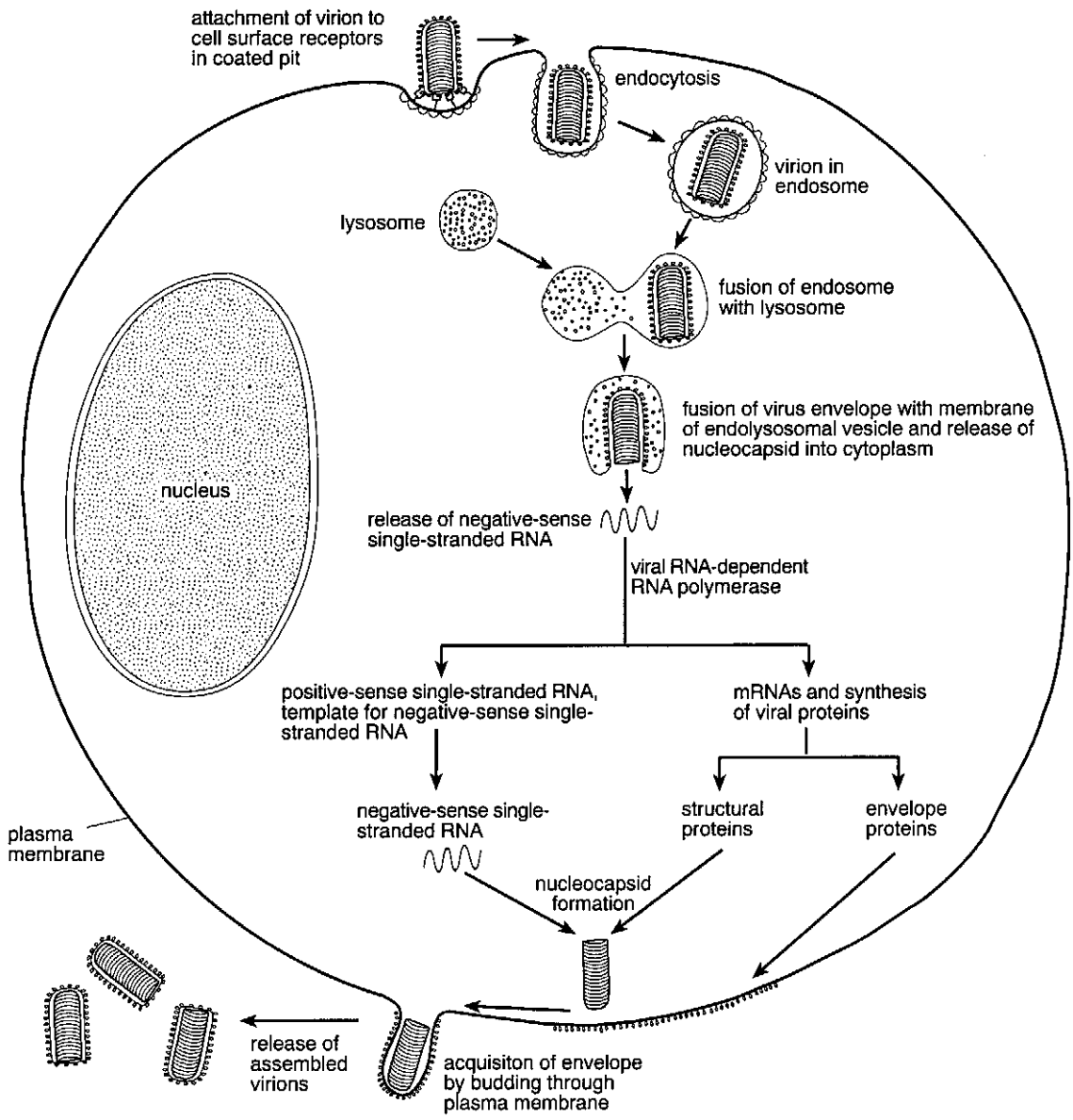


Figure 49.4 Stages in the replication of a rhabdovirus, an enveloped, negative-sense, single-stranded RNA virus.

produced by RNA-dependent DNA polymerase (reverse transcriptase) using the viral RNA as a template. As the second strand of DNA is formed, the parental RNA is removed from the RNA-DNA hybrid molecule. The double-stranded DNA is integrated into the host cell genome as a provirus (Fig. 49.5). The integrated DNA provirus, which may be incorporated into cellular chromosomes at a number of sites, can be transcribed to new viral RNA.

Protein synthesis

Within the cell, the sites at which particular proteins are synthesized relate to the type and function of the protein. Membrane proteins and glycoproteins are synthesized on membrane-bound ribosomes while soluble proteins including enzymes are synthesized on ribosomes free in the cytoplasm. Short specific amino acid sequences, known as sorting sequences, facilitate the incorporation of proteins at various cellular locations where they are required for metabolic activity. Most viral proteins undergo post-translational modification including proteolytic cleavage, phosphorylation and glycosylation. During glycosylation, sugar side-chains are added to viral proteins in a programmed manner as the proteins are being transferred from the rough endoplasmic reticulum to the Golgi apparatus preparatory to final assembly of intact virions and their release from the cell.

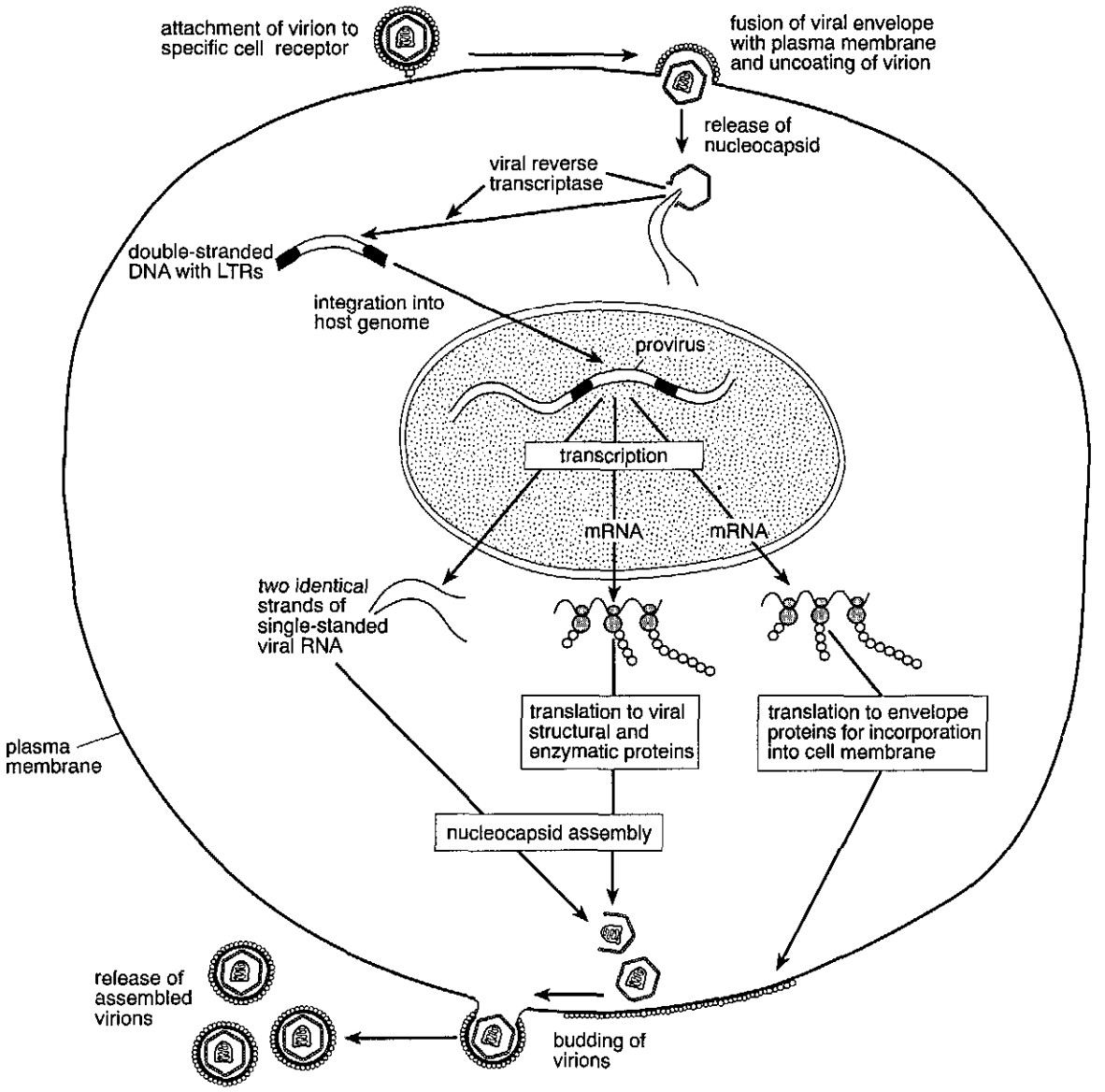


Figure 49.5 Stages in the replication of a retrovirus.
LTRs: long terminal repeats.

Assembly and release of virions

The mechanisms for the assembly and release of enveloped and non-enveloped viruses are distinct. Non-enveloped viruses of animals have an icosahedral structure. The structural proteins of these viruses associate spontaneously in a symmetrical and stepwise fashion to form procapsids. Subsequently, viral nucleic acid is incorporated into the procapsid. Proteolytic cleavage of specific procapsid polypeptides may be required for the final formation of infectious particles. Non-enveloped viruses are usually released following cellular disintegration. The assembly of picornaviruses and reoviruses occurs in the cytoplasm of the cell whereas parvoviruses, adenoviruses and papovaviruses are assembled in the nucleus.

In enveloped viruses, the final step in the process of virion assembly involves acquisition of an envelope by budding from cell membranes. Prior to budding, cell membranes are modified by the insertion of virus-specified transmembrane glycoproteins, which aggregate in patches in the plasma membrane. The presence of viral glycoproteins alters the antigenic composition of infected cells which become targets for cytotoxic T lymphocytes. Togavirus nucleocapsids bind to the hydrophilic domains of the virus-specified membrane proteins, which project slightly into the cytoplasm, and become surrounded by the altered portion of membrane. The nucleocapsids of helical viruses bind to a virus-specified matrix protein which lines the cytoplasmic side of membrane patches.

Budding of viruses through the plasma membrane does

not usually breach the integrity of the membrane and, as a result, many enveloped viruses are non-cytopathic and may be associated with persistent infections. Unlike most other enveloped viruses, togaviruses, paramyxoviruses and rhabdoviruses are cytolytic. Flaviviruses, coronaviruses, arteriviruses and bunyaviruses acquire their envelopes inside cells by budding through the membranes of the rough endoplasmic reticulum or the Golgi apparatus. These viruses are then transported in vesicles to the cell surface where the vesicle fuses with the plasma membrane releasing the virion by exocytosis. Herpesviruses, which replicate in the nucleus, are unique in that they bud through the inner lamella of the nuclear membrane and accumulate in the space between inner and outer lamellae, in the cisternae of the endoplasmic reticulum and in cytoplasmic vesicles (Fig. 49.2). Release from the cell can occur either by exocytosis or by cytolysis. The assembly and release of poxviruses is a complex process taking several hours. Although replication occurs entirely in the cytoplasm of the host cell at discrete sites, termed viroplasms or 'viral factories', nuclear factors may be involved in transcription and assembly. Maturation proceeds to the formation of infectious intracellular mature virus, which can be detected following deliberate lysis of infected host cells *in vitro*. Following assembly, virus particles move out of the assembly area and become enveloped in a double membrane derived from the trans-Golgi network. At the periphery of the cell, fusion with the plasma membrane results in loss of the outer layer of the double membrane and release of extracellular enveloped virus.

References

Baltimore, D. (1971). Expression of animal virus genomes. *Bacteriological Reviews*, **35**, 235–241.

Further reading

Fields, B.N., Knife, D.M. and Hewley, P.M. (1996). *Fundamental Virology*. Third Edition. Lippincott-Raven, Philadelphia.

Chapter 50

Genetics and evolution of viruses

Viruses exhibit enormous genetic diversity. These minute pathogens are capable of infecting not only vertebrates but also invertebrates, plants, fungi, protozoa, algae and bacteria. Viruses utilize a variety of molecular mechanisms to compensate for their limited genetic capability. They may encode genes in different reading frames. These reading frames may overlap, may be encoded in opposite directions or may be read by frameshifting. In recent years, the development of techniques in molecular biology including molecular cloning, nucleotide sequencing and the polymerase chain reaction have greatly expanded knowledge of viral genomes. Viral genetics is concerned not only with understanding the detailed structure of the viral genome and the extent to which it determines the biological and disease-producing potential of viruses but also with elucidation of the mechanisms of genetic change and virus evolution. Genomic change is responsible for alterations in antigenicity and pathogenicity which, in turn, may influence the course of viral diseases of animals and humans. Mutation is the most frequent cause of genetic change in viruses. Less commonly, genetic interactions termed recombinations, which can occur between different viruses or between a virus and its host cell, may account for alteration in virus characteristics.

Mutation

Spontaneous and random errors in the copying of viral nucleic acid, termed mutations, can occur during the replication of viruses. The rate of mutation in the genome of DNA viruses ranges from 10^{-8} to 10^{-11} per incorporated nucleotide whereas the mutation rate in the genomes of RNA viruses is much higher, ranging from 10^{-3} to 10^{-4}. When replicating in the nucleus of host cells, the genomes of DNA viruses are subjected to error-correction carried out by cellular exonucleases. The lower fidelity of genome replication in RNA viruses is attributed to poor error correction by RNA replication enzymes. In an infected cell, the genomes of the progeny of an RNA virus, such as vesicular stomatitis virus with an 11 kb genome, may differ both from the parental genome and from each other by at least one nucleotide. Exposure to X-rays, UV-irradiation or chemical mutagens increases the frequency of mutation. Point mutations, resulting from single nucleotide substitutions, are the most common type of mutation. Less common types of mutation result from the deletion or insertion of one or more nucleotides. Additive effects of point mutations occurring over several generations may account for phenotypic variations which are influenced by selection processes. Because they do not result in alteration of the amino acid composition of the coded protein, most point mutations are silent. In other instances the mutation may be lethal and virus strains containing such lethal mutations are rapidly eliminated. Occasionally, mutations conferring a selective advantage allow positive selection of the mutant virus. Non-lethal mutations may accumulate rapidly in the genomes of RNA viruses giving rise to quasispecies, genetically diverse populations of viruses centered around consensus sequences. Phenotypic expression of a mutation may be reversed by a back-mutation at the nucleotide responsible for the original mutation. Alternatively, a suppressor mutation may prevent phenotypic expression of the mutant gene. The phenotypic expression of a particular mutation may give rise to clearly distinguishable mutants including conditional-lethal, antibody-escape and defective-interfering mutants.

Conditional-lethal mutants can replicate only under defined permissive conditions. Examples of such mutants include temperature-sensitive mutants, which can multiply most efficiently at temperature ranges different from parental wild-type viruses, and host-range mutants which can infect host species different from those of parent viruses. The temperature-sensitive mutants, which replicate at temperatures slightly below core mammalian body temperature, have been particularly useful in the development of live intranasal vaccines. Such vaccines stimulate local immunity without systemic spread. When first isolated, viruses may grow poorly in cell cultures and in laboratory animals. However, adaptation can usually be achieved by serial passage, resulting in selection of rapidly growing mutants. This selection process is dependent on spontaneous mutations particularly in genes encoding surface proteins, which determine binding efficiency to host cell receptors. Host-range mutants, while often replicating more readily *in vitro*, tend to be less virulent for natural host species. This has been exploited as a method for attenuation in the production of many modified live vaccines.

Selection for antibody escape mutants can occur when viruses replicate in the presence of antibody. Because of altered antigenic surface determinants, the mutants are unaffected by neutralizing antibodies induced by the wild-type virus. Such a selection process may facilitate persistent or recurrent infections.

Defective-interfering mutants require the presence of a helper virus, usually the wild-type virus, for replication. Most defective-interfering mutants are deletion mutants. Such mutants can interfere with the replication of wild-type helper viruses and may, therefore, become progressively more numerous on serial passage. Defective-interfering mutants may play a role in disease by promoting the establishment and maintenance of persistent infections.

Viral recombination

The exchange or transfer of genetic material between different but closely related viruses infecting the same cell is termed genetic recombination. This type of genetic exchange can also occur between virus and host cells. The genomes of the recombinants contain new genetic information. The alteration of genetic information may result from intramolecular recombination, copy-choice recombination, reassortment or genetic reactivation.

Intramolecular recombination usually occurs in DNA viruses and involves dissociation and re-establishment of covalent bonds within the nucleic acid. Copy-choice recombination between positive-sense single-stranded RNA viruses occurs through a template switching mechanism. The RNA polymerase switches between template strands during synthesis of the complementary negative-sense strand. This process can occur in picornaviruses, togaviruses and coronaviruses. An exceptionally high frequency of genetic recombination has been observed in mixed infections of coronaviruses. It is suggested that western equine encephalitis virus arose through copy-choice recombination as a heterologous recombinant of the two togaviruses, eastern equine encephalitis virus and a Sindbis-like virus.

Reassortment can occur randomly in RNA viruses with segmented genomes, such as orthomyxoviruses, reoviruses and bunyaviruses. In this type of recombination, genome segments of two or more related viruses infecting the same cell are exchanged. The process is an important source of genetic variability in nature, permitting rapid adaptation of viruses to new hosts, the development of viruses with new antigenic characteristics and changes in virulence. Reassortment is generally restricted to taxonomically related viruses, either viruses of the same species, such as orthomyxoviruses, or viruses belonging to the same serogroup within a species, such as reoviruses and bunyaviruses. Genetic reassortment plays a major role in the epidemiology of human influenza A virus infections. Periodically, viruses with novel antigenic properties emerge, facilitating spread throughout the human population. There is convincing evidence that dual infection of pig populations with avian and human influenza viruses accounts for the emergence of new virulent subtypes, which can spread to the human population by close contact with pigs. In genetic reactivation, infectious progeny are produced from parental viruses, of which one or both are non-infectious, following mixed infection of a cell. When infectious progeny are produced from related viruses inactivated by lethal mutations at different loci in their genomes, the phenomenon is referred to as multiplicity reactivation. Cross-reactivation or genome rescue occurs when an inactivated virus becomes capable of replicating after acquiring genetic material from an infective virus.

Virus-host cell recombination

It is known that recombination between viral and cellular genetic material occurs. Such recombination may be important for virus evolution and virulence. Several retroviruses have become potentially oncogenic by incorporating cellular oncogenes into their genomes. Integration of the cellular DNA into the viral genome often results in a concomitant loss of viral genetic material resulting in replication-defective progeny viruses which require helper viruses for replication. The non-cytopathic biotype of bovine virus diarrhoea virus, which causes persistent infection in cattle, may become cytopathic, through mutation or recombination and cause mucosal disease. The specific change is cleavage of the non-structural fusion protein, NS2-3, and the subsequent separate expression of NS3. One mechanism which allows this change is insertion of one or more cellular ubiquitin gene sequences into a key region of the viral genome.

Other interactions involving viruses

Viruses may interact in a number of ways at the level of gene products. These types of interaction, which can result in phenotypic alteration of viral activity, include complementation and phenotypic mixing.

In cells with dual infections, complementation can occur if the defective gene product of one virus is substituted by the gene product of another virus. This results in survival or increased yields of the recipient virus. There is no lasting effect from this type of interaction because the genome remains unchanged. Complementation can occur between related and unrelated viruses; the defective viruses are termed satellite viruses. Adeno-associated viruses, which are members of the *Parvoviridae*, can replicate only in the nucleus of cells simultaneously infected with an adenovirus. Hepatitis delta virus, a satellite virus of humans, requires co-infection with hepatitis B virus for replication, especially for the provision of envelope proteins.

Structural proteins may be exchanged following infection of a cell usually by two related viruses, a process

known as phenotypic mixing. One form of this type of interaction, transcapsidation, involves complete or occasionally partial exchange of capsids between non-enveloped viruses. Phenotypic mixing also occurs in enveloped viruses which acquire envelopes when budding from host cells. The progeny nucleocapsids of one virus may be released from a cell surrounded by an envelope containing glycoproteins specified by another virus. Defective oncogenic retroviruses, which are dependent on helper viruses for replication, may derive envelopes from host cell membrane which contains glycoproteins encoded by their helper viruses. As phenotypic mixing does not involve genetic change, it is a transient event.

Viral genomic sequence analysis

Cell culture techniques are no longer required for producing sufficient quantities of virus for detailed studies. It is now possible to produce large amounts of viral nucleic acid using either molecular cloning techniques or the polymerase chain reaction following extraction of viral nucleic acid from clinical samples. The nucleic acid of bovine papillomavirus and rabbit haemorrhagic disease virus has been analysed and sequenced without cultivation of these viruses *in vitro*. This has provided valuable information about both viruses, allowing alternative approaches to diagnosis and vaccination procedures. In recent years, automated methods for sequencing viral genomes have become simpler and cheaper. Sequences, both partial and complete, of a wide range of microorganisms are retrievable from international databanks such as GenBank (http://www. ncbi.nlm.nih.gov/GenBank).

Two rapid sequencing techniques are currently used, the enzymatic method of Sanger *et al.* (1977) and the chemical degradation method of Maxam and Gilbert (1977). The Sanger method is preferred for simple DNA sequencing. This method is based on the enzymatic synthesis of a strand of DNA complementary to the sample DNA which is used in single-stranded form as the template (Fig. 50.1). A primer sequence is annealed to the single-stranded DNA and the reaction mixture is divided into four aliquots. The four deoxynucleotide triphosphates (dNTPs) are added to each aliquot. These normal deoxynucleotides are radiolabelled, usually with ^{32}P or ^{35}S. A different dideoxynucleotide triphosphate (ddNTP) is then added to each of the four reaction mixtures. These deoxynucleotide analogues, ddAPT, ddCTP, ddGTP and ddTTP, contain the sugar dideoxyribose which lacks a hydroxyl group present in deoxyribose. Incorporation of a ddNTP into a growing complementary DNA strand by DNA polymerase blocks further synthesis of the chain. Incorporation of a ddNTP is random because of competition with its dNTP analogue, resulting in the production of chains of various lengths in each reaction mixture. The use of four reaction mixtures, each with a different ddNTP provides four populations of radiolabelled oligonucleotides which can be separated by polyacrylamide gel electrophoresis and identified by autoradiography in four adjacent lanes of the gel (Fig. 50.1).

The Maxam-Gilbert method involves chemical breaking of the sample DNA into fragments using four separate base-specific cleavage reactions (G, A+G, C, C+T). The fragments are radiolabelled. The four reaction mixtures contain molecules of various lengths determined by the location of a particular base along the original DNA molecule which can be determined by electrophoresis and radiography.

Viral genome sequencing has been used to great effect in the field of molecular epidemiology. The genotyping of isolates provides information on the source and epidemiology of viral diseases of veterinary importance such as foot-and-mouth disease and Venezuelan equine encephalitis. Functional genomics is concerned with the linking of phenotypic traits to specific genes. The identification of open reading frames (ORFs) is relatively easy in viruses because translatable sequences start with the methionine codon (AUG). This facilitates the identification of genes encoding viral structural and non-structural proteins. Information of this type is of considerable value in the development of diagnostic reagents and antiviral drugs, and also in the design of novel vaccines.

Restriction fragment length polymorphism

Restriction endonucleases are sequence-specific bacterial DNases that recognize a unique palindromic sequence, four to nine nucleotide pairs in length. Several hundred such enzymes, many of which are commercially available, have been identified and purified from bacteria. Restriction fragment length polymorphism, also known as DNA fingerprinting, involves cleavage of DNA into fragments. The fragments can be separated by gel electrophoresis and the pattern evaluated. The number and size of the fragments produced by the action of a specific restriction endonuclease on a particular DNA molecule are characteristic, allowing differentiation of the viral species or strain from which the nucleic acid derives.

Molecular cloning

Advances in the understanding of nucleic acid replication and the enzymatic cleavage of the molecule have paved the way for several practical applications used in genetic engineering. It is now possible to introduce foreign nucleic acid at specific sites in the genomes of viruses, bacteria, yeasts and vertebrate cells. The foreign DNA is amplified by replicating it with host DNA and, if inserted in frame with the appropriate regulatory sequences, acts as part of the host genome. Commonly, the foreign DNA is incorporated into a vector such as a plasmid or virus, especially bacteriophages. The recombinant vector can be

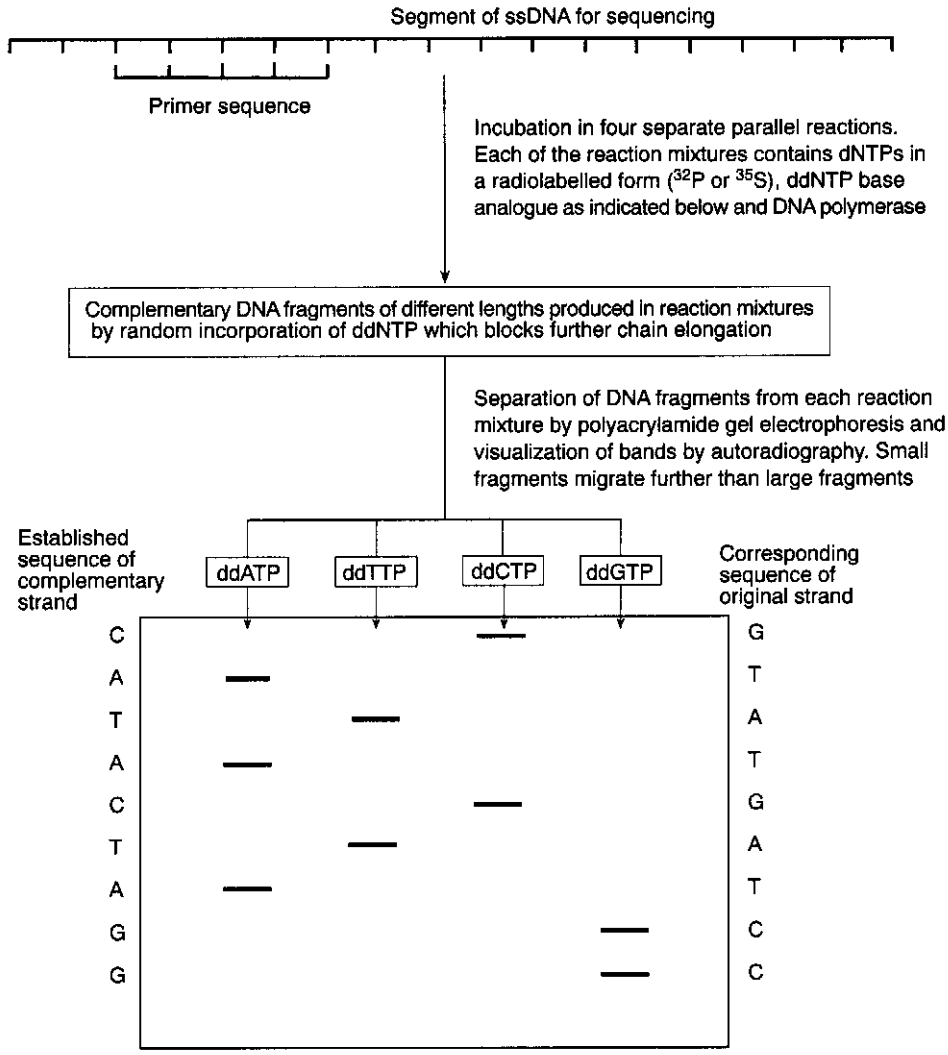


Figure 50.1 Outline of the Sanger method for DNA sequencing using dideoxynucleotide triphosphate.
ssDNA: single-stranded DNA; dNTP: deoxyribonucleotide triphosphate; ddNTP: dideoxyribonucleotide triphosphate; ddATP: dideoxyadenosine triphosphate; ddTTP: dideoxythymidine triphosphate; ddCTP: dideoxycytidine triphosphate; ddGTP: dideoxyguanosine triphosphate

delivered to bacterial, fungal, insect or mammalian cells either by infection or insertion. The vector may serve as a cloning vector, for the amplification of the foreign nucleic acid, or an expression vector, for the production of proteins encoded by the foreign DNA. Poxviruses have large DNA genomes permitting the incorporation of large amounts of foreign DNA into the non-essential areas of the genome. Vaccinia virus, in particular, has been used successfully as an expression vector. A vaccinia virus-rabies glycoprotein G recombinant has been developed and used effectively to vaccinate red foxes, the principal wildlife reservoir of rabies in Europe. This virus-vector vaccine, which is safe, relatively cheap and stable in the environment, is delivered in bait form. Other virus-vector vaccines are currently being developed.

Evolution of viruses

Genetic change, arising from mutation and recombination, occurs at different rates among families of viruses. Some genetically-controlled traits may confer a selective advantage which relates to prevailing conditions and selection pressures. Periodically, genomic change may contribute to the emergence of major outbreaks of new diseases. A point mutation in the genome of feline panleukopenia virus or a closely related virus has been proposed as the event responsible for the emergence of canine parvovirus, which causes a serious disease in dogs, first described during the late 1970s. Genomic change may also account for infection of horses with Hendra virus and of pigs with Nipah virus. Both of these paramyxoviruses have wildlife

reservoirs, probably fruit bats. Subsequent transmission of these viruses to humans through contact with infected domestic animals illustrates the important unforeseen consequences of viral genomic alteration.

Influenza A virus and myxoma virus provide important insights into the evolution of viruses. Influenza A viruses can infect a wide range of animal species. Isolates of the virus from birds and humans can infect pigs and, in this species, reassortment can occur with the emergence of new subtypes. Genetic change of this type, which takes place at intervals of more than ten years, are most likely to originate in southeast Asia because of high population densities and close contact between humans and domestic pigs and ducks. Virus variants produced in this way may escape neutralization by antibodies and, by becoming the dominant circulating strain, cause pandemics. During the twentieth century, pandemics of human influenza occurred in 1900, 1918, 1957 and 1968.

Myxoma virus is a poxvirus which causes a mild infection in *Sylvilagus* species, the native American rabbit. It is transmitted mechanically by fleas and mosquitoes feeding on infected animals. The virus causes serious, often fatal, disease in European rabbits, *Oryctolagus cuniculus*. The European rabbit which had been introduced into Australia in the mid-19th century, became a major agricultural pest. Myxoma virus was released in 1950 into the Australian rabbit population in an attempt to control numbers. Case fatality rates of more than 99% during dramatic summertime epizootics were initially recorded. The disease tended to disappear during winter months due to the greatly reduced rabbit numbers and decreased mosquito activity. In such circumstances less virulent viral mutants, which caused a more prolonged disease course with a greater opportunity for transmission, were selected. Subsequently, attenuated strains of the virus became dominant. This resulted in a recovery rate in infected rabbits exceeding 10%, with the selection of genetically more resistant rabbits. The case fatality rate for a particular strain of virus in serial infection studies on rabbits from areas where repeated outbreaks occurred declined from 90% to 50%. However, virulent strains of the virus became re-established in the genetically resistant populations ensuring further efficient transmission of the virus. In 1968, the European rabbit flea was released to improve transmission of the virus in areas where mosquitoes had not been effective vectors. In rabbit populations in which this flea flourished, the seasonal incidence of myxomatosis changed from sharp summer outbreaks to protracted winter and spring outbreaks. A dynamic equilibrium currently exists between virulence of the virus and genetic resistance of the rabbit population. However, rabbit numbers are substantially reduced when compared with the population size prior to the introduction of myxomatosis.

References

Maxam, A.M. and Gilbert, W. (1977). A new method for sequencing DNA. *Proceedings of the National Academy of Sciences*, **74**, 560–564.

Sanger, F., Nicklen, S. and Coulson, A.R. (1977). DNA sequencing with chain terminating inhibitors. *Proceedings of the National Academy of Sciences*, **74**, 5463–5467.

Further reading

Fenner, F. and Fantini, B. (1999). *Biological control of vertebrate pests. The history of myxomatosis, an experiment in evolution.* CABI Publishing, Wallingford.

Chapter 51

Propagation of viruses and virus-cell interactions

Viruses exhibit considerable diversity both in their ability to survive outside the host and in their cultural requirements. Some viruses, such as those causing enteric disease in animals, tolerate wide pH ranges and are relatively stable in the environment. In contrast, other viruses are labile and only survive for short periods outside the host. Enveloped viruses are readily inactivated by lipid solvents, such as chloroform and ether, and by various detergents such as sodium deoxycholate. Viruses are sensitive to ultraviolet and gamma irradiation. Because of their diploid genome, retroviruses are more resistant than other viruses to these forms of radiation. Enveloped viruses are more thermolabile than those without envelopes. As a rule, the rate of inactivation can be measured in seconds at 60°C, in minutes at 37°C, in hours at 20°C and in days at 4°C. Ice crystal formation during freezing damages viruses, especially enveloped viruses. The infectivity of viruses is retained for long periods when they are stored at -70°C or when they are freeze-dried. Storage at -20°C in conventional freezers is an unsatisfactory method for virus preservation. Long-term storage in liquid nitrogen at -196°C can be achieved by rapid freezing of small aliquots of high titred viral suspensions in a medium containing a high concentration of protective protein or a cryoprotective agent such as dimethyl sulphoxide. Freeze-drying involves dehydration of frozen viral suspensions under vacuum in glass ampoules which are then sealed to preserve the vacuum. Lyophilization is employed for the preservation of valuable seed stocks of viruses and for storage of modified live viral vaccines.

Propagation of viruses

As viruses replicate only in living cells, a source of viable cells is required for their propagation. Tissue culture is widely used for virus propagation; inoculation of chick embryos and experimental animals is employed for the isolation and propagation of particular viruses. Propagation is required for the isolation and identification of viruses involved in disease, for the titration of viruses for vaccine production and for the provision of stocks for research purposes.

Tissue culture

Techniques for tissue culture, the growth and maintenance of living tissue *in vitro*, can be grouped under two headings: explant cultures and cell cultures. Originally, methods for growing cells involved using small fragments or explants of tissue. This technique is still used for isolating viruses from animals with persistent diseases such as caprine arthritis-encephalitis. In a special form of explant culture, termed organ culture, the size and type of the tissue fragment is sufficient to retain tissue architecture. Explant tracheal cultures are required for the isolation of some coronaviruses.

Digestion of tissue into individual cells is used in cell culture preparation. Dispersion of cells from tissue involves mechanical cutting or chopping of the tissue into small pieces followed by digestion using trypsin or other proteolytic enzymes. The liquid or semi-solid culture medium in which cells are grown must supply the required environmental conditions and nutrients. It must be isotonic and be maintained at physiological pH values. There must also be a supply of inorganic ions, carbohydrate (usually glucose), amino acids, vitamins, growth factors, peptides and proteins. Some cells grow in chemically defined media but usually natural products such as foetal calf serum, yeast extract or embryo extract are incorporated in the medium. Phenol red is commonly added as a pH indicator. A bicarbonate buffer is often used to maintain the correct pH. Exposure to oxygen causes the pH to rise due to loss of dissolved CO_2 and bicarbonate. Cells can be grown in sealed containers or exogenous CO_2 may be provided at a rate of 5 to 10%.

Cell cultures may be primary, semi-continuous or continuous. Primary cell cultures are derived directly from tissues and contain many cell types. Tissues from foetuses or from neonatal animals are more suitable for tissue culture preparation than those from mature animals. For the isolation of a specific virus, the most sensitive systems are primary cell lines derived from the target tissue of the virus in a susceptible animal species. The cells in these cultures retain many of the characteristics of cells in the intact organ. However, the sensitivity of a given cell culture for cultivating viruses does not appear to depend solely on the organ of origin as it is frequently possible to isolate viruses that infect one body system in cells derived from a different system. The number of cell divisions occurring in primary cultures are relatively few. Accordingly, primary cultures must be prepared at

frequent intervals. Primary cultures can be passaged by dispersing the primary monolayer with trypsin, usually with added ethylenediamine tetra-acetic acid, and distributing the cells onto fresh surfaces to form secondary cell cultures. The number of further passages is finite and an end point, termed the Hayflick limit, is eventually reached. The Hayflick limit is related to the longevity of the animal species from which the cells derived. Cells which are nurtured may survive beyond the Hayflick limit, continue to grow and constitute a cell line. The cells of semi-continuous or diploid cell lines retain their characteristic diploid chromosomal constitution and can support the growth of a wide range of viruses. Such cell lines, most of which are predominantly fibroblastic, tend to die out between the 30th and 50th passage.

Continuous cell lines are derived from either normal or neoplastic tissue and can be passaged indefinitely. In these cell lines, referred to as heteroploid cell lines, the cells have an abnormal number of chromosomes. In general, they are not as sensitive as either primary or semi-continuous cell lines for virus isolation. However, viruses can usually be adapted to grow in continuous cell lines. Established cell lines facilitate large-scale growth of viruses for vaccine production or research purposes. Well characterized continuous cell lines can be obtained from commercial organizations which specialize in preserving and distributing stock. Samples can be purchased from the American Type Culture Collection and the European Collection of Cell Cultures. Continuous cell lines with recognized names include Madin Darby bovine kidney (MDBK) and Crandell feline kidney (CRFK). Following prolonged passaging, continuous cell lines may become contaminated with *Mycoplasma* species or viruses and may acquire altered characteristics such as increased or decreased susceptibility to viruses. Most virology laboratories freeze early stocks of cells and, periodically, revive cells from the frozen stock in order to ensure maintenance of characteristics. Cells can be stored in a viable state for long periods at temperatures below −130°C, typically at −196°C using liquid nitrogen. Dimethyl sulphoxide or glycerol is usually added to the medium in which cells are suspended as cryoprotective agents.

Detection of viral growth in cell cultures

Viral growth in cell culture, which results in damage to infected cells, may be detectable by light microscopy. Microscopic changes in infected cells, termed cytopathic effect (CPE), include change in shape, cell detachment, fusion leading to syncytium formation, the presence of inclusion bodies and cell death. On primary isolation, some viruses do not produce CPE. Following passage onto fresh monolayers, they may become cytopathic. The full range of the effects of viral infection on cultured cells, especially inclusion body formation, may be demonstrable only by staining. Inclusion bodies are intracellular structures which have characteristic staining features. In virus-infected cells they may be composed of viral nucleic acid, viral protein or altered cellular material. Inclusion bodies, which may be single or multiple, may be located in the cytoplasm or the nucleus. They are described as acidophilic, when stained by eosin, and basophilic, when stained by haematoxylin. Intracytoplasmic inclusions may be found in cells infected with poxviruses, reoviruses, rabies virus and paramyxoviruses while intranuclear inclusions occur in cells infected with adenoviruses, herpesviruses and parvoviruses. Infections with some viruses such as canine distemper virus may produce both intranuclear and intracytoplasmic inclusion bodies.

Non-cytopathic viruses require alternative detection methods. Enveloped viruses belonging to some families insert viral glycoproteins into the plasma membrane of infected cells. These glycoproteins may induce cell fusion, producing syncytia, or promote haemadsorption, the binding of erythrocytes to the surface of infected cells. Syncytia may be formed in cell cultures infected with lentiviruses, paramyxoviruses and some herpesviruses. Cells infected with orthomyxoviruses, paramyxoviruses and togaviruses tend to exhibit haemadsorption at sites of virion budding. Haemadsorption can be detected early in the replication cycle. The glycoproteins responsible for haemadsorption, referred to as haemagglutinins, also cause clumping of erythrocytes following mixing with free virions. Many viruses, both enveloped and non-enveloped, can cause haemagglutination with erythrocytes from particular animal species. Feline panleukopenia virus haemagglutinates porcine erythrocytes while porcine parvovirus haemagglutinates chick, guinea-pig, rat, monkey, human and cat erythrocytyes. Supernatants from infected cell cultures, lysed by freezing and thawing, can be examined for the presence of viral particles by electron microscopy. This is an insensitive technique and particle concentrations in excess of 10^6 per ml are required for detection.

Serological tests may be required for the specific identification of a viral species. A useful method for the detection and specific identification is infection of susceptible cells grown as monolayers on coverslips in Leighton tubes or in flat-bottomed vials. Following fixation of infected cells, virus can be identified employing a fluorescein-labelled antibody. Monoclonal or polyclonal conjugates may be used depending on availability and the level of specificity required. This technique is independent of any cytopathic or other property of the test virus and contributes to the speed and sensitivity of the isolation procedure. However, a separate conjugate is required for the specific detection of each virus species. Specific antibodies against viruses can also be utilized in virus neutralization tests to block cytopathic effects and in haemadsorption/haemagglutination inhibition assays to inhibit the erythrocyte-binding activity of viruses. Neutralization assays are the definitive methods for distinguishing many viruses and defining serotypes.

Although immunoelectron microscopy is more sensitive and specific than direct electron microscopy, it cannot be used as a general screening method for viruses in clinical samples.

Embryonated eggs

Formerly, embryonated eggs were used extensively for virus isolation. Decline in the use of embryonated eggs has resulted from improvements in tissue culture techniques. However, inoculation of embryonated eggs remains the preferred method for isolation of influenza A viruses and for many avian viruses. In order to exclude maternally-derived anti-viral antibodies, the source of the eggs requires careful selection, preferably eggs from specific pathogen-free flocks. Passage in embryonated eggs has proved useful for attenuating certain viruses used in the production of modified live virus vaccines.

Embryonated eggs may be inoculated using a number of defined routes (Fig. 51.1). Viruses can be inoculated into the allantoic cavity, the amniotic cavity or the yolk sac. They can also be inoculated onto the chorioallantoic membrane (CAM). In addition, well-developed chick embryos can be inoculated intravascularly. The route selected for inoculation is largely determined by the tissue affinity of each particular virus. Evidence of the effect of viral growth includes death or dwarfing of the embryo and the formation of pocks on the CAM. Haemagglutination or immunofluorescence may be used as additional tests for the detection of virus.

Experimental animals

For ethical reasons, experimental infection of animals is now used less frequently than formerly. A number of alternative methods for the growth and study of viruses is available. However, for several virus families, the use of cell culture for isolation is of limited value and animal inoculation remains the preferred procedure. Suckling mice are used for the detection of arthropod-borne viruses and for rabies virus. Inoculation of the natural host species is a requirement for the isolation of some viruses. Challenge experiments in the natural host species may be necessary to evaluate vaccines. The production of polyclonal antisera requires inoculation of laboratory animals with virus. Investigation of the pathogenetic mechanisms relating to viral infections and the subsequent immune response of the host frequently require inoculation of either laboratory animals or natural hosts.

Determination of virus concentration

When evaluating the efficiency of vaccines, when standardizing virus neutralization or haemagglutination-inhibition assays or when producing whole virus vaccines, accurate standardization of virus concentration is essential. It is also necessary when determining the minimum dose of virus required to produce clinical disease. Titration may be used to measure the total number of virions present (employing electron microscopy) or to calculate the number of infectious virions by inoculating a susceptible

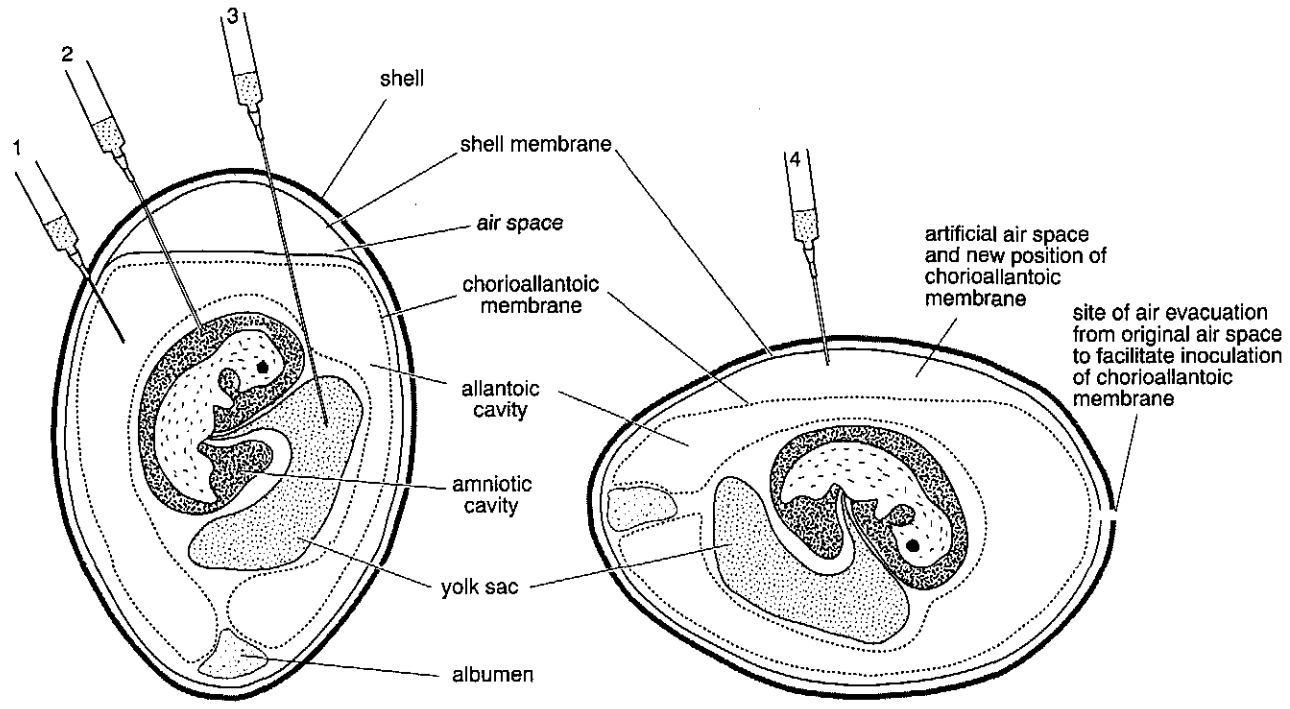


Figure 51.1 Routes for inoculation of viruses into embryonated eggs. 1: into allantoic cavity 2: into amniotic cavity 3: into yolk sac 4: onto chorioallantoic membrane

host or tissue culture system. Alternatively, a measure of a specific viral activity such as haemagglutination may be used or the concentration of viral antigen may be determined. Some titration procedures are quantal, providing an approximation of the number of virions in a preparation. Estimation of viral concentration by quantal techniques is adequate for most experimental or diagnostic purposes. Typically, these assays are carried out using serial ten-fold dilutions of a given virus preparation to inoculate several replicate host or cell culture systems per dilution. The end point in mice or embryonated eggs is usually death and the unit is expressed as the 50% lethal dose (LD_{50}). In cell culture systems, the end point is expressed as the 50% tissue culture infective dose ($TCID_{50}$). The results are usually expressed as a $\log_{10}$ value per ml of the original undiluted specimen. The statistical methods devised by Reed-Muench and by Kärber are frequently employed in these calculations.

The concentration of haemagglutinating viruses is often expressed in haemagglutinating (HA) units. The HA unit is equivalent to the highest dilution of the virus suspension causing complete haemagglutination. This simple titration method is insensitive because many thousands of virions are required to agglutinate sufficient numbers of erythrocytes for visual detection.

The plaque assay is a common and accurate method for quantifying viral infectivity. Serial ten-fold dilutions of the test virus preparation are inoculated onto monolayer cell cultures and allowed to adsorb for approximately one hour. The cell sheets are then overlaid with tissue culture medium containing agar. The solidified agar prevents spread of virus throughout the cell culture while permitting cell-to-cell transfer. As a result, cytopathic viruses produce foci of cell death around the replication sites, which then appear as clear areas called plaques. Based on the number of plaques and the dilution and size of the inoculum, the titration end point is calculated and expressed as plaque forming units (pfu) per ml. Although it is possible for a single virion to produce one plaque, in practice the virion:pfu ratio for most animal viruses is seldom less than 10:1. Even within a single virus preparation, there appears to be substantial microheterogeneity as not all virus particles in a preparation can form plaques.

Virus-host cell interactions

Virus infections may produce effects ranging from latency to cell death (Table 51.1). Infections may be productive or non-productive, consequences influenced by the ability of the virus to replicate effectively in a given cell type. When a host cell allows virus replication to proceed, it is said to be permissive. This state may not be static as occurs when latent infections are reactivated following a period of quiescence or non-productive interaction. Reactivation tends to occur when latently-infected animals are subjected to stressful environmental conditions.

Mechanisms of cell injury

Cytopathic viruses kill cells which they infect. Frequently, cell necrosis is the result of the cumulative effects of a number of biochemical changes induced by viral replication, resulting in the production of ultrastructural lesions. Cell death due to necrosis tends to occur late in the viral replication cycle after progeny virus production is complete and cytolysis may in fact facilitate the release of virions. Reoviruses, poxviruses, picornaviruses, paramyxoviruses and rhabdoviruses can inhibit the transcription of host cell RNA to permit synthesis of viral mRNA. This alteration of host cell metabolism may relate to interference with protein synthesis due to the production of virus-encoded factors. Herpesviruses, influenza viruses and vesicular stomatitis virus inhibit processing of host cell mRNA by interfering with the splicing of the primary cellular mRNA transcripts. Rapid and pronounced shut down of protein synthesis occurs in cells infected with picornaviruses, poxviruses and herpesviruses. Other viruses such as adenoviruses cause a gradual shut down of cellular activity at a late stage in the viral replicative cycle. The viral mechanisms which block protein synthesis include competition for ribosomes from large amounts of viral mRNA, degradation of cellular mRNA by viral enzymes, interference with cellular mRNA translation and the alteration of the intracellular ionic environment in favour of viral mRNA translation. Viral proteins of lentiviruses and adenoviruses inhibit the processing of cellular proteins and their transport from the endoplasmic reticulum. Accumulations of viral structural proteins occurring late in infection may be directly toxic for host cells. *In vivo* humoral and cellular immune responses directed against viral proteins incorporated into the plasma membrane of the host cell, may be cytotoxic.

Programmed cell death, apoptosis, can be induced by viral infection. In apoptosis, activation of cellular endonuclease leads to fragmentation of cellular DNA. The resulting DNA fragments produce a characteristic pattern of evenly spaced bands when electrophoresed in agarose gel. Apoptosis may be triggered by certain viruses in the early stages of infection, resulting in the death of individual cells prior to virus replication. This event could

Table 51.1 Virus-host cell interactions.

Virus	Production of progeny virus	Outcome
Cytopathic	*Productive*	*Necrosis*
	Non-productive	Apoptosis
Non-cytopathic	Productive	Persistence Transformation
	Non-productive	Latency Transformation

be an important host defence mechanism. Some viruses produce substances which block apoptosis, thus prolonging cell survival.

Non-cytopathic viral infections

Viruses such as retroviruses, which are usually non-cytopathic, do not interfere with protein synthesis in host cells. These viruses frequently produce persistent infections with progressive changes which eventually lead to cell death. Members of the sub-family *Alphaherpesvirinae* typically produce productive lytic infections, usually in epithelial cells. At the sites of productive infection, progeny virus enters sensory nerve fibres and is transported in the axoplasm to sensory nerve ganglia. Latent infection develops within the neuronal perikarya. Viral replication is restricted because stimuli for replication are not produced in the non-dividing neurons. The intracellular viral DNA is present in a circular episomal form in association with nucleosomes and transcription is limited to a few latency-associated transcripts (LATs). The function of LATs is not known. They do not appear to be translated into proteins and they are not essential for the maintenance of the latent state. The lack of viral protein expression allows infected neurons to remain undetected by the immune system. Multiple copies of the viral DNA may be demonstrable in latently infected host cells. Episodes of reactivation with production of infectious progeny virus can occur periodically. The newly produced virions are transported in the sensory nerve fibres to the superficial sites of the primary infection where they may again induce lytic lesions. The mechanism of reactivation is not fully understood. Certain stimuli or stress factors, such as trauma, immunosuppression, hormonal changes and intercurrent disease, can trigger reactivation. This frequently coincides with circumstances in which transmission of virus to susceptible animals may readily occur.

Retroviruses integrate a DNA copy of their RNA genome, termed provirus, into the host chromosome. This allows propagation of the provirus along with host chromosomes. Integration of provirus, while not destroying the cell, may alter the host genotype and the expression of host genes. The genomes of DNA viruses such as papillomaviruses are maintained in cells as circular episomal molecules. Viral replication, which is promoted by viral proteins, is synchronized with host cell division.

Viral oncogenesis

A number of DNA and RNA viruses cause neoplastic transformation of cells (Table 51.2). Transformation results from interference with growth signals in cells. Central to the understanding of the mechanisms involved in tumour production by viruses was the discovery of oncogenes. Originally recognized in retroviruses, more than 60 oncogenes, termed v-*onc* genes, have been identified. Cellular genes, c-*onc* genes or proto-oncogenes, corresponding to most v-*onc* genes, are present in normal cells in which they regulate cell division and differentiation. Cellular oncogenes encode proteins which function as growth factors, growth factor receptors, transcription factors and intracellular signal transducers. It is generally accepted that v-*onc* genes of retroviruses were acquired during the evolution of these viruses following virus-host cell interaction. Central to this process is the possession of a unique enzyme, reverse transcriptase, and the incorporation of a DNA copy of viral RNA into the host genome. Retroviruses may exert an oncogenic effect by carrying v-*onc* genes or alternatively by causing over-expression or inappropriate expression of c-*onc* genes. The v-*onc* genes carried by rapidly transforming retroviruses differ in certain respects from the c-*onc* genes from which they were derived. They are under the control of strong viral promoters known as long terminal repeats (LTRs), carry mutations as a result of the high error rate of reverse transcriptase and may be joined to other viral genes in a way that modifies their function. As a result they are outside of the control of normal cellular gene regulation and oncogene proteins may be over-produced or may function in an abnormal way leading to uncontrolled cell division. In contrast, slowly transforming retroviruses lack a v-*onc* gene and give rise to tumours in a random manner. Insertion of provirus (insertional mutagenesis) with its strong LTR promoter sequences close to a c-*onc* gene may greatly increase the production of the normal c-*onc* gene protein.

The oncogenes of DNA viruses are not usually derived from cellular genes. The DNA viruses are generally present in cells as circular episomal nucleic acid and their oncogenes encode proteins necessary for viral replication. If the replication cycle of the virus is curtailed, over-expression of the gene products may occur, inducing uncontrolled replication of the host cell. The genome of the herpesvirus, Marek's disease virus, appears to contain a number of v-*onc* genes similar to those of retroviruses.

Table 51.2 Some oncogenic viruses of veterinary importance.

Virus classification	Family	Virus
DNA	*Herpesviridae*	Marek's disease virus
	Papillomaviridae	Bovine papillomavirus Equine papillomavirus Canine oral papillomavirus
RNA	*Retroviridae*	Avian leukosis virus Feline leukaemia virus Jaagsiekte sheep retrovirus

Chapter 52

Pathogenesis of viral diseases

The mechanisms whereby viruses produce disease relate not only to direct interference with cellular functions but also to indirect effects caused by host responses to infection. Virulence refers to the relative ability of viruses to produce disease in a host. It is dependent on the attributes of both the virus and its host, and is influenced by factors such as the dose and route of entry of the virus and the species, age and immune status of the host. Frequently, comparisons are made between related viruses as to their virulence, and a particular isolate may be described as more virulent or less virulent than others. Because of the complex interactions between viruses and their hosts, clinical disease does not necessarily follow exposure to a virulent virus. Viral infections may result in subacute, acute or chronic disease. Although present in the tissues, some viruses, which exhibit latency, produce disease only under defined environmental conditions. In slow viral infections, lesions develop gradually and clinical signs emerge only after long incubation periods. The stages in the development of a typical viral infection are illustrated in Figure 52.1.

Routes of infection

Viruses can infect vertebrate hosts by a variety of routes. In many instances, viruses must first attach to and infect cells on the skin and other body surfaces. Viruses can be introduced into the body through breaks in surface epithelia such as scratches, abrasions or wounds. They may also be introduced by parenteral inoculation either through bites or following the use of contaminated needles. When intact, the stratified squamous epithelium of the epidermis usually presents an effective barrier to infection. Because they have an affinity for epidermal cells, papillomaviruses replicate in the epidermis producing local proliferative lesions. In contrast, viruses such as myxomavirus of rabbits and sheeppox virus which also replicate in the dermis, may spread to other tissues through blood vessels and lymphatics.

Mucous membranes, which present a minimal physical barrier to infection, are coated with a layer of mucus. This viscous secretion provides some physical protection and, in addition, often contains protective immunoglobulins. In the respiratory tract, infectious agents may be moved towards the oral cavity by the coordinated activity of ciliated epithelium. Peristalsis in the intestinal tract and the flushing action of urine in the urinary tract assist in the removal of infectious agents from these anatomical locations. Some viruses, which are spread venereally, may invade through abrasions, frequently resulting from sexual activity, in the urethral or vaginal mucosa.

In order to establish infection in the gastrointestinal tract (GIT), enteric viruses must evade local mucosal immune defenses and either attach to specific receptors on epithelial cells or be engulfed by M cells, specialized epithelial cells overlying Peyer's patches in the ileum. In the GIT, prevailing environmental conditions are generally unfavourable for virus survival. Successful viral

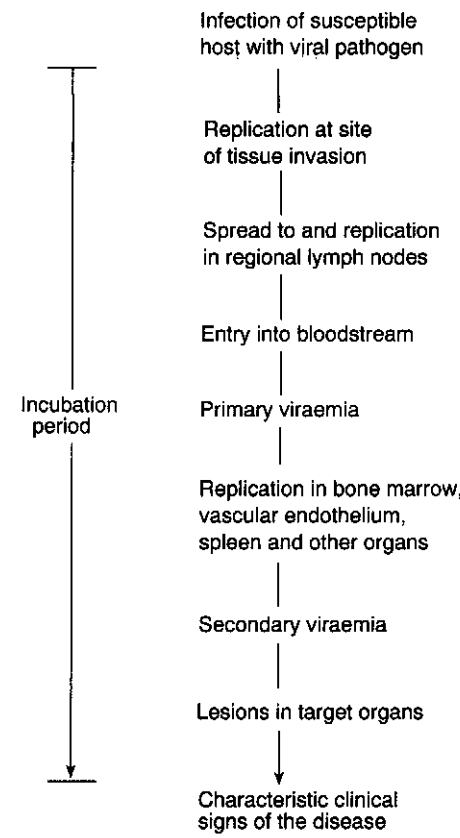


Figure 52.1 Stages in the spread of a typical systemic viral infection. The incubation period is the interval from invasion of the body to the development of clinical signs.

pathogens of the GIT, which damage intestinal cells, are usually unaffected by gastric acid, are tolerant to bile salts and are resistant to inactivation by proteolytic enzymes. Remarkably, the infectivity of some viruses such as coronaviruses and rotaviruses is enhanced by exposure to proteolytic enzymes. With the notable exception of coronaviruses, enteric viral pathogens are non-enveloped. Bile salts generally have an adverse effect on viral envelopes. The basis for the resistance of the envelopes of coronaviruses to bile salts is unknown.

Arboviruses

Arboviruses are defined as viruses maintained in nature through biological transmission between vertebrate hosts by haematophagous arthropods. The viruses multiply in the tissues of the arthropod vector. The most important arthropod vectors are mosquitoes, ticks, sandflies and midges. The vector remains infected for life. The term 'arbovirus' has no taxonomic status. It is applied to viruses belonging to several viral families including *Togaviridae*, *Flaviviridae*, *Reoviridae*, *Rhabdoviridae*, *Arenaviridae* and *Bunyaviridae*. Most arboviruses are maintained in complex sylvatic life cycles involving a primary vertebrate host and a primary arthropod host. Such cycles usually remain undetected unless domestic animals and man encroach. Alternatively the virus may escape its primary cycle by means of a secondary vector or vertebrate host due to ecological change. As a result the virus is brought into the peridomestic environment. Domestic animals and man are generally 'dead-end' hosts as they do not develop sufficient viraemia to contribute to the transmission of the virus. The majority of arbovirus infections are zoonoses and are found in tropical developing countries and have a distinct geographical distribution. Ecological factors limiting the distribution of particular arboviruses include temperature, rainfall and distribution of both vertebrate reservoir host and arthropod vector.

Dissemination in the host

Following infection, local spread from cell to cell frequently occurs. When viruses bud from cells dissemination may be influenced by the manner and site of budding. Release of viral progeny from the apical surface of mucosal cells favours localized infection in tubular structures such as the air passages and the intestine. In contrast, release of virus from the basal surface into subepithelial tissues facilitates systemic infection. Factors which may limit the ability of some viruses to spread systemically include the absence of suitable cell receptors and lack of permissive cells. In addition, optimal viral replication may occur only in dividing cells or at a temperature lower than core body temperature. These virus and host factors may play an important role in determining the tropism of a virus, and account for the selective infection of certain cells in particular organs. In subepithelial tissues, viruses often enter the lymphatic network and may be transported to the regional lymph nodes either as free virions or in infected macrophages. From the nodes, virus may pass to the efferent lymphatics and the thoracic duct, eventually entering the blood stream, the most important route of dissemination. Some viruses are transported along peripheral nerves. The preferential use of one route of spread does not necessarily exclude spread by other routes.

Haematogenous spread

Primary replication at the site of entry is frequently followed by transitory low-titred primary viraemia which results in infection of various organs including those of the reticuloendothelial system and the vascular endothelium. Further multiplication at these locations is followed by a sustained, high-titred secondary viraemia. In the bloodstream, virus may be free in the plasma or associated with cellular elements.

The body employs a number of mechanisms for clearing virus from the circulation. These include complement and antibody, and the phagocytic cells of the reticuloendothedial system in the liver, spleen, lung and lymph nodes. The magnitude and duration of a viraemia are determined by the amount of virus entering the bloodstream and the effectiveness of the clearance mechanisms. Large viruses are cleared more rapidly from the circulation than small viruses. There is rapid removal of viruses opsonized by antibody or by complement. Viraemias in which virus remains free in the plasma, such as those associated with parvoviruses, flaviviruses, togaviruses and picornaviruses, are usually of short duration, with clearance usually coinciding with the appearance of neutralizing antibodies in the serum. Prolonged viraemias are features of infections with viruses such as canine distemper virus, feline leukaemia virus and Marek's disease virus, which are associated with circulating cells. These viruses are often unaffected by the action of antibodies and complement. Some viruses like lentiviruses can replicate in monocytes or lymphocytes and, in many instances, produce persistent viraemias.

Invasion of tissues and organs from the blood can occur in a number of ways, which may relate to virus interaction with macrophages or vascular endothelial cells. Viruses such as picornaviruses, retroviruses, togaviruses and parvoviruses can infect endothelial cells and, following replication, are released into the tissues of target organs. In other instances, after endocytosis by endothelial cells, viruses may be translocated to the basal surface and released into tissues by exocytosis. In some anatomical locations such as the choroid plexuses, virions may pass from the bloodstream into surrounding tissues through fenestrated endothelium. Viruses can be transported into tissue spaces inside lymphocytes or monocytes as these cells migrate from the circulation. Engulfment and destruction of viruses by the phagocytic cells of the

reticuloendothelial system is an important defence mechanism which limits viraemia. In some instances, phagocytosis of viruses by these cells may result in transfer of virions to adjoining tissues.

Neural spread

Neurotropic viruses such as rabies virus, Aujeszky's disease virus and Borna disease virus can invade the CNS through peripheral nerves. Enveloped viruses are usually transported as naked nucleocapsids by axoplasmic flow. Within the CNS, dissemination often involves spread across synaptic junctions. In addition, spread within peripheral nerves from the CNS to other locations may occur. Alphaherpesviruses can spread in peripheral nerves from the site of infection to ganglia, causing latent infections. Reactivation of infection may result in recrudescence of superficial lesions following transport of virus from the ganglia along nerve fibres.

Clinical signs

The signs of viral infections reflect both virus replication in tissues and host responses. Some viruses kill the cells in which they replicate, producing clinical signs which relate to the anatomical locations of the affected cells. Because of the considerable reserve and regenerative capacity of the liver, the loss of large numbers of hepatocytes may not result in significant clinical disturbance. In contrast, loss of relatively few neurons may have severe clinical consequences. In some viral infections, loss of specialized functions or reduction in functional efficiency in infected cells may induce clinical signs. Viral infections of the respiratory and intestinal tracts are frequently complicated by secondary bacterial infections, which may contribute to the development of clinical signs. Bacterial species present in the normal flora may contribute to these opportunistic infections. Denuded epithelial surfaces, impaired clearance mechanisms or increased availability of bacterial nutrients may promote secondary infections.

A number of viruses, including infectious bursal disease virus of poultry and feline immunodeficiency virus, target cells of the immune system. Progressive depletion of lymphocytes can lend to an immunodeficient state. As a consequence, affected animals may present with a variety of clinical signs due to secondary bacterial infection. In most viral infections, the immune system has an important protective role. However, there are viral infections in which the principal lesions result from hypersensitivity reactions and subsequent immunopathological changes. Immune complexes formed in persistent viral infections such as feline infectious peritonitis and equine infectious anaemia are responsible for vasculitis and glomerulonephritis.

Abortion following viral infection is usually indicative of substantial damage to the tissues of the placenta or foetus. The effect on the foetus is often influenced by the stage of gestation and by the virulence of the virus. Infection with virulent virus early in gestation generally results in foetal death with resorption or abortion. Infection of pregnant cattle with bovine viral diarrhoea virus (BVDV) before 100 days gestation may result in abortion, congenital defects or immunotolerance. If infection occurs at a later stage during pregnancy calves may be born with congenital defects whereas infection towards the end of pregnancy induces a protective immune response.

When viral epidemics occur in susceptible populations, the outcome in individual animals ranges from asymptomatic infection to fatal infection. Host factors which may influence the outcome include age, immune status, inheritance and nutrition. Young animals are generally more susceptible to viral infections than older animals whose increased resistance can usually be attributed to maturation of the immune system and to immunological memory. Reverse age resistance, although uncommon, is well documented in rabbit haemorrhagic disease. In this condition, rabbits less than 5 weeks of age do not develop disease because the target organ, the liver, does not appear to be susceptible early in life. Severe malnutrition may exacerbate certain viral diseases; a possible consequence of which may be depressed cell-mediated immunity. Success in the breeding of birds with enhanced resistance to avian leukosis and Marek's disease illustrates the influence of genetic factors on disease susceptibility.

Virus shedding and patterns of infection

The shedding of infectious virions from surfaces or orifices allows transfer to other susceptible hosts. Although shedding usually coincides with the onset of clinical signs, it may begin earlier in some viral infections. Respiratory viruses are usually transmitted in aerosols generated by coughing and sneezing. Enteric viruses are frequently shed in enormous quantities in faeces and can generally survive in harsh environments. Viruses which produce generalized infections may be shed by a number of routes. Body fluids such as saliva, semen, urine and milk may contain particular viruses. Arboviruses usually produce high titred viraemias of short duration and they rely on appropriate haematophagous vectors for transmission. Surgical procedures or blood sampling may facilitate spread of blood-borne viruses. Some viruses can be transmitted vertically by the transplacental route. Endogenous retroviruses are transmitted in the DNA of germ cells.

The maintenance of infection in a population requires continuous infection of susceptible animals. Two main strategies have been adopted by viruses for this purpose, namely acute infection and long-term persistence.

Acute infection is characterized by a short clinical course with rapid elimination of virus from the tissues. In these infections, there may be shedding of large amounts

of virus over a short period. To maintain infection in a susceptible population viruses which cause acute infections must either be highly contagious, such as influenzaviruses, or be capable of surviving for long periods in the environment, such as parvoviruses.

Persistent viral infections are characterized by a prolonged course with constant or intermittent shedding. These infections, which may be acute initially, can persist either in latent or chronic forms. Latent infections are characterized by persistence of the virus in a non-productive form. Periodic reactivation of productive infection with shedding of infectious virus may occur. This type of infection is best exemplified by alpha-herpesviruses, which produce productive infections in epithelial cells and latent infections in sensory neurons. In chronic infections, virus is constantly present and may be shed intermittently or continuously. This type of infection can occur when the immune response of the host fails to eliminate virus from the tissues.

Infections characterized by long incubation periods of months or years are referred to as slow infections. Infections of this type, produced by lentiviruses, jaagsiekte sheep retrovirus and prions, have a progressive clinical course usually resulting in death.

Mechanisms of persistence

Viruses which persist in the body usually employ a number of strategies to ensure prolonged infections. For a virus to persist in host tissues, some infected cells must survive. Viruses have evolved a number of different strategies to reduce pathogenic effects on host cells. Viruses which produce non-lytic infections are likely to cause chronic infections. A number of viruses that are normally lytic have been shown to be capable of establishing persistent infections. Alphaherpesviruses, by exploiting the fact that sensory neurons are only partly permissive, have the ability to set up latent infections in such cells. In these non-dividing cells, the viruses persist as circular episomal DNA until adverse environmental factors induce immunosuppression which permits virus replication. Adenoviruses may persist in the body by producing low-grade, cycling infection with small numbers of cells infected at any given time. The evolution of defective-interfering or less cytolytic variants may permit this type of infection to occur.

Viruses have evolved several strategies to evade the immune response and so avoid elimination from an immunocompetent host. Certain tissues in the body, referred to as immunologically privileged sites, are exempt from immune surveillance. The blood-brain barrier restricts contact between lymphocytes and CNS tissues. In addition, neurons do not express the major histocompatibility complex (MHC) class I or class II molecules which are required for T cell recognition of virus-infected cells. During latent infection of host cells by herpesviruses, viral gene expression is diminished and viral proteins are not expressed on the cell surface. Such infected cells are not easily recognized by the immune system. Viruses, particularly RNA viruses, undergo frequent mutation. Variants with altered epitopes at sites important for antibody neutralization or T cell recognition can emerge. These variants may evade immune detection and become the predominant infecting strain. This type of antigenic variation has been described in influenzavirus infections as a consequence of alterations in the surface glycoproteins, haemagglutinin and neuraminidase. Although these viruses do not persist in individual infected animals, they may persist in animal populations. Lentiviruses, such as equine infectious anaemia (EIA) virus, exhibit significant antigenic variation during replication in infected animals. Recurrent episodes of clinical disease can occur in EIA, in association with the emergence of antibody-escape variants and subsequent viraemia. It has been shown that some viruses can down-regulate expression of host cell surface markers such as MHC class I and class II molecules. Some viral proteins appear to act defensively by interfering with the function of anti-viral cytokines. Immune tolerance of the type associated with congenital BVDV infection permits persistent infection in calves. Affected calves have a lifelong viraemia and are the principal source of infection for other cattle.

Further reading

Tyler, K.L., Fields, B.N. (1996). Pathogenesis of viral diseases. In *Fields Virology*. Volume 1. Third Edition. Eds. B.N. Fields, D.M. Knipe and P.M. Howley. Lippincott-Raven, Philadelphia. pp. 173–218.

Chapter 53

Laboratory diagnosis of viral infections

Many viral diseases of animals can be diagnosed on the basis of clinical signs together with postmortem findings and histopathological changes. However, confirmation of the involvement of specific viral pathogens often requires special laboratory procedures. Surveillance for particular viruses is an important aspect of the management of valuable animals such as bulls used for artificial insemination and stallions, which have the potential to spread infection to many other animals. As part of international trade regulations, certification of freedom from certain viral diseases must accompany animals exported to countries in which the diseases are exotic. Moreover, rapid and accurate laboratory confirmation of exotic viral diseases, including those with zoonotic potential, is essential for the successful implementation of eradication policies and for the protection of human health. Surveillance of animal populations for new or emerging viral diseases is an important responsibility of national veterinary services.

More than 200 major viral diseases affect animal species of veterinary importance. Because of the considerable resources required for the provision of comprehensive diagnostic services in virology, national diagnostic services are usually concentrated on those diseases prevalent in a country. Moreover, laboratories often provide a diagnostic service for particular animal species. Special laboratory containment facilities are mandatory for some viruses which cause highly contagious diseases such as foot-and-mouth disease. The Office International des Épizooties (OIE) in Paris monitors and publishes details of significant animal disease outbreaks worldwide. This work is possible only through international cooperation and a network of laboratories dealing with viral diseases of international importance.

Collection, preservation and transportation of samples

Ideally, specimens for laboratory examination should be collected as early as possible from affected animals before secondary bacterial or fungal infections become established. It is advisable to collect samples from apparently normal in-contact animals because some of these animals may be actively shedding virus. The specimens selected for laboratory examination should relate to the clinical signs or to lesion distribution postmortem. Swabs from the oropharynx or nasopharyngeal aspirates are suitable specimens for investigation of respiratory diseases. In enteric viral diseases, large numbers of virus particles are shed in faeces. In those diseases characterized by viraemia, virus may be demonstrable in cells of the buffy coat.

Preservation of the infectivity or antigenicity of viruses may be required for particular tests. As many viruses are labile, specimens for virus isolation should be collected into transport medium, refrigerated and transmitted to the laboratory without delay. Samples should be frozen at -70°C if delay in delivery is anticipated. Freezing in a domestic freezer at -20°C decreases the infectivity of most viruses. Transport medium consists of buffered isotonic saline containing a high concentration of protein, such as bovine albumin or foetal calf serum, which prolongs virus survival. Antibiotics and antifungal drugs are added in order to inhibit the growth of contaminants. Samples for electron microscopy, in which demonstration of virion morphology is the primary objective, require less exacting conditions for storage and transportation. Air-dried smears for fluorescent antibody (FA) staining should be fixed in either acetone or methanol for up to 10 minutes in order to preserve viral antigens. This fixation process allows penetration of FA conjugates into cells. A similar fixation procedure is required for cryostat sections of frozen tissues prior to FA staining. Formalin-fixed tissue samples embedded in paraffin wax can be stored for many years and used to demonstrate the presence of viral antigen by immunohistochemical techniques.

Guidance from clinicians regarding the possible aetiology of the disease under investigation is essential for deriving maximum benefit from laboratory tests. This requires an accurate assessment of the history and clinical signs together with a tentative clinical diagnosis. In some instances, postmortem and histopathological examination of tissues may be sufficient for diagnostic purposes, particularly if specific inclusion bodies are found in infected tissues.

Detection of virus, viral antigens or nucleic acid

The presence of virus in tissues can be confirmed by

isolation of live virus, by demonstration of virus particles or viral antigen and by detecting viral nucleic acid.

Isolation of live virus

Virus isolation using cell culture, fertile eggs or experimental animals is the standard against which other diagnostic methods are usually compared. A monolayer composed of a particular cell type cannot be expected to support the growth of the many viruses which cause animal diseases. Laboratories usually have a limited range of cell lines most often used for virus isolation and appropriate to the range of samples received. Embryonated eggs are widely used for the isolation of influenza A virus and avian viruses. Because of ethical considerations and cost, virus isolation in experimental animals is now seldom employed.

Virus isolation is a sensitive procedure when cultural conditions are optimal for a particular virus and also generates a supply of virus for further studies. However, it is labour-intensive, slow and expensive. A number of blind passages may be required before a virus becomes adapted to a particular cell line and, as a consequence, a test result may not be available for some weeks. Because some viruses do not produce a cytopathic effect, additional detection procedures such as haemadsorption and FA staining may be needed to demonstrate their presence in cell cultures. Even when a virus produces a pronounced cytopathic effect, additional tests are often required for definitive identification.

Electron microscopy

Viruses can be demonstrated in diagnostic specimens by electron microscopy. This method can be used not only to recognize mixed viral infections but also to detect viruses which cannot be grown *in vitro*. Although this technique is particularly useful for identifying enteric viruses, it has serious limitations. Large numbers of viral particles (usually greater than 10^7 per ml) must be present in clinical samples. Sufficient concentrations of virus may be found in samples of faeces, scabs, vesicular fluids and wart tissues to allow detection. Moreover, viruses with similar morphology may be difficult to distinguish by electron microscopy. Members of a viral family usually have identical morphology and cannot be distinguished on this basis alone. In a few viral families, such as *Poxviridae* and *Reoviridae*, morphological differences do exist at the generic level.

A number of methods are used for the preparation of samples for examination. When preparing fluid samples, low speed centrifugation to remove large particulate debris is followed by ultracentrifugation to sediment virus particles. Negative staining with heavy metal compounds such as phosphotungstic acid or uranyl acetate increases contrast so that the brighter virions stand out against a dark background. The addition of antiserum to the specimen in immmunoelectron microscopy enhances the sensitivity of the procedure by clumping virus particles and facilitating their recovery by centrifugation. Alternatively, antiserum, when applied to the copper grid used for examining the specimen, can aggregate virions in the specimen.

Immunofluorescence and immunohistochemistry

Antiviral antibodies labelled with fluorochromes can be used to detect viral antigens in clinical specimens. Enzyme-labelled antiviral antibodies can also be used for this purpose. When using procedures involving immunofluorescence, a microscope with a special light source is required in order to visualize fluorescing material. Specific antibodies labelled with fluorochromes, usually fluorescein isothiocyanate or rhodamine isothiocyanate, are employed to detect virus-infected cells in specimens. The fluorochrome, exposed to light of a particular wavelength, emits light at a slightly longer wavelength allowing demonstration of the labelled antibody bound to virus particles. The technique is used to detect virus-infected cells in fixed smears, cryostat sections or monolayers.

Direct and indirect immunofluorescent techniques can be used to demonstrate virus or viral antigen in specimens (Fig. 53.1). In direct immunofluoresence, conjugated antibody specific for the particular virus is applied to the fixed specimen. After incubation, the specimen is washed to remove unbound antibody before microscopic examination. The indirect method employs unlabelled antiviral antiserum and a labelled antiglobulin specific for the species of animal from which the unlabelled antiserum was derived. Immunofluorescence techniques are rapid and sensitive but require careful interpretation as non-specific fluorescence can occur in certain specimens.

Enzyme-labelled antibodies can also be used to identify

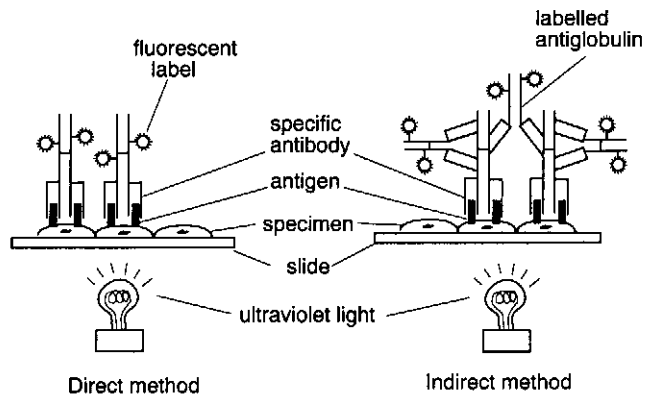


Figure 53.1 Direct and indirect immunofluorescent techniques for demonstrating viral antigen in cryostat (frozen tissue) sections, cell smears or monolayers. In the direct method, specific antibody is labelled with fluorescein isothiocyanate whereas in the indirect method, specific antibody bound to the viral antigen is demonstrated by the addition of labelled antiglobulin for the species of origin of the antibody.

viruses or viral antigen in clinical specimens. Horseradish peroxidase is the most common enzyme employed for conjugation with specific antiserum. After binding to viral antigen, antibody is identified by the addition of hydrogen peroxide and a benzidine derivative to the preparation. During the ensuing reaction, the colourless soluble benzidine derivative is converted to a coloured insoluble precipitate. For this technique, paraffin-embedded and resin-embedded tissues can be used and stained preparations do not fade when stored for long periods. Moreover, the stained preparations can be examined by conventional light microscopy. Endogenous tissue peroxidases, however, may produce misleading reactions unless suitable controls are included.

An avidin-biotin system can be used for enhancing immunohistochemical reactions. Biotin can be covalently linked to antibody without interfering with its antigen-binding capacity. Avidin bound either to a fluorochrome or an enzyme has a high affinity for the biotin molecules linked to the antibody.

Solid-phase immunoassays

In these assays, either antigen or antibody is immobilized on a surface. Suitable surfaces include polystyrene or synthetic membranes for enzyme immunoassays and radioimmunoassays; latex beads are often employed for agglutination tests. These assays are sensitive and relatively uncomplicated. Commercial tests based on these methods have been developed for particular viruses.

Radioimmunoassays employ antibodies labelled with radioisotopes and the bound antibody is measured using a gamma counter. These types of assays have been largely superseded by safer immunoassay procedures. Enzyme immunoassays, usually termed enzyme-linked immunosorbent assays (ELISA), are currently widely used for the immunodiagnosis of viral infections. In these assays, antibodies are labelled with enzymes which produce a colour change when they react with appropriate substrates. This colour change can be assessed visually or spectrophotometrically. For the detection of virus, wells in polystyrene plates are coated with specific anti-viral antibody and test material is added (Fig. 53.2). If present in the test material, viral antigen binds to the antibody during incubation and is not removed by washing. Enzyme-labelled antibody specific for the viral antigen is then added. Following incubation and washing, substrate for the enzyme is added; a colour change indicates a positive reaction.

Large numbers of samples can be processed quickly using ELISA. Rapid one-step assays suitable for use in veterinary practices are commercially available. Many of these kits utilize a membrane as the solid phase in order to increase the surface area on which antigen-antibody reactions can occur, thereby reducing incubation and washing times.

Latex particles, coated with anti-viral antibodies, agglutinate in the presence of viral antigen. No special equipment is required for these tests, which are simple and inexpensive. However, factors which decrease the reliability of these tests include non-specific reactions and prozone effects due to high antigen concentrations.

Immunodiffusion

This procedure is carried out in agar. The technique involves placing a fluid sample containing the virus under test in a well in the agar opposite a well containing antiserum. As the fluids diffuse out of the wells, a line of precipitate forms if the sample under test contains viral antigen. Although this test is easy to perform and inexpensive, it is relatively insensitive.

Complement fixation test for antigen detection

When antigen binds to antibody to form immune complexes, complement becomes activated and fixes to the complexes. In the complement-fixation test, the test sample is added to a known antiserum which has been heat-inactivated to destroy complement. Following incubation, a precise amount of guinea-pig complement is added. If viral antigen is present in the test sample, immune complexes formed during incubation fix the guinea-pig complement. Sheep red cells treated with specific rabbit antibody are added as an indicator to detect residual complement activity. If the guinea-pig complement is not bound, indicating the absence of viral antigen

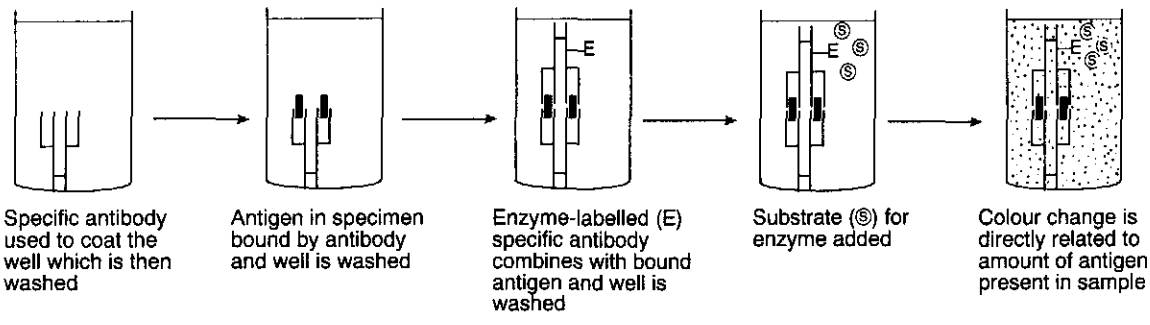


Figure 53.2 Steps in ELISA for detecting antigen in a test sample (antigen capture method).

from the test sample, red cell lysis occurs. If viral antigen is present in the test sample, red cell lysis does not occur. The CFT requires rigorous standardization of reagents and careful interpretation.

Haemagglutination and haemadsorption

Viruses belonging to several families including *Orthomyxoviridae*, *Paramyxoviridae*, *Adenoviridae*, *Parvoviridae* and *Togaviridae* can interact with erythrocytes of many animal species causing haemagglutination. This unique ability derives from viral glycoproteins (haemagglutinins) which attach to receptors on erythrocytes resulting in the formation of aggregates which settle out of suspension. The haemagglutinins of orthomyxoviruses and paramyxoviruses allow these viruses to bind to neuraminic acid-containing receptors on erythrocytes. These viruses also possess neuraminidases which can destroy the erythrocyte receptors, causing dissociation. The surface structures (spikes) of influenza viruses which contain haemagglutinin and neuraminidase activity can be used to type influenza virus A isolates using specific antibody. As large numbers of virus particles are required to produce visible haemagglutination, this test is relatively insensitive.

The term haemadsorption is used to describe binding of erythrocytes to cells infected with haemagglutinating viruses. The haemagglutinating glycoproteins of these viruses are incorporated into cell membranes during viral replication.

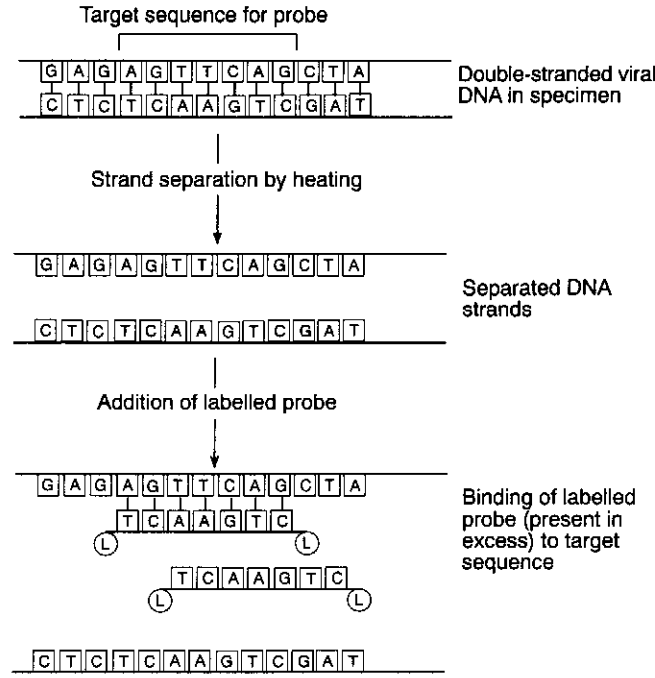


Figure 53.3 Outline of method for detecting viral DNA in clinical specimens using a labelled DNA probe specific for a target sequence. Suitable labels (L) include enzymes or radioactive isotopes. Detection of labelled probe by appropriate methods confirms the presence of viral nucleic acid.

Nucleic acid detection

The sensitivity and versatility of methods for the detection of viral nucleic acids has greatly improved in recent years and they are now becoming the method of choice for viral identification. These methods are particularly valuable when dealing with viruses which are either difficult to grow or cannot be grown *in vitro*. They are useful for latent infections in which infectious virus is absent and also for specimens containing inactivated virus. Cloned viral DNA is available for probing of samples and tissues by nucleic acid hybridization. This technique, however, has been largely replaced in recent years by the polymerase chain reaction which has the advantage of amplifying the target gene sequences.

Hybridization methods

Single-stranded DNA can hybridize to a complementary single strand of DNA or RNA by hydrogen-bonded base pairing. Under defined conditions of ionic concentration and elevated temperature, the strands of a target nucleic acid molecule can be dissociated and, on cooling, can hybridize with labelled DNA or RNA oligonucleotide probes when these are present in excess in the reaction mixture (Fig. 53.3). The temperature and ionic concentration employed largely determine the degree of discrimination of the test. Under stringent conditions, a high degree of specificity is achieved. Under conditions of low stringency, a probe containing mismatched base pairs can anneal to the dissociated nucleic acid strand. The length of the probe and the target region of the genome influence both the specificity of the test and the interpretation of the result. The oligonucleotide probes are synthesized or produced by cloning. Radioactive isotopes, which were extensively used as labels for probes, have been largely replaced by non-radioactive labels including alkaline phosphatase and horseradish peroxidase. Affinity labelling with biotin or digoxigenin and chemoluminescent labelling with acridinium esters are also employed.

Hybridization can be detected in a liquid phase or solid phase. Reactions in solution allow the target nucleic acid and the probe to interact freely. A modification of this hybridization procedure is the hybridization protection assay which employs a chemoluminescent probe. In this method, the chemoluminescent ester of the probe is protected from alkaline hydrolysis if the probe forms a stable molecular hybrid with the target molecule; unbound esterified probes are degraded giving negative results. A useful solid phase assay, termed dot-blot hybridization employs charged nylon or nitrocellulose membranes. Nucleic acid extracted from a sample, spotted directly onto the membrane, binds firmly after baking. The labelled probe is added, allowed to hybridize and unbound material

is washed off. Another method based on solid phase assay is Southern blotting. In this method, extracted DNA is digested with restriction endonucleases and the resulting fragments are separated by gel electrophoresis. The separated fragments are then transferred by blotting onto a membrane for evaluation by hybridization.

Hybridization procedures can also be applied to intact cells or tissue sections. This *in situ* hybridization method is based on liberation of target nucleic acid while retaining cellular integrity and allows identification of the exact site of virus location.

Polymerase chain reaction

This *in vitro* method is based on the amplification of a particular nucleic acid sequence using a thermostable DNA polymerase and two oligonucleotide primers (Fig. 53.4). The polymerase was originally isolated from the thermophilic bacterium *Thermus aquaticus* and is known as *Taq* polymerase. In order to amplify the nucleic acid sequence of interest, primers complementary to sequences at each end of that particular portion of nucleic acid must be synthesized. Reactions, involving target DNA, primers, DNA polymerase and a mixture of individual nucleotides, are carried out under carefully controlled conditions in a programmable thermocycler. In most polymerase chain reaction (PCR) procedures, 30 to 50 thermal cycles take place. Each thermal cycle consists of a step in which the double-stranded DNA is separated at high temperature, an annealing step which allows primers to hybridize with complementary target sequences during a cooling phase, and an extension step, at an intermediate temperature, during which primer products are extended by DNA polymerase. These primer products act as new templates for the production of nucleic acid in each subsequent cycle. The number of target DNA copies approximately doubles with every cycle. The amplified DNA fragments can be separated by agarose gel electrophoresis and stained with ethidium bromide. Exact information on the DNA fragments requires the use of restriction endonuclease analysis, direct sequencing or probe analysis. An important modification of the PCR is the reverse transcriptase polymerase chain reaction (RT-PCR) which has extended the application of the technique to RNA viruses.

Quantitative PCR methods, which can be used to measure the amount of target DNA or RNA in a sample, have been developed. In addition, retrospective PCR analyses are now possible with the development of *in situ* PCR which can be applied to paraffin-embedded tissues and museum specimens. Multiplex PCR involves the use of multiple primer pairs in the same amplification reaction to increase the number of detectable target sequences. This has allowed the development of assays capable of detecting multiple pathogens or virulence-associated genes in a single sample. Nested PCR is a popular modification of PCR involving two sets of primers. The second pair of primers is designed to amplify an internal region of the sequence amplified by the first pair. It is possible to design the first primer set to be less specific, capable of detecting a range of related pathogens, while the second primers set is species- or strain-specific. Because of their exquisite sensitivity, PCR techniques require rigorous standardization to exclude cross-contamination and ensure reproducibility and reliability. When collecting clinical specimens, contamination from the environment or from other sources must be avoided. The possible presence of viral 'passengers' and degraded viral genomes becomes an important issue as the sensitivity of detection tests increases.

Diagnostic serology

Serological procedures can be used for the retrospective diagnosis of viral diseases and for epidemiological surveys. These procedures can be automated and diagnostic reagents for many viral pathogens are commercially available. Single blood samples from animals in susceptible populations suffice for establishing the prevalence of a disease. When using serological procedures for the diagnosis of endemic disease in flocks or herds, paired serum samples taken at an interval of at least three weeks, are required to demonstrate rising antibody titres. Initial samples should be collected when clinical signs are first evident and the second samples during convalescence. A single blood sample may be adequate for diagnosis if reagents are available for demonstrating IgM antibodies, which are indicative of a primary immune response. Difficulties with the interpretation of serological tests may arise due to cross-reactions with antigenically related viruses. In young animals, passively acquired maternal antibodies which may persist for several months, can lead to misinterpretation of results.

Enzyme-linked immunosorbent assay

The procedure used for carrying out ELISA to detect antibodies is different from that already described for the demonstration of virus in specimens. The essential difference is that viral antigen is bound to the solid phase (polystyrene wells or membranes). Dilutions of test serum are added and allowed to react with the antigen. After washing, enzyme-conjugated antiglobulin is added and, following further incubation and washing, appropriate substrate is added. The intensity of the colour change is proportional to the amount of antibody in the test serum (Fig. 53.5). Either polyclonal or monoclonal labelled conjugates may be used; the choice depends on whether or not the immunoglobulin isotype is required for estimating the duration of the infection.

Immunofluorescence for antibody detection

Using indirect immunofluorescence, test serum is added to known viral antigen fixed on a microscope slide

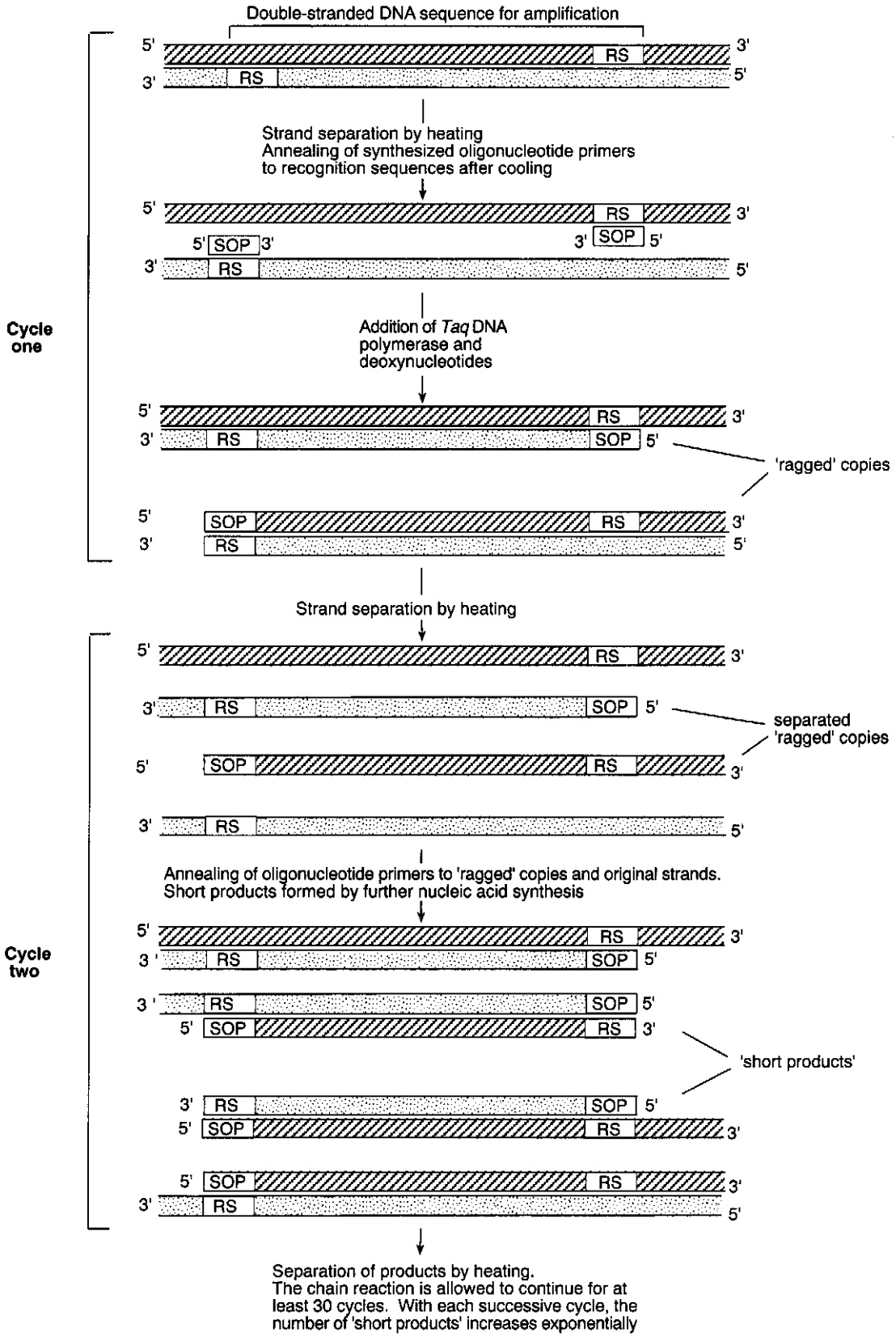


Figure 53.4 Outline of the steps involved in the polymerase chain reaction.
RS, recognition sequence; SOP, synthesized oligonucleotide primer;
Taq DNA polymerase, heat-stable enzyme derived from the bacterium *Thermus aquaticus*.

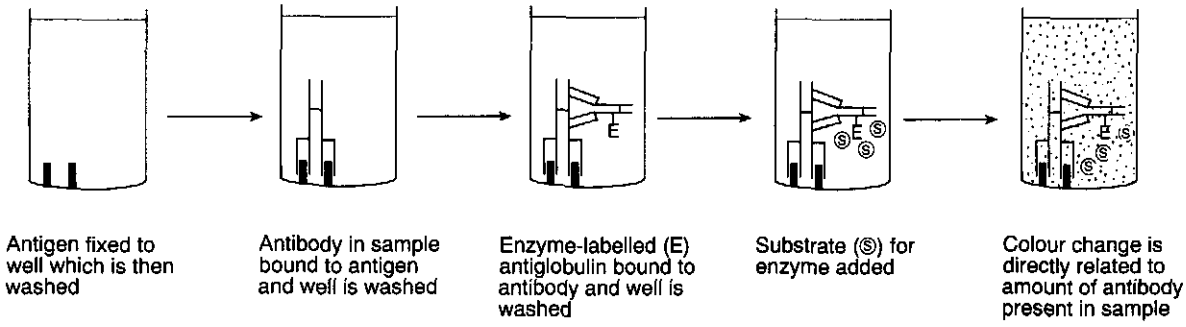


Figure 53.5 Steps in ELISA for detecting antibody in a test sample (indirect method).

(Fig. 53.6). Following incubation, the slide is washed and, after the addition of fluorescein isothiocyanate-conjugated antiglobulin, incubated for a further period. The slide is again washed before examination under UV light. The test is both sensitive and rapid but requires careful interpretation. Antibody production by plasma cells can be demonstrated in cryostat sections by the sandwich method (Fig. 53.7).

Serum neutralization test

This test is highly specific and sensitive for viruses which produce cytopathic effects (CPE). It is considered to be the definitive standard against which other serological tests are compared. Neutralizing antibodies usually correlate closely with immune protection. In this test, which is typically carried out in microtitre plates, a constant amount of stock virus is added to doubling dilutions of a test serum. Cells susceptible to the virus are added to the wells. The presence of neutralizing antibodies in the serum prevents infection of the cells and CPE. The titre of the serum is the highest dilution at which the virus is neutralized. The neutralizing effect of test serum can also be evaluated in susceptible experimental animals and in chick embryos. Neutralizing antibodies tend to persist in recovered animals for long periods, often for many years.

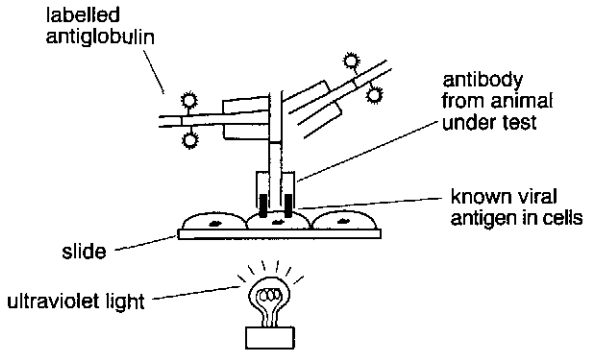


Figure 53.6 Indirect immunofluorescence technique for demonstrating antibodies in serum.

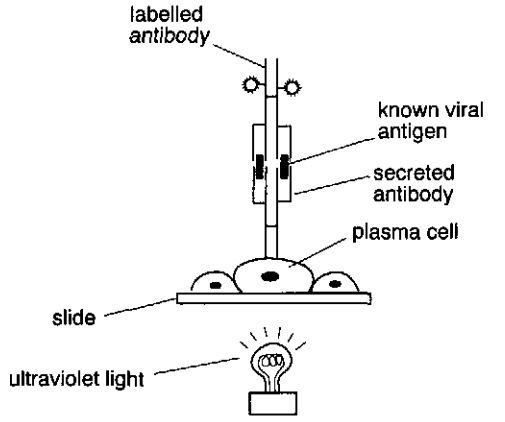


Figure 53.7 Sandwich method for demonstrating antibody production in tissue specimen.

Haemagglutination inhibition test

Because viruses in certain families have the ability to haemagglutinate, the inhibitory effect of antibodies on haemagglutination can be used for the diagnosis of infection with these viruses. The haemagglutination inhibition (HAI) test is specific, reliable and easily performed. The test, usually carried out in microtitre plates, involves serial twofold dilutions of serum to which a known concentration (four haemagglutinating units) of virus is added. The highest dilution which inhibits erythrocyte agglutination is the HAI titre of the test serum. Non-specific inhibitors of haemagglutination, sometimes present in sera, can be inactivated by heating or by treatment with kaolin, trypsin, periodate or bacterial neuraminidase.

Complement fixation tests

Due to difficulties with standardization, complement fixation tests have been superseded by more convenient diagnostic tests such as ELISA. Complement-fixing antibodies tend to appear before neutralizing antibodies but do not persist. Sera from some species are difficult to

titrate by this method due to the presence of anti-complementary activity.

Western blotting (immunoblotting) technique

This test, which was primarily developed as a research procedure for the identification of antigenic proteins, can also be used for the diagnosis of viral disease. Purified virus, solubilized with an anionic detergent such as sodium dodecyl sulphate, is electrophoresed in a polyacrylamide gel. Separated proteins are transferred electrophoretically onto a nitrocellulose membrane, which is then washed, dried and cut into longitudinal strips. Following incubation in test sera, the strips are washed and incubated with enzyme-labelled antiglobulin. Addition of substrate produces an insoluble coloured product where test antibody is bound to the separated viral proteins.

Interpretation of test results

Because false-positive and false-negative results can occur in many test procedures, inclusion of positive and negative controls is essential. The sensitivity and specificity of a particular diagnostic test should be established. The sensitivity of a diagnostic test, expressed as a percentage, is the number of animals identified as positive out of the total number of animals with the disease. The specificity of a test is the percentage of uninfected animals in which the result is negative. In order to detect all animals with an important viral infection, a test with high sensitivity is required. For laboratory confirmation of a viral infection in an individual animal, a test with high specificity is essential.

The isolation of virus or the demonstration of antibody to a specific virus does not necessarily confirm an aetiological link with a disease state. For the conclusive confirmation of test results, it may be necessary to demonstrate a correlation between the site of virus recovery and the nature and extent of lesions. Circumstantial evidence for the aetiological involvement of a virus in a clinically affected animal is supported by the recovery of the same virus from susceptible in-contact animals. Moreover, a rising antibody titre to the putative causal virus, demonstrated by the use of paired serum samples, is of diagnostic importance. Published reports on the potential importance of a similar disease syndrome and its aetiology may point to the requirement for particular laboratory investigations.

Further reading

Mullis, K.B. (1990). The unusual origin of the polymerase chain reaction. *Scientific American*, **262**, 56–65.

Saiki, R.K., Scharf, S., Faloona, F. *et al.* (1985). Enzymatic amplification of β-globulin genomic sequences and restriction site analysis for the diagnosis of sickle-cell anaemia. *Science*, **230**, 1350–1354.

Section V

Viruses and Prions

Chapter 54

Herpesviridae

The family *Herpesviridae* contains more than 100 viruses. Fish, amphibians, reptiles, birds and mammals including humans are susceptible to herpesvirus infection. These viruses are of special importance because of their widespread occurrence, their evolutionary diversity and their involvement in many important diseases of domestic animals and humans. The name, herpesvirus (Greek *herpein*, to creep), refers to the sequential appearance and local extension of lesions in human infection. Herpesviruses are enveloped and range from 120 nm to 200 nm in diameter. They contain double-stranded DNA within an icosahedral capsid (Fig. 54.1). A layer of amorphous material, the tegument, lies between the envelope and the capsid. Herpesviruses enter cells by fusing with the plasma membrane. Replication occurs in the cell nucleus. The envelope, which derives from the nuclear membrane of the host cell, incorporates at least eight viral encoded glycoproteins. Enveloped virions accumulate in the endoplasmic reticulum prior to final processing of the glycoproteins in the Golgi apparatus and release by exocytosis. Active infection results in cell death. Intranuclear inclusions are characteristic of herpesvirus infections. Extension of viral infection occurs through points of cell contact without exposure of virus to neutralizing antibodies in blood or interstitial fluids. Protective antibody responses are usually directed against the envelope glycoproteins. Herpesvirus virions, which are fragile and sensitive to detergents and lipid solvents, are unstable in the environment.

Key points

- Enveloped DNA viruses with icosahedral symmetry
- Replicate in the nucleus
- Labile in the environment
- Three subfamilies of veterinary importance: *Alphaherpesvirinae, Betaherpesvirinae, Gammaherpesvirinae*
- Cause diseases of the respiratory, reproductive and nervous systems; may cause cell transformation in some species
- Latency is a common outcome of infection with these viruses

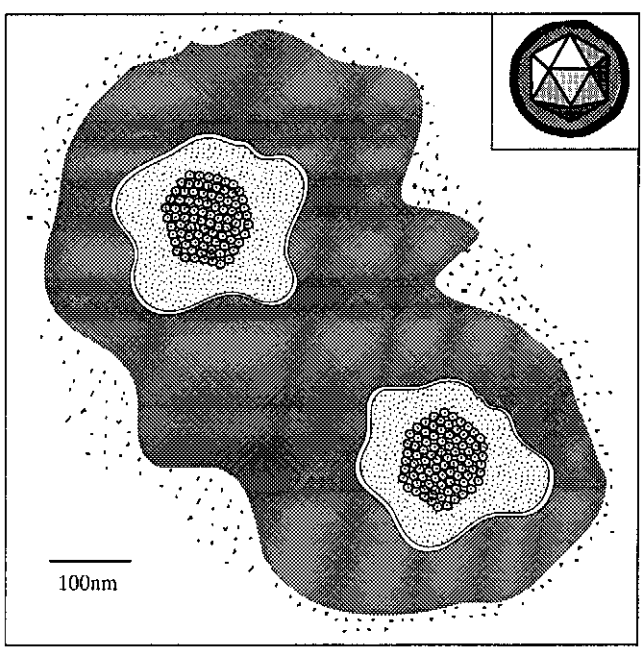


Figure 54.1 Herpesvirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

The family is divided into three subfamilies (Fig. 54.2), *Alphaherpesvirinae*, *Betaherpesvirinae* and *Gammaherpesvirinae*. The subfamilies comprise nine genera. A tenth unnamed genus Ictalurid herpes-like viruses contains herpesviruses of fish and is currently unassigned to a subfamily. Alphaherpesviruses replicate and spread rapidly, destroying host cells and often establishing latent infections in sensory ganglia. Betaherpesviruses, which replicate and spread slowly, cause infected cells to enlarge, hence their common name cytomegaloviruses. They may become latent in secretory glands and lymphoreticular cells. Gammaherpesviruses, which infect T or B lymphocytes, can produce latent infections in these cells. When lymphocytes become infected, there is minimal expression of viral antigen. Some gammaherpesvirus species also replicate in epithelial and fibroblastic cells causing cytolysis. A number of gammaherpesviruses are implicated in neoplastic transformation of lymphocytes.

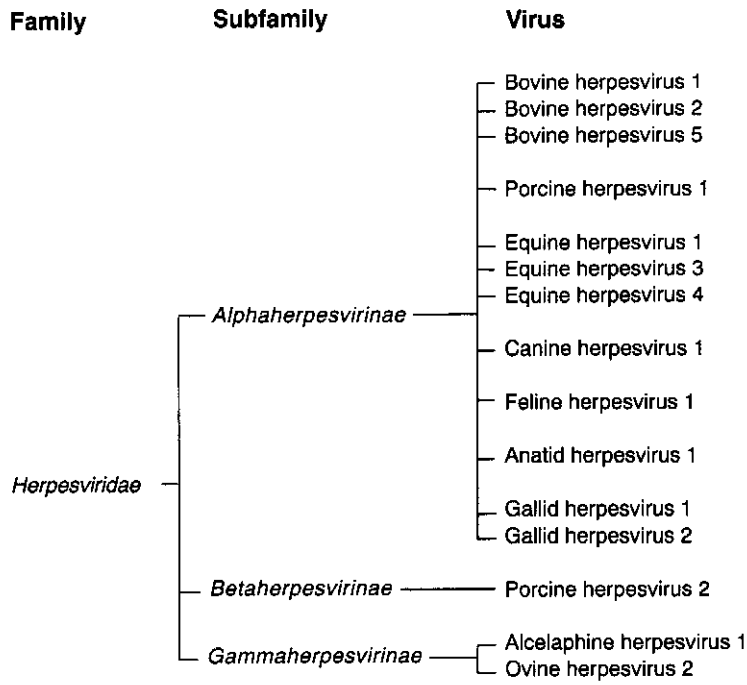


Figure 54.2 Classification of herpesviruses of domestic animals.

Clinical infections

Herpesviruses establish life-long infections with periodic reactivation resulting in bouts of clinical disease. Shedding of virus may be periodic or continuous. During latency, the episomal viral genome becomes circular and gene expression is limited. Reactivation of infection is associated with various stress factors including transportation, adverse weather conditions, overcrowding and intercurrent infection. Natural infections with particular herpesviruses are usually restricted to defined host species. Because these viruses are highly adapted to their natural hosts, infections may be inapparent or mild. However, in very young or immunosuppressed animals, infection can be life-threatening. Some herpesviruses such as Marek's disease virus and Epstein-Barr virus are implicated in neoplastic transformation of cells.

Herpesviruses can cause respiratory, genital, mammary and CNS diseases in cattle (Table 54.1). Aujeszky's disease, which affects pigs and other domestic species, is the major porcine herpesvirus infection (Table 54.2). In horses, herpesviruses can cause respiratory, neurological and venereal diseases, and abortion along with neonatal infection (Table 54.3). The herpesviruses of domestic carnivores are presented in Table 54.4 and those of birds in Table 54.5.

Infectious bovine rhinotracheitis and pustular vulvovaginitis

Bovine herpesvirus 1 (BHV-1), which infects domestic and range cattle, is associated with several clinical conditions including infectious bovine rhinotracheitis, infectious pustular vulvovaginitis, balanoposthitis, conjunctivitis and generalized disease in newborn calves. Infection with BHV-1 is an important cause of losses in livestock worldwide. A single antigenic type of BHV-1 which contains subtypes 1.1, 1.2a and 1.2b, has been

Table 54.1 Herpesvirus infections of ruminants.

Virus	Genus	Comments
Bovine herpesvirus 1	*Varicellovirus*	Causes respiratory (infectious bovine rhinotracheitis) and genital (infectious pustular vulvovaginitis, balanoposthitis) infections. Occurs worldwide
Bovine herpesvirus 2	*Simplexvirus*	Causes ulcerative mammillitis in temperate regions and pseudo-lumpy-skin disease in tropical and subtropical regions
Bovine herpesvirus 5	*Varicellovirus*	Causes encephalitis in calves; described in several countries
Ovine herpesvirus 2	*Rhadinovirus*	Causes subclinical infection in sheep and goats worldwide. Causes malignant catarrhal fever in cattle and in some wild ruminants
Alcelaphine herpesvirus 1	*Rhadinovirus*	Causes subclinical infection in wildebeest in Africa and in zoos. Causes malignant catarrhal fever in cattle, deer and in other susceptible ruminants

Table 54.2 Herpesvirus infections of pigs.

Virus	Genus	Comments
Porcine herpesvirus 1 (Aujeszky's disease virus)	*Varicellovirus*	Causes Aujeszky's disease (pseudorabies) primarily in pigs. Encephalitis, pneumonia and abortion are features of the disease. In many species other than pigs, pseudorabies manifests as a neurological disease with marked pruritis. Occurs worldwide
Porcine herpesvirus 2	Unassigned	Causes disease of the upper respiratory tract in young pigs (inclusion body rhinitis)

recognized using restriction enzyme analysis. Subtype 1.1, implicated in respiratory disease, is included in most vaccines. Subtypes 1.2a and 1.2b cause mild respiratory disease and are implicated in the infectious balanoposthitis/infectious pustular vulvovaginitis (IBP/IPV) syndrome. Subtypes 1.1 and 1.2a rarely cause abortion; the less virulent subtype 1.2b has not been associated with abortion.

Pathogenesis and pathology

The virus which causes infectious bovine rhinotracheitis is usually acquired through aerosols. Replication occurs in the mucous membranes of the upper respiratory tract and large amounts of virus are shed in nasal secretions. Virus also enters local nerve cell endings and is transported intra-axonally to the trigeminal ganglion where it remains latent. In most instances, infection is contained within two weeks by a strong immune response. However, tissue necrosis may facilitate secondary bacterial infection with severe systemic effects and, possibly, death. Rarely, a lymphocyte-associated viraemia in pregnant cows may produce foetal infection and abortion. Necrotic foci may be present in various organs of aborted foetuses, particularly in the liver.

Table 54.3 Herpesvirus infections of horses.

Virus	Genus	Comments
Equine herpesvirus 1	*Varicellovirus*	Causes abortion, respiratory disease, neonatal infection and neurological disease. Occurs worldwide
Equine herpesvirus 3	*Varicellovirus*	Causes mild venereal infection in both mares and stallions
Equine herpesvirus 4	*Varicellovirus*	Causes rhinopneumonitis in young horses and sporadic abortion. Occurs worldwide

Following genital infection with subtypes 1.2a and 1.2b, virus replicates in the mucosa of the vagina or prepuce, and latent infection may become established in the sacral ganglia. Focal necrotic lesions on genital mucosae may eventually coalesce to form large ulcers. An intense inflammatory reaction can develop in the reproductive tract with secondary bacterial infection leading to endometritis. Viraemia is not a feature of genital infection with BHV-1 subtypes and infected pregnant cows rarely abort.

Clinical signs

In outbreaks of disease, either the respiratory or the genital form usually predominates. The incubation period is up to four days. The severity of the clinical signs in the respiratory form of the disease are largely determined by the extent of secondary bacterial infection. Affected animals develop a high temperature and nasal discharge accompanied by anorexia. The nares are inflamed ('red nose'), and conjunctivitis, lacrimal discharge and corneal opacity are often present. In uncomplicated infections, animals recover after about a week. If bacterial infection becomes established, animals develop dyspnoea, coughing and open-mouth breathing. Death may ensue. In severe outbreaks in feedlot cattle, morbidity may approach 100% and mortality may be as high as 10%. Cows with infectious pustular vulvovaginitis exhibit vaginal discharge and frequent urination. Animals usually recover within two weeks. However, secondary bacterial infection may result in metritis, temporary infertility and purulent vaginal discharge persisting for several weeks. Infected bulls have lesions on penile and preputial mucosae.

A fatal generalized disease with fever, oculonasal discharge, respiratory distress, diarrhoea, incoordination and convulsions has been described in young calves.

Diagnosis

- Swabs collected from the nares and genitalia of several affected animals during the early acute phase of the disease are suitable for virus isolation. Because the virus is fragile, specimens for transport to the laboratory should be placed in viral transport medium and kept refrigerated. The virus produces a rapid cytopathic effect in bovine cell lines.
- Smears from nasal or genital swabs and frozen sections of tissues from aborted foetuses can be used for the rapid demonstration of viral antigen using immunofluorescence. Viral antigen can also be detected using ELISA.
- The presence of characteristic gross and microscopic lesions in aborted foetuses is suggestive of infection with BHV-1.
- The polymerase chain reaction has been adapted for detection of BHV-1 DNA in appropriate samples.
- The evidence of a rising antibody titre in paired serum samples using virus neutralization or ELISA is

Table 54.4 Herpesvirus infections of domestic carnivores.

Virus	Genus	Comments
Canine herpesvirus 1	*Varicellovirus*	Causes a fatal generalized infection in neonatal pups
Feline herpesvirus 1	*Varicellovirus*	Causes feline viral rhinotracheitis in young cats

indicative of active infection.

- As part of a surveillance programme, bulk milk samples can be tested for antibodies using ELISA.

Control

Inactivated, subunit and modified live vaccines are available for the control of BHV-1 (van Oirschot *et al.*, 1996). Vaccination reduces the severity of clinical signs but may not prevent infection. Modified live vaccines may cause abortion and should not be administered to pregnant animals. Vaccine strains, which are temperature-sensitive, do not replicate at temperatures above 37°C and should be administered by the intranasal route. Appropriate serological tests permit differentiation of vaccinated animals from those infected with field virus. Marker vaccines, based either on genetically engineered viruses lacking one or more genes encoding surface glycoproteins or on subunit preparations containing one or more glycoproteins, are available. A serological test, usually a blocking ELISA, is used to detect antibodies against a glycoprotein not present in the marker vaccine. The availability of marker vaccines enables either eradication or reduction of infection in national herds to proceed economically. Successful eradication programmes based on test and slaughter policies have been carried out in Denmark and Switzerland.

Table 54.5 Herpesvirus infections of birds.

Virus	Genus	Comments
Gallid herpesvirus 1	Infectious laryngotracheitis-like viruses	Causes infectious laryngo-tracheitis. Present in many countries
Gallid herpesvirus 2 (Marek's disease virus)	Marek's disease-like viruses	Causes Marek's disease, a lymphoproliferative condition in 12 to 24 week old chickens. Occurs worldwide
Anatid herpesvirus 1	Unassigned	Causes acute disease in ducks, geese and swans characterized by oculonasal discharge, diarrhoea and high mortality. Occurs worldwide

Bovine herpes mammillitis and pseudo-lumpy-skin disease

Infection with bovine herpesvirus 2 (BHV-2) is associated with outbreaks of a severe ulcerative condition of the teats of dairy cows. The condition has been reported in many countries around the world. Bovine herpesvirus 2 infection resulting in a generalized mild skin infection, termed pseudo-lumpy-skin disease to distinguish it from the more serious lumpy skin disease caused by a poxvirus, has been reported in tropical and subtropical regions.

Epidemiology

In temperate regions, outbreaks of herpes mammillitis are sporadic and usually occur in autumn or early winter. Latent infection and subsequent reactivation are important factors in the spread and perpetuation of infection within herds. In cows calving for the first time, lesions appear a few days after parturition Serous exudate *from lesions* contains large quantities of virus and transmission to other cows in the herd occurs through direct and indirect contact during milking. Infection occurs through small abrasions in the skin. Insects may transmit the virus from animal to animal mechanically. Calves suckling affected cows can become infected and may transmit the virus. In Africa, a wide range of wild animal species appear to act as subclinical reservoirs of infection. Insect transmission is considered to be important in warm climates and may account for the occurrence of the generalized skin form of the disease in those regions.

Pathogenesis

The virus replicates optimally at a temperature lower than normal body temperature. Following intradermal or subcutaneous inoculation, BHV-2 replicates without dissemination to other sites. In contrast, generalized infection with widespread skin nodules develops in experimental animals following intravenous inoculation of BHV-2.

Clinical signs

The number of animals displaying clinical signs during outbreaks is variable and subclinical infection is common. First-lactation cows which have recently calved, particularly animals with udder oedema, are most severely affected. The incubation period is up to eight days. Lesions appear as thickened plaques on one or more teats. Ulceration of the skin leads to scar formation. The lesions are painful and there is a reduction in milk yield due to difficulty with milking. In severe cases, lesions may also appear on the skin of the udder. Circular ulcers may be present on the lips, nostrils or muzzle of calves suckling affected cows. In pseudo-lumpy-skin disease, a variable number of nodules appear on the skin over the neck, shoulders, back and perineum. The nodules, which are circular and hard, have depressed centres. They heal without scar formation within a couple of weeks.

Diagnosis

Diagnosis is based on demonstration of virus in scrapings or in vesicular fluid by direct electron microscopy or by virus isolation in tissue culture. The optimal incubation temperature for inoculated cell cultures is 32°C.

Control

Commercial vaccines are not available. Affected animals should be isolated and milked separately. Teat dipping and disinfection of milking machine clusters between cows is advisable. The most susceptible animals, first-lactation cows, should be milked first. Insect control measures may help to limit the spread of disease within a herd.

Malignant catarrhal fever

This severe, sporadic disease of cattle, deer and other ruminants is frequently fatal. Malignant catarrhal fever (MCF) is caused by two related but distinct viruses, alcelaphine herpesvirus 1 (ALV-1) and ovine herpesvirus 2 (OHV-2). As the wildebeest is the natural host of ALV-1, infection with this virus is confined to Africa. Sheep are the natural hosts of OHV-2 and infection occurs worldwide in sheep and goats. In these species infection is common and subclinical.

Epidemiology

Alcelaphine herpesvirus 1 is transmitted vertically and horizontally in wildebeest populations. There is evidence that latency occurs in lymphoid cells. Some wildebeest calves are infected transplacentally but most acquire infection shortly after birth through nasal secretions from their mothers or from other calves. Viraemia persists in young wildebeest for the first few months of life. This facilitates shedding of large quantities of virus in nasal and ocular secretions. In-contact cattle may become infected. The pattern of virus shedding is thought to be similar in sheep especially at lambing, when transmission to cattle and farmed deer is most likely to occur. Infection in cattle and deer is thought to be acquired through contact with young lambs. Some studies suggest that most lambs do not become infected before seven months of age (Li *et al.*, 1998). Cattle and deer are considered to be 'end-hosts' because they do not appear to transmit virus.

Pathogenesis

The pathogenesis of MCF is poorly understood. It is presumed that virus enters the body through the upper respiratory tract. A cell-associated viraemia occurs. However, virus is virtually absent from the sites of lesions and it is thought that tissue changes in MCF have an immunopathological basis. Cell-mediated reactions have been implicated in lesion development. It has been proposed that stimulation of T lymphocytes through overproduction of IL-2 by deregulated natural killer (NK) cells contributes to lesion development.

Clinical signs

The incubation period, although variable, generally lasts three to four weeks. The most common clinical presentation is characterized by sudden onset of fever, oculonasal discharge, enlarged lymph nodes, conjunctivitis, corneal opacity and erosive mucosal lesions in the upper respiratory tract. Profuse mucopurulent nasal discharge leads to encrustation of the muzzle. Some animals display neurological signs including muscle tremors, incoordination and head pressing. An intestinal form of the disease presents with diarrhoea or dysentery.

The course of this usually fatal disease is up to seven days. Some animals may linger for weeks or months and may recover (O'Toole *et al.*, 1997). In peracute disease, particularly in deer, death may occur without premonitory signs.

Diagnosis

Diagnosis is based on clinical presentation along with extensive vasculitis characterized histologically by fibrinoid degeneration and marked lymphoid infiltration. Ulceration of surface epithelia is a prominent feature of MCF. Viral DNA can be detected in circulating leukocytes using PCR (Muller-Doblies *et al.*, 1998). Although a competitive inhibition ELISA has been developed for the detection of serum antibodies to ALV-1 and OHV-2 (Li *et al.*, 1994), it is not as reliable as PCR or histopathological examination. This ELISA test can be used to determine the prevalence of antibodies in sheep populations. Although ALV-1 can be isolated from the buffy coat of animals with wildebeest-associated MCF, ovine herpesvirus 2 has not yet been isolated.

Control

As no vaccine is available, control depends on the separation of susceptible species from reservoir hosts. Identification and elimination of sheep harbouring OHV-2 using the PCR assay in order to establish a virus-free flock may be worthwhile in defined circumstances.

Aujeszky's disease

This disease, also called pseudorabies, is caused by Aujeszky's disease virus (ADV). A single serotype of the virus is recognized. The pig, in which subclinical and latent infections can occur, is the natural host of the virus. Other domestic animals are susceptible and infection in these incidental hosts is usually fatal.

Epidemiology

Infection is endemic in the pig populations of most countries. The disease has been eradicated from Denmark and the United Kingdom. Outbreaks of disease in naive herds can be devastating with rapid spread to pigs of all ages. The virus is shed in oronasal secretions, milk and semen. Transmission usually occurs by nose-to-nose

contact or by aerosols. Transplacental transmission occurs and aborted foetuses are a source of virus. Although the virus is not stable in the environment, it may remain infectious for a few days under suitable conditions. Windborne transmission over distances of a few kilometres has been recorded. Sheep, which are highly susceptible, may acquire infection following direct contact with pigs or when sharing the same airspace. Scavenging carnivores may become infected after eating pig meat; cats are particularly susceptible to infection. Because the incubation period and clinical course in these incidental hosts are short, the opportunity for transmission of virus from these animals is limited.

Pathogenesis

Following infection, the virus replicates in the epithelium of the nasopharynx and tonsils. Virus spreads from these primary sites to regional lymph nodes and to the CNS along axons of the cranial nerves. Virulent strains of ADV produce a brief viraemia and become widely distributed around the body, particularly in the respiratory tract. Virus replication which occurs in alveolar macrophages interferes with their phagocytic function. Transplacental transfer results in generalized infection of foetuses. Infected animals excrete virus for up to three weeks following infection. Latency occurs in a high percentage of infected animals with virus localized in the trigeminal ganglia and tonsils.

Clinical signs

The age and susceptibility of infected pigs and the virulence of the infecting strain influence the severity of the clinical signs. Young pigs are most severely affected; mortality may approach 100% in suckling piglets. In neonatal piglets, the incubation period may be as short as 36 hours compared with five days in older pigs. Neurological signs including incoordination, tremors, paddling and convulsions predominate in young pigs. Affected animals usually die within two days. Mortality is much lower in weaned pigs although neurological and respiratory signs are often present. Fever, weight loss, sneezing, coughing, nasal discharge and dyspnoea may be evident in fatteners. Neurological signs are uncommon in these older animals and they usually recover within a week. Infection in sows early in pregnancy usually results in resorption of foetuses and return to oestrus. Later in pregnancy, infection frequently produces abortion; full-term piglets may be still-born, or weak. In herds with endemic ADV infection, neonatal animals are protected by maternally-derived antibody.

Disease in other domestic animals occurs sporadically and is characterized by neurological signs resembling those of rabies. Intense pruritis ('mad itch') leading to self-mutilation is a feature of the disease, particularly in ruminants. The clinical course is short with most affected animals dying within a few days.

Diagnosis

The history, clinical signs and lesions may suggest ADV infection. Laboratory confirmation is based on virus isolation, detection of viral antigen, serology and histopathological findings.

- Specimens of brain, spleen and lung from acutely affected animals are suitable for virus isolation. Nasal swabs, collected from live animals, may also be used. In the event of a delay between collection and tissue culture inoculation, samples should be refrigerated.
- Cryostat sections of tonsil or brain are suitable for detection of viral antigen by immunofluorescence.
- Serological tests, including virus neutralization, ELISA and latex agglutination, are available for detecting ADV antibodies. In young pigs, maternally-derived antibodies may be present up to the age of four months. Differential ELISA methods have been introduced for the detection of antibodies to the surface glycoproteins gC, gE and gG. These assays are designed to differentiate animals infected with field virus from those vaccinated with a gene-deleted mutant lacking a surface glycoprotein.

Control

If used strategically, vaccination can prevent the development of clinical disease. Modified live, inactivated and gene-deleted vaccines are available. Vaccines have been produced in which the thymidine kinase (TK) gene has been deleted together with a gene encoding one of the non-essential surface glycoproteins. Because endogenous levels of TK are low in neurons, this gene-deleted virus cannot replicate in neurons and as a consequence is significantly reduced in virulence. However, such vaccinal strains may fail to prevent latent infection caused by field virus in the trigeminal ganglia. In addition, the possibility of recombination of field virus with a gene-deleted virus and the subsequent generation of a virulent ADV has been suggested (Maes *et al.*, 1997). Eradication of Aujeszky's disease can be achieved by depopulation, by test and removal, or by segregation of litters. The availability of effective marker vaccines makes eradication feasible and economically worthwhile.

Equine rhinopneumonitis and equine herpesvirus abortion

Equine herpesvirus 1 (EHV-1) and equine herpesvirus 4 (EHV-4), which are endemic in horse populations worldwide, are responsible for outbreaks of respiratory disease in young horses and for abortion. Before 1981, it was considered that a single virus, EHV-1 composed of two subtypes, was responsible for both clinical syndromes. However, restriction endonuclease analysis showed that the subtypes were two distinct viruses. Infection with EHV-1 is associated with respiratory disease, abortion,

fatal generalized disease in neonatal foals and encephalomyelitis. Although infection with EHV-4 is primarily associated with respiratory disease, sporadic abortions have also been attributed to this virus.

Epidemiology

Close contact facilitates transmission of these fragile viruses. Transmission usually occurs by the respiratory route following contact with infected nasal secretions, aborted foetuses, placentae or uterine fluids. Both EHV-1 and EHV-4 can occur as latent infections. Serological studies using a type-specific ELISA have shown that there is a high prevalence of antibody to EHV-4 approaching 100% in some surveys (Gilkerson *et al.*, 1999b). It is thought that episodes of reactivation of latent EHV-4 infection without coincident clinical disease occur in adult horses, resulting in transmission to foals. The prevalence of antibody to EHV-1 is about 30% in adult horses; it is lower in foals (Gilkerson *et al.*, 1999b). It appears that foals become infected with EHV-1 from their dams or from other lactating mares in the group. Foal-to-foal spread may occur both before and after weaning (Gilkerson *et al.*, 1999a). Mares infected with EHV-1 are potential sources of infection even without aborting. When reactivation of infection in a latent carrier mare occurs in a stud farm, exposure of non-immune, in-contact pregnant mares may lead to an abortion storm.

Pathogenesis

These viruses replicate initially in the upper respiratory tract and regional lymph nodes with spread, in some cases, to the lower respiratory tract and lungs. Latent infection of the trigeminal ganglia with both EHV-1 and EHV-4 may occur. Infections with EHV-4 appear to be restricted to the respiratory tract and viraemia is uncommon. In contrast, local replication of EHV-1 may be followed by a cell-associated viraemia which can result in abortion or neurological disease. The virus can spread directly from infected leukocytes to contiguous cells, thus avoiding neutralization by circulating antibody. Equine herpesvirus 1 has a predilection for vascular endothelium. Vasculitis and thrombosis in the placenta along with transplacental infection of the foetus result in abortion. Vasculitis and thrombosis, associated with EHV-1 infection, may be present in the CNS especially in the spinal cord. Neurological changes appear to be related to infection with particular strains of EHV-1.

Clinical signs

Respiratory disease caused by EHV-4 occurs in foals over two months of age, in weanlings and in yearlings. Following an incubation period of two to ten days, there are signs of fever, pharyngitis and serous nasal discharge. Secondary bacterial infection is common, giving rise to mucopurulent nasal discharge, coughing and, in some cases, bronchopneumonia. In the absence of serious secondary infection, recovery usually occurs within two weeks. Respiratory disease associated with EHV-1 is clinically indistinguishable from that caused by EHV-4. Outbreaks of disease caused by EHV-1 are less common. Immunity, following primary respiratory tract infection, lasts only a few months and is restricted to infection with antigenically similar viruses. Multiple infections result in significant cross-protection against the heterologous herpesvirus. Mares which abort following infection with EHV-1 rarely show premonitory signs. Abortion occurs several weeks or months after exposure, usually during the last four months of gestation. Such infected mares rarely abort during subsequent pregnancies and their fertility is unaffected. Infection close to term may result in the birth of an infected foal which usually dies due to interstitial pneumonia and viral damage in other tissues, sometimes complicated by secondary bacterial infection. Although neurological signs associated with EHV-1 infection are relatively uncommon, they may present in several horses during an outbreak of abortion or respiratory disease on a farm. The signs range from slight incoordination to paralysis, recumbency and death.

Diagnosis

- Virus isolation and identification are used routinely for the laboratory confirmation of herpesvirus infection in horses. Nasopharyngeal swabs should be collected during the early stages of respiratory infections and dispatched in suitable transport medium to the laboratory. Viral antigen may be demonstrated in cryostat sections of lung, liver and spleen from aborted foetuses using immunofluorescence.
- The characteristic gross and microscopic lesions, particularly intranuclear inclusions, may be sufficient for confirmation of herpesvirus abortion. Vasculitis is commonly found in the myeloencephalopathy caused by herpesvirus.
- The polymerase chain reaction has been adapted for the detection of viral DNA in clinical specimens.
- Demonstration of a fourfold rise in antibody titre in paired serum samples is useful for confirmation of a recent outbreak. Most serological tests do not distinguish infection with EHV-1 from infection with EHV-4 because of antigenic cross-reactivity. Recent ELISA-type assays employing monoclonal antibody or recombinant glycoprotein G antigens can discriminate between the two viruses.

Control

Effective management practices and vaccination are essential for control. Animals returning from sales, races or other events should be segregated for up to four weeks. On large stud farms, horses should be kept in small, physically-separated groups. It is essential that pregnant mares be segregated in a stress-free environment. Following an outbreak of disease, affected animals should

be isolated. Premises should be disinfected and movement should be restricted until animals on the premises have been clear of the disease for at least one month.

Modified live and inactivated virus vaccines are commercially available. Vaccination with live vaccines is not permitted in some countries. Many vaccine preparations contain both EHV-1 and EHV-4. As vaccination is not considered to be fully protective, frequent boosters are recommended. Vaccination appears to reduce the severity of clinical signs and to decrease the likelihood of abortion.

Equine coital exanthema

This benign venereal disease of horses, caused by equine herpesvirus 3 (EHV-3), is thought to occur worldwide.

Epidemiology

Serological surveys indicate that the prevalence of infection in breeding animals is about 50%. The reported incidence of the disease is much lower, presumably because many infections are subclinical. The principal mode of transmission is venereal but transfer of EHV-3 may also occur through contaminated instruments.

Pathogenesis

Although latent infection with EHV-3 has not been conclusively demonstrated, it is thought that it probably occurs in sacral ganglia and that outbreaks of disease are initiated by reactivation of latent infection. The virus has a tropism for keratinized epithelium and is temperature sensitive with replication restricted at core body temperature. Viraemia and abortion are not associated with EHV-3 infection.

Clinical signs

The incubation period is up to ten days. Lesions on external genitalia appear initially as red papules which develop into vesicles and pustules. The pustules rupture leaving ulcers that may coalesce. Lesions are occasionally encountered on the teats, lips and nares. Secondary bacterial infection is common. In uncomplicated cases, lesions heal within two weeks. On pigmented skin, sites of healed lesions appear as white spots. Infection can affect the fertility of stallions as they may refuse to serve mares when penile lesions are severe.

Diagnosis

Clinical diagnosis is based on the distribution and appearance of the lesions. Electron microscopy of lesion scrapings or virus isolation in tissue culture at 33°C to 35°C can be used to confirm infection. Virus neutralization and ELISA are suitable assays for demonstrating a rising antibody titre in paired serum samples.

Control

Affected horses should be isolated and should not be used for breeding until lesions have completely healed. Disposable gloves should be used for genital examination and equipment should be thoroughly disinfected after use. An effective vaccine is not available.

Canine herpesvirus infection

Infection in domestic and wild *Canidae* caused by canine herpesvirus 1 (CHV-1) is common worldwide. Clinical disease caused by the virus, which is characterized by high mortality following generalized infection in neonatal pups, is uncommon.

Epidemiology

Prevalence rates, based on serological surveys of dogs are 88% in England and 42% in the Netherlands (Reading and Field, 1998; Rijsewijk *et al.*, 1999). Infection usually occurs by the oronasal route following direct contact between infected and susceptible animals. During periods of stress latent infections may be reactivated with shedding of virus. The sites of latency include sensory ganglia (Burr *et al.*, 1996; Miyoshi *et al.*, 1999). Virus is shed in oronasal and vaginal secretions. Newborn pups, which can acquire infection either during parturition or *in utero*, may transmit infection to littermates.

Pathogenesis

Following infection, CHV-1 replicates in the nasal mucosa, pharynx and tonsils. The virus replicates most effectively at temperatures below normal adult body temperature with the result that infection in adults is usually confined to the upper respiratory tract or to the external genitalia. Because the hypothalamic regulatory centre is not fully operational in pups under four weeks of age, they are particularly dependent on ambient temperature and maternal contact for maintenance of normal body temperature. A cell-associated viraemia and widespread viral replication in visceral organs can occur in infected neonatal animals with subnormal body temperatures.

Clinical signs

In adult dogs and pups over four weeks of age, infection is usually asymptomatic. Occasionally, vesicular lesions on external genitalia and mild vaginitis or balanoposthitis may be observed. Primary infection of pregnant bitches may result in abortion, stillbirths and infertility. Pups infected during parturition or shortly after birth develop clinical signs within days. Affected pups stop suckling, show signs of abdominal pain, whine incessantly and die within days. Morbidity and mortality rates in affected litters are high. Bitches whose pups are affected tend to produce healthy litters subsequently. Pups which receive colostral antibodies may become infected without developing clinical signs.

Diagnosis

Diagnostically significant postmortem findings include focal areas of necrosis and haemorrhage particularly, in the kidneys. Intranuclear inclusions are usually present. Virus isolation can be carried out in canine cell lines from fresh specimens of liver, kidney, lung and spleen.

Control

No commercial vaccine is available although administration of inactivated virus to pregnant bitches has been shown to be protective (Poulet *et al.*, 2001). Affected bitches and their litters should be isolated to prevent infection of other whelping bitches. Heating lamps and pads, which raise the body temperature of pups to 39°C, may help to reduce the severity of infection if initiated prior to or at the time of exposure to the virus.

Feline viral rhinotracheitis

This acute upper respiratory tract infection of young cats is caused by feline herpesvirus 1 (FHV-1). The virus, which is distributed worldwide, accounts for about 40% of respiratory infections in cats. Both domestic and wild species of *Felidae* are susceptible.

Epidemiology

Close contact is required for transmission. The prevalence of infection is higher in cats in colonies than in cats reared individually. Virus is shed in oral and oculonasal secretions. Due to its relative lability, the virus survives for short periods in the environment. Most recovered cats are latently infected. Reactivation with virus replication and shedding is particularly associated with periods of stress such as parturition, lactation or change of housing. Several days elapse between exposure to stress and shedding of virus. The kittens of carrier queens may become infected subclinically while protected by maternally-derived antibody. These kittens may become carriers and, as adults, perpetuate the infection. In common with many herpesvirus infections in other species, the trigeminal ganglia are important sites of latency.

Pathogenesis

Initially, FHV-1 replicates in oronasal or conjunctival tissues before infecting the epithelium of the upper respiratory tract. Viraemia and generalized infection does not appear to occur except in animals at the extremes of age and in immunocompromized cats. Secondary bacterial infections, which commonly occur, exacerbate the clinical signs.

Clinical signs

The incubation period is usually short, about 2 days, but may be up to six days. Young cats display signs of acute upper respiratory tract infection including fever, sneezing, inappetence, hypersalivation, conjunctivitis and oculonasal discharge. Crusts form around the eyes, sometimes causing the eyelids to stick together. In more severe disease, pneumonia or ulcerative keratitis may be evident. The mortality rate is low except in young or immunosuppressed animals. Rarely, cats may present with facial and nasal dermatitis, which has been linked to reactivation of latent infection (Hargis and Ginn, 1999).

Diagnosis

Clinical differentiation of feline viral rhinotracheitis from feline calicivirus infection is difficult.

- Virus can be isolated in feline cell lines from oropharyngeal or conjunctival swabs.
- Specific viral antigen can be demonstrated in acetone-fixed nasal and conjunctival smears using immunofluorescence.
- The virus neutralization test on paired serum samples can be used to demonstrate a rising titre for confirmation of diagnosis.

Treatment and control

Treatment is non-specific and supportive. Antibiotics are used to control secondary bacterial infections. The protection provided by vaccination is incomplete as vaccinated cats can become infected but clinical signs tend to be much reduced. Inactivated vaccines are suitable for use in pregnant queens and help to boost levels of maternal antibody available for kittens. Intranasal vaccines may give rise to mild upper respiratory signs. When cats are at low risk of exposure to infection, booster vaccinations at three year intervals may be sufficient (Elston *et al.*, 1998; Scott and Geissinger, 1999). Commercial vaccine preparations also contain feline calicivirus. Recent developments in vaccine production include the use of avirulent deletion mutant virus strains and, in addition, the insertion of the feline calicivirus capsid gene into the modified FHV-1 genome (Gaskell and Willoughby, 1999). Good husbandry practices and disease control procedures should be implemented in catteries in conjunction with regular vaccination schedules to minimize the impact of clinical disease.

Infectious laryngotracheitis

This highly contagious respiratory disease of chickens and sometimes pheasants is caused by gallid herpesvirus 1 (GaHV-1). Infectious laryngotracheitis (ILT) occurs in many countries. Although strains of GaHV-1 vary in virulence, they are antigenically homogeneous. In areas of intensive poultry production, the disease is usually well-controlled by a combination of vaccination and biosecurity. However, the virus tends to persist in small enterprises and specialized chicken flocks as an endemic infection.

Epidemiology
Infection is acquired through aerosols, especially in intensively reared birds. Latency occurs in the trigeminal ganglia and carrier birds may shed virus intermittently after periods of stress such as the onset of laying or when groups of birds are mixed. Indirect transmission from one production unit to another may occur through contaminated fomites.

Pathogenesis
Following inhalation, virus replicates locally in the upper respiratory tract. Spread along sensory nerves results in localization in the trigeminal ganglia.

Clinical signs
The incubation period is up to 12 days. The epidemic form of the disease, caused by virulent strains of GaHV-1, is characterized by coughing, gasping, moist rales, oculonasal discharge, expectoration of bloodstained mucus and head shaking. Mortality may reach 70%. Death is often due to severe obstructive haemorrhagic laryngotracheitis. Mild respiratory signs, conjunctivitis and decreased egg production are features of infection with strains of low virulence.

Diagnosis
In severe outbreaks of ILT, the clinical signs and postmortem findings may be sufficiently characteristic for diagnosis. In outbreaks of the mild form of the disease, laboratory confirmation is necessary. Virus isolation can be carried out on the chorioallantoic membrane of embryonated eggs or in avian cell cultures. Rapid methods of diagnosis include demonstration of herpesvirus particles in tracheal samples by electron microscopy and detection of viral antigen in smears or frozen sections by immunofluorescence. Viral antigen can be detected in tracheal samples by ELISA or AGID. Antibodies to GaHV-1 can be demonstrated by virus neutralization, ELISA or AGID.

Control
Flock management systems and vaccination protocols form the basis of control methods. In broiler flocks, short production cycles and all-in all-out management systems ensure that they remain free of the disease. In layer flocks, vaccination is usually carried out using live vaccines administered by aerosols or in drinking water. Vaccination protects against clinical disease but is ineffective against infection by field virus and the establishment of latency. Genetically engineered vaccines are currently being evaluated.

Marek's disease

This contagious lymphoproliferative disease of chickens is caused by gallid herpesvirus 2 (Marek's disease virus), which is cell-associated and oncogenic. The disease, which is of major economic significance in the poultry industry, occurs worldwide. Herpesviruses of chickens and turkeys in the genus Marek's disease-like viruses can be divided into three serotypes or species. Serotype 1 (Gallid herpesvirus 2) includes all pathogenic strains and the attenuated variants derived from these strains; serotype 2 (Gallid herpesvirus 3) contains avirulent and non-oncogenic strains; the third serotype, meleagrid herpesvirus 1, is an avirulent herpesvirus of turkeys. Serotype 1 strains can be categorized as mildly virulent, virulent and highly virulent.

Epidemiology
Productive replication with release of infective virus occurs only in the epithelium of the feather follicle. Cell-free virus is released from the follicles along with desquamated cells. This dander can remain infective for several months in dust and litter in poultry houses. Infected birds remain carriers for life and their chicks, which are protected initially by maternally-derived antibody, acquire infection within a few weeks, usually by the respiratory route. In addition to the virulence of the infecting strain of herpesvirus, host factors which contribute to the severity of the disease include the sex, age at the time of infection and genotype. Two relevant genetic loci have been identified. One, associated with the major histocompatibility complex allele, influences the immune response to Marek's disease, and the other influences the susceptibility of T lymphocytes to transformation. Female birds are more susceptible to the disease than male birds; the reason for this difference is unknown. Resistance to the development of disease increases with age. Transportation, vaccination, handling and beak trimming are stress factors which increase susceptibility to disease.

Pathogenesis and pathology
Following inhalation, virus replicates locally before transfer, probably within macrophages, to the major lymphoid organs where it causes cytolysis, primarily of B cells. Latent infection occurs in T cells activated by the cytolytic process involving B cells. A persistent cell-associated viraemia results in dissemination of the virus throughout the body. Infection of epithelial cells in feather follicles occurs about two weeks after infection. Cytolysis of these epithelial cells results in shedding of virus particles into the environment. Genetically susceptible chickens are prone to tumour development and immunosuppression, associated with apoptosis of T cells and thymocytes along with downregulation of CD8 molecule expression. Lymphomatous lesions may become evident in these birds from two weeks to several months after infection. Transformation of T cells probably relates to oncogenes in certain serotype 1 strains. Multiple copies of the virus genome are found in transformed cells, both as episomal DNA and as a form integrated into host cell

DNA. Several genomic regions with potential roles in transformation have been identified (Venugopal, 2001) including *meq* (which resembles the *Jun/Fos* family of oncogenic transcription factor proteins), a basic-leucine zipper gene (Calnek, 1998). Defective immune surveillance mechanisms can allow transformed cells to form lymphoid tumours. The peripheral nerves are often affected, exhibiting proliferative (type A), inflammatory (type B) or minor infiltrative (type C) changes. Demyelination, which is found in type A and type B lesions, causes paralysis. The acute form of Marek's disease is characterized by diffuse infiltration of many internal organs with neoplastic lymphoid cells. A surface antigen, expressed on transformed lymphocytes, formerly called Marek's disease tumour-associated antigen (MATSA) is now considered to be merely a marker for activated T cells.

Clinical signs

Birds between 12 and 24 weeks of age are most commonly affected. Clinically, Marek's disease presents as partial or complete paralysis of the legs and wings. The mortality rate rarely exceeds 15% with deaths occurring over a number of weeks or months. In the acute form of the disease, birds are severely depressed before death or may die without clinical evidence of disease. The mortality rate in the acute form of the disease is usually between 10% and 30%; outbreaks with mortality as high as 70% have been reported.

Diagnosis

The diagnosis of Marek's disease is based on clinical signs and pathological findings.

- Paralysis of legs and wings in conjunction with thickening of the peripheral nerves is typical of Marek's disease.
- Nerve involvement is not always evident in adult birds. In these circumstances, differentiation from lymphoid leukosis is particularly important.
- Differentiation from lymphoid leukosis is based on the age of affected birds, the incidence of clinical cases and the histopathological findings.
- Virus can be isolated from the buffy coat of blood samples from infected birds. Chicken kidney cells or duck embryo fibroblasts can be used for isolation of the virus.
- Viral antigen can be detected in preparations of skin or feather tips, using a radial precipitin test.
- Serum antibodies to GaHV-2 may be demonstrated using AGID, ELISA, immunofluorescence or virus neutralization.
- Primers, which can distinguish attenuated and wild-type strains, have been developed for PCR assays.
- In the absence of characteristic clinical signs, infection with gallid herpesvirus 2 is not indicative of Marek's disease in a flock.

Control

The use of appropriate management strategies, genetically resistant stock and vaccination have reduced losses from Marek's disease. Disinfection, all-in all-out policies, and rearing young chicks away from older birds for the first two or three months of life reduce exposure to infection, decreasing the likelihood of serious disease. A range of modified live vaccines containing the three avian herpesvirus serotypes are commercially available. Although a single dose of virus injected into day-old chicks provides good lifelong protection, it does not prevent superinfection with virulent field viruses. A synergistic protective effect is obtained by using two or three strains in a vaccine. Consequently meleagrid herpesvirus 1 is commonly incorporated into bivalent or trivalent vaccines. Novel vaccines, based on recombinant DNA technology are being developed because of a reduction in the efficacy of conventional vaccines due to emergence of virulent mutants (Witter, 1998). Relatively recently the introduction of automated *in ovo* vaccination at the eighteenth day of incubation has replaced conventional vaccination methods in large commercial units (Ricks *et al.*, 1999).

References

Burr P.D., Campbell, M.E.M., Nicolson, L. and Onions, D.E. (1996). Detection of canine herpesvirus 1 in a wide range of tissues using the polymerase chain reaction. *Veterinary Microbiology*, **53**, 227–237.

Calnek, B.W. (1998). Lymphomagenesis in Marek's disease. *Avian Pathology*, **27**, S54–S64.

Elston, T., Rodan, I., Flemming, D., Ford, R.B. *et al.* (1998). Feline vaccine guidelines: from the Advisory Panel on Feline Vaccines. *Feline Practice*, **26**, 14–16.

Gaskell, R. and Willoughby, K. (1999). Herpesviruses of carnivores. *Veterinary Microbiology*, **69**, 73–88.

Gilkerson, J.R., Whalley, J.M., Drummer, H.E. *et al.* (1999a). Epidemiological studies of equine herpesvirus 1 (EHV-1) in thoroughbred foals: a review of studies conducted in the Hunter Valley of New South Wales between 1995 and 1997. *Veterinary Microbiology*, **68**, 15–25.

Gilkerson, J.R., Whalley, J.M., Drummer, H.E. *et al.* (1999b). Epidemiology of EHV-1 and EHV-4 in the mare and foal populations on a Hunter Valley study farm: are mares the source of EHV-1 for unweaned foals. *Veterinary Microbiology*, **68**, 27–34.

Hargis, A.M. and Ginn, P.E. (1999). Feline herpesvirus 1-associated facial and nasal dermatitis and stomatitis in domestic cats. *Veterinary Clinics of North America: Small Animal Practice*, **29**, 1281–1290.

Li, H., Shen, D.T., Knowles, D.P. *et al.* (1994). Competitive inhibition enzyme-linked immunosorbent assay for antibody in sheep and other ruminants to a conserved epitope of malignant catarrhal fever virus. *Journal of Clinical Microbiology*, **32**, 1674–1679.

Chapter 55

Papillomaviridae

Viruses in the family *Papillomaviridae* are non-enveloped with icosahedral capsids which contain a single molecule of circular double-stranded DNA. The family contains a single genus, *Papillomavirus*. Papillomaviruses are 55 nm in diameter (Fig. 55.1). Replication takes place in the nucleus and release of new virions occurs by lysis of the infected cell. Members of the genus are resistant to lipid solvents, acid, and heating at 60°C for 30 minutes. Infections are often persistent and usually established early in life.

Formerly papillomaviruses were grouped with polyomaviruses in the family *Papovaviridae*. The name *Papovaviridae* is derived from the first letters of the names of important members of the family: **pa**pillomaviruses, **po**lyomaviruses and **va**cuolating agent (originally designated simian virus 40). Infections with polyomaviruses are of minor veterinary significance. Although they produce no clinical effects in natural hosts, most polyomaviruses (*polyoma*, many tumours) are oncogenic when inoculated into newborn rodents. Budgerigar fledgling disease, caused by an avian polyomavirus, is characterized by outbreaks of acute generalized infection with high mortality in young budgerigars.

Key points

- Non-enveloped, double-stranded DNA viruses
- Icosahedral symmetry
- Contains one genus, *Papillomavirus:*
  - —Have not been cultured *in vitro*
  - —Cause papillomas and fibropapillomas in domestic animals
  - —Malignant transformation of alimentary and urinary tract papillomas may occur in cattle ingesting bracken fern
  - —Bovine papillomas type 1 and 2 are aetiologically involved in equine sarcoids

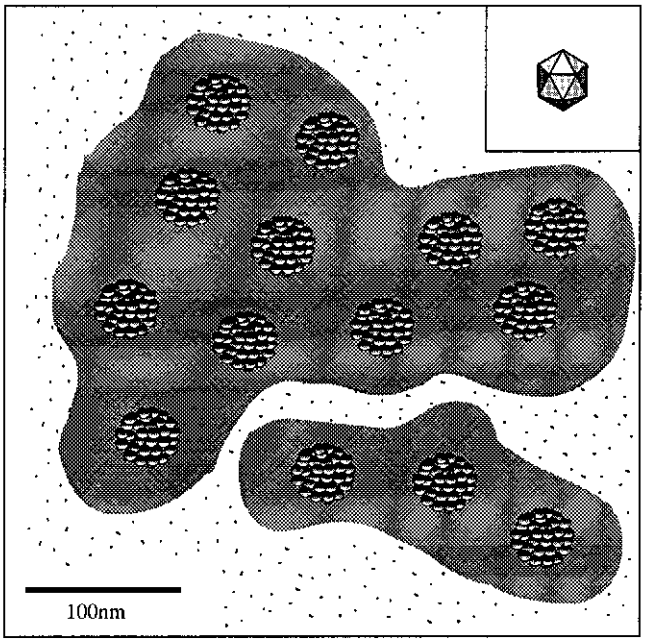


Figure 55.1 Papillomavirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Clinical infections

The epitheliotropic, host-specific papillomaviruses (Latin *papilloma*, nipple) cause proliferative lesions (warts) in many mammalian and avian species. Although they have not been grown in cell culture, the DNA sequences of several papillomaviruses have been elucidated, allowing specific detection in lesions. In infected cells, the viral DNA is usually episomal. Papillomaviruses are used experimentally for inserting foreign DNA into cultured cells.

The clinical conditions associated with papillomavirus infections in domestic animals are presented in Table 55.1. Each papillomavirus tends to be host-specific and to produce proliferative lesions in specific anatomical sites. Although infections with papillomaviruses occur in many animal species, only those which affect cattle, horses and dogs are of clinical significance. Lesions are most commonly observed in young animals and usually regress spontaneously after weeks or months. Regression is attributed to the development of cell-mediated immunity. Typical papillomas are composed of finger-like projections of proliferating epithelium supported by a thin core of

mature fibrous tissue. In fibropapillomas, the fibrous tissue component predominates. In some host species, several types of papillomaviruses can cause neoplastic change. More than 80 types have been identified in humans, while in cattle six types are recognized. Individual types of virus share less than 50% sequence homology and exhibit differences in reciprocal serological assays. The six types of bovine papillomavirus (BPV), types 1, 2 and 5 (group A) and types 3, 4 and 6 (group B), are categorized on the basis of size of genome and the type and location of associated lesions. Progression of papillomas to malignant tumours has been documented in humans, cattle and rabbits.

Pathogenesis

Papillomaviruses usually infect the basal cells of squamous epithelium as a result of minute abrasions. They may also gain entry at vulnerable sites such as junctions between different types of epithelia. Infected cells proliferate and differentiation is delayed. Viral gene expression is restricted during this proliferative phase. Full gene expression results in the production of viral capsids only after cellular differentiation begins in the upper layers of the epithelium. New virions can be visualized by electron microscopy in the nuclei of differentiated keratinized cells. The release of virus occurs during desquamation of infected cells from the surface of lesions.

Diagnosis

- The clinical appearance of papillomas (warts) is distinctive. Laboratory confirmation is not usually required for papillomatous lesions.
- Histopathological examination may be required to determine the nature of some lesions, especially equine sarcoids.
- Electron microscopic examination of specimens from the epidermis may reveal characteristic virus particles.
- Hybridization assays and PCR methods are available for the detection of papillomavirus DNA, but are not used routinely. Isolates can be typed by extraction of DNA and restriction endonuclease analysis or by Southern blotting.

Bovine cutaneous papillomatosis

Bovine cutaneous papillomatosis is caused by several types of bovine papillomavirus. Cutaneous fibropapillomas are generally associated with BPV types 1, 2 or 5, whereas BPV types 3 or 6 are commonly linked to papillomas in which the fibrous tissue component is minimal. The different gross appearances of the two types of proliferative response are exemplified by lesions produced by types of papillomavirus on teats of cows. Teat fibropapillomas, associated with BPV-5 infection, have smooth surfaces and are described as 'rice grain' type. In contrast, 'frond' type teat papillomas arise from infection with BPV-6. The localization of the proliferative lesions caused by these two bovine papillomaviruses on the teats demonstrates the predilection of viruses in this group for particular anatomical sites.

Table 55.1 Papillomaviruses of domestic animals and associated clinical conditions.

Virus	Clinical conditions
Bovine papillomavirus	
Types 1 and 2	Fibropapillomas in young cattle; occur mainly on the head and neck and occasionally on the penis. Implicated in the pathogenesis of equine sarcoids. Type 2 is implicated in bladder neoplasia and enzootic haematuria
Type 3	Cutaneous papillomas with a tendency to persist
Type 4	Papillomas in the alimentary tract; malignant transformation may result from ingestion of bracken fern
Type 5	Fibropapillomas on the teats ('rice grain' type)
Type 6	Papillomas on the teats ('frond' type)
Equine papillomavirus	Papillomas in young horses; occur mainly around lips and on nose
Canine oral papillomavirus	Irregularly-shaped papillomas in the oral cavity of young dogs
Ovine papillomavirus	Papillomas and fibropapillomas (rare)

Fibropapillomas arising from infection with BPV types 1 or 2 are often found on the head and neck of cattle under two years of age. Spontaneous regression of the lesions generally occurs within 1 year. Cutaneous papillomas caused by BPV-3 tend to persist. Because infection with BPV is usually self-limiting, treatment is seldom required. Surgical removal of large lesions on teats may be necessary because of interference with milking. Although inactivated autogenous vaccines are used therapeutically, their efficacy is unproven. Inactivated vaccines can be used prophylactically.

Bovine alimentary papilloma-carcinoma complex

Papillomas of the oesophagus, rumen and reticulum are associated with BPV-4 infection. The lesions, which are often solitary and relatively small, are found incidentally at postmortem examination. Epidemiological and experimental studies have demonstrated that there is an increased frequency in the occurrence of malignant transformation of virus-induced alimentary papillomas to squamous cell

carcinomas when animals are ingesting bracken fern (Jarrett *et al.*, 1978; Campo *et al.*, 1994). Such malignant lesions, which are found in anatomical sites identical to those of the papillomas, may cause difficulty in swallowing, ruminal tympany and loss of condition. Nodular fibropapillomas caused by BPV-2, which are occasionally found in similar upper alimentary tract locations, do not appear to become malignant.

Enzootic haematuria

Enzootic haematuria is encountered worldwide in cattle on poor pastures with abundant bracken fern growth. The haemorrhage originates from tumours in the bladder wall. Individual neoplastic lesions derive from either epithelial or mesenchymal tissues. Experimental studies suggest that BPV-2 and toxic compounds from bracken contribute to oncogenesis (Campo *et al.*, 1992). It is probable that immunosuppression following ingestion of bracken may allow activation of latent BPV-2 in bladder tissues and this effect, together with the action of carcinogens also present in bracken, is responsible for the induction and progression of neoplastic lesions.

Equine papillomatosis

Papillomas are commonly encountered in horses between 1 and 3 years of age. Two types of equine papillomavirus have been identified based on DNA studies. Type 1 is associated with papillomas on the muzzle and legs while type 2 is associated with papillomas of the genital tract. Spread may occur by direct or indirect contact. The lesions usually regress spontaneously after several months and recovered animals are immune to reinfection.

Equine sarcoid

The equine sarcoid, a locally invasive fibroblastic skin tumour, is the most common neoplasm of horses, donkeys and mules (Marti *et al.*, 1993). Bovine papillomavirus types 1 and 2 or closely related viruses are implicated in sarcoid development. Experimental inoculation with these viruses results in fibromatous lesions which resemble sarcoids but which regress spontaneously. Viral DNA with a high degree of homology to BPV has been identified in tissue from sarcoids using both *in situ* hybridization (Lory *et al.*, 1993) and PCR (Otten *et al.*, 1993).

Lesions usually develop in horses between three and six years of age. Multiple cases can occur in families or groups of horses in close proximity. However, the incidence of equine sarcoid (estimated at 0.5 to 2%) is comparatively low for a viral disease. This may indicate that the horse is a non-permissive host (Marti *et al.*, 1993).

Sarcoids can occur on any part of the body, either singly or in clusters. The most commonly affected sites are the head, ventral abdomen and limbs. They are highly variable in appearance but can be arbitrarily categorized as verrucous or fibroblastic. Clinical diagnosis should be confirmed histologically. Surgical removal is the usual form of treatment. Recurrence is common following conventional surgery and cryosurgery is more successful. Radiation therapy, CO_2 laser surgery and chemotherapy have also been used with varying degrees of success (Knottenbelt *et al.*, 1995). Immunotherapy, aimed at stimulating cell-mediated immunity, may be effective in some cases. This involves intralesional injection of BCG or cell wall extract of *Mycobacterium bovis* into horses previously sensitized to tuberculoprotein.

Canine oral papillomatosis

Multiple transmissible papillomas in the oropharyngeal region of dogs are often encountered. The disease, which is caused by canine oral papillomavirus, is common in young dogs and is readily transmitted. While the aetiology of this oral condition is well established, the cause of papillomas occurring at other sites in the dog is uncertain (Narama *et al.*, 1992).

Canine oral papillomavirus is transmitted by direct and indirect contact. The incubation period is up to 8 weeks. Lesions are usually multiple and although generally confined to the oral mucosa, are sometimes found on the conjunctiva, eyelids and muzzle. The papillomas initially appear as smooth, white, raised lesions but later they become rough and cauliflower-like. Spread may occur inside the oral cavity. There is spontaneous regression within months. Surgical removal is generally unnecessary unless the papillomas persist or cause physical discomfort. Inactivated vaccines have been used but do not appear to be effective. Live, unattenuated vaccines, which are effective, may produce neoplastic lesions at the injection site (Bregman *et al.*, 1987).

References

Bregman, C.L., Hirth, R.S., Sundberg, J.P. and Christensen, E.F. (1987). Cutaneous neoplasms in dogs associated with canine oral papillomavirus vaccine. *Veterinary Pathology*, **24**, 477–487.

Campo, M.S., Jarrett, W.F.H., Farron, R., O'Neill, B.W. and Smith, K.T. (1992). Association of bovine papillomavirus type 2 and bracken fern with bladder cancer in cattle. *Cancer Research*, **52**, 6898–6904.

Campo, M.S., O'Neill, B.W., Barron, R.J. and Jarrett, W.F.H. (1994). Experimental reproduction of the papilloma-carcinoma complex of the alimentary canal in cattle. *Carcinogenesis*, **15**, 1597–1601.

Jarrett, W.F.H., McNeill, P.E., Grimshaw, W.T.R., *et al.* (1978). High incidence area of cattle cancer with a possible interaction between an environmental carcinogen and a papilloma virus. *Nature*, **274**, 215–217.

Chapter 56

Adenoviridae

Adenoviruses (Greek *adenos*, gland), first isolated from explant cultures of human adenoids, are icosahedral (70 to 90 nm in diameter) and contain a single linear molecule of double-stranded DNA. Fibres project from each of the twelve vertices of the capsid (Fig. 56.1). Agglutination of rat or monkey erythrocytes, a property of many adenoviruses, is dependent on the fibre proteins which possess type-specific immunodeterminants. The family *Adenoviridae* is composed of two genera, *Mastadenovirus* and *Aviadenovirus*, and a group of unassigned viruses. Mammalian adenoviruses, assigned to the genus *Mastadenovirus*, infect mammals only, share a common antigen and are serologically distinct from those that infect birds. Serogroups and serotypes are defined on the basis of neutralization assays. Haemagglutination inhibition is used to confirm serospecificity. Adenoviruses are moderately stable in the environment in which they can survive for many weeks. They can withstand freezing, mild acid and lipid solvents. Infectivity is abolished by heating at 56°C for more than 10 minutes.

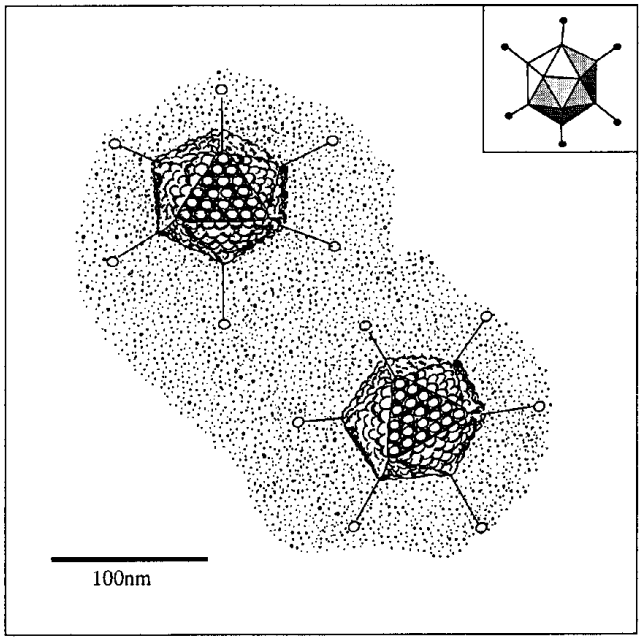


Figure 56.1 Adenovirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Key points

- Non-enveloped, double-stranded DNA viruses,
- Icosahedral symmetry
- Replicate in nuclei, forming intranuclear inclusion bodies
- Moderately stable in the environment
- Two genera:
  —*Aviadenovirus*, avian adenoviruses
  —*Mastadenovirus*, mammalian adenoviruses
- Systemic and respiratory diseases in dogs
- Systemic diseases in poultry

Adenoviruses replicate in cell nuclei. Newly assembled virions form crystalline aggregates, demonstrable as intranuclear basophilic inclusions in stained tissue sections. Adenoviruses have a natural host range generally confined to a single species or to closely related species. Infection is common in animals and man. Fifty one human serotypes have been recognized, grouped into six species (human adenovirus A to F). Most human infections appear to be subclinical or mild although immunodeficient individuals may develop severe clinical disease. In contrast to human infections, certain animal infections result in severe illness. Genera and species belonging to the *Adenoviridae* are listed in Fig. 56.2 (adapted from van Regenmortel *et al.*, 2000). Adenoviruses of veterinary importance are presented in Table 56.1.

Clinical infections

Adenovirus infections can be particularly severe in dogs and domestic fowl. Two serotypes of canine adenovirus are recognised. Infection with canine adenovirus type 1 (CAV-1) causes infectious canine hepatitis, a severe generalized disease, whereas infection with canine adenovirus type 2 (CAV-2) is commonly linked to localized respiratory disease. In other domestic mammals, adenovirus infections are associated occasionally with enteric or respiratory problems. Pulmonary infection with equine adenovirus A is invariably fatal in Arabian foals with combined immunodeficiency disease.

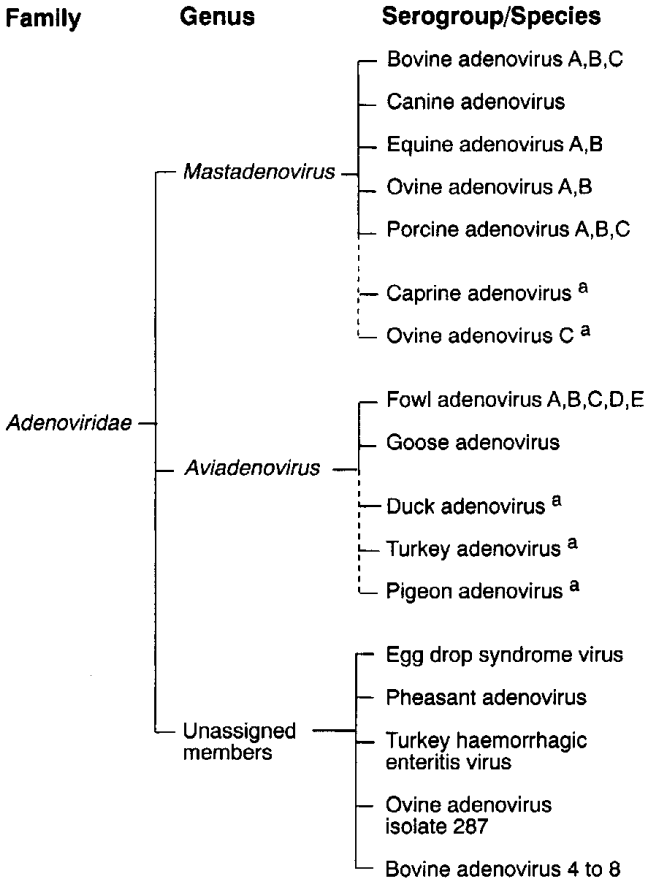


Figure 56.2 Genera and species of veterinary importance in the *Adenoviridae*. The species name is complemented by a letter if more than one species exist in the same host. Viruses currently unassigned to a genus are included.

a species tentatively assigned to genus

Avian adenoviruses are distributed worldwide and infection is extremely common in poultry flocks. Most of these infections are either subclinical or associated with relatively mild disease. However, severe disease may follow infection with certain avian adenoviruses, egg-drop syndrome virus and turkey haemorrhagic enteritis virus. Fowl adenovirus A, also known as chicken embryo lethal orphan virus (CELO virus, fowl adenovirus 1), is the type species for avian adenoviruses and causes quail bronchitis.

Infectious canine hepatitis

This worldwide, generalized viral disease of dogs, caused by CAV-1, principally affects the liver and vascular endothelium. The complete DNA sequence of CAV-1 has been determined (Morrison *et al.*, 1997). Infectious canine hepatitis has become relatively uncommon because of the widespread use of effective vaccines. Although dogs are the most commonly affected species, foxes, wolves, coyotes, skunks and bears are also susceptible. Transmission can occur following ingestion of urine, faeces or saliva from infected animals. The immune response usually eliminates virus from host tissues by 14 days after initial infection. However, virus may persist in the kidneys and, in some instances may be excreted in urine for more than six months.

Pathogenesis

Following ingestion, CAV-1 localizes in the tonsils and Peyer's patches (Fig. 56.3). As viraemia develops, replication in vascular endothelium results in rapid distribution of virus throughout the body. Virus replication also occurs in the parenchymal cells of the liver and kidneys. Clinical recovery in most dogs coincides with the production of neutralizing antibodies at about ten days after infection. Glomerulonephritis, corneal oedema and anterior uveitis, attributable to immune complex deposition, may develop in some infected animals.

Clinical signs

The incubation period is up to seven days. Dogs of all

Table 56.1 Adenoviruses of veterinary importance.

Virus	Comments
Canine adenovirus 1	Causes infectious canine hepatitis, with lesions arising from direct cytopathic effects and immune complex formation
Canine adenovirus 2	Involved in infectious tracheobronchitis (kennel cough), a highly contagious respiratory disease
Equine adenovirus A	Usually a subclinical or mild respiratory infection; associated with pneumonia in Arabian foals with combined immunodeficiency disease
Bovine adenoviruses	Associated with occasional outbreaks of respiratory and enteric disease
Ovine adenoviruses	Associated with occasional outbreaks of respiratory and enteric disease
Porcine adenoviruses	Usually subclinical infections, occasionally cause diarrhoea
Fowl adenoviruses	Frequently isolated from healthy birds or following respiratory disease. Associated with quail bronchitis and inclusion body hepatitis
Egg drop syndrome virus	Causes egg drop syndrome in laying hens
Pheasant adenovirus	Causes marble spleen disease with sudden deaths in 2-8 month old birds; acute pulmonary oedema and splenic necrosis
Turkey haemorrhagic enteritis virus	Causes turkey haemorrhagic enteritis; dysentery in 4-12 week-old poults; mortality rate up to 60%

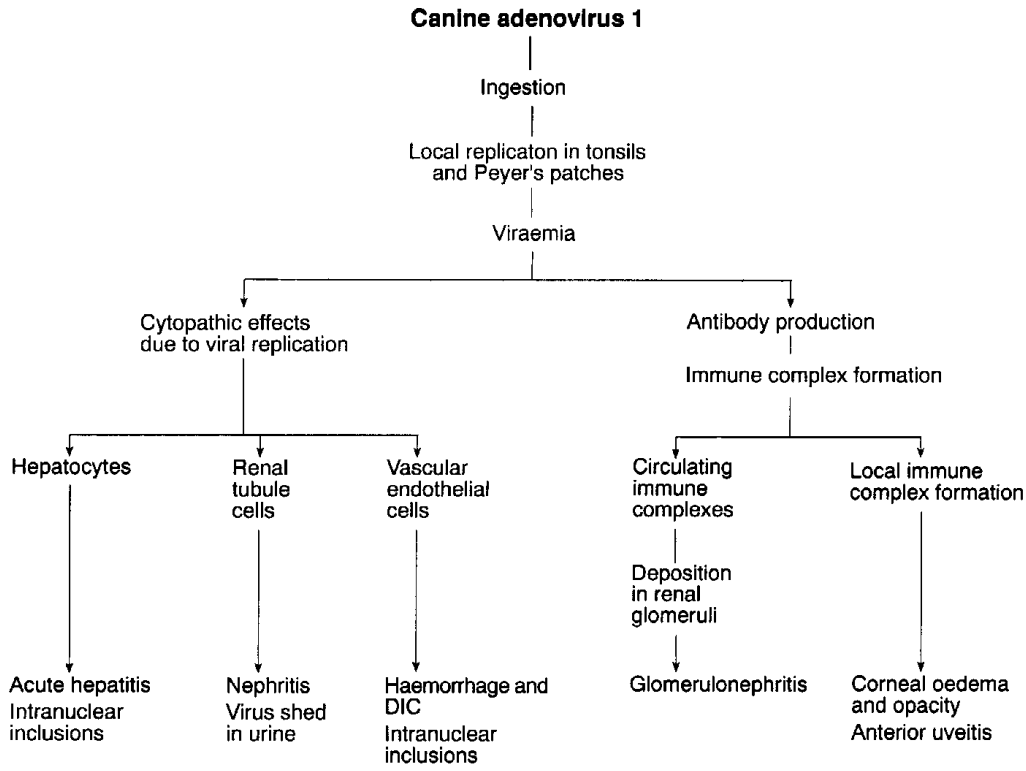


Figure 56.3 Mechanisms involved in the development of localized and generalized lesions following infection with canine adenovirus 1.

DIC: disseminated intravascular coagulation

ages are susceptible and subclinical infection is common. Clinical disease is most frequently encountered in young dogs. The mortality rate ranges from 10 to 30% in mature dogs and up to 100% in young pups. In peracute disease, death occurs so rapidly that poisoning may be suspected. In acute disease, affected dogs present with fever, depression, anorexia, increased thirst, vomiting and diarrhoea. Abdominal palpation may elicit pain and, although hepatomegaly may be detected, jaundice is uncommon. Corneal opacity, either unilateral or bilateral, which may occur within weeks of clinical recovery in about 20% of affected animals, usually resolves spontaneously. Recovered animals have life-long immunity.

Diagnosis

- A history of fever, sudden collapse and abdominal pain in young, unvaccinated dogs may suggest infectious canine hepatitis.
- Haematological examination may reveal a marked reduction in neutrophils and lymphocytes during the febrile stage. Clotting time may be prolonged.
- Basophilic intranuclear inclusion bodies in hepatocytes, Kupffer cells and endothelial cells are confirmatory.
- Viral antigen can be demonstrated by immunofluorescence in cryostat sections of liver.
- A PCR method for detecting viral DNA in clinical specimens has been described (Kiss *et al.*, 1996).
- Suitable specimens for virus isolation in canine kidney cells include oropharyngeal swabs, blood, urine and faeces from affected animals sampled during the febrile stage of the disease. Samples of spleen, lymph nodes and kidney obtained at postmortem are also suitable for virus isolation. Because of high levels of arginase activity, liver samples are unsuitable as this enzyme inhibits viral replication in tissue culture.
- A rising antibody titre, using virus neutralization or haemagglutination inhibition tests, is indicative of active CAV-1 infection.

Treatment and control

Supportive treatment may allow time for hepatocellular regeneration. Fluids should be administered intravenously to counteract dehydration and control shock. In severely ill animals, blood transfusions may be required to prevent or control haemorrhage.

Inactivated and modified live vaccines are available. Occasionally, vaccination with modified live CAV-1 vaccines results in mild nephropathy with shedding of virus in urine and, in some instances, corneal opacity. These side effects do not occur with modified live CAV-2 vaccines which stimulate effective long lasting immunity to CAV-1. Pups with unknown levels of maternally-derived antibody should be vaccinated twice, at 8 to 10

weeks and at 12 to 14 weeks of age. Booster injections, either annually or at intervals of two years, are recommended. Inactivated CAV-1 vaccines, which do not induce obvious side effects, require boosters at more frequent intervals in order to maintain adequate antibody levels.

Infection with canine adenovirus type 2

Canine adenovirus type 2, which is readily transmitted by aerosols, replicates in both upper and lower respiratory tract. Clinical signs are generally mild or inapparent. Affected dogs may present with clinical signs similar to those of canine infectious tracheobronchitis (kennel cough). Most dogs recover and are immune to subsequent challenge. Occasional cases of bronchopneumonia may develop due to secondary bacterial infection. Virus shedding continues for about 9 days after infection.

Inclusion body hepatitis

Although inclusion body hepatitis (IBH) mainly affects broilers, it sometimes occurs in pullets. There is uncertainty about the cause of IBH and a number of fowl adenovirus serotypes have been aetiologically implicated. In affected flocks, a sudden increase in mortality is a feature of the disease and it may reach 30% if there is immunosuppression due to infectious bursal disease or chicken anaemia virus infection. Lesions include an enlarged liver with scattered haemorrhages and necrosis, intramuscular haemorrhage and anaemia. Intranuclear inclusions in hepatocytes are prominent. Diagnosis is based on characteristic hepatic lesions. Because apparently healthy birds excrete adenoviruses and may be serologically positive, the significance of positive serological tests is questionable. As vaccines are not readily available and the aetiology of IBH is uncertain, specific control measures cannot be recommended.

Egg drop syndrome

Egg drop syndrome caused by an adenovirus prevalent in ducks and probably introduced into chickens through a contaminated vaccine, was first described in 1976. In flocks the disease is characterized by a drop in egg production or by failure to reach peak production. Infected hens may lay abnormal eggs. Laying hens up to 36 weeks of age are those most commonly affected. Inflammatory lesions are found in the oviduct, particularly the pouch shell gland. Intranuclear inclusions may be present in the epithelial cells of this gland. Samples of oviduct, including material from the pouch shell gland, are suitable specimens for virus isolation in avian cell lines, especially duck kidney or fibroblast cell lines. As the virus agglutinates avian red cells, haemagglutination inhibition is the preferred method for serological screening of flocks. Control of egg drop syndrome relies on the use of inactivated vaccines before laying begins. Appropriate hygiene combined with disinfection can be used to limit the spread of infection. Because of the high risk of cross-infection, chickens and ducks should be housed separately.

References

Kiss, I., Matiz, K., Bajmoci, E., Rusvai, M. and Harrach, B. (1996). Infectious canine hepatitis: detection of canine adenovirus type 1 by polymerase chain reaction. *Acta Veterinaria Hungarica*, **44**, 253–258.

Morrison, M.D, Onions, D.E. and Nicolson, L. (1997). Complete DNA sequence of canine adenovirus type 1. *Journal of General Virology*, **78**, 873–878.

van Regenmortel, M.H.V., Fauquet, C.M., Bishop, D.H.L., *et al.*, (2000). *Virus Taxonony, Seventh Report of the International Committee on Taxonomy of Viruses.* Academic Press, San Diego.

Further reading

McCracken, R.M. and Adair, B.M. (1993). Avian adenoviruses. In *Virus Infections of Birds.* Eds. J.B. McFerran and M.S. McNulty. Elsevier Science Publishers, Amsterdam. pp. 121–144.

Chapter 57

Poxviridae

The family *Poxviridae* contains the largest viruses which cause disease in domestic animals. Poxvirus symmetry is complex. The virions in this family are either brick-shaped (220 to 450 nm x 140 to 260 nm) with a surface membrane composed of tubular or globular proteins or ovoid (250 to 300 nm x 160 to 190 nm) with a surface membrane composed of a regular spiral filament (Figs. 57.1 and 57.2). They contain more than 100 proteins including several virus-encoded enzymes. A biconcave core or nucleoid contains linear double-stranded DNA and one or two lateral bodies within a surface membrane (Fig. 57.3). A cell-derived envelope encloses some of the mature extracellular virions.

The family is divided into two subfamilies, *Chordopoxvirinae*, the poxviruses of vertebrates, and *Entomopoxvirinae*, the poxviruses of insects (Fig. 57.4). The subfamily *Chordopoxvirinae* is comprised of eight genera, namely *Orthopoxvirus*, *Parapoxvirus*, *Avipoxvirus*, *Capripoxvirus*, *Leporipoxvirus*, *Suipoxvirus*, *Molluscipoxvirus* and *Yatapoxvirus*. Genetic recombination within genera results in extensive serological cross-reactions and cross-protection. Replication in the cytoplasm of host cells takes place within defined areas (viral factories). Virus envelopes, derived from host cell membranes, contain host cell lipids and virus-encoded proteins such as the haemagglutinin protein of the orthopoxviruses. Both enveloped and non-enveloped forms of the virus are infectious. Virions are stable at room temperature under dry conditions but sensitive to heat, detergents, formaldehyde and oxidizing agents. The genera differ in ether sensitivity.

Key points

- Enveloped DNA viruses
- Complex symmetry
- Replicate in cytoplasm
- Stable in the environment
- Skin lesions a prominent feature
- Individual poxviruses tend to infect particular host species; some poxviruses are not species-specific

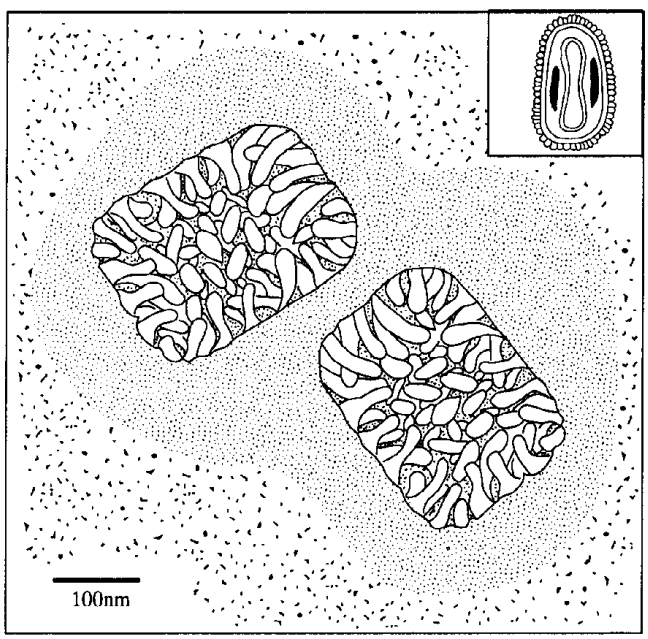


Figure 57.1 Orthopoxvirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Infections with poxviruses, which can affect many vertebrate and invertebrate hosts, usually result in vesicular skin lesions (Table 57.1). Smallpox, caused by variola virus, was formerly a human disease of major international significance. The use of vaccinia virus for the prevention of smallpox, first introduced by Jenner in the late 18th century, eventually led to the eradication of this highly contagious disease at the close of the twentieth century.

Clinical infections

Transmission of poxviruses can occur by aerosols, by direct contact, by mechanical transmission through arthropods and through fomites. Skin lesions are the principal feature of these infections. Several virus-encoded proteins are released from infected cells including a homologue of epidermal growth factor which stimulates cell proliferation. Typically, pox lesions begin as macules and progress through papules, vesicles and pustules to scabs which detach leaving a scar. In generalized infections there is a cell-associated viraemia and recovered

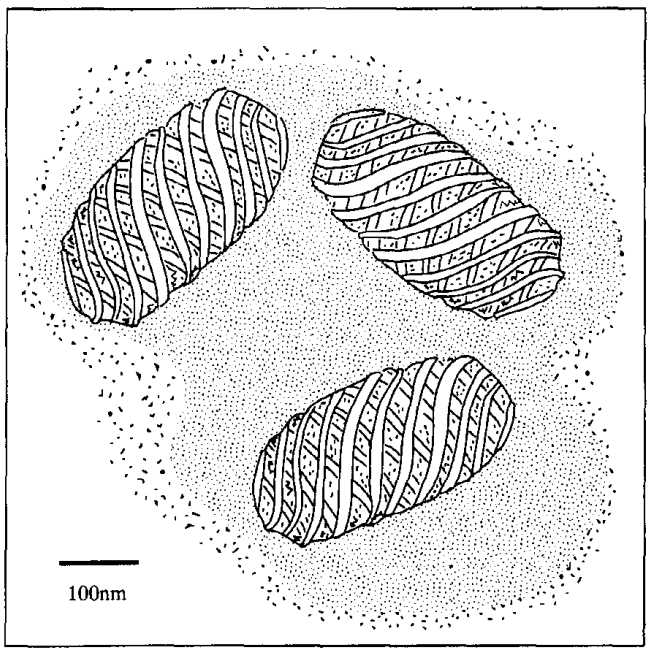


Figure 57.2 Parapoxvirus particles as they appear in an electron micrograph.

animals have solid immunity. Some localized pox infections may induce transient immunity and reinfection can occur.

Three closely related parapoxviruses, namely pseudocowpox virus, bovine papular stomatitis virus and orf virus, infect ruminants. These viruses are transmissible to humans producing lesions which are clinically similar. Moreover, the three viruses are morphologically indistinguishable and identification of the causal agent relies on nucleic acid analysis.

Capripoxviruses are economically important viruses producing generalized infections with significant mortality in domestic ruminants. Sheeppox virus, goatpox virus and lumpy skin disease virus are closely related and share a group-specific structural protein (p32), which allows the same vaccine to be used against each virus.

Many avian species are susceptible to infection with members of the genus *Avipoxvirus*. Although antigenic relationships exist among avian poxviruses, this relatedness is variable. Virus species within the genus, named in accordance with their affinity for particular host species, include fowlpox virus, canarypox virus, pigeonpox virus and turkeypox virus. The type species of the genus is fowlpox virus.

Infections caused by vaccinia virus

Although the natural host species of vaccinia virus is unknown, mild infections have been described in a wide range of species including sheep, cattle, horses and humans. Formerly, the virus was used to vaccinate against smallpox. Buffalopox virus and rabbitpox virus are considered to be subspecies of vaccinia virus. Spread of vaccinia virus among cattle and transmission to humans occurs at milking. The lesions on the teats of cows resemble those caused by cowpox virus. A clinical condition similar to horsepox or equine papular dermatitis can be produced experimentally using vaccinia virus (Studdert, 1989). In recent years, vaccinia virus has been used as a recombinant virus vector for vaccination against several diseases including rabies, canine distemper and measles.

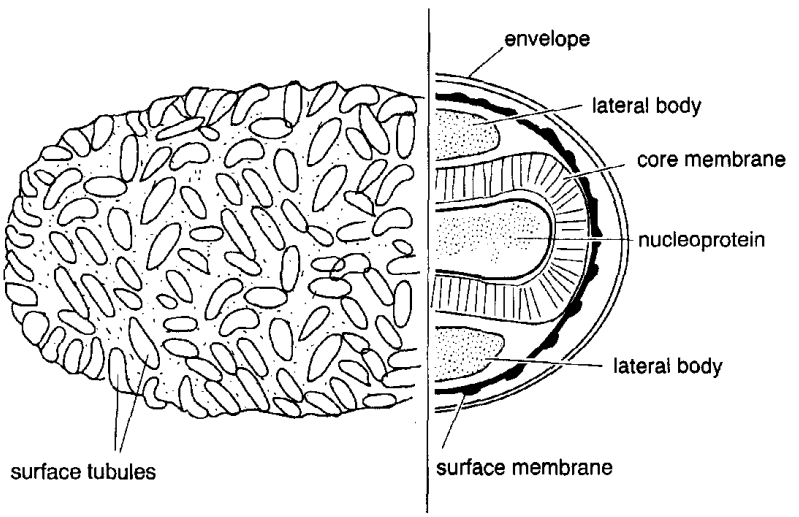


Figure 57.3 Diagrammatic representation of an orthopoxvirus displaying the surface structure of an unenveloped virion (left) and a cross-section of an enveloped virion (right).

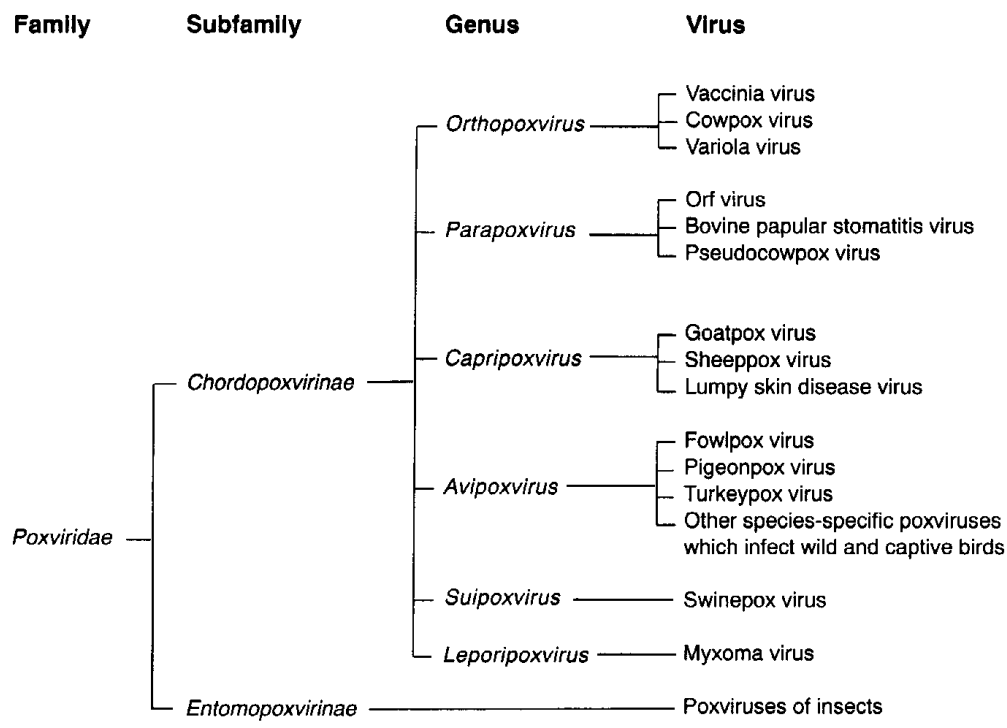


Figure 57.4 Classification of poxviruses with emphasis on those which affect domestic animals.

Infections caused by cowpox virus

Cowpox is endemic in parts of Europe. Although infection and disease have been described in cattle, cats, humans and a range of mammals in zoos, these species are considered to be incidental hosts. The reservoir hosts are probably wild rodents. There is evidence that voles and woodmice are the principal reservoir hosts in western Europe (Bennett *et al.*, 1997). In milking cows, lesions are usually confined to the teats. The domestic species in which the disease is most often recognized is the cat. Affected cats usually come from rural areas and are described as good hunters. Moreover, infections in cats tend to peak in the autumn when small rodent populations are high, suggesting rodent-derived infections. Although cat-to-cat transfer occurs, this mode of transmission is rare. Small papules on the head or forelimbs, the first recognizable signs of the infection, eventually ulcerate. Scab formation is usually followed by complete resolution in about six weeks. Secondary skin lesions develop in some cats and a few animals may show signs of coryza or conjunctivitis. Rarely, pneumonia and pleural exudation have been described. Diagnosis can be confirmed by histopathology, electronmicroscopy or virus isolation. No specific control measures are warranted. Human infections, which are usually contracted from infected cats, are uncommon.

Infections caused by pseudocowpox virus

Pseudocowpox, also known as milker's nodule, is caused by pseudocowpox virus, a parapoxvirus with worldwide distribution. It is a common mild condition affecting the teats of lactating cows. Infection spreads slowly through milking herds with variation in the number of affected animals at any particular time. Transmission is by direct or indirect contact. Transfer of infection can occur through teat cups and on milkers' hands. Transmission can also occur mechanically by flies or when calves are being suckled. Small red papules on teats or udder develop into ulcers with overlying scab formation. Healing at the centre of the lesions produces characteristic ring- or horseshoe-shaped scabs. Typical parapoxvirus particles can be demonstrated in scab material using electron microscopy. Control is based on appropriate hygienic measures at milking including the use of effective teat dips. In human infections, lesions are usually confined to the hand, forearm or face.

Bovine papular stomatitis

This mild viral disease of young cattle occurs worldwide. It is caused by a parapoxvirus, bovine papular stomatitis virus, which is transmitted by direct or indirect contact.

Table 57.1 Members of the *Poxviridae* of veterinary significance.

Virus	Genus	Host species	Significance of infection
Vaccinia virus	*Orthopoxvirus*	Wide host range	Infections in sheep, water buffaloes, rabbits, cattle, horses and humans. Used as a recombinant virus vector for rabies vaccine
Cowpox virus	*Orthopoxvirus*	Rodents, cats, cattle	Small species of rodents are the likely reservoir hosts. Cats are the principal incidental hosts; infection results in skin lesions. Rare cause of teat lesions in cattle. Transmissible to humans
Uasin gishu virus	*Orthopoxvirus*	Unknown wildlife reservoir, horses	Rare disease, reported in Kenya and neighbouring African countries. Causes papilloma-like skin lesions in horses
Camelpox virus	*Orthopoxvirus*	Camel	Widely distributed in Asia and Africa. Causes systemic infection with typical pox lesions; severe infection in young camels
Pseudocowpox virus	*Parapoxvirus*	Cattle	Common cause of teat lesions in milking cows; causes milker's nodule in humans
Bovine papular stomatitis virus	*Parapoxvirus*	Cattle	Produces mild papular lesions on the muzzle and in the oral cavity of young cattle. Transmissible to humans
Orf virus	*Parapoxvirus*	Sheep, goats	Primarily affects young lambs; causes proliferative lesions on the muzzle and lips. Transmissible to humans
Sheeppox / Goatpox virus	*Capripoxvirus*	Sheep,goats	Endemic in Africa, Middle East and India. Causes generalized infection with characteristic skin lesions and variable mortality
Lumpy skin disease virus	*Capripoxvirus*	Cattle	Endemic in Africa. Causes generalized infection with severe lesions and variable mortality
Swinepox virus	*Suipoxvirus*	Pigs	Causes mild, skin disease. Occurs worldwide. Transmitted by the pig louse (*Haematopinus suis*)
Fowlpox virus	*Avipoxvirus*	Chickens, turkeys	Causes lesions on the head and on the oral mucous membrane. Occurs worldwide. Transmitted by biting arthropods
Myxoma virus	*Leporipoxvirus*	Rabbits	Causes mild disease in cottontail rabbits, the natural host, and severe disease in European rabbits. Introduced into Europe, Australia and Chile as a biological control measure

Infection is common and usually subclinical. Mature cattle are considered to be reservoirs of infection.

Affected calves commonly develop lesions in the buccal cavity and on the muzzle. These lesions are characterized by hyperaemic foci that develop into papules with concentric zones of inflammation. Affected animals usually recover within three weeks. A more severe chronic form of the disease, which may be associated with concurrent infections or immunosuppression, has been described (Yeruham *et al.*, 1994). Virions can be demonstrated in skin scrapings by electron microscopy. The virus is transmissible to humans.

Orf

This important disease of sheep, also known as contagious pustular dermatitis or contagious ecthyma, occurs worldwide and is caused by a parapoxvirus. Goats, camels and humans are also susceptible to infection.

Epidemiology

The virus is transmitted through direct or indirect contact. Under dry environmental conditions, it is stable and can survive in scab material for months. Infectivity is substantially reduced after exposure to adverse climatic conditions. In most flocks, infection is maintained by sheep with chronic lesions (McKeever and Reid, 1986).

Pathogenesis

The virus, which is epitheliotropic, produces proliferative wart-like lesions following entry through skin abrasions. The virus replicates in epidermal keratinocytes and infected cells release an endothelial growth factor which is implicated in epithelial cell proliferation (Haig and Mercer, 1998). Papular lesions progress to vesicles, pustules and eventually to scab formation. Proliferation of cells underlying scabs produces verrucose masses. In the absence of secondary bacterial infection, lesions usually heal within four weeks.

Clinical signs

The disease primarily affects young sheep. The incubation period is up to seven days. Although lesions most often occur on the commissures of the lips and on the muzzle, they may also develop in the mouth and on the feet, genitalia and teats. Mild lesions may go unnoticed. Severely affected lambs with lesions in the buccal cavity often fail to eat, lose condition and may die. Outbreaks last for some months and vary in severity from farm to farm and from year to year. The disease does not usually recur until the birth of new susceptible lambs in the flock. Although isolates from individual flocks can differ in genotype, there is no evidence that the severity of the disease correlates with the strain of the virus involved (Gilray *et al.*, 1998). Environmental management factors may influence the outcome of infection (Gumbrell and McGregor, 1997).

Immunity following natural infection may not confer complete protection. However, lesions in animals previously affected are usually less severe and heal more rapidly than those developing after first exposure. In chronically infected sheep, lesions may be either mild or proliferative. Neonatal lambs are susceptible to infection despite receipt of colostrum from previously infected ewes.

Diagnosis

Lesions of orf are readily recognized by their characteristic appearance and distribution. Virus present in lesion material can be identified by electron microscopy.

Treatment and control

There is no specific treatment for infection with orf virus. Antibiotic therapy reduces the effect of secondary bacterial infection in young lambs.

In endemically infected flocks, control is based on the use of a fully virulent live vaccine derived from scab material or cell culture. Ewes should be vaccinated by scarification in the axilla at least eight weeks before lambing. When close to lambing, they must be moved to a new grazing area in order to minimize exposure of lambs to infectious vaccinal scab material. Lambs should be vaccinated only if an outbreak occurs in a flock. If carried out effectively, thorough cleaning and disinfection of surfaces and equipment between periods of housing may reduce the amount of virus in the buildings.

Humans are susceptible to infection with orf virus. Typically, a single lesion occurs on hands, forearm or face. Care should be exercised when handling affected sheep and when using live vaccines.

Sheeppox and goatpox

Both of these diseases are endemic diseases in south eastern Europe, the Middle East, Africa and Asia. The viruses of sheeppox and goatpox are members of the genus *Capripoxvirus*. A range of capripoxvirus strains have been isolated from sheep and goats and there is evidence of recombination between strains (Gershon *et al.*, 1989). Although some strains are extremely pathogenic in both sheep and goats, other strains produce severe disease in only one of these species.

Epidemiology

Virus particles are shed from skin lesions and in ocular and nasal discharges during the acute stages of the disease. Infection occurs through skin abrasions or by aerosol. Biting insects may also transmit the virus mechanically. Housing or stockading animals facilitates transmission of the virus. Following infection, capripoxvirus strains induce immunity. In endemic areas, where indigenous animals frequently have a high level of naturally acquired immunity, generalized disease and mortality are rare. In isolated flocks, outbreaks of severe disease may occur.

Pathogenesis and pathology

The virus replicates locally either in the skin or in the lungs. Spread to the regional lymph nodes is followed by viraemia and replication in various internal organs. Skin lesions, typical of poxvirus infection, appear about seven days post infection. Lung lesions often present as areas of consolidation and haemorrhage.

Clinical signs

Following an incubation period of about one week, infected animals develop fever, oedema of the eyelids, conjunctivitis and nasal discharge. Within a few days, macules which rapidly develop into papules appear on the skin and external mucous membranes. Scabs form over necrotic papules. The severity of the clinical signs depends on the breed of host animal and the strain of capripoxvirus. Lesions in mild afebrile disease may be minimal and confined to the skin beneath the tail.

Mortality rates with infections from some strains of capripoxvirus may be up to 50% even in indigenous breeds. The disease is most severe in young animals and in imported breeds. Some European breeds are extremely susceptible and mortality rates may approach 100%. Secondary bacterial infection or dissemination of the virus to other sites may result in a more severe form of the disease.

Diagnosis

Diagnosis can often be made solely on clinical grounds. Skin biopsies or postmortem specimens may be used for laboratory confirmation.

- Eosinophilic intracytoplasmic inclusions may be demonstrable histologically in epidermal cells.
- Electron microscopy can be used for the rapid identification of poxvirus particles in material from

lesions. Capripoxviruses can be readily distinguished from parapoxviruses.
- Virus may be isolated in lamb testis or kidney cell monolayers.
- An antigen-trapping ELISA has been developed for the detection of capripoxvirus antigen (Carn, 1995).
- Several serological methods including virus neutralization, Western blot analysis and the indirect fluorescent antibody test are available.

Control

In endemic areas, control is based on annual vaccination. Modified live and inactivated vaccines are available. A subunit vaccine has also been developed (Carn *et al.*, 1994). Inactivated vaccines are less effective than modified live vaccines because cell-mediated immunity is the predominant protective response.

Capripoxviruses are being employed as vectors for important ruminant viral vaccines (Romero *et al.*, 1993). These vector vaccines may provide protection against diseases caused by capripoxviruses as well as diseases such as rinderpest and peste des petits ruminants.

Lumpy skin disease

This acute disease, which is endemic in sub-Saharan Africa, the Malagasy Republic and Egypt, is caused by lumpy skin disease virus (Neethling virus), a capripoxvirus.

Epidemiology

Although virus is present in the saliva of infected animals and transmission may occur through environmental contamination, lumpy skin disease is not particularly contagious (Carn and Kitching, 1995). The principal method of transmission is by mechanical transfer through biting insects. As a consequence, disease outbreaks usually occur during the rainy season when insect activity is high and epidemics are often associated with heavy rains. New outbreaks may appear in areas far removed from an initial outbreak. It is unclear how the virus persists between epidemics but subclinically infected cattle are probably important. There is some evidence for a wildlife reservoir, possibly the African Cape buffalo.

Pathogenesis and pathology

Virus, which is transmitted mechanically by biting insects, rapidly disseminates through a leukocyte-associated viraemia. Many cell types including keratinocytes, myocytes, fibrocytes and endothelial cells become infected. Damage to endothelial cells, which results in vasculitis, thrombosis, infarction, oedema and inflammatory cell infiltration, accounts for the nodular skin lesions.

Clinical signs

The incubation period is up to 14 days. There is a persistent fever accompanied by lacrimation, nasal discharge and a drop in milk yield. Superficial lymph nodes become enlarged and there is oedema of the limbs and dependent tissues. Circumscribed skin nodules develop particularly on the head, neck, udder and perineum. Nodules also develop on the mucous membranes of the mouth and nares. Some skin lesions may develop into 'sit-fasts'. These are composed of a central plug of necrotic tissue which sloughs producing a deep ulcer. Secondary bacterial infection or myiasis can exacerbate the condition. Recovery may take several months. Affected animals are often debilitated and pregnant cows may abort. The severity of the disease relates to the strain of virus and the breed of cattle. Domestic breeds (*Bos taurus*) are more susceptible than zebu (*Bos indicus*). Some animals have few skin lesions and no systemic reaction while others display the full spectrum of clinical signs. Although the mortality rate is usually less than 5%, the economic impact of the disease can be considerable.

Diagnosis

- Generalized skin nodules in cattle in an endemic area are highly suggestive of lumpy skin disease.
- Intracytoplasmic inclusions may be demonstrable histologically in recently developed lesions.
- Capripoxvirus particles in biopsy material or desiccated crusts can be identified using electron microscopy.
- The virus can be isolated in lamb testis cell monolayers.
- An antigen trapping ELISA is available for the detection of capripoxvirus antigen (Carn, 1995).
- Serological assay methods include virus neutralization, Western blot analysis and the indirect fluorescent antibody test.

Control

In endemic regions vaccination is the method of control. Two modified live vaccines, one based on a South African strain of lumpy skin disease virus and the other on a Kenyan strain of sheepox virus, are available. A recombinant vaccine providing protection against lumpy skin disease and rinderpest has been developed (Romero *et al.*, 1993). Imported cattle should be vaccinated before introduction into high risk areas. Surveillance and eradication policies are appropriate control measures in countries bordering on endemic regions (Yeruham *et al.*, 1995).

Swinepox

This disease, which occurs worldwide, is mild and often goes unrecognized. Swinepox virus is the sole member of the genus *Suipoxvirus*. The virus is transmitted mechanically by the pig louse, *Haematopinus suis*. Following an

incubation period of about one week, infected animals display a slight fever and rash. Papules and pustules with scab formation resolve within three to four weeks. These skin lesions are similar to those which occur in pigs infected with vaccinia virus. Virus particles can be demonstrated in material from the lesions by electron microscopy. No vaccine is available. Control within a herd can be achieved by improved hygiene together with louse eradication.

Fowlpox

This disease, affecting domestic poultry, including chickens and turkeys, is caused by infection with fowlpox virus. The infection is slow-spreading and characterized by proliferative skin lesions and diphtheritic lesions in the upper digestive and respiratory tracts. Fowl pox has a worldwide distribution.

Epidemiology

Fowlpox, pigeonpox and turkeypox viruses are closely related and are not strictly host-specific. Several avian species are susceptible to infection with fowlpox virus. Transmission occurs by contact and by mechanical transfer on the mouthparts of biting arthropods, particularly mosquitoes. Virus enters the body through abrasions on unfeathered skin, on oral mucosa or on respiratory mucosa. Aerosols generated from scab material may result in transmission by inhalation. There is some evidence that persistence as latent infection and reactivation by stress may occur in a few birds.

Pathogenesis

Virus multiplication occurs at the site of introduction and may be confined to that site when the strain of infecting virus is of low virulence. Infections caused by virulent strains result in viraemia with replication in internal organs. The route of introduction influences the distribution and severity of lesions. Factors such as malnutrition, debilitation and stress may contribute to the severity of the disease.

Clinical signs

The incubation period is up to 14 days. Two forms of fowlpox have been described, a cutaneous form (dry pox) and a diphtheritic form (wet pox). In the cutaneous form, nodular lesions develop on the comb, wattles and other unfeathered areas of skin. Progression to vesicle formation is followed by ulceration and scab formation. Healing occurs within two weeks. In severely affected birds, lesions may involve both feathered and unfeathered areas of skin, and involvement of the eyelids may lead to complete closure. In the diphtheritic form of the disease, yellowish necrotic lesions (cankers) develop on the mucous membranes of the mouth, oesophagus and trachea. Oral lesions may interfere with eating. Tracheal involvement may lead to laboured breathing and rales.

The mortality rate, which is higher in birds with the diphtheritic form, may approach 50% in severe outbreaks, particularly when accompanied by secondary bacterial or fungal infection. Economic losses are largely due to a transient drop in egg production in laying birds and reduced growth in young birds.

Diagnosis

- Large intracytoplasmic inclusions (Bollinger bodies) containing small elementary bodies (Borrel bodies) may be demonstrable in epithelial cells. Immunofluorescence and immunoperoxidase techniques can be used to identify viral antigen in intracytoplasmic inclusions.
- Typical poxvirus particles can be demonstrated by electron microscopy in material from lesions.
- Virus may be isolated on the chorioallantoic membrane of nine to 12 day old embryonated eggs.
- Nucleic acid probes can be used for diagnosis.
- Suitable serological tests include ELISA, virus neutralization, agar gel precipitation and passive haemagglutination.

Treatment and control

There is no specific treatment. Control of secondary bacterial infection is desirable. In endemic areas, improved management and hygiene along with regular vaccination have reduced the effect of the disease on commercial poultry production. Modified live fowlpox or pigeonpox virus vaccines, produced in tissue culture or chick embryo, are available commercially. Chickens are usually vaccinated at about one month of age. Recombinant vaccines employing fowlpox and canarypox viruses are being developed.

Myxomatosis

This severe generalized disease of European rabbits is caused by myxoma virus, the type species of the genus *Leporipoxvirus*.

Epidemiology

The natural hosts of myxoma virus are New World species of rabbits, *Sylvilagus brasiliensis* in South America and *S. bachmani* in California. Infection has long been endemic in South America and western North America. In natural hosts, myxoma virus infection causes a benign cutaneous fibroma. In contrast, infection in the European rabbit (*Oryctolagus cuniculus*) is lethal. South American isolates of myxoma virus were introduced during the 1950s into populations of *O. cuniculus* in Europe, Chile and Australia as a method for controlling rabbit numbers. More than 99% of infected rabbits died and the disease is

now endemic in these regions. Both attenuated virus strains and resistant rabbit populations have emerged. Virus is transmitted mechanically on the mouthparts of mosquitoes and fleas. Epidemics, which may occur annually, relate to the presence of arthropod vectors and large numbers of young susceptible rabbits.

Pathogenesis and pathology

Virus replicates at the site of inoculation and in regional lymph nodes and the subsequent viraemia is mainly cell-associated, with most virus particles in lymphocytes. Gelatinous, myxoma-like swellings are evident in the skin about one week after infection.

Clinical signs

Subcutaneous gelatinous swellings are particularly prominent in the head and anogenital areas. Blepharo-conjunctivitis, accompanied by an opalescent ocular discharge, develops. Affected animals are febrile and listless, and some may die within 48 hours. Activation of *Pasteurella multocida* infection can result in nasal discharge. The mortality rate, which ranges from 25% to 90%, is influenced by the genetic resistance of the rabbit population and the virulence of the virus strain. Low ambient temperatures increase the severity of the disease.

Diagnosis

The clinical signs are characteristic. Isolation of virus or detection of poxvirus particles in exudate or in material from lesions by electron microscopy is confirmatory.

Control

Commercial and laboratory rabbit stocks can be protected by vaccination with modified live myxoma virus or with rabbit fibroma virus, a related *Leporipoxvirus*. Control of flea infestation in colonies, and insect proofing of accommodation to reduce transmission of infection by other arthropods, may be necessary in endemic areas.

References

Bennett M., Crouch, A.J., Begon, M., *et al.* (1997). Cowpox in British voles and mice. *Journal of Comparative Pathology*, **116**, 35–44.

Carn, V.M. (1995). An antigen trapping ELISA for the detection of capripoxvirus in tissue culture supernatant and biopsy samples. *Journal of Virological Methods*, **51**, 95–102.

Carn, V.M. and Kitching R.P. (1995). An investigation of possible routes of transmission of lumpy skin disease virus (Neethling). *Epidemiology and Infection*, **114**, 219–226.

Carn, V.M., Timms, C.P., Chand, P. *et al.* (1994). Protection of goats against capripox using a subunit vaccine. *Veterinary Record*, **135**, 434–436.

Gershon, P.D., Kitching, R.P. Hammond, J.M. and Black, D.N. (1989). Poxvirus genetic recombination during natural virus transmission. *Journal of General Virology*, **70**, 485–489.

Gilray, J.A., Nettleton, P.F., Pow, I., *et al.* (1998). Restriction endonuclease profiles of orf virus isolates from the British Isles. *Veterinary Record*, **143**, 237–240.

Gumbrell R.C. and McGregor, D.A. (1997). Outbreak of severe fatal orf in lambs. *Veterinary Record*, **141**, 150–151.

Haig, D.M. and Mercer, A.A. (1998). Orf. *Veterinary Research*, **29**, 311–326.

McKeever, D.J. and Reid, H.W. (1986). Survival of orf virus under British winter conditions. *Veterinary Record*, **118**, 613–614.

Romero, C.H., Barrett, T., Evans, S.A., *et al.* (1993). Single capripoxvirus recombinant vaccine for the protection of cattle against rinderpest and lumpy skin disease. *Vaccine*, **11**, 737–742.

Studdert, M.J. (1989). Experimental vaccinia virus infection of horses. *Australian Veterinary Journal*, **66**, 157–159.

Yeruham, I., Abraham, A. and Nyska, A. (1994). Clinical and pathological description of a chronic form of bovine papular stomatitis. *Journal of Comparative Pathology*, **111**, 279–286.

Yeruham, I., Nir, O., Braverman, Y., *et al.* (1995). Spread of lumpy skin disease in Israeli dairy herds. *Veterinary Record*, **137**, 91–93.

Chapter 58

Asfarviridae

African swine fever virus (ASFV), formerly assigned to the family *Iridoviridae*, has recently been reassigned to a newly created family, *Asfarviridae*, containing the single genus *Asfivirus*. African swine fever virus is the type species of this genus. The virus has similarities in genome structure and method of replication to poxviruses but is different in other respects. Virions are 175 to 215 nm in diameter and consist of a membrane-bound nucleoprotein core inside an icosahedral capsid surrounded by an outer lipid-containing envelope (Fig. 58.1). This complex virus contains more than 50 proteins, including a large number of structural proteins and several virus-encoded enzymes required for transcription and post-translational modification of mRNA. The genome consists of a single molecule of linear double-stranded DNA. Following replication in the cytoplasm of host cells, virus is released either by budding through the plasma membrane or following cellular disintegration. African swine fever virus is stable in the environment over a wide range of temperature (4°C to 20°C) and pH values. The virus may persist for months in meat. Infectivity can be destroyed by heating and by treating with lipid solvents and some disinfectants such as orthophenylphenol.

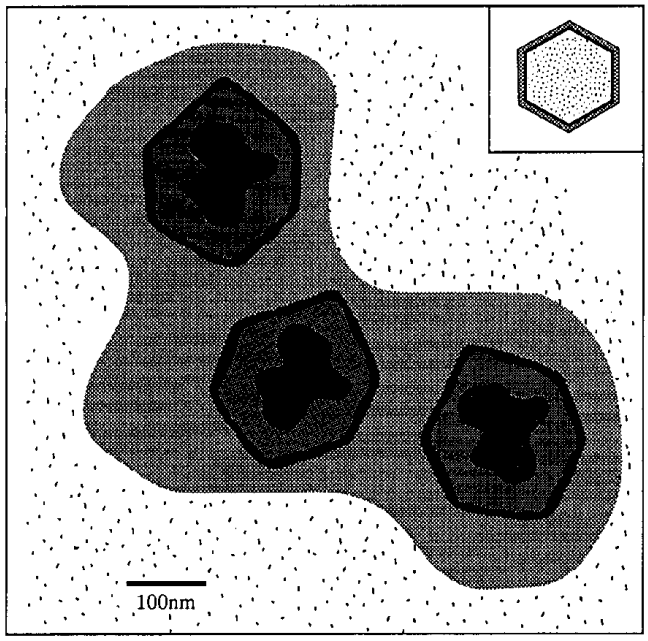


Figure 58.1 African swine fever virus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Key points

- Enveloped DNA virus
- Icosahedral symmetry
- Replicates in cytoplasm of host cells and in soft ticks of the *Ornithodorus* species
- Causes African swine fever

African swine fever

African swine fever (ASF) is an economically important viral disease of pigs, characterized by fever, haemorrhages in many tissues and a high mortality rate. It is endemic in sub-Saharan Africa and Sardinia. Outbreaks have occurred in Belgium, Italy, Malta, Brazil, Cuba, Haiti and the Dominican Republic. The Iberian Peninsula was declared free of the disease in 1995, almost 30 years after initial introduction into the region, but ASFV reappeared in 1999 in Portugal.

Epidemiology

Domestic and wild pigs are the only species susceptible to infection. In Africa, ASFV is maintained in a sylvatic cycle involving soft ticks of the genus *Ornithodorus* and inapparent infection of warthogs and bushpigs (Fig. 58.2). Adult warthogs with persistent inapparent infection rarely develop viraemia. In contrast, young warthogs develop viraemia and are a major source of virus for soft ticks. Replication of virus occurs in the ticks and both trans-ovarial and trans-stadial transmission have been described. Soft ticks feed for short periods on hosts before dropping off and sheltering in crevices in walls or cracks in the ground. The presence of ticks in a particular region makes the eradication of ASF difficult. The principal tick species involved in transmission are *O. moubata* in Africa and *O. erraticus* in Spain and Portugal. Experimentally, several other *Ornithodorus* species support virus replication. Virulent strains of ASFV, producing high mortality in infected animals, are widely distributed in

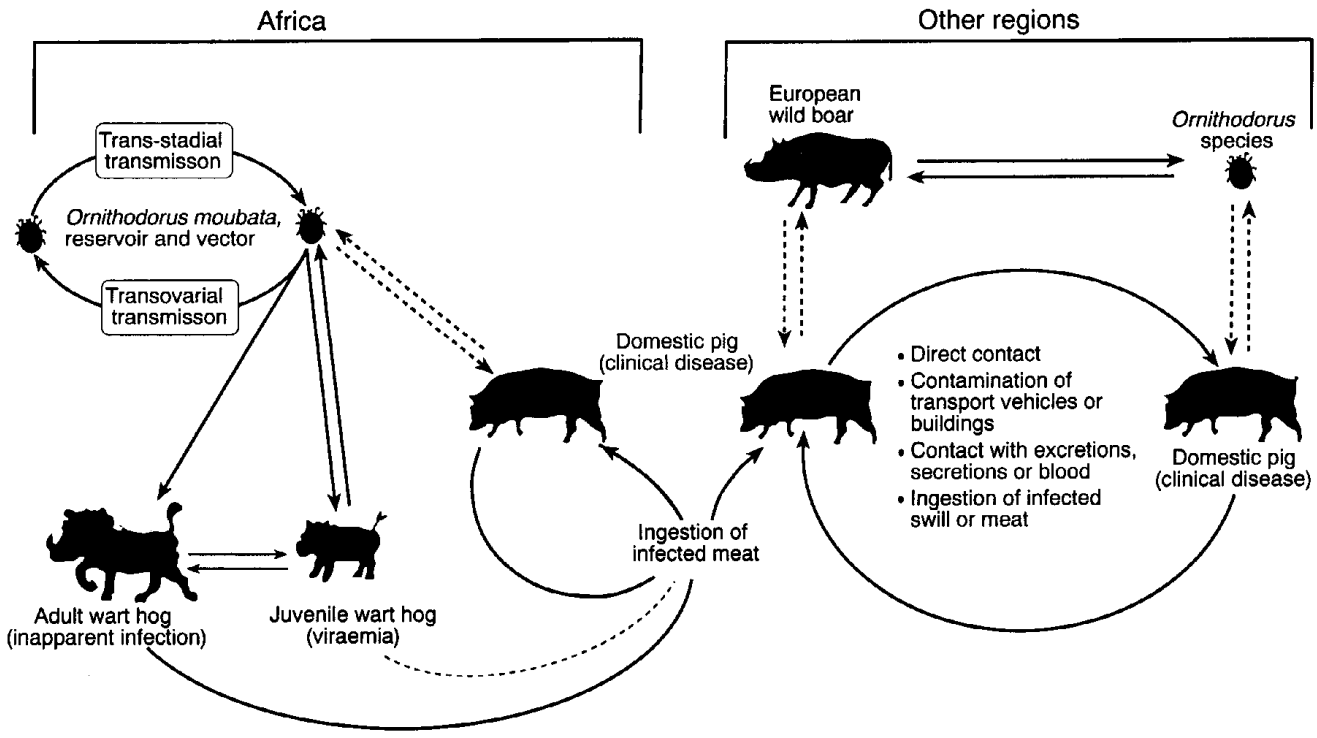


Figure 58.2 The maintenance and transmission of African swine fever virus in wild and domestic pig populations and in tick vectors.

Africa. Many isolates from other parts of the world are less virulent and mortality rates are usually below 50%.

Following infection of domestic pigs with virulent virus, body fluids and tissues contain large quantities of virus until death or recovery occurs. Ingestion of uncooked meat from infected warthogs or domestic pigs is a major method of transmission. Spread can also occur by direct contact usually through oral or nasal secretions. Occasionally, animals become infected by contact with blood shed as a result of fighting. Indirect transmission can occur through contaminated transport vehicles, fomites and footwear. Feeding uncooked swill is an important mechanism of spread of ASF internationally, with outbreaks often starting in herds close to airports and harbours.

Pigs which have recovered from clinical disease may remain infected for long periods. Carrier pigs are considered to be important sources of virus dissemination. Although recovered pigs are clinically unaffected by challenge with genotypically related ASFV, the challenge virus may replicate and spread to other pigs.

Pathogenesis and pathology

Infection in domestic pigs is usually acquired via the oronasal route. The virus replicates initially in pharyngeal mucosa, tonsils and the regional lymph nodes. Infection then spreads by the bloodstream to other lymph nodes, bone marrow, spleen, lung, liver and kidneys. Secondary replication in these sites results in prolonged viraemia. Although the virus replicates primarily in cells of the lymphoreticular system, it can also infect megakaryocytes, endothelial cells, renal epithelial cells and hepatocytes. Lesions include splenic enlargement, swollen haemorrhagic gastrohepatic and renal lymph nodes, subcapsular petechiation in the kidneys, petechial and ecchymotic haemorrhages on serosal surfaces, oedema of the lungs and hydrothorax. The widespread haemorrhages result from disseminated intravascular coagulation, endothelial damage and destruction of megakaryocytes (Rodriguez *et al.*, 1996). Leukopenia is marked. Because the virus does not appear to replicate in T and B lymphocytes, it has been suggested that the lymphopenia follows apoptosis of lymphocytes and necrosis of lymphoid organs (Carrasco *et al.*, 1996). Lesions in chronic disease include pneumonia, fibrinous pleuritis and pericarditis, pleural adhesions and hyperplasia of lymphoreticular tissues.

Clinical signs

The clinical signs of ASF, which range from inapparent to peracute, relate to the challenge dose and virulence of the virus and to the route of infection. The incubation period, which may extend from four to 19 days, is typically five to seven days in acute cases. Animals with peracute disease die suddenly without premonitory clinical signs. Fever, inappetence, depression and recumbency are features of acute disease. Cutaneous hyperaemia and, in some cases, haemorrhages may be evident. Other signs include dyspnoea, conjunctivitis, diarrhoea, bleeding from the nose

and rectum, and abortion. The mortality rate is high. Subacute disease has a course of three to four weeks. Clinical signs include pneumonia, swollen joints, emaciation, depression and inappetence. Mortality rates, which are variable, depend on the age and general health of infected pigs. Animals may recover clinically or may develop a chronic form of the disease, which usually occurs in regions where ASFV is endemic. The immune mechanisms responsible for recovery and protection from ASFV are poorly understood. Neutralizing antibodies are not demonstrable in the sera of recovered animals. Cell-mediated immunity is considered to be an important component of the immune response.

Diagnosis

Laboratory confirmation of ASF is mandatory because the clinical signs and lesions, which occur in some other important pig diseases such as classical swine fever, erysipelas and septicaemic salmonellosis, are similar.

- Suitable samples for laboratory examination include blood, serum, tonsil, spleen and lymph nodes.
- The most convenient and frequently used tests for detection of ASFV are direct immunofluorescence and haemadsorption (Sanchez-Vizcaino, 1999). Direct immunofluorescence, which is fast and economical, can be carried out on impression smears or cryostat sections. However, the sensitivity of the test is only 40% in pigs with subacute or chronic ASF because of the blocking action of bound antibody in antigen- antibody complexes. Most field strains of ASFV induce haemadsorption. Pig erythrocytes adhere to the surface of infected monocytes and macrophages, forming a characteristic rosette. The haemadsorption test can be carried out by employing leukocytes from blood samples of pigs under investigation or by inoculating primary blood leukocyte cultures with blood or homogenized tissue from suspect pigs.
- A challenge experiment, involving the inoculation of suspect material into pigs vaccinated against classical swine fever and into unvaccinated pigs, may be used to differentiate the two diseases.
- The polymerase chain reaction can be used to detect DNA from ASFV in tissues unsuitable for virus isolation or antigen detection.
- Antibodies persist for long periods in recovered animals. Serological testing may be the only means of detecting animals infected with strains of low virulence. Techniques for detecting antibodies to ASF include ELISA, immunoblotting, indirect immunofluorescence, complement fixation and radioimmunoassay.

Control

A successful vaccine is not yet available. Inactivated vaccines do not induce protection. Although live attenuated vaccines induce protection against challenge with homologous virus strains in some pigs, a proportion of these animals become carriers and may develop chronic lesions.

Countries maintain disease-free status by prohibiting importation of pigs and pig products. Waste food scraps from aircraft and ships must be boiled before inclusion in pig feed. In the face of an outbreak of ASF in countries free of infection, an eradication policy is implemented. The occurrence of low virulence strains renders eradication difficult.

Restriction of pig movement, serological monitoring of carrier pigs and prevention of contact between domestic pigs and warthogs or ticks are important control measures in countries where the disease is endemic. Eradication of tick species which act as vectors of ASFV is an essential part of a control programme.

References

Carrasco. L., Chacon, M-L., Martin de las Mulas, J. *et al.* (1996). Apoptosis in lymph nodes in acute African swine fever. *Journal of Comparative Pathology*, **115**, 415–428.

Rodriguez, F., Fernandez, A., Perez, J. *et al.* (1996). African swine fever: morphopathology of a viral haemorrhagic disease. *Veterinary Record*, **139**, 249–254

Sanchez-Vizcaino, J.M. (1999). African swine fever. In *Diseases of Swine*. Eighth Edition. Eds. B.E. Straw, S. D'Allaire, W.L. Mengeling, and D.J. Taylor. Blackwell Science, Oxford. pp. 93–102.

Chapter 59

Parvoviridae

Viruses belonging to the family *Parvoviridae,* referred to as parvoviruses, are small (Latin *parvus*, small) and range in size from 18 to 26 nm in diameter (Fig. 59.1). These non-enveloped, icosahedral viruses possess a linear genome of single-stranded DNA. There are two sub-families: *Parvovirinae* which includes viruses of vertebrates and *Densovirinae,* members of which infect arthropods. Of the three genera within the subfamily *Parvovirinae* (Fig. 59.2) only the genus *Parvovirus* contains viruses of veterinary importance.

Parvoviruses replicate only in the nuclei of dividing host cells, a feature which determines the tissues targeted. After entering a cell, the virion is uncoated and its single-stranded DNA genome is converted to double-stranded DNA by DNA polymerases in the nucleus. Following viral replication, cell lysis occurs as virions are released.

Parvoviruses are stable in the environment. They are resistant to many factors including lipid solvents, a wide range of pH values (pH 3 to 9) and heating at 56°C for more than 60 minutes. They are inactivated by formalin, β-propiolactone, sodium hypochlorite and oxidizing agents. With the exception of Aleutian mink disease virus and goose parvovirus, parvoviruses of vertebrates agglutinate erythrocytes. Haemagglutination inhibition (HAI) by specific antisera is widely used for their identification. Mink enteritis virus, canine parvovirus and racoon parvovirus are considered to be host-range mutants of feline panleukopenia virus.

Key points

- Small, non-enveloped, single-stranded DNA viruses
- Icosahedral symmetry
- Replicate in the nucleus, form intranuclear inclusion bodies
- Require rapidly-dividing cells for replication
- Stable in the environment
- Resistant to heat, solvents, disinfectants and pH changes
- Genus *Parvovirus:*
  - —Many have haemagglutinating activity
  - —Shed in large numbers in faeces
  - —Enteric and systemic diseases in dogs and cats
  - —Reproductive failure, SMEDI syndrome, in pigs

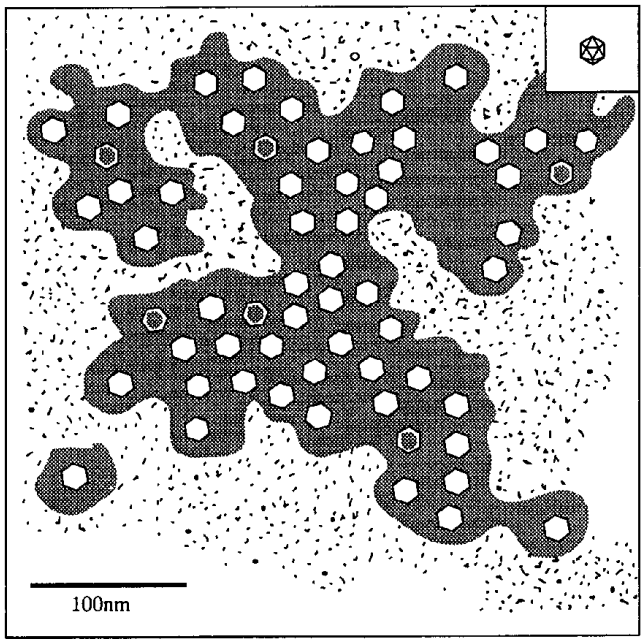


Figure 59.1 Parvovirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Clinical infections

Parvoviruses can infect many domestic and wild animals (Table 59.1). Although most members of the group produce acute systemic diseases, some such as canine minute virus and bovine parvovirus are of uncertain pathogenic significance. Two distinct parvoviral diseases of mink, Aleutian mink disease and mink enteritis, are recognized. Mink enteritis, first described in the 1940s, affects kits and clinically resembles feline panleukopenia. Aleutian mink disease is a persistent infection, primarily affecting animals which are homozygous for pale coat colour. The disease is characterized by B lymphocyte stimulation leading to plasmacytosis, hypergammaglobulinaemia and immune complex-related lesions in the kidneys and other organs. Aleutian mink disease can also

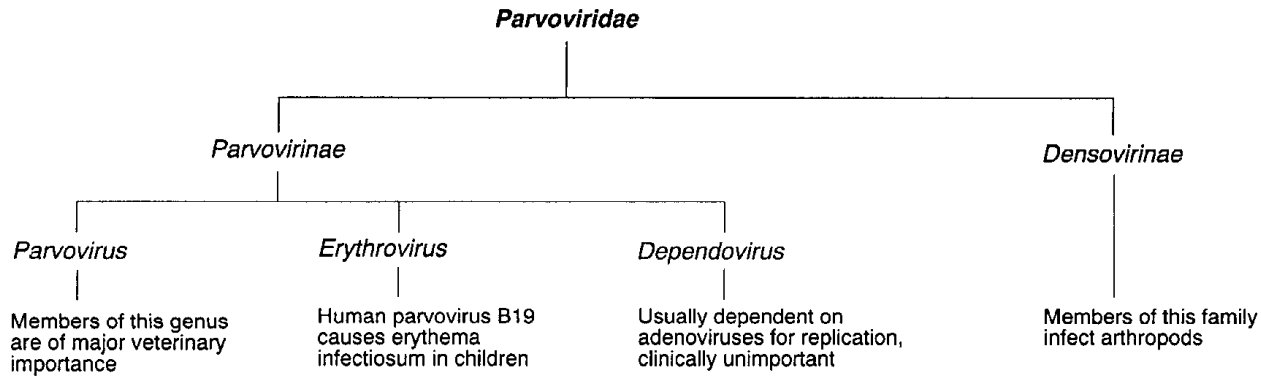


Figure 59.2 The family *Parvoviridae*, subfamilies *Parvovirinae* and *Densovirinae* and the genera of *Parvovirinae* which can infect mammalian species.

Table 59.1 Parvoviruses of veterinary significance.

Virus	Hosts	Consequences of infection
Feline panleukopenia virus	Domestic and wild cats	Highly contagious systemic and enteric disease most common in weaned kittens, manifested as depression, vomiting, diarrhoea. Intrauterine infection : abortion or cerebellar ataxia in neonatal kittens
Canine parvovirus (Canine parvovirus 2)	Dogs	Highly contagious enteric disease with depression, vomiting, dysentery and immunosuppression. Intrauterine or perinatal infection: myocarditis in pups (now rare)
Porcine parvovirus	Pigs	Major cause of stillbirths, mummified foetuses, embryonic deaths and infertility (SMEDI syndrome)
Mink enteritis virus	Mink	Generalized disease of mink kits, analagous to feline panleukopenia
Aleutian mink disease virus	Mink, ferrets	Chronic, progressive disease of mink homozygous for pale coat colour. Persistent viraemia, plasmacytosis, hypergammaglobulinaemia and immune complex-related lesions
Goose parvovirus (goose plague virus)	Geese	Highly contagious, fatal disease of 8-30 day old goslings (Derzsy's disease): hepatitis, myositis, including myocarditis
Canine minute virus (Canine parvovirus 1)	Dogs	Role of virus in disease is uncertain; serological surveys suggest the virus is widespread
Bovine parvovirus	Cattle	Associated with sporadic outbreaks of diarrhoea in calves

occur in domestic ferrets (Welchman *et al.*, 1993). Goose parvovirus infection, which causes hepatitis and myositis in goslings, is highly contagious and often fatal. The most important parvoviral diseases of domestic animals are feline panleukopenia, canine parvovirus infection and porcine parvovirus infection.

Feline panleukopenia

Feline panleukopenia, also known as feline infectious enteritis or feline distemper, is a highly contagious generalized disease of domestic and wild cats caused by feline panleukopenia virus. Only one serotype of this virus has been identified. The disease, which is worldwide in distribution, is one of the most common feline viral infections.

Epidemiology

Most species of *Felidae* are highly susceptible to infection which is generally endemic in unvaccinated cat populations. Some species of *Mustelidae*, *Procyonidae* and *Viverridae* can also become infected but they seldom develop clinical disease. Although cats of all ages are susceptible to infection, disease occurs predominantly in young recently-weaned kittens, as maternally-derived antibody levels wane. Many infections are subclinical, particularly in older cats and in kittens partially protected by maternally-derived immunity. The disease may have a cyclical or seasonal pattern which is related to the births of kittens. Transplacental infection occurs in fully susceptible queens.

High rates of virus excretion occur during the acute stage of the disease, mainly in faeces but also in saliva, urine, vomitus and blood. Faecal shedding usually continues for some weeks following clinical recovery. Although long-term, low-grade shedding of virus by some subclinical carriers may occur, the stability of the virus in the environment is of greater significance in its persistence and spread. In cool, moist, dark environments,

infectivity may last for more than a year. Fleas and humans may act as mechanical vectors.

Pathogenesis and pathology

Following ingestion or inhalation, replication occurs in the mitotically active lymphoid tissues of the oropharynx and associated lymph nodes. Viraemia develops within 24 hours, producing infection of mitotically active cells in other tissues, particularly the cells of the intestinal crypts and the lymphopoietic cells of the bone marrow, thymus, lymph nodes and spleen. Destruction of these target tissues results in panleukopenia and villous atrophy. The crypts of Lieberkühn are dilated and contain necrotic epithelial cells. Intranuclear inclusions are sometimes evident in crypt cells. Intestinal villi become blunted and may fuse. The effects of transplacental infection on foetuses generally relate to the stage of gestation at the time of viral invasion and range from cerebellar hypoplasia and retinal dysplasia to foetal death.

Clinical signs

The incubation period of feline panleukopenia ranges from two to ten days but is typically four to five days. Subclinical infection is common and results in a mild fever and leukopenia, usually followed by life-long immunity. Subacute disease presents as depression, fever and diarrhoea lasting one to three days, followed by rapid recovery. The disease is most severe in young, unvaccinated kittens between 6 and 24 weeks of age and is characterized by sudden onset of pronounced depression, anorexia and fever. Vomiting, sometimes accompanied by diarrhoea or dysentery, follows within two days and can result in severe dehydration and electrolyte imbalance. Abdominal pain may be evident. The mortality rate ranges from 25 to 90% with most deaths occurring within three to five days of the onset of illness. Animals which survive require several weeks for full recovery. Immunity is strong and long-lasting. Subnormal temperatures are followed by death within 24 hours. Although intrauterine infection of the developing foetuses often occurs, infected pregnant queens usually show no signs of illness. Foetal infections early in gestation may result in resorption or abortion. Because of developing foetal immune competence, infections after mid-gestation are usually less severe. However, stillbirths, early neonatal death and teratological changes such as cerebellar hypoplasia and retinal dysplasia may occur in the litters of queens infected during late pregnancy. Kittens with cerebellar hypoplasia exhibit cerebellar ataxia manifested as hypermetria, incoordination and, frequently, intention tremors. These signs persist for life.

Diagnosis

- Feline panleukopenia should be considered when unvaccinated cats present with diarrhoea.
- A white cell count of less than 7 x 10^9/L is often encountered in acutely affected animals. Neutropenia is more common than lymphopenia. Cell counts return to normal after a few days in those cats which survive.
- Specimens for virus isolation in primary feline cell lines include oropharyngeal swabs, faeces, spleen, mesenteric lymph nodes and ileum.
- Large numbers of virus particles may be demonstrated by electron microscopy in faecal samples from cats with acute disease.
- Typical histopathological changes may be present in sections of the ileum and jejunum. Intranuclear inclusion bodies may be detected in crypt cells.
- Viral antigen can be detected in faeces using ELISA or haemagglutination employing pig or Rhesus monkey red cells. Commercially available canine parvovirus kit sets can be used for the detection of feline panleukopenia virus antigen (Addie *et al.*, 1998).
- A rising antibody titre may be detected in serum samples by a number of tests including the haemagglutination-inhibition (HAI) or virus neutralization (VN) tests.

Treatment

No specific treatment is available.

- Intensive supportive therapy is usually necessary:
  - — Appropriate fluid therapy for dehydration should be given.
  - — Whole blood or plasma from immune donors may be beneficial in cats with anaemia or hypoproteinaemia.
  - — Parenterally administered broad-spectrum antibiotics can be used to combat secondary bacterial infections.
- Affected animals should be housed in a clean, warm environment and maintained on an optimal diet supplemented with B complex vitamins.

Control

Vaccination is the principal control measure. There is only one serotype of feline panleukopenia virus and immunity following natural infection is strong and long-lasting.

- Modified live and inactivated vaccines are commercially available:
  - — Inactivated vaccines are less effective than modified live vaccines and require booster inoculations. They are safe for pregnant queens and might be considered for vaccination of Siamese and Burmese kittens which can have adverse reactions to modified live vaccines (Carwardine, 1990).
  - — Modified live vaccines can be used to immunize kittens at 8 to 10 weeks of age, with a booster dose at 12 to 14 weeks of age. Annual booster vaccinations are recommended by vaccine manufacturers. These vaccines should not be used in pregnant queens because replicating virus may cause cerebellar hypoplasia in developing foetuses.
  - — Cats should have completed a vaccination schedule

or should be given a booster injection at least two weeks before introduction to premises where feline panleukopenia has recently occurred.

- Clinical infections cause heavy environmental contamination. Premises should be thoroughly disinfected with 1% sodium hypochlorite or 2% formalin (Scott, 1980). Cats on such premises should be vaccinated without delay.

Canine parvovirus infection

Infection with canine parvovirus (CPV) emerged in the late 1970s as a worldwide disease in dogs with high morbidity and mortality. Acute or subacute heart failure in pups infected *in utero* or during the perinatal period, was a common manifestation of the disease. With the gradual development of immunity in the adult dog population, as a consequence of both natural exposure and vaccination, the clinical pattern of the disease changed. The most common clinical presentation now encountered is acute enteric disease in young dogs between weaning and 6 months of age. Since the appearance of CPV in 1978, further mutations affecting the genome and antigenicity of the virus have occurred. Three subtypes of the virus are currently recognized. Infection or vaccination with one subtype confers immunity against the other subtypes (Greenwood *et al.*, 1995). Canine parvovirus is now considered to be a host-range mutant of feline panleukopenia virus or a closely related parvovirus.

Epidemiology

Many canine species are susceptible to infection and transmission is predominantly by the faecal-oral route. Infected dogs shed large numbers of viruses in their faeces. The number of viruses in faeces may be 10^9/g from the fifth or sixth day after infection. Persistent shedding is uncommon and the continued presence of the disease in dog populations depends mainly on the stability of the virus in the environment. The low dose of virus required to establish infection and the ease with which mechanical transfer can take place are important additional factors contributing to the spread of infection.

Pathogenesis and pathology

The virus replicates initially in pharyngeal lymphoid tissues and Peyer's patches. Viraemia develops and the main target tissues are those with rapidly multiplying cell populations. During the first two weeks of life there is active cardiac myocyte division allowing viral replication with resultant necrosis and myocarditis. In older pups, the virus invades the actively dividing epithelial cells of the crypts in the small intestine. Loss of cells from the intestinal crypts leads to blunting of villi, with resultant reduced absorptive and digestive capacity leading to diarrhoea. There may be extensive haemorrhage into the intestinal lumen in severely affected pups. Destruction of lymphoid tissues of the intestinal mucosa and mesenteric lymph nodes contributes to immunosuppression which allows proliferation of Gram-negative bacteria with secondary invasion of damaged intestinal tissues. Endotoxaemia, leading to endotoxic shock, may follow.

Clinical signs

The age and the immune status of the animal largely determine the form and severity of the disease. After a short incubation period of four to seven days, animals with enteric disease show sudden onset of vomiting and anorexia. Depression and fever may also be observed. Diarrhoea, often blood-stained, develops within 48 hours and in severe cases, there may be frank haemorrhage. The faeces have a foetid smell. Concurrent intestinal parasitism and viral or bacterial infections may exacerbate the condition. Affected dogs deteriorate rapidly due to dehydration and weight loss. Prolonged illness is uncommon; severely affected animals die within three days. Animals which survive the disease develop a long-lasting immunity.

In the myocardial form of the disease, which is now rare, affected pups usually show signs of acute heart failure before eight weeks of age. Some pups may develop congestive heart failure months after the initial infection, as a result of extensive fibrosis following myocardial necrosis.

Diagnosis

- Samples for laboratory examination should include faeces, blood and other tissues, particularly affected portions of intestines and myocardium.
- The nature and distribution of the gross and microscopic enteric lesions may point to a parvoviral infection. Immunocytochemical staining can confirm the presence of viral antigen in tissue sections.
- The presence of basophilic intranuclear inclusions in cardiac myocytes is confirmatory.
- A leukopenia may be detected, particularly in severely affected animals.
- Definitive diagnosis in clinically affected animals early in the course of the disease relies on the demonstration of virus or viral antigen in faeces:
  - —Numerous virus particles can be demonstrated by electron microscopy.
  - —ELISA or HA may be used to demonstrate viral antigen.
- Virus can be isolated in a number of suitable canine and feline cell lines.
- Serological tests, including HAI, VN and indirect immunofluorescence, may confirm the diagnosis.

Treatment

No specific treatment is available.

- Intensive supportive therapy including anti-emetics and fluid administration is required for the treatment of parvoviral enteritis.
- Broad-spectrum antibiotics, administered parenterally, reduce the risk of secondary bacterial infections.
- Dogs with subacute or chronic heart failure may improve with rest and diuretic therapy.

Control

Vaccination alone usually cannot control the cycle of endemic parvovirus infection in kennels and therefore it is important to minimize exposure of pups to the virus until they reach 20 weeks of age (Pollock and Coyne, 1993). Non-protective, low levels of maternal antibody can interfere with the efficacy of some modified live vaccines. In many pups these critical levels last until eight to twelve weeks of age, but in some instances interference with vaccination continues until 18 weeks of age (O'Brien, 1994).

- Inactivated and modified live vaccines are commercially available:
  - — Inactivated vaccines usually provide protection for up to one year and are safe to use in pregnant bitches.
  - — Although modified live vaccines generally give good, long-lasting protection, annual boosters may be required. These vaccines should not be used for pregnant bitches. Modified live vaccines vary in their degree of viral attenuation. The less attenuated strains of vaccinal viruses can replicate in pups despite the presence of residual maternal antibodies (Churchill, 1987; Burtonboy *et al.*, 1991) and vaccination of pups at 12 weeks of age is often claimed by vaccine manufacturers to be protective.
- Thorough disinfection of premises must be carried out following a disease outbreak:
  - — Effective disinfectants include 1% sodium hypochlorite and 2% formalin.
  - — Fumigation with formaldehyde gas, where feasible, is the most efficient disinfection procedure.

Porcine parvovirus infection

Porcine parvovirus is an important cause of reproductive failure in pigs. The virus, which occurs as a single serotype, is found worldwide and infection is endemic in many conventional pig herds.

Epidemiology

On farms where the disease is endemic, many sows are immune. They remain seropositive for up to four years and transmit passive protection through colostrum to their piglets. Maternally-derived immunity usually persists for about four months, but it can persist in some pigs until they are six to nine months of age. During this period, the maternally-derived antibodies may interfere with the development of active immunity and consequently some gilts can be seronegative and susceptible to infection at mating. Infected pigs shed virus in their faeces and other secretions, including semen, for only a few weeks. However, pens may remain contaminated for several months because of the exceptional stability of the virus.

Pathogenesis

After infection by the oronasal route and occasionally through semen, local replication of virus is followed by viraemia. The virus has a predilection for the mitotically active cells of foetal tissues. Transplacental infection in pregnant sows occurs 10 to 14 days after exposure to the virus. The major damage to foetuses arises before onset of immunocompetence, at about 60 to 70 days of gestation (Huysman *et al.*, 1992). Infection of embryos in the first weeks of life results in death and resorption. When infection occurs later in gestation, but before day 70, foetuses die and become mummified. Infection after 70 days of gestation usually results in the birth of healthy seropositive piglets.

Clinical signs

Porcine parvovirus infection is a major cause of SMEDI, an acronym used to describe porcine reproductive failure in which stillbirths, mummified foetuses, early embryonic death and infertility occur. Abortion and neonatal deaths have been reported occasionally. Generally, small litters with mummified piglets of different sizes are produced following transplacental infection and subsequent sequential exposure of foetuses by intrauterine spread. If the number of viable embryos is reduced below four, the entire litter is usually lost. Infection with porcine parvovirus does not appear to damage the male reproductive tract.

Diagnosis

- When reproductive failure is detected in young or recently introduced sows, particularly if associated with mummified foetuses, infection with porcine parvovirus must be considered.
- Several foetuses should be submitted for laboratory examination.
- Demonstration of viral antigen in cryostat sections of foetal tissues, particularly in the lungs, by immunofluorescence is reliable and sensitive.
- Agglutination of guinea-pig erythrocytes by homogenates of foetal tissue indicates the presence of viral haemagglutinin.
- Swine kidney cell lines may be used for virus isolation. However, viral infectivity is gradually lost following death of the foetus and isolation from mummified tissues may be unsuccessful.
- Serological techniques include HAI and VN tests. Antibodies may be detected in sera or body fluids of

older foetuses or aborted piglets. However, serological testing is usually of little diagnostic value in endemically infected herds.

Control

Control in herds in which the disease is endemic is based on exposure of gilts and susceptible sows to porcine parvovirus prior to mating, thereby inducing immunity. Vaccination can be used to enhance immunity in herds with endemic disease. It can also be used for male or female breeding stock introduced into these herds.

- Natural exposure can be achieved by increasing the contact between susceptible gilts and older seropositive sows. Methods of stimulating immunity include exposing animals to contaminated faeces or to placental or foetal tissue from infected sows.
- Modified live and inactivated vaccines against the single serotype of porcine parvovirus have been developed experimentally, but only inactivated vaccines are available commercially. Gilts and susceptible seronegative sows and boars should be vaccinated two to four weeks before mating. Vaccination prevents intrauterine infection for a limited period. Most vaccination strategies rely on subsequent natural exposure to the virus to reinforce immunity (Huysman *et al.*, 1992).

References

Addie, D.D., Toth, S., Thompson, H. *et al.* (1998). Detection of feline parvovirus in dying pedigree kittens. *Veterinary Record*, **142**, 353–356.

Burtonboy, S., Charlier, P., Hertoghs, J., *et al.* (1991). Performance of high titre attenuated canine parvovirus vaccine in pups with maternally derived antibody. *Veterinary Record*, **128**, 377–381.

Carwardine P. (1990). Adverse reactions to vaccine. *Veterinary Record*, **127**, 243.

Churchill, A.E. (1987). Preliminary development of a live attenuated canine parvovirus vaccine from an isolate of British origin. *Veterinary Record*, **120**, 334–339.

Greenwood, N.M., Chalmers, W.S.K., Baxendale, W. and Thompson, H. (1995). Comparison of isolates of canine parvovirus by restriction enzyme analysis, and vaccine efficacy against field strains. *Veterinary Record*, **136**, 63–67.

Huysman, C.N., van Leengoed, L.A.M.G., de Jong, M.C.M. and van Osta, A.L.M. (1992). Reproductive failure associated with porcine parvovirus in an enzootically infected pig herd. *Veterinary Record*, **131**, 503–506.

O'Brien, S.E. (1994). Serologic response of pups to the low-passage modified live canine parvovirus-2 component in a combination vaccine. *Journal of the American Veterinary Medical Association*, **204**, 1207–1209.

Pollock, R.V.H. and Coyne, M.J. (1993). Canine parvovirus. *Veterinary Clinics of North America: Small Animal Practice*, **23**, 555-568.

Scott, F.W. (1980). Virucidal disinfectants and feline viruses. *American Journal of Veterinary Research*, **41**, 410–414.

Welchman, D. de B., Oxenham, M. and Done, S.H. (1993). Aleutian disease in domestic ferrets: diagnostic findings and survey results. *Veterinary Record*, **132**, 479–484.

Further reading

Tijssen, P. (1990). *CRC Handbook of Parvoviruses*, Volumes. 1 and 2. CRC Press, Florida.

Chapter 60

Circoviridae

Viruses in the recently established family *Circoviridae* cause disease in vertebrate animals and plants. Circoviruses (17 to 22 nm in diameter), are non-enveloped with icosahedral symmetry (Fig. 60.1). They are stable in the environment at pH 3 to pH 9 and are resistant to heating at 60°C for 30 minutes.

The genome consists of a molecule of circular single-stranded DNA. Replication occurs in the nuclei of dividing cells. Recent genetic sequencing studies suggest that circoviruses fall into three groups (Niagro *et al.*, 1998) and this is reflected in recent classification changes. Chicken anaemia virus (22 nm in diameter), the type species, has been assigned to the newly created genus, *Gyrovirus*. The animal circoviruses (17 nm in diameter), porcine circovirus and beak and feather disease virus have been assigned to the genus *Circovirus*. The third grouping contains plant viruses which have now been removed from the family and placed in the unassigned genus *Nanovirus*.

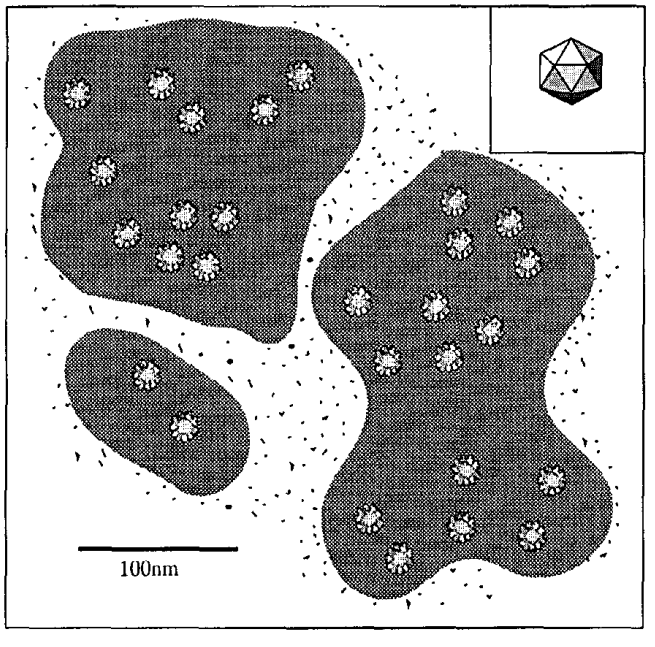


Figure 60.1 Circovirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Key points

- Small, non-enveloped single-stranded DNA viruses with icosahedral symmetry
- Replicate in the nucleus of dividing cells
- Stable in the environment
- Circoviruses cause infections in chickens, pigs and plants

Clinical infections

Circoviruses, which are host-specific and have a world-wide distribution, infect cells of the haemolymphatic system. Infections with chicken anaemia virus and with porcine circovirus are of veterinary interest. Beak and feather disease virus is associated with a debilitating, immunosuppressive disease of young pscittacine birds, particularly cockatoos.

Chicken anaemia virus infection

Young birds infected with chicken anaemia virus (CAV) develop aplastic anaemia and generalized lymphoid atrophy. This virus, which infects only chickens, is present in poultry flocks worldwide. All of the field isolates of the virus appear to be equally pathogenic and belong to a single serotype.

Epidemiology

Both horizontal and vertical transmission occur. Infection is by the faecal-oral route. Vertical transmission through the egg occurs during the one to three week viraemic period following infection in laying hens. Once infection is established in a breeder flock, most birds develop antibody before laying begins. Maternally-derived antibodies do not prevent chicks from becoming infected and shedding the virus. However, they prevent the development of clinical disease. An age resistance to disease but not to infection develops in chicks at about two weeks of age. In breeding flocks containing many serologically-positive adult birds, subclinical infection is common in chickens. However, age resistance and the

protective effect of maternally-derived antibodies can be overcome if infections with other immunosuppressive viruses, such as infectious bursal disease virus or gallid herpesvirus 2, are present.

Pathogenesis and pathology

Viraemia develops following infection of susceptible day old chicks and virus can be recovered from most organs and from faeces for about three or four weeks. The principal target cells are precursor T cells in the thymus and haemocytoblasts in the bone marrow. Destruction of these cells results in immunosuppression and anaemia. Gross postmortem findings include atrophy of the thymus tissues and bursa, pale bone marrow and haemorrhages under the skin and in skeletal muscle.

Clinical signs

Chickens develop clinical signs at about two weeks of age. Affected birds are depressed, anorexic and pale. The mortality rate, which is usually about 10%, may be up to 50%. Birds which survive the acute phase of the disease recover slowly. Subclinical infection in broilers from breeder flocks can adversely affect weight gains.

Diagnosis

A presumptive diagnosis is based on the clinical signs and gross lesions at postmortem. Laboratory confirmation relies on detection of viral antigen by immunocytochemical techniques. Viral DNA can be demonstrated in bone marrow and thymus by *in situ* hybridization, by dot-blot hybridization or by PCR. Virus isolation, although possible, is exacting and expensive. Serum antibodies can be detected using virus neutralization, indirect immunofluorescence and ELISA. A commercial ELISA is available and can be used to identify seronegative breeder flocks before laying begins.

Control

Because infection with CAV is common, it is difficult to maintain breeder flocks free of infection. Breeding birds should be exposed to infection before laying commences. Seronegative flocks can be deliberately exposed to virus by transfer of old litter from CAV-positive farms or by the addition of crude tissue homogenates from affected birds to drinking water but such exposure is unreliable and inherently unsafe.

A live vaccine is available in some European countries. Vaccination does not prevent economic losses in broilers due to subclinical infection. Control of other immunosuppressive viruses in CAV-positive flocks is essential due to the additive effects of these infections.

Pig circovirus infection

Porcine circovirus was first described as a picornavirus-like contaminant of the continuous pig kidney cell line PK/15. Experimental challenge suggests that this virus, termed porcine circovirus 1, is of doubtful pathogenicity. Recently, an antigenically and genomically distinct circovirus, porcine circovirus 2 (PCV 2) was isolated from piglets with wasting disease. Sero-epidemiological studies indicate that infection with PCV 2 is widespread in pig populations worldwide. Post-weaning multisystemic wasting syndrome (PMWS), a progressive wasting condition with lesions in several organ systems, was first described in Canada in 1991 in specific pathogen-free herds (Allan and Ellis, 2000). This syndrome is believed to be caused by infection with PCV 2. Affected animals, which are usually about six weeks of age, present with weight loss, dyspnoea and enlarged lymph nodes. The mortality rate may reach 10% in severe outbreaks. Adverse environmental factors or concurrent infections with other agents appear to be essential for the development of clinical disease. The diagnosis of PMWS is based on clinical signs and pathological findings. Antibodies to PCV 2 may be detected using indirect immunofluorescence or ELISA. Virus isolation in pig cell lines is also indicative of infection. A definitive diagnosis requires demonstration of PCV 2 antigen or viral nucleic acid in association with lesions. No vaccine is currently available. The resistance of the virus to detergents makes decontamination of infected premises difficult. Control is based on good husbandry, rapid removal of affected animals and the elimination of other infectious agents. It has been suggested that PCV 2 may be aetiologically involved in the porcine dermatitis and nephropathy syndrome.

References

Allan, G.M. and Ellis, J.A. (2000). Porcine circoviruses: a review. *Journal of Veterinary Diagnostic Investigation*, **12**, 3–14.

Niagro, F.D., Forsthoefel, A.N., Lawther, R.P. *et al.* (1998). Beak and feather disease virus and porcine circovirus genomes: intermediates between the geminviruses and plant circoviruses. *Archives of Virology*, **143**, 1723–1744.

Further reading

Done, S., Gresham, A., Potter, R. and Chennells, D. (2001). PMWS and PDNS – two recently recognised diseases of pigs in the UK. *In Practice*, **23**, 14–21.

Chapter 61

Retroviridae

Retroviruses (Latin *retro*, backwards) are labile enveloped RNA viruses, 80 to 100 nm in diameter. Seven genera are currently assigned to the family: *Alpharetrovirus*, *Betaretrovirus*, *Gammaretrovirus*, *Deltaretrovirus*, *Epsilonretrovirus*, *Lentivirus* and *Spumavirus* (Fig 61.1). The family name refers to the presence, in the virion, of a reverse transcriptase which is encoded in the viral genome.

The envelope, which is acquired from the plasma membrane of the host cell, surrounds an icosahedral capsid containing two linear, positive-sense, single strands of RNA and core proteins including the enzymes reverse transcriptase and integrase (Fig. 61.2). Historically, based on electron microscopy, retroviruses were categorized as A-type, B-type, C-type and D-type particles.

Reverse transcriptase acts as an RNA-dependent DNA polymerase, which transcribes from RNA to DNA. Four major genes, contained in the RNA genome of members of the *Retroviridae*, are *gag*, *pro*, *pol* and *env*. The *gag* (group specific antigen) gene encodes internal structural proteins. The *pro* (protease) gene encodes the enzyme protease while the *pol* (polymerase) gene encodes the enzymes reverse transcriptase and integrase. The *env* (envelope) gene encodes surface (SU) and transmembrane (TM) envelope glycoproteins. Cell entry follows attachment of an envelope glycoprotein to specific cell receptors. Under the influence of the reverse transcriptase, double-stranded DNA copies of the viral genome are synthesized in the cytoplasm of the host cell. During this process, repeat base sequences, containing several hundred base-pairs and called long terminal repeats (LTR), are added to the ends of the DNA transcripts. Transcripts are integrated into the chromosomal DNA, as provirus, at random sites through the action of viral integrase. The sites of proviral integration determine the extent and nature of cellular changes. The LTR contain important promoter and enhancer sequences, which are involved in the transcription of mRNA and virion RNA from provirus. Release of mature virions often occurs by budding from cell membranes (Figs. 61.3 and 61.4). If the provirus of certain retroviruses is inserted close to the host genes which regulate cell division, the proviral LTR may increase the rate of mitosis resulting in neoplasia (insertional mutagenesis).

A high mutation rate is a feature of retroviral replication because errors are relatively frequent during reverse transcription. In addition, recombination between retroviral genomes in doubly-infected cells can occur because reverse transcriptase can transfer from the RNA template of one virus to that of another. Consequently, antigenically different retroviruses frequently emerge and classification of species and subtypes often proves difficult.

Key points

- Enveloped, spherical labile viruses
- Diploid, containing two linear positive-sense single strands of RNA
- Icosahedral capsid surrounds helical nucleocapsid
- Members of this family are unique in possessing a reverse transcriptase which transcribes viral RNA to double-stranded DNA
- Double-stranded DNA is inserted as a provirus into the host genome
- Mutation and recombination occur with high frequency
- The family is composed of seven genera:
  - —*Lentivirus* contains viruses which usually cause immunodeficiency diseases
  - —*Spumavirus* contains viruses which cause cell vacuolation *in vitro* but do not produce disease
  - —Viruses in the remaining five genera can induce neoplastic change in specific cell types

Retroviruses can be categorized as endogenous or exogenous. Endogenous retroviruses occur widely among vertebrates. They resulted at some time from infection of germline cells and are transmitted only as provirus in germ cell DNA from parent to offspring. They are regulated by cellular genes and are usually silent. Endogenous retroviral genomes may contribute *env* genes to produce recombinant feline leukaemia viruses and avian leukosis viruses. Occasionally, they can be activated by irradiation, mutagens or carcinogens with the production of new virions. Endogenous retroviruses of pigs may be potentially dangerous in humans receiving xenotransplants.

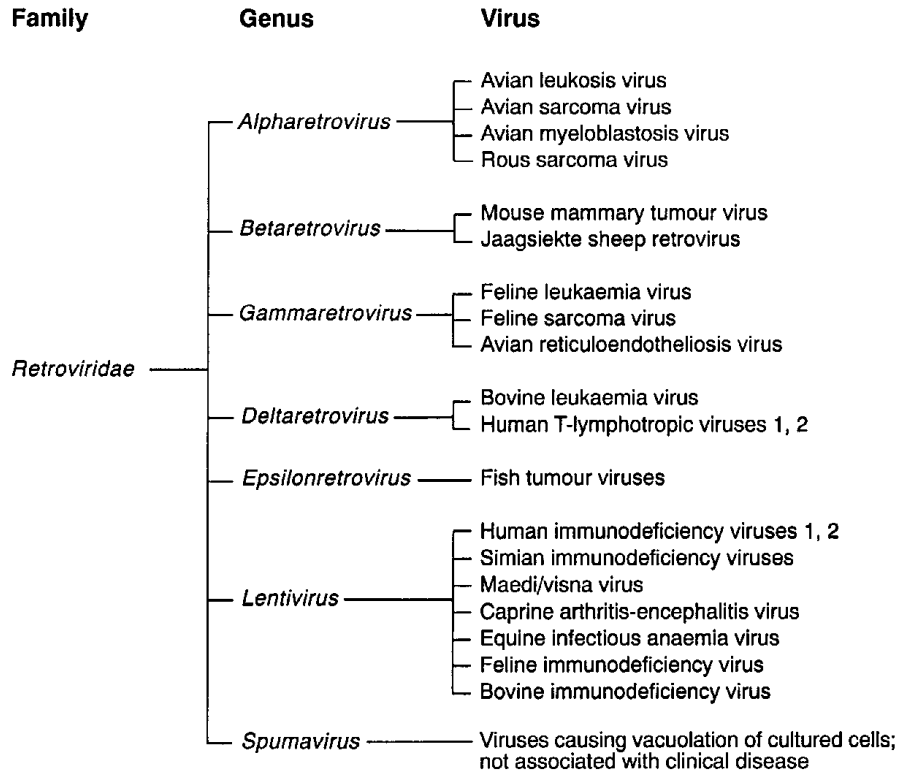


Figure 61.1 Classification of retroviruses with emphasis on those which produce disease or induce cellular changes *in vitro*.

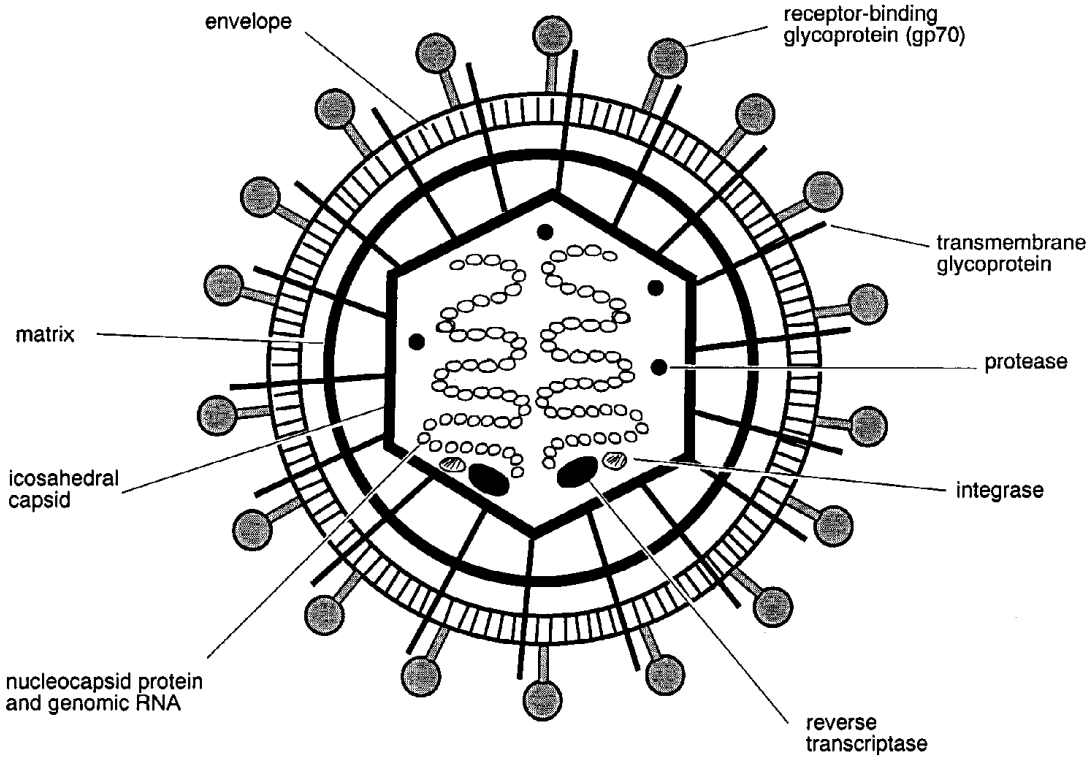


Figure 61.2 Schematic representation of the structures and components of a retrovirus virion.

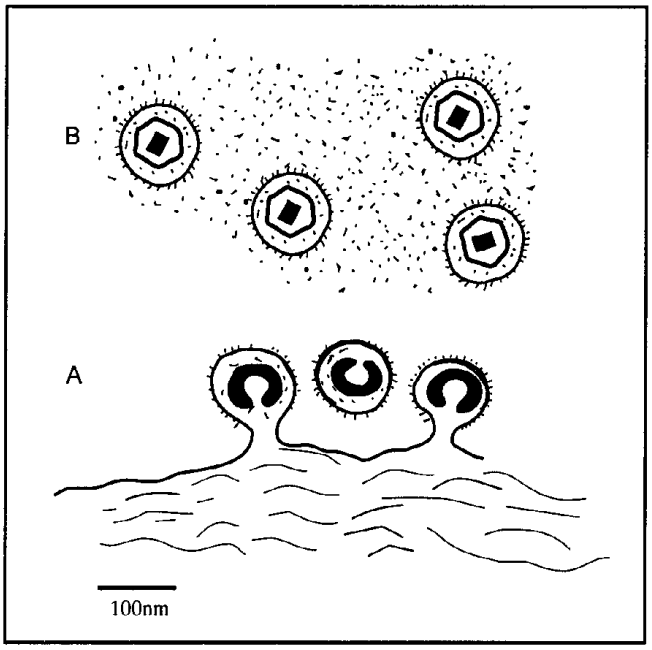


Figure 61.3 A. Budding of a typical type C retrovirus, with crescent-shaped nucleocapsid, from a cell membrane; B. Mature, extracellular type C retrovirus virions.

Retroviruses are sensitive to heat, lipid solvents and detergents. Because of their diploid genomes, they are relatively resistant to UV light.

Clinical infections

Retroviruses in the genera *Alpharetrovirus*, *Betaretrovirus*, *Gammaretrovirus*, *Deltaretrovirus* and *Epsilonretrovirus* are frequently referred to as oncogenic retroviruses because they can induce neoplastic transformation in cells which they infect.

On the basis of the interval between exposure to the virus and tumour development, exogenous oncogenic retroviruses are designated either as slowly transforming (*cis*-activating) viruses or as rapidly transforming (transducing) viruses. Slowly transforming retroviruses induce B cell, T cell or myeloid tumours after long incubation periods. For malignant transformation to occur, the provirus must be integrated into the host cell DNA close to a cellular oncogene (c-*onc*, proto-oncogene), resulting in interference with the regulation of cell division. Rapidly transforming retroviruses, which can induce tumour formation after short incubation periods, contain viral oncogenes (v-*onc*). Viral oncogenes are considered to be cellular oncogenes acquired by recombination during virus evolution. If the oncogene is integrated into the viral genome without loss of replicative virus genes, as in Rous sarcoma virus, the retrovirus is described as replication-competent. Frequently, as a consequence of cellular oncogene integration, existing viral sequences necessary for replication are deleted. Such replication-defective retroviruses, which cannot multiply without helper viruses, are rarely transmitted under normal field conditions. Occasionally, they may cause rapidly-developing neoplastic disease. A third method of tumour induction is exemplified by bovine leukaemia virus which depends on the *tax* gene encoding for a protein capable of up-regulating both viral LTR and cellular promoter sequences, even when the provirus is integrated into a different chromosome (*trans*-activation). A schematic representation of the important genes in various categories of oncogenic viruses is presented in Fig. 61.5.

The newly-established *Epsilonretrovirus* genus contains viruses associated with neoplasia in fish. The oncogenic retroviruses of poultry are presented in Table 61.1 and those of domestic mammals in Table 61.2.

Lentiviruses (Latin *lentus*, slow) cause diseases with long incubation periods and insidious protracted courses. Important animal and human diseases caused by lentiviruses include acquired immunodeficiency syndrome (AIDS), feline immunodeficiency, equine infectious anaemia and maedi/visna. Lentiviruses of domestic animals are presented in Table 61.3. Although spumaviruses (Latin *spuma*, foam) cause vacuolation of cultured cells, they are not associated with clinical disease. Spumaviruses have been identified in humans, cats and cattle.

Avian leukosis

The avian leukosis virus (ALV) group includes both replication-competent and replication-defective retroviruses. Neoplastic conditions in chickens including the

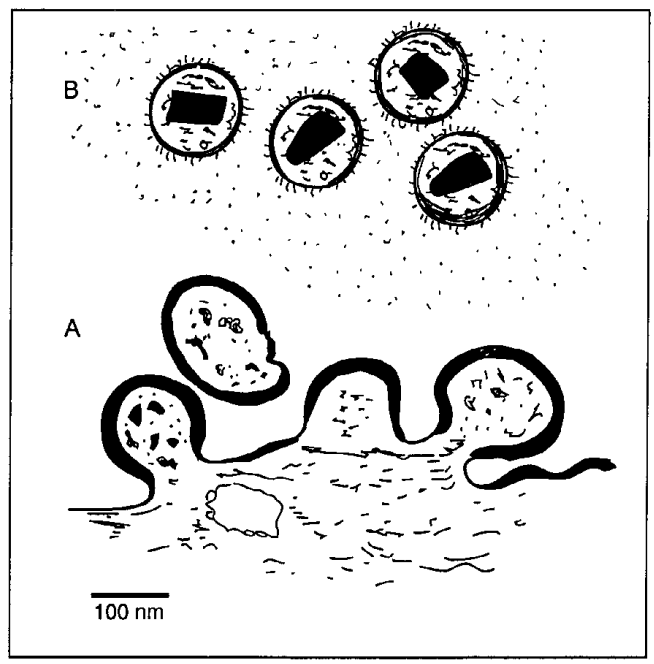


Figure 61.4 A. Budding of lentivirus particles from a cell membrane; B. Mature, extracellular lentivirus virions.

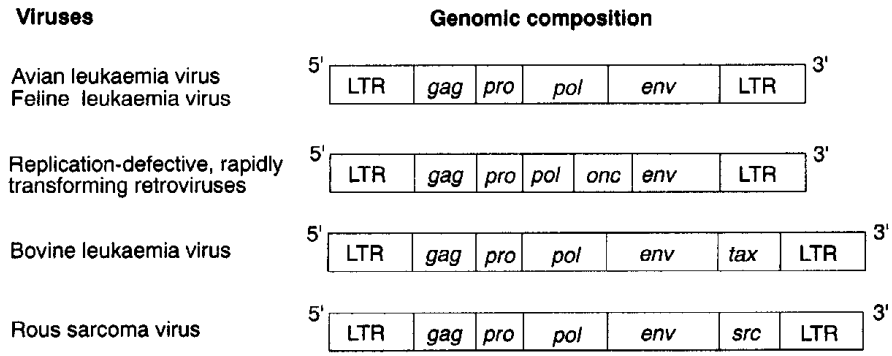


Figure 61.5 Schematic representation of the important genes present in oncogenic retroviruses.
LTR: long terminal repeat sequences; *gag*: gene encoding group-specific antigen (core and capsid proteins); *pro*: gene encoding protease; *pol*: gene encoding polymerase (reverse transcriptase); *env*: gene encoding envelope glycoproteins; *onc*: oncogene; *tax*: transactivating gene; *src*: sarcoma gene

lymphoid, erythroid and myeloid leukoses, fibrosarcoma, haemangiosarcoma, and nephroblastoma are caused by viruses in the group. Lymphoid leukosis, a B cell lymphoma, is the most common of these neoplastic conditions and is economically the most important.

Avian leukosis viruses are divided into ten subgroups (A to J) on the basis of differences in viral envelope glycoproteins. Isolates from chickens belong to subgroups A, B, C, D, E and J; viruses in the other subgroups infect other avian species. Most isolates from outbreaks of disease in chickens belong to subgroup A. Endogenous avian leukosis viruses, which are commonly present in chickens and are transmitted vertically in the germ line cells, usually belong to subgroup E. Members of subgroup J, a group recently recognized in broilers, are associated with myeloid leukosis and have arisen from recombination of a novel family of endogenous viruses (ev/J) and exogenous avian leukosis viruses (Benson *et al.*, 1998).

There is usually an incubation period of months to years between natural infection with ALV and the development of neoplasia, because of the time required for the genetic events to occur that lead to transformation of cells to malignancy. Neoplastic conditions associated with ALV include lymphoid leukosis, myeloid leukosis, sarcomas and renal tumours. Avian leukosis virus is also associated with osteopetrosis. One of the final steps in oncogenesis can be the generation of recombinant, rapidly transforming viruses, which have incorporated a cellular oncogene into their genome. Viruses isolated from such tumours include avian erythroblastosis virus, avian myeloblastosis virus and Rous sarcoma virus, all of which rapidly cause tumours when inoculated experimentally into susceptible chickens. These viruses are usually defective and require a helper ALV for replication. Rous sarcoma virus is exceptional in that it has an oncogene (*src*) in addition to a complete ALV genome, and consequently is replication competent as well as capable of rapid cell transformation *in vivo* and *in vitro*. This type of acutely transforming virus is very seldom, if ever, transmitted under natural conditions. The endogenous ALVs carried by chickens in most flocks do not directly cause tumours.

Epidemiology

Exogenous ALV is transmitted both vertically, through virus present in egg albumen, and horizontally, by close contact. Chicks which hatch from infected eggs are usually immunotolerant and exhibit persistent viraemia. They are the principal source of virus in a flock. Virus is transmitted in saliva and faeces to in-contact birds. Viral shedding into oviducts results in transmission to chick embryos. Chicks infected after hatching develop a transient viraemia before they produce neutralizing

Table 61.1 Oncogenic retroviruses of poultry.

Genus	Virus	Hosts	Comments
Alpharetrovirus	Avian leukosis virus	Chickens, pheasants, partridge, quail	Endemic in commercial flocks. Exogenous and endogenous transmission of virus can occur. Causes lymphoid leukosis in birds between 5 and 9 months of age
Gammaretrovirus	Reticuloendotheliosis virus	Turkeys, ducks, chickens, quail, pheasants	Infection usually subclinical. Sporadic disease may present with anaemia, feathering defects, impaired growth or neoplasia. Disease outbreaks have occurred following use of vaccine contaminated with reticuloendotheliosis virus

Table 61.2 Oncogenic retroviruses of domestic mammals.

Genus	Virus	Hosts	Comments
Betaretrovirus	Jaagsiekte sheep retrovirus (ovine pulmonary adenocarcinoma virus)	Sheep	Causes jaagsiekte, a slowly progressive neoplastic lung disease of adult sheep which is invariably fatal. Occurs worldwide except in Australasia
	Enzootic nasal tumour virus	Sheep	Closely related to Jaagsiekte sheep retrovirus. Causes adenocarcinoma of low grade malignancy which affects the nares
Gammaretrovirus	Feline leukaemia virus	Cats	Important cause of chronic illness and death in young adult cats. Causes immunosuppression, enteritis, reproductive failure, anaemia and neoplasia. Worldwide in distribution
Deltaretrovirus	Bovine leukaemia virus	Cattle	Causes enzootic bovine leukosis in adult cattle. A small percentage of infected cattle develop lymphosarcoma

antibodies. These birds often become carriers, shed virus intermittently and may produce infected chicks. Natural exposure of adult birds to infection does not usually result in virus shedding. Neoplasms develop most often in persistently-viraemic birds. Virus-neutralizing antibodies are passed from antibody-positive hens in the yolk sac to their chicks, providing passive immunity to infection for the first few weeks of life.

Pathogenesis

Following infection, virus spreads throughout the body, replicating in most tissues. Avian leukosis virus transforms B cells after integration of provirus close to the c-*myc* gene which induces cellular replication under the influence of the viral LTR promoter. Less commonly, ALV has been associated with erythroblastosis when the c-*erb*B gene in an erythroid cell is activated. Subgroup J isolates are associated with late onset myeloid leukosis in broilers (Benson *et al.*, 1998). Rapidly-transforming viruses are formed rarely in individual birds by transfer of a c-*onc* (proto-oncogene) into the ALV provirus during its integration. Multiple insertions of the provirus into the host cell genome results in exaggerated gene expression and over-production of a transformation-associated protein. More than a dozen different oncogenes have been identified in transforming avian retroviruses. The protein products of oncogenes may act as hormone or growth factor receptors, transcription control factors and kinases in signal transduction pathways. Although a particular virus strain may be capable of producing neoplasia of more than one cell type, usually one cell type predominates in an affected animal.

Table 61.3 Lentiviruses of domestic animals.

Genus	Virus	Hosts	Comments
Lentivirus	Feline immunodeficiency virus	Cats	Causes life-long infection with persistent viraemia and immunosuppression in cats over 5 years of age. Worldwide distribution
	Equine infectious anaemia virus	Horses, mules, donkeys	Causes life-long infection with recurring febrile episodes. Anaemia is a prominent clinical sign
	Maedi/visna virus	Sheep	Causes life-long infection with progressive respiratory disease and indurative mastitis in older sheep. Clinical signs develop in a small percentage of infected animals. Some infected sheep develop progressive neurological disease
	Caprine arthritis-encephalitis virus	Goats	Causes life-long infection. Associated with polyarthritis and indurative mastitis in adults and progressive nervous disease in kids. Common in dairy goat herds. Worldwide distribution
	Bovine immunodeficiency virus	Cattle	Widely distributed; pathogenicity currently uncertain
	Jembrana disease virus	Cattle	Closely related to bovine immunodeficiency virus but disease described only in Bali cattle in Indonesia. Acute disease which occurs within days of infection is characterized by fever, anorexia, enlarged lymph nodes and, in some instances, death. Animals which recover remain viraemic

Clinical signs

The incubation period for lymphoid leukosis is usually more than four months. Disease is generally sporadic in infected flocks but occasional epidemics have been described. Affected birds become inappetent, weak and emaciated. They have pale wattles and the liver and bursa of Fabricius may be enlarged. Osteopetrosis, in which the long bones of the legs become visibly thickened, sometimes accompanies lymphoid leukosis. Subclinical infections with ALV are associated with depressed egg production and fertility, decreased hatchability and growth rate, and increased death rates. Economic loss from lymphoid leukosis is mainly due to deaths in egg-laying and breeding birds between five and nine months of age.

Diagnosis

- Postmortem findings and histopathological determination of tumour type are usually diagnostic.
- Differentiation from Marek's disease is important and is based on the age of affected birds, the presence of bursal tumours, absence of thickening of peripheral nerves and histological assessment of neoplastic cell types.
- Virus isolation is difficult and not usually attempted.
- Commercial ELISA kits for the detection of ALV group-specific antigen are available. The presence of flock infection can be demonstrated by detecting antibodies in serum or egg yolk. Suitable assays include virus neutralization, ELISA and indirect immunofluorescence.
- The polymerase chain reaction has been adapted for the detection of ALV nucleic acid.

Control

The eradication of exogenous subgroup A ALV infection has been successfully achieved in most commercial chicken flocks. The cycle of vertical transmission is interrupted by hatching and rearing infection-free chicks in isolation. Ongoing monitoring for the presence of infection is essential. Because autosomal genes encode subgroup-specific cell surface receptors through which ALV gains entry to the cell, genetically resistant birds can be bred with decreased numbers of specific cell surface receptors. Attention has focussed largely on resistance to subgroup A viruses. Selection for genetic resistance is an ongoing process because mutant viruses capable of overcoming host resistance frequently arise. Control of ALV infection in commercial flocks is based on high standards of hygiene and effective management aimed at reducing levels of infection. Birds from disease-free or genetically-resistant stock should be used for breeding. Because the virus is labile, all-in all-out management systems along with thorough washing and disinfection programmes between batches of birds are effective. Chicks of uncertain status should not be mixed with disease-free chicks. Chick rearing should take place remote from older birds. Vaccination with inactivated or modified live ALV vaccines has not been successful. Recombinant avian leukosis and fowlpox viruses expressing subgroup A envelope glycoproteins have been shown to have potential as effective vaccines.

Feline leukaemia and associated clinical conditions

Infection with feline leukaemia virus (FeLV) not only results in feline leukaemia but is also associated with a variety of other clinical conditions. Isolates of FeLV, a gammaretrovirus, are assigned to three subgroups (A, B and C) on the basis of differences in the gp70 envelope glycoprotein. Feline leukaemia virus A (FeLV-A), the predominant subgroup, is isolated from all FeLV-infected cats. Viruses of subgroup B, which arise through recombination between the *env* genes of FeLV-A and endogenous FeLV-related proviral DNA, are present in about 50% of isolates. FeLV-B is only transmitted together with FeLV-A. In a proportion of cats exposed to a mixture of FeLV-A and FeLV-B, the FeLV-B component is lost. Therefore, the continuing survival of FeLV-B depends upon the generation of new recombinants in cats persistently infected with FeLV-A. Cats that are infected with both FeLV-A and FeLV-B have a higher risk of developing tumours than those infected with FeLV-A alone. Each FeLV-C isolate is unique, arising *de novo* in a FeLV-A infected cat through mutations in the receptor-binding region of the FeLV-A *env* gene. Once generated, FeLV-C viruses rapidly cause a fatal anaemia and consequently are not transmitted to other cats.

Like the ALVs, FeLV causes tumours by several means, including insertional mutagenesis and recombination with a variety of cellular proto-oncogenes to produce acutely transforming, replication-defective viruses. Examples of the latter are FeLVs isolated from thymic lymphomas, and feline sarcoma viruses (FeSV) that are isolated from rare multicentric fibrosarcomas in young cats. These viruses are not transmitted under natural conditions.

Epidemiology

Infection with feline leukaemia virus, which occurs in domestic cats worldwide, is an important cause of mortality. Close contact is required for transmission of this labile virus and the incidence of infection is related to population density. Highest infection rates are found in catteries and multicat households. Large amounts of virus are shed in saliva with smaller quantities present in tears, urine, milk and faeces. Infection is usually acquired by licking, grooming and through bite wounds. Young kittens are more susceptible to infection than adults. Although maternally-derived antibody is protective in kittens up to six weeks of age, a significant proportion of those exposed before 14 weeks of age become persistently infected. Such

animals constitute the main reservoir of FeLV and they are prone to develop an FeLV-related disease. Most cats, which are exposed after four months of age, develop immunity and eliminate the virus. Kittens born to persistently-infected queens develop persistent infection, acquired either transplacentally or through ingestion of milk.

Pathogenesis

Following oronasal exposure, virus replicates in the lymphoid tissues of the oropharyngeal region. The virus spreads in infected mononuclear leukocytes to other lymphoreticular tissues and bone marrow. In most cats, cell mediated immunity and neutralizing antibodies to the gp70 envelope glycoprotein are produced at this stage, usually resulting in virus elimination. However, a latent bone marrow infection, which is eliminated after several months, is present in about 50% of cats. Failure to contain the infection results in extensive virus production in the bone marrow and persistent viraemia. The virus, present in both leukocytes and the plasma, is disseminated to glandular and mucosal epithelia. Large quantities of virus are shed from the salivary glands and the upper respiratory tract. Because the production of virus particles requires cellular DNA synthesis, tissues with high mitotic activity such as bone marrow and epithelia are targeted. Prolonged periods of viral replication in haemolymphatic tissues can lead to depletion of lymphoid and myeloid cells producing immunosuppression and anaemia. Severe immunosuppression is caused by infection with certain strains of FeLV-A. Isolates of FeLV-C are associated with severe non-regenerative anaemia.

Neoplastic changes in lymphoid or myeloid cells follow insertion of provirus close to a cellular oncogene with activation or deregulation of the gene, or generation of acutely transforming viruses.

Clinical signs

The incubation period ranges from months to years. The majority of persistently-infected cats die within three years of infection. About 80% of these cats die from non-neoplastic FeLV-associated disease; the remaining 20% of infected cats succumb to neoplasia, particularly lymphosarcoma. During the early phase of infection, cats may develop fever, malaise and lymphadenopathy, which may not be detected clinically. A variable period as asymptomatic carriers follows. Clinical signs, which are often non-specific and chronic, are usually seen in young adult cats between two and four years of age. Anaemia, reduction in reproductive performance, enteritis and a variety of secondary infections due to the immunosuppressive effects of the virus are important features of the disease. Immune complex formation initiated by circulating antigen may give rise to glomerulonephritis.

Lymphosarcoma, the most commonly occurring feline tumour, is usually linked to infection with FeLV. Thymic, alimentary, multicentric and leukaemic forms of lymphosarcoma are described and clinical signs relate to the anatomical sites involved. Fibrosarcoma and myeloid tumours, which have also been associated with FeLV infection, occur less frequently than lymphosarcoma.

Diagnosis

Detection of viral antigen in blood or saliva is the method commonly used for the laboratory diagnosis of feline leukaemia. Virus isolation, which is expensive and time-consuming, is used as a confirmatory test.

- Commercial ELISA and rapid immunomigration tests, designed to detect the major capsid protein (p27), are available.
- The immunofluorescent antibody test can be used to detect viral antigen in the cytoplasm of leukocytes in blood smears. It is commonly employed as a confirmatory test because it is more sensitive and specific than ELISA.
- Serological testing for antibodies is not used for diagnosis since viraemic cats are immunotolerant and do not have anti-FeLV antibodies. However, the demonstration of virus neutralizing antibodies indicates that a cat is immune and resistant to infection.
- An antigen termed feline oncovirus-associated cell membrane antigen (FOCMA) is expressed in all FeLV- and FeSV-transformed cells. The development of antibodies to FOCMA provides protection against FeLV-associated neoplasia.

Control

A test and removal policy has been shown to be effective in eradicating infection from catteries. The status of individual cats should be confirmed by retesting after 12 weeks. Infected cats, which must be separated from susceptible animals, should be excluded from breeding programmes. Serological testing at intervals of six months is recommended and cats about to be introduced into a cattery should be isolated until test results are known. Several commercial vaccines, including killed whole virus, recombinant canarypox virus, subunit and recombinant subunit types, are available. Vaccination does not alter the course of infection in persistently-infected cats. Accordingly, cats should be tested prior to vaccination. Vaccination does not provide complete protection and other appropriate control measures should be implemented.

Enzootic bovine leukosis

This retroviral disease of adult cattle is characterized by persistent lymphocytosis, the presence of circulating antibody to the causal agent bovine leukaemia virus (BLV) and the development of B cell lymphosarcoma in a number of infected animals. Enzootic bovine leukosis (EBL) has a

worldwide distribution. Some countries have eradicated the disease; other countries are embarking on eradication programmes.

Epidemiology

Transmission, which can occur by direct contact or transplacentally, usually takes place through transfer of blood or secretions such as colostrum and milk containing infected lymphocytes. Infection with BLV is lifelong. The labile virus is intimately cell-associated.

Less than 10% of calves born to infected dams are infected at birth. Calves are protected from contact infection for several months by maternally-derived antibody. Animals are usually infected between six months and three years of age (Hopkins and DiGiacomo, 1997). Iatrogenic transmission is important and has been linked to reuse of needles, multidose injectors, contaminated surgical instruments and rectal examination procedures. While biting flies may transmit the virus mechanically, their importance as vectors is uncertain. The prevalence of infection is higher in dairy cattle than in beef cattle. Susceptibility to infection is influenced by genotype and related to the bovine major histocompatability antigen type.

Pathogenesis

The primary target cell is the B lymphocyte. Bovine leukaemia virus does not possess an oncogene. Nucleic acid sequences at the 3' end of the *env* gene termed the X region, encode for the regulatory proteins Tax and Rex which are central to neoplastic transformation. The Tax protein interacts with cellular transcription factors resulting in transactivation of the promoter in the LTR of integrated BLV provirus. Upregulation of some cellular genes including those coding for IL-2 and its receptor may also occur.

Clinical signs

Although infections are lifelong, most animals remain subclinically infected. About 30% of infected animals develop persistent lymphocytosis, an increase in lymphocyte numbers in the blood without clinical signs of disease. A small percentage of BLV-seropositive cattle eventually develop lymphosarcoma. Clinical disease usually occurs in adult animals between four and eight years of age. The presenting signs, which relate to the sites of tumour formation, include enlargement of superficial lymph nodes, digestive disturbance, inappetence, weight loss and general debility.

Diagnosis

Enzootic bovine leukosis (EBL) must be differentiated from sporadic bovine leukosis which usually affects calves and young adult cattle. Formerly, blood lymphocyte counts were employed for laboratory diagnosis of infected animals and for the eradication of EBL. However, all infected cattle do not develop lymphocytosis. Serological testing for virus-specific antibody is now used for diagnosis and eradication.

- Several serological tests including AGID, ELISA and radioimmune assay are suitable for the detection of antibodies to BLV. Antibodies detected in calves less than six months of age may be of colostral origin.
- Although virus can be isolated by cultivation of peripheral blood lymphocytes, this technique is not performed routinely.
- The polymerase chain reaction has been developed as a sensitive research tool for the detection of provirus in peripheral blood lymphocytes.

Control

No commercial vaccine is currently available. Test and removal strategies have been successfully used for both national and individual herd eradication programmes. Serological testing at six month intervals is recommended (Brunner *et al.*, 1997). In countries in which the prevalence of BLV infection is too high to permit removal of all seropositive animals from herds, management practices aimed at reducing the spread of infection should be adopted. Such practices include separating infected and susceptible animals, rearing calves on milk from non-infected cows and serological testing of replacement animals.

Jaagsiekte

This lentiviral disease, also called ovine pulmonary adenomatosis, is a slowly-progressing neoplastic disease of adult sheep. It is caused by jaagsiekte sheep retrovirus (JSRV), also known as ovine pulmonary adenocarcinoma virus. Jaagsiekte is an Afrikaaner word meaning 'panting sickness'. With the exception of Australasia, jaagsiekte has a wide geographical distribution. The infection occurs rarely in goats. Multiple copies of endogenous retroviruses related to JSRV have been found in the genomes of both sheep and goats. Jaagsiekte sheep retrovirus is not considered to be endogenous.

Epidemiology

Respiratory exudates from affected sheep are infectious and transmission occurs by the respiratory route. Close contact facilitates spread of infection with the incidence of disease highest in housed animals. Within an infected flock, disease incidence may be up to 20%; it is influenced by breed and type of flock management.

Pathogenesis

The virus replicates in two types of pulmonary cells, type II alveolar cells and non-ciliated bronchial cells. Tumours arising from these cell types progressively replace normal lung tissue leading to death from asphyxia. About 10% of

the tumours metastasize to regional lymph nodes. Metastasis to heart or skeletal muscle occurs rarely. As the presence of an oncogene has not been demonstrated in the viral genome, the mechanism of neoplastic transformation is unclear. Recent studies indicate that the envelope protein has transformation potential (Maeda *et al.*, 2001).

Clinical signs

The incubation period may range from several months up to two years. Tumour nodules have been detected in lambs as early as ten days after experimental inoculation. Affected animals are usually three to four years of age and in poor bodily condition. They display respiratory embarrassment and mouth breathing, particularly after exercise. By raising the hindlegs and lowering the head (wheelbarrow test), a clear fluid flows from the nostrils. Moist rales may be heard. Often, only a single animal in a flock may be clinically affected. The course of the disease may extend over weeks or months. Secondary pasteurellosis is a frequent complication.

Diagnosis

The characteristic clinical signs may be masked in individual cases by secondary infection. Histopathological confirmation is desirable. Attempts to culture the virus in monolayers have been unsuccessful. It is possible to detect virus in lung exudates or washings by ELISA and viral nucleic acid may be detected by PCR. As infected animals do not appear to develop a specific humoral immune response (Ortin *et al.*, 1998), it is not possible currently to confirm infection by serology.

Control

Jaagsiekte was successfully eradicated from Iceland in 1952 after drastic depopulation procedures. The incidence of disease in a flock can be reduced by strict isolation and elimination of suspect animals immediately after clinical or laboratory confirmation.

Feline immunodeficiency virus infection

This condition was first reported in 1987 and infection with feline immunodeficiency virus (FIV), a lentivirus, is now recognised worldwide as an important cause of disease in cats. The infection in cats is sometimes referred to as 'feline AIDS' on account of similarities to acquired immunodeficiency syndrome (AIDS) caused by human immunodeficiency virus. Five subtypes of FIV have been identified based on diversity in the envelope gene amino acid sequences. This diversity may account for differences in the pathogenesis and clinical progression of the disease associated with different isolates.

Epidemiology

Infection with FIV occurs in domestic cats. Related lentiviruses have been isolated from a number of wild *Felidae*, including pumas and lions. Animals remain infected for life. Virus is shed mainly in the saliva and transmission usually occurs through bites. Accordingly, infection rates are highest in free-roaming, adult male cats. Non-aggressive intimate contact may also be important in transmission under natural conditions. Queens may transmit infection to kittens *in utero*, during parturition or in milk, particularly during the acute phase of infection.

Pathogenesis

The virus replicates principally in $CD4^+$ (helper) T lymphocytes. Replication also occurs in macrophages, astrocytes and microglial cells. Infected cats remain persistently viraemic. Viraemia, the level of which increases rapidly after infection, peaks at seven to eight weeks before gradually declining and increasing again during the terminal stage of the disease. Humoral responses are normal, or occasionally enhanced, with antibodies appearing two weeks after infection. However, there is progressive deterioration in cell-mediated immunity due to depletion of $CD4^+$ T lymphocytes. Lymphocyte depletion is attributed to the cytopathic effect of virus in addition to decreased production of lymphocytes and apoptosis. Other immunological abnormalities associated with the infection include reduced interleukin-2 responsiveness and production, impaired lymphocyte blastogenesis in response to mitogens and reduced antibody response to T cell-dependent antigens. Reduction in $CD4^+$ lymphocyte numbers, increased production of virus, the emergence of variants with increased virulence and infection with opportunistic pathogens contribute to the development of clinical immunodeficiency.

Clinical signs

The prevalence of clinical disease is highest in cats over six years of age. The course of the disease may be divided into an acute phase, an asymptomatic phase, a phase characterized by vague clinical signs and a terminal phase with marked immunodeficiency (Hartmann, 1998). The acute phase which may last several weeks or months is manifested by pyrexia, generalized lymphadenopathy and neutropenia. A prolonged period during which infected cats appear clinically normal follows. The third phase is marked by recurrent fever, leukopenia, anaemia, weight loss, lymphadenitis, chronic gingivitis and behavioural changes. Opportunistic infections are frequent in the terminal phase of the disease. Chronic stomatitis and gingivitis are common findings. Other manifestations include chronic respiratory, enteric and skin infections. Many cats present with weight loss, anaemia and leukopenia. Neurological signs, usually due to direct viral damage, develop in a small number of infected cats. Concurrent FeLV infection may exacerbate the immunodeficiency and accelerate the appearance of clinical signs. An increased incidence of neoplasia,

particularly B cell lymphomas, is recorded in FIV-infected cats. Not all infected cats develop disease.

Diagnosis

- Serological testing for antibodies to FIV is the principal method for confirming infection. Commercial ELISA and immunoconcentration kit-sets are available. Alternative tests include immunoblotting and indirect immunofluorescence. Some cats fail to produce antibodies for several months following infection; antibody levels may become undetectable in terminally-ill cats. The kittens of infected queens may remain seropositive for up to five months due to ingestion of colostral antibodies.
- Although virus isolation from blood or saliva is possible, it is not considered realistic for routine diagnostic purposes.
- Proviral DNA can be detected using the polymerase chain reaction.

Treatment and control

Treatment is primarily aimed at the control of secondary infections. A number of antiviral drugs, such as azidothymidine, directed against viral reverse transcriptase have a beneficial effect in clinically ill cats but do not eliminate infection. A commercial vaccine is not currently available. Development of an effective vaccine is complicated by several factors including the existence of multiple virus subtypes (Torres *et al.*, 1997). Control is based on prevention of exposure by separating infected and non-infected cats in multicat households, by preventing cats from roaming freely, by using seronegative queens for breeding and by screening all cats before introduction into seronegative populations.

Equine infectious anaemia

This disease, also called swamp fever, affects horses, mules and donkeys in many countries. It is caused by a lentivirus, equine infectious anaemia virus (EIAV). Infected *Equidae* remain viraemic for life.

Epidemiology

Virus is transmitted mechanically by haematophagous insects particularly *Tabanus* species and *Stomoxys* species. The virus survives for only short periods on the mouthparts of biting flies. These haematophagous insects usually obtain a complete blood meal from a single host. If interrupted during feeding, they may transfer virus to another host when they resume feeding. Transmission occurs most often in the summer, during periods of high insect activity, in low-lying swampy areas close to woodlands, the preferred habitat of tabanids. Iatrogenic transmission can occur through contaminated needles or surgical instruments. Although *in utero* transmission occurs, it is uncommon.

Pathogenesis

Virus replicates in macrophages, monocytes and Kupffer cells. A cell-associated viraemia develops with dissemination throughout the body (Oaks *et al.*, 1998). Infected horses fail to eliminate the virus despite mounting a strong immune response. They become persistently-infected following insertion of provirus into the genome of host cells. With the continuous production of virus particles, many target cells become infected. In the course of further provirus production by reverse transcription in infected cells, mutations frequently arise due to errors during the transcription process. This can result in the emergence of new virus strains exhibiting antigenic variation in envelope glycoproteins (antigenic drift). Febrile episodes and marked immune stimulation signal the emergence of these new strains. Non-neutralizing antibodies produced against virus early in the course of infection lead to the formation of immune complexes. Such immune complexes activate complement contributing to fever, anaemia and thrombocytopenia, and initiating glomerulonephritis. Haemolysis, enhanced erythrophagocytosis and depressed erythropoiesis are responsible for the anaemia in chronically affected horses. In most animals, clinical episodes eventually cease, probably as a consequence of a broad based neutralizing response against a wide range of viral epitopes.

Clinical signs

The majority of infected horses display mild signs which may go undetected. Most of the clinical signs are attributed to the host's immune response rather than to direct viral damage. Following an incubation period of up to three weeks, infected animals may present with fever, depression and petechiae on mucous membranes and conjunctivae. Rarely, severe epistaxis and ventral oedema may be followed by death. However, most horses recover from this phase and remain clinically normal for several weeks when recrudescence of signs may recur. The number and severity of recurring disease episodes varies widely. Most occur during the first year after infection and decline in number thereafter. Many horses which appear clinically normal remain carriers. Some exhibit a chronic form of the disease characterized by weight loss, anaemia, ventral oedema and debilitation, leading eventually to death.

Diagnosis

Laboratory confirmation of infection is based on the demonstration of serum antibodies to the core virus protein p26.

- The serological test recognized for international trade is the AGID test (Coggins test). Although the ELISA is a suitably sensitive assay, positive results should be confirmed by the more specific AGID test. Results can also be confirmed by immunoblotting.

- Antibodies may not be detectable early in the course of the disease.
- False-positive results may be encountered in foals up to six months of age due to the presence of colostral antibodies.
- The presence of virus in blood can be demonstrated by inoculating a susceptible horse.
- Virus can be isolated in leukocyte cultures prepared from the blood of susceptible horses. Due to the time and expense involved, virus isolation is rarely attempted.
- Proviral DNA may be detected by PCR and viral RNA can be detected using RT-PCR.

Control

Commercial vaccines are not available in western countries and control measures are aimed at reducing the risk of infection. In many countries, legislation requires certification of freedom from EIA prior to importation. Restriction of animal movement is also used to minimize the risk of disease spread. Management practices including detection and removal of seropositive animals, insect control and testing of animals prior to introduction onto a property are worthwhile control measures. Care should be taken to ensure that chemicals used for disinfecting surgical instruments inactivate EIAV.

Small ruminant lentivirus group

Two distinct lentiviruses have been described in small ruminants. Maedi/visna virus (MVV) affects sheep and caprine arthritis-encephalitis virus (CAEV) affects adult goats and kids. These viruses are closely related and cause persistent infections and comparable disease syndromes. Each virus can infect both species. Genomic analyses of these ovine and caprine lentivirus isolates suggest that they evolved from a common ancestral genotype. The current view is that they comprize a heterogeneous group with a variable host range and different pathogenic capabilities (Pasick, 1998).

Maedi/visna

This lentiviral disease, also referred to as ovine progressive pneumonia, la bouhite and zwoegersiekte, occurs in many countries. Maedi/visna virus causes a life-long infection in sheep and is associated with chronic progressive disease in adult sheep. Maedi and visna, Icelandic words meaning 'laboured breathing' and 'wasting' respectively, refer to the clinical features encountered in the respiratory form and the rare nervous form of the disease. Maedi/visna caused significant losses in Icelandic sheep prior to its eradication in 1965. Apart from Iceland, Australia and New Zealand, ovine lentiviruses are present in most countries.

Epidemiology

Infection is frequently subclinical. The clinical severity of disease is influenced by viral virulence, the age of the host when exposed and other host factors. Although virus is widely distributed in infected animals, it is transmitted principally in pulmonary exudates and milk. Horizontal transmission occurs by inhalation of aerosols and is facilitated by close confinement of animals during winter housing. In flocks with endemic disease, lambs are usually exposed to aerosol infection. Other methods of transmission such as ingestion of faecally-contaminated water may occur but are of minor importance.

Pathogenesis

Clinical signs take several months or years to develop. The chronic, progressive inflammation is characterized by mononuclear cell infiltration and lymphoproliferation particularly in the lungs and mammary glands. Lesions are also present in synovial membranes and in the brain. Persistent infection and the interaction between viral antigens and cellular and humoral components of the immune system are responsible for the development of lesions. Provirus integrated into the genome of monocytes and their precursors is activated only when the monocytes develop into macrophages. Restricted virus production in the monocytes permits insidious spread of virus around the body with minimal immune stimulation ('Trojan horse' mechanism). The occurrence of antigenic variants during the course of an infection as a result of genetic instability also facilitates persistence. For some weeks after infection, there is a period of viraemia in most animals. A vigorous immune response, sufficient to restrict virus production to low levels, does not eliminate infection. Both humoral and cell-mediated responses occur in infected sheep but neither is fully effective and may, in fact, contribute to the pathogenesis of the disease. The period from infection to seroconversion is usually up to eight weeks but may take several months or years. This delay in antibody response reflects a low level of viral antigen production (Brodie *et al.*, 1998).

Clinical signs

Clinical disease is rarely observed in animals less than two years of age. A small percentage of infected animals develop clinical signs. Respiratory distress which becomes more severe as the disease progresses is the most common clinical presentation. Affected sheep are usually afebrile until the later stages of the disease when secondary bacterial infection often occurs. Death is due to anoxia or to secondary infection. Indurative mastitis with reduction in milk production, which results in poor growth of lambs, is a relatively common finding. Lameness and swelling of one or more joints, particularly the carpus and tarsus, has been described in some flocks. Neurological signs, which are relatively rare, may occur in association with other signs of the disease.

Diagnosis

A presumptive clinical diagnosis of maedi/visna can be confirmed serologically and should be conducted on a flock basis. The time required for seroconversion may be long and unpredictable but, once antibody production is initiated, it continues. Commonly used serological assays include AGID, ELISA and Western blotting. Virus isolation is possible but time-consuming and expensive. Viral nucleic acid can be detected in peripheral blood and in tissues by PCR but the procedure is not widely used.

Control

Eradication of MVV infection requires regular serological monitoring and removal of seropositive animals. Newborn lambs should be removed at birth from infected dams and reared separately. Colostrum and milk from certified MVV-free ewes should be used for feeding lambs. An effective vaccine is not available.

Caprine arthritis-encephalitis

This lentivirus disease of goats, characterized by polyarthritis in adults and, rarely, by leuko-encephalomyelitis in kids, occurs worldwide. It is caused by persistent infection with caprine arthritis-encephalitis virus (CAEV).

Epidemiology

Although infection is common among dairy goats in most countries, clinical disease is uncommon. The virus is usually acquired by kids during the neonatal period through ingestion of colostrum or milk from infected does. Close contact between susceptible and infected goats over an extended period is required for horizontal transmission.

Pathogenesis

The pathogenesis is similar to that of maedi/visna with persistent infection and non-protective immune mechanisms responsible for lesion development. Virus production occurs after infected monocytes develop into macrophages. Viral antigens provoke a vigorous cell-mediated reaction which is responsible for the characteristic changes in target tissues.

Clinical signs

The most common presentation in adult animals is arthritis which is insidious in onset and slowly progressive. Swelling of affected joints particularly the carpus occurs but the degree of lameness is variable. Affected animals gradually lose weight. Reduced milk production relates to chronic mastitis. A rapidly progressive neurological disease, which can affect kids up to four months of age, manifests as posterior paresis progressing to quadriplegia.

Diagnosis

Laboratory confirmation relies on detection of virus-specific antibodies. The most commonly used assays are AGID and ELISA. Virus isolation can be achieved by co-cultivating leukocytes from blood or milk with synovial membrane cells. Nucleic acid detection by PCR is also possible.

Control

This is based on test and segregation programmes. Infected flocks should be tested regularly and seropositive animals should be culled; kids should be separated from their dams at birth and reared on heat-treated colostrum or pasteurized milk. An effective vaccine is not available.

References

Benson, S.J., Ruis, B.L., Fadly, A.M. and Conklin, K.F. (1998). The unique envelope gene of the subgroup J avian leukosis virus derives from ev/J proviruses a novel family of avian endogenous viruses. *Journal of Virology*, **72**, 10157–10164.

Brodie, S.J., de la Concha-Bermejillo, A., Snowder, G.D. and DeMartini, J.C. (1998). Current concepts in the epizootiology, diagnosis and economic importance of ovine progressive pneumonia in North America: a review. *Small Ruminant Research*, **27**, 1–17.

Brunner, M.A., Lein, D.H. and Dubovi, E.J. (1997). Experiences with the New York State bovine leukosis eradication and certification programme. *Veterinary Clinics of North America: Food Animal Practice*, **13**, 143–150.

Hartmann, K. (1998). Feline immunodeficiency virus infection: an overview. *Veterinary Journal*, **155**, 123–137.

Hopkins, S.G. and DiGiacomo, R.F. (1997). Natural transmission of bovine leukaemia virus in dairy and beef cattle. *Veterinary Clinics of North America: Food Animal Practice*, **13**, 107–128.

Maeda, N., Palmarini, M., Murgia, C. and Fan, H. (2001). Direct transformation of rodent fibroblasts by jaagsiekte sheep retrovirus DNA. *Proceedings of the National Academy of Sciences, USA*, **98**, 4449–4454.

Oaks, J.L., McGuire, T.C., Ulibarri, C. and Crawford, T.B. (1998). Equine infectious anaemia virus is found in tissue macrophages during subclinical infection. *Journal of Virology*, **72**, 7263–7269.

Ortin, A., Minguijor, E., Dewar, P. *et al.* (1998). Lack of a specific immune response against a recombinant capsid protein of jaagsiekte sheep retrovirus in sheep and goats naturally affected by enzootic nasal tumour or sheep pulmonary adenomatosis. *Veterinary Immunology and Immunopathology*, **61**, 229–237.

Pasick, J. (1998). Maedi-visna virus and caprine arthritis-encephalitis virus: distinct species or quasispecies and its implications for laboratory diagnosis. *Canadian Journal of Veterinary Research*, **62**, 241–244.

Torres, B., Elyar, J.S., Okada, S. and Yamamoto, J.K. (1997). Fundamentals of FIV infection: is a vaccine possible? *Feline Practice*, **25**, 6–12.

Chapter 62

Reoviridae

Viruses in the family *Reoviridae* were originally isolated from respiratory and enteric sources without any associated disease, namely orphan. These icosahedral viruses, 60 to 80 nm in diameter, are non-enveloped and possess a layered capsid which is composed of concentric protein shells (Fig. 62.1). The genome of the virion is composed of ten to twelve segments of double-stranded RNA. Genetic reassortment readily takes place in cells co-infected with viruses of the same species. Replication occurs in the cytoplasm of host cells often with the formation of intracytoplasmic inclusions. The family contains nine genera. Members of the genera *Orthoreovirus*, *Rotavirus* and *Orbivirus* infect animals and man (Fig. 62.2). Members of the genus *Coltivirus*, which primarily infect rodents and humans, occasionally cause clinical disease in domestic animals. The genera *Fijivirus*, *Phytoreovirus* and *Oryzavirus* contain viruses of plants. The genus *Cypovirus* contains viruses of arthropods while members of the genus *Aquareovirus* infect fish. Viruses in the family are moderately resistant to heat, organic solvents and non-ionic detergents. Orthoreoviruses and rotaviruses are stable over a wide range of pH values unlike orbiviruses which lose infectivity at low pH values.

Key points

- Non-enveloped viruses with double or triple layered capsid and icosahedral structure
- Segmented double-stranded RNA
- Replicate in cytoplasm
- Three genera of veterinary importance *Orthoreovirus, Orbivirus, Rotavirus:*
  - —Orthoreoviruses cause arthritis and tenosynovitis in poultry
  - —Rotaviruses cause enteritis in neonatal farm animals
  - —Orbiviruses are arthropod-borne infections that cause African horse sickness in horses and bluetongue in sheep and in other domestic and wild ruminants

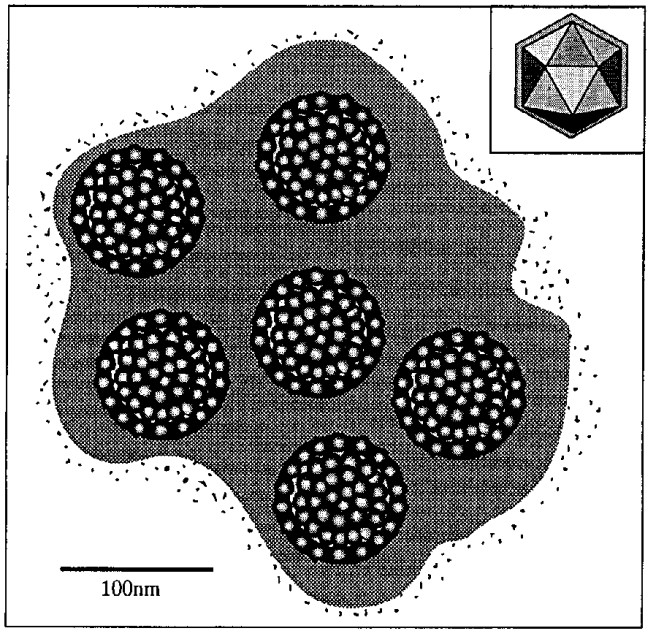


Figure 62.1 Rotavirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Clinical infections

Orthoreoviruses, which are widespread in nature, have been isolated from many animal species (Table 62.1). Mammalian and avian orthoreoviruses possess distinct group antigens. Avian orthoreoviruses have been implicated in arthritis, tenosynovitis, chronic respiratory disease and enteritis. Rotaviruses cause acute diarrhoea in young intensively-reared farm animals. Transmission of orthoreoviruses and rotaviruses is through contact with contaminated faeces.

Within the 19 currently recognized serogroups (species) of orbiviruses, there are defined serotypes and, in addition, antigenic complexes. The main serogroup-specific antigen is the immunodominant outer core protein VP7. Individual serotypes are distinguished by serum neutralization assays utilizing antibodies against outer capsid

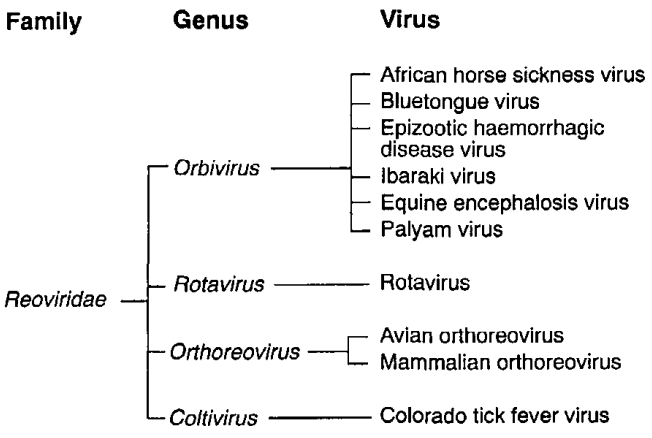


Figure 62.2 Viruses of veterinary importance in the family *Reoviridae*.

proteins. African horse sickness and bluetongue are particularly important diseases caused by orbiviruses. Epizootic haemorrhagic disease of deer and Ibaraki disease in cattle, both caused by closely-related orbiviruses, have clinical effects in these species similar to those of bluetongue in sheep. Infection with equine encephalosis virus has been recognized only in South Africa. Serological evidence suggests that this infection is widespread but acute disease occurs only sporadically. African horse sickness, bluetongue and epizootic haemorrhagic disease of deer are transmitted by arthropods, especially by *Culicoides* species.

Diseases caused by avian orthoreoviruses

Infections caused by avian orthoreoviruses are usually inapparent. In certain circumstances, however, they may either cause primary disease or contribute to the severity of mixed infections. Using serum neutralization tests, at least nine serotypes are recognized. Although transmission is mainly by the faecal-oral route, transfer to developing chicks *in ovo* can occur. Arthritis/tenosynovitis, caused by orthoreoviruses in chickens between four and 16 weeks of age, has been reported worldwide. Lameness is a prominent feature of this disease and rupture of the tendon of the gastrocnemius muscle may occur. Affected birds have limited mobility and may die of starvation. Morbidity is usually less than 10%. The synovial lesions resemble those caused by infection with *Mycoplasma synoviae* or with *Staphylococcus aureus*. Orthoreovirus involvement can be confirmed by virus isolation. Specimens suitable for virus isolation are affected articular cartilage and tendon sheaths. Synovial fluid is not a reliable source for virus isolation. Suspensions of macerated tissues are inoculated into the yolk sac of embryonated eggs or onto monolayers of chick embryo

Table 62.1 Viruses of veterinary importance in the family *Reoviridae*.

Genus	Virus	Comments
Orbivirus	African horse sickness virus	Arthropod-borne infection of *Equidae*, principal vector *Culicoides* species. Endemic in Africa. High mortality rate
	Bluetongue virus	Arthropod-borne infection of sheep, cattle, goats and wild ruminants. Principal vector *Culicoides* species. Severe disease in some species of deer. Teratogenic effects. Clinical disease rare in cattle
	Epizootic haemorrhagic disease virus	Arthropod-borne infection of deer, cattle and buffalo. Principal vector *Culicoides* species. Clinically similar to bluetongue. Important disease of deer in North America. Subclinical infection occurs in cattle. Eight serotypes recognized
	Ibaraki virus	Member of the epizootic haemorrhagic disease virus serogroup. Acute febrile disease of cattle similar to bluetongue. Probably arthropod-borne. Present in south-east Asia
	Equine encephalosis virus	Reported in South Africa. Majority of infections subclinical. Sporadic cases of acute fatal disease. Cerebral oedema, fatty liver and enteritis are prominent features
	Palyam virus	Arthropod-borne disease of cattle. Causes abortion and teratogenic effects. Recorded in southern Africa, southeast Asia and Australia. Many viruses in the serogroup
Rotavirus	Rotaviruses	Occur in intensively-reared neonatal animals. Mild to severe diarrhoea, severity influenced by virulence of viral strain, age, colostral intake and management factors
Orthoreovirus	Avian orthoreoviruses	Important cause of viral arthritis/tenosynovitis in chickens. Multiple serotypes described. Turkeys and other avian species susceptible
	Mammalian orthoreoviruses	Associated with mild enteric and respiratory disease in many species, severity dependent on secondary infections. Three serotypes recognized
Coltivirus	Colorado tick fever virus	Rodent species act as reservoirs. Arthropod-borne, mainly ticks and also mosquitoes. Primarily of significance in humans, may cause encephalitis in children

liver cells. Viral antigen may be detectable by immunofluorescence in cryostat sections of tissues. Although serological testing is not particularly useful because of the high prevalence of subclinical infections, it may be employed to determine the immune status of a flock. Both inactivated and modified live vaccines have been used in parent flocks to stimulate high levels of maternally-derived antibody in chicks. However, vaccines may induce protection only against homologous serotypes (Meanger *et al.*, 1997). Control measures include total depopulation at the end of a production cycle followed by thorough cleansing and disinfection of premises.

Enteric disease caused by rotaviruses in young animals

Rotaviruses cause diarrhoea in intensively-reared young farm animals worldwide. Isolates are divided into seven antigenically-distinct serogroups (A to G), also termed species, based on reactions with the major capsid protein, VP6. Most isolates belong to serogroup A. Fourteen serotypes (G1-G14) are recognized within serogroup A on the basis of the antigenicity of VP7, an outer capsid glycoprotein which is highly immunogenic and induces type-specific neutralizing antibodies. Field infections with rotaviruses are considered to be species specific. However, virus isolates from one species can be transmitted experimentally to other species.

Epidemiology

High titres of virus (10^9 virus particles per gram of faeces) are excreted by clinically affected animals. Horizontal transmission occurs following ingestion of contaminated feed. Because the virus is stable in the environment, premises may be heavily contaminated and intensively-reared animals are those most often affected. Buildings can remain contaminated for long periods if thorough cleansing and disinfection procedures are not implemented.

Pathogenesis

The severity of infection is largely determined by the virulence of the infecting viral strain, the amount of virus ingested and the level of maternally-derived immunity. Other factors which influence the outcome of infection include age of the animal at the time of exposure, overcrowding and the presence of other enteric pathogens. The virus, which can survive gastric acidity, passes through the stomach and infects enterocytes at the tips of villi in the small intestine. Because the rate of enterocyte replacement is relatively slow in young animals, affected villi become stunted and covered by cuboidal cells. These immature replacement cells have reduced levels of disaccharidases and defective glucose-coupled sodium transport. Undigested lactose provides an ideal substrate for bacterial proliferation in the intestinal lumen. In addition, it exerts an osmotic effect which results in the retention of fluid in the lumen and, along with impaired fluid absorption, contributes to the development of diarrhoea.

Clinical signs

The incubation period is short, usually less than 24 hours. Affected animals are anorexic and depressed, and produce light-coloured, semi-liquid or pasty faeces. In uncomplicated cases, animals frequently recover within four days without treatment. Concurrent infection with other enteric pathogens such as *Escherichia coli*, *Salmonella* species and *Cryptosporidium* species may add to the severity of the diarrhoea and deaths may occur.

Diagnosis

- Specimens suitable for laboratory examination include faeces and intestinal contents.
- Although negative-contrast electron microscopy is rapid, large numbers of virus particles (10^6 per gram of faeces) must be present for reliable confirmation. Immunoelectron microscopy increases the sensitivity of the procedure. Mixed viral infections can be detected by negative-contrast electron microscopy.
- Viral antigen can be demonstrated in faeces by ELISA and latex agglutination. The antisera employed in these tests is usually specific for serogroup A rotaviruses. Reagents for these assays are commercially available. Immunofluorescence can be used to detect viral antigen in smears or in cryostat sections of affected small intestines.
- Sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE) has been used successfully to demonstrate RNA segments of rotaviruses in clinical samples. The sensitivity of this procedure is comparable with electron microscopy. The electrophoretic patterns permit differentiation of rotavirus serogroups.
- Rotaviruses are difficult to isolate in tissue culture from clinical samples. The addition of low concentrations of trypsin to the growth medium facilitates viral uncoating and improves viral replication.

Treatment

In mildly affected animals, water should be substituted for milk in the diet. Oral electrolyte solutions may be beneficial in some cases. Intravenous fluid replacement and antibiotic administration are required in severe cases complicated by bacterial infection.

Control

Measures aimed at reducing the levels of virus challenge in young animals are essential. These must be combined with management procedures which ensure that neonatal animals receive adequate amounts of colostrum. Local immunity is more important than circulating antibody; ingestion of colostrum provides protective antibodies in

the intestinal lumen. Vaccination of pregnant dams enhances antibody levels in mammary secretions. Oral vaccination of newborn animals using modified live vaccine is of questionable value. Stressful environmental conditions should be minimized.

African horse sickness

This is a non-contagious disease of horses, mules and donkeys caused by African horse sickness virus (AHSV). Nine serotypes of this orbivirus, which can be distinguished by neutralization tests, constitute the African horse sickness serogroup. The disease is endemic in subtropical and tropical Africa. Although serious outbreaks have occurred in the Middle East, India and Pakistan, the disease has not persisted in these regions. Outbreaks have been recorded in Spain, Portugal and Morocco in recent years. African horse sickness is classified as a list A disease by the Office International des Epizooties.

Epidemiology

The virus is transmitted by haematophagous insects. The principal vector is *Culicoides imicola*, a species of Afro-Asian midge, which remains infected for life. This midge prefers a warm climate; it aestivates at temperatures below 10°C and replication of the virus in the midge ceases (Mellor *et al.*, 1998). *Culicoides imicola* is distributed as far north as latitude 41°N. Endemic disease occurs only in regions where *C. imicola* is constantly present. Epidemics of African horse sickness periodically occur outside these regions following climatic conditions which allow wind-borne transfer of infected midges for up to 700 km. Outbreaks of the disease are seasonal, usually occurring in late summer. The virus may be isolated from clinically normal maintenance hosts such as the zebra and African donkey.

Pathogenesis and pathology

The primary sites of viral replication are believed to be regional lymph nodes, spleen and lungs. Viraemia persists throughout the febrile period. Endothelial cells are important sites of secondary viral replication, resulting in increased vascular permeability, oedema, haemorrhage and intravascular coagulation. Postmortem findings include diffuse pulmonary oedema, hydrothorax, ascites and hydropericardium

Clinical signs

The incubation period is up to seven days. Four forms of this febrile disease, all of which can occur in a particular outbreak, are recognized. A peracute pulmonary form is characterized by depression and nasal discharge with rapid progression to severe respiratory distress. Mortality rate may approach 100%. A subacute cardiac form manifests as conjunctivitis, abdominal pain and progressive dyspnoea. Subcutaneous oedematous swellings of the head and neck are most obvious in the supraorbital fossae, palpebral conjunctiva and intermandibular space. In this form of the disease, the mortality rate is up to 70%. A third form of African horse sickness presents with both cardiac and pulmonary features. A mild or subclinical form, termed horse sickness fever, may be observed in zebras and donkeys.

Diagnosis

- Characteristic clinical signs, such as oedema of the supraorbital fossae, may allow a clinical diagnosis. Postmortem findings, including pericardial and pleural effusions, are consistent with a diagnosis of African horse sickness.
- Suitable samples for laboratory examination include blood, lymph node and spleen. Inoculation of embryonated eggs or cell cultures may be used to demonstrate the presence of virus. Inoculation of newborn mice intracerebrally may also be used for this purpose. Virus can be identified by immuno-fluorescence and typed using virus neutralization with monovalent antiserum or competitive ELISA.
- Viral antigen can be detected by ELISA in samples.
- Viral RNA can be detected by RT-PCR (Zientara *et al.*, 1998). This test may provide results within 24 hours.
- Suitable serological methods include CFT, AGID, ELISA and serum neutralization tests. In acute disease, infected animals may die before antibodies are produced. Seroconversion in donkeys, used as sentinel animals outside endemic areas, confirms the presence of the disease.

Control

Vector control, quarantine of affected animals and vaccination are the main methods of control. Insect vector control includes the use of repellents and insecticides, the elimination of insect breeding areas and housing of animals in insect-proof buildings at dawn and at dusk when insect activity is maximum. Attenuated vaccines, both monovalent and polyvalent containing up to four serotypes, are available. However, these vaccines fail to prevent viraemia. Moreover, the vaccine virus may revert to virulence and be transmitted by vectors. In addition, vaccinated animals cannot be differentiated serologically from those with field infections. Inactivated vaccines based on serotype 4 are effective in preventing both clinical disease and viraemia. A polyvalent vaccine must be used if there is a risk of exposure to different serotypes. Protective immune responses may be generated using recombinant expressed structural proteins as subunit vaccines (Roy and Sutton, 1998). Such vaccines should be safe and permit differentiation of vaccinated from infected animals. Adequate biocontainment facilities are mandatory for vaccine production outside endemic regions.

Bluetongue

This non-contagious viral disease of sheep and other domestic and wild ruminants is transmitted by biting insects, principally *Culicoides* species. Isolates of the causal agent, bluetongue virus (BTV) belong to a distinct serogroup in the *Orbivirus* genus. Twenty four serotypes of BTV have been described. Bluetongue (BT) is of greatest significance in sheep and deer. The severity of the disease is influenced by the serotype of the virus, the breed of sheep and prevailing environmental conditions. Bluetongue is classified as a list A disease by the Office International des Epizooties.

Epidemiology

Bluetongue is widely distributed between latitudes 40°N and 35°S, reflecting the distribution of *Culicoides* species. *Culicoides imicola* is the principal vector in Africa and the Middle East. In Australia, *C. fulvus*, *C. wadai* and *C. brevitarsis* are involved in transmission. Other *Culicoides* species of importance in transmission are *C. varipennis* var. *sonorensis* in North America and *C. insignes* in South America. Female midges feeding on viraemic animals become infected and virus replicates in their tissues. *Culicoides* species can transmit virus in saliva within seven to ten days and they remain infected for life. Temperatures of 18°C to 29°C along with high humidity favour insect activity, accounting for the seasonal occurrence of the disease in many parts of the world. *Culicoides* species are most active at dawn and dusk. In localized areas within endemic regions there may be an increased frequency of BT outbreaks. These areas are particularly suitable for the breeding of *Culicoides* species because of the accumulation of animal faeces in marshland. Extension of disease to contiguous areas occurs through the movement of viraemic animals or insect vectors. Although the flight range of *Culicoides* species is limited, they may be transported over long distances by wind movement resulting in BT outbreaks in susceptible ruminant populations outside endemic regions. Such events may precipitate epidemics which are usually self-limiting unless the climate is suitable for vector activity throughout the year.

In endemic areas, infection of cattle is common and usually inapparent. The viraemia in cattle commonly lasts several weeks facilitating acquisition of virus by insect vectors. Consequently, cattle are considered to be important reservoirs of virus (Barratt-Boyes and MacLachlan, 1995). During the viraemic phase, virus can be detected in the semen of a proportion of rams and bulls. Venereal transfer of infection is uncommon. Embryos collected from infected ewes may transmit infection to recipient ewes but this can be prevented by washing embryos (Singh *et al.*, 1997).

Pathogenesis and pathology

After experimental infection, the virus replicates initially in regional lymph nodes. It is then carried in blood or lymph to other lymphoid tissues where further replication takes place. Virus localizes and multiplies in the endothelium of small blood vessels producing vascular damage with stasis, exudation and tissue hypoxia. The initiation and development of surface lesions in areas of tissue hypoxia relate to minor trauma and may be complicated by secondary bacterial infection. Lesions are particularly evident in the oral cavity, around the mouth and on the coronet of the hoof. In the bloodstream the virus is highly cell-associated, particularly with erythrocytes. It has been suggested that this may protect virus from antibody. Sporadic cases of clinical disease in cattle are thought to involve type I hypersensitivity reactions with participation of IgE as a result of previous exposure to BTV or related orbiviruses.

Clinical signs

The clinical presentation is highly variable, from subclinical to severe disease with high mortality. Severe disease is generally confined to Merinos and European mutton breeds. Nutritional status, exposure to sunlight and age also appear to influence the severity of lesions. The incubation period in sheep is up to ten days. Affected animals are febrile and depressed with vascular congestion of the lips and muzzle. Oedema of the lips, face, eyelids and ears develops. Erosions and ulcers are evident on the oral mucosa. There is excessive salivation and a watery discharge that subsequently becomes mucopurulent and dries to form crusts around the nares. The tongue may be swollen and cyanotic. Lameness may result from coronitis and laminitis. Some animals develop torticollis. Abortion may occur and lambs may be weak or deformed at birth. Mortality rate may be up to 30% and, in some outbreaks, may be higher. Animals recovering may lose part of the fleece some weeks after infection. Rare clinical cases in susceptible cattle are characterized by fever, stiffness, ulceration of the oral mucosa, 'burnt muzzle' and dermatitis. Cattle infected during pregnancy may abort or give birth to malformed calves.

Diagnosis

A presumptive diagnosis of BT may be based on clinical findings and postmortem lesions. Confirmation requires isolation and identification of the virus or demonstration of BTV-specific antibodies.

- Samples suitable for virus isolation include unclotted blood from febrile animals or fresh spleen and lymph node collected at postmortem. Virus may be isolated by intravenous inoculation of embryonated eggs.
- A highly sensitive nested PCR has been developed for detecting BTV nucleic acid in clinical samples (Aradaib *et al.*, 1998).

- Antigen detection ELISA systems have also been described (Stanislawek *et al.*, 1996; Hamblin *et al.*, 1998).
- Serological tests for the detection of antibodies to the BTV serogroup include CFT, AGID, indirect immunofluorescence and competitive ELISA. Neutralization or HAI assays are used for demonstrating type-specific antibodies. In animals from endemic regions, a rising antibody titre must be demonstrated using paired serum samples.

Control

As bluetongue is a list A disease, it is subject to international regulations controlling trade. The discovery of a number of serotypes of BTV in the Northern Territory of Australia resulted in disruption of trade in animals, semen and embryos although clinical disease was not present (Muller, 1995). Populations of insect vectors may be reduced by the use of larvicides at breeding sites. Insecticides applied to susceptible animals may temporarily halt feeding by vectors. Live attenuated vaccines have been used successfully for many years and provide protection against virulent virus of homologous serotype. Polyvalent vaccines are essential in regions where a number of serotypes are present. Attenuated vaccines may be teratogenic when used in ewes during the first half of gestation. These vaccines should not be used during periods of vector activity because of the risk of transferring vaccinal virus to pregnant ewes and the possibility of genetic reassortment with field virus and reversion to virulence (Osburn *et al.*, 1996). Killed adjuvanted vaccines can induce protection but are more expensive to produce and require two inoculations. Recombinant virus-like particles, capable of inducing protective immunity, have been produced in insect cells infected with recombinant baculoviruses expressing BTV proteins. However, vaccines produced by this method are not yet available commercially (Murray and Eaton, 1996).

References

Aradaib, I.E., Schore, C.E., Cullor, J.S. and Osburn, B.I. (1998). A nested PCR for detection of North American isolates of bluetongue virus based on NS1 genome sequence analysis of BTV-17. *Veterinary Microbiology*, **59**, 99–108.

Barratt-Boyes, S.M. and MacLachlan, N.J. (1995). Pathogenesis of bluetongue virus infection of cattle. *Journal of the American Veterinary Medicine Association*, **206**, 1322–1329.

Hamblin, C., Salt, J.S., Graham, S.D. *et al.* (1998). Bluetongue virus serotypes 1 and 3 infection in Poll Dorset sheep. *Australian Veterinary Journal*, **76**, 622–629.

Meanger, J., Wickramasinghe, R., Enriquez, C.E. and Wilcox, G.E. (1997). Immune response to avian reovirus in chickens and protection against experimental infection. *Australian Veterinary Journal*, **75**, 428–432.

Mellor, P.S., Rawlings, P., Baylis, M. and Wellby, M.P. (1998). Effect of temperature on African horse sickness virus infection in *Culicoides*. In *African Horse Sickness*. Eds. P.S. Mellor, M.Baylis, C. Hamblin, C.H. Calisher and P.P.C. Mertens. Springer-Verlag, Wien. pp. 156–163.

Muller, M.J. (1995). Veterinary arbovirus vectors in Australia – a retrospective. *Veterinary Microbiology*, **46**, 101–116.

Murray, P.K. and Eaton, B.T. (1996). Vaccines for bluetongue. *Australian Veterinary Journal*, **73**, 207–210.

Osburn, B.I., De Mattos, C.A., De Mattos, C.C. and MacLachlan, N.J. (1996). Bluetongue disease and the molecular epidemiology of viruses from the western United States. *Comparative Immunology and Microbiology of Infectious Disease*, **19**, 181–190.

Roy, P. and Sutton, G. (1998). New generation of African horse sickness virus vaccines based on structural and molecular studies of the virus particles. In *African Horse Sickness*. Eds. P.S. Mellor, M. Baylis, C. Hamblin, C.H. Calisher and P.P.C. Mertens. Springer-Verlag, Wien. pp. 177–202.

Singh, E.L., Dulac, G.C. and Henderson, J.M. (1997). Embryo transfer as a means of controlling the transmission of viral infections. XV. Failure to transmit bluetongue virus through the transfer of embryos from viraemic sheep donors. *Theriogenology*, **47**, 1205–1214.

Stanislawek, W.L., Lunt, R.A., Blacksell, S.D. (1996). Detection by ELISA of bluetongue antigen directly in the blood of experimentally infected sheep. *Veterinary Microbiology*, **52**, 1–12.

Zientara, S., Sailleau, C., Moulay, S. *et al.* (1998). Use of reverse transcriptase-polymerase chain reaction (RT-PCR) and dot-blot hybridization for the detection and identification of African horse sickness virus nucleic acids. In *African Horse Sickness*. Eds. P.S. Mellor, M. Baylis, C. Hamblin, C.H. Calisher and P.P.C. Mertens. Springer-Verlag, Wien. pp. 317–327.

Chapter 63

Birnaviridae

Birnaviruses are so named because their genomes contain two segments of linear, double-stranded RNA. The icosahedral virions are about 60 nm in diameter (Fig. 63.1). Five polypeptides, designated VP1, VP2, VP3, VP4 and VP5, have been identified. The major capsid protein (VP2) contains epitopes which induce neutralizing antibodies. Replication occurs in the cytoplasm of host cells and involves a virion-associated RNA-dependent RNA polymerase. The family *Birnaviridae* contains three genera: *Avibirnavirus*, *Aquabirnavirus* and *Entomobirnavirus*, which infect chickens, fish and insects respectively. Virions are stable over a wide pH range and at a temperature of 60°C for 1 hour. They are resistant to treatment with ether and chloroform.

Key points

- Double-stranded RNA viruses with icosahedral symmetry
- Replicate in cytoplasm
- Stable in the environment
- The family is composed of three genera:
  - —*Avibirnavirus* contains viruses which cause infectious bursal disease
  - —*Aquabirnavirus* contains viruses which cause infectious pancreatic necrosis in salmonids
  - —*Entomobirnavirus* contains viruses which infect insects

Clinical infections

Two economically important diseases associated with birnaviruses are infectious bursal disease of chickens and infectious pancreatic necrosis of salmonids. These diseases occur worldwide and cause considerable losses in poultry units and in farmed salmon.

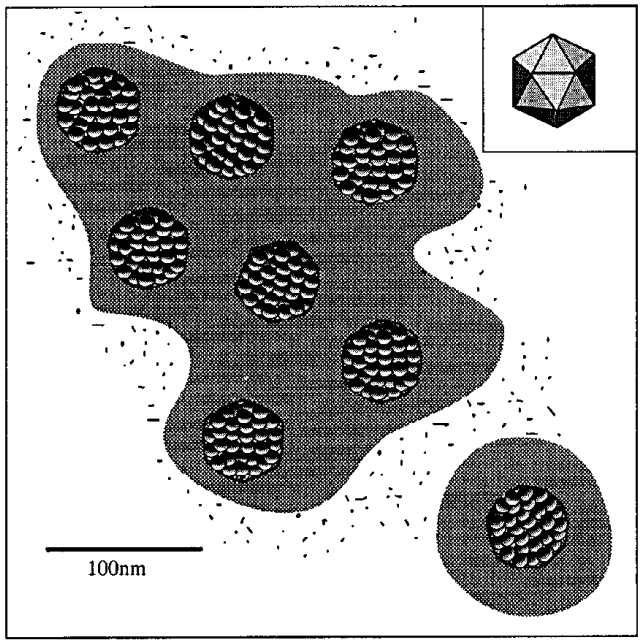


Figure 63.1 Birnavirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Infectious bursal disease

This condition is a highly contagious disease of young chickens which is caused by infectious bursal disease virus (IBDV). The causal agent was first isolated in Gumboro, Delaware and the disease was originally known as Gumboro disease. Although turkeys and ducks are susceptible to infection, clinical disease occurs only in chickens. Based on neutralization tests, isolates of IBDV are assigned to two serotypes. There is considerable variation in the virulence of isolates in serotype 1. Very virulent (VV) strains were first reported in Europe and Asia in the late 1980s. These strains, although antigenically similar to classical serotype 1 strains, can cause disease even when maternally-derived antibody against the classical vaccine strains is present. Serotype 2 isolates are not associated with clinical disease. A variety of antigenic variants are recognized within each serotype. In the United States, in recent years, variant serotype 1 isolates have been detected in flocks that had been vaccinated with classical strain vaccines. These antigenic variants, which are highly immunosuppressive in young chicks, cause rapid bursal atrophy.

Epidemiology

Infection, which is usually acquired by the oral route, occurs when maternally-derived antibody levels are waning at two to three weeks of age. Virus is shed in the faeces for up to two weeks after infection and can remain infectious in the environment of a poultry house for several months. Spread to other poultry units by fomites occurs. Neither a carrier state nor vertical transmission has been demonstrated.

Pathogenesis

Within hours of ingestion, virus can be detected in macrophages and lymphoid cells in the caeca, duodenum and jejunum. Virus reaches the liver via the portal circulation and infects Kupffer cells. Infection spreads to the bursa of Fabricius where rapid replication results in a pronounced secondary viraemia and dissemination to other tissues. The main target cells are B lymphocytes and their precursors in the bursa. The major antigenic protein VP2 has been shown to induce apoptosis in infected cells. Depletion of B lymphocytes in early life results in impaired immune responses, lowered resistance to infectious diseases and ineffective responses to vaccines. Bursal damage in chickens older than three weeks of age only marginally affects immune competence because many B lymphocytes are distributed peripherally before damage occurs.

Clinical signs

The severity of clinical signs is influenced by the virulence of the virus, the age of chicks at the time of infection, the breed of the chicks and the level of maternally-derived antibody. Chicks develop an acute form of the disease between three and six weeks of age following a short incubation period. Affected birds are depressed and inappetent and show evidence of diarrhoea and vent pecking. Morbidity ranges from 10% to 100% with a mortality rate up to 20% or, occasionally, higher. The course of the disease is short with surviving birds recovering in about four days. Many outbreaks are mild, detectable only by impaired weight gains. Although infections before three weeks of age are usually subclinical, severe depression of the humoral antibody response may result. Clinical signs in these birds are usually vague. Suboptimal growth, predisposition to secondary infections and poor response to vaccination may be encountered.

Diagnosis

- In acute disease, clinical signs and a swollen oedematous bursa at postmortem are often sufficient for diagnosis. Confirmation and identification of subclinical infection requires laboratory tests.
- Viral antigen can be detected in smears or frozen sections of the bursa using immunofluorescence. Macerated bursal tissue is suitable for detection of viral antigen by ELISA or by gel diffusion tests.
- Specimens of bursa, spleen or faeces are suitable for virus isolation. Most strains grow on the chorioallantoic membrane of embryonated eggs.
- Recovered birds develop high antibody titres as mature peripheral B lymphocytes are unaffected. Suitable serological assays include ELISA and virus neutralization.

Control

Depopulation, thorough cleaning and effective disinfection programmes are required following an outbreak of disease in a unit. Most commercial units rely for control on vaccination. Both modified live and inactivated serotype 1 vaccines are available. Live vaccines can be administered by aerosol or in drinking water. To ensure high levels of maternally-derived antibody in chicks, breeding birds are usually vaccinated at four to ten weeks of age with a live vaccine and again, close to laying, with an inactivated oil-adjuvanted vaccine. Vaccines used in parent stock should contain both classical and variant strains of IBDV. Chicks can be actively immunized after maternally-derived antibody levels decline at about four weeks of age. In high risk flocks, vaccination may begin at one day of age to protect birds with little or no maternally-derived antibody, followed by booster inoculations at two and three weeks of age. Partially attenuated vaccines termed 'intermediate' and 'intermediate plus' ('hot') are generally used in this way in broilers and commercial layer replacements as they are capable of overcoming low levels of maternally-derived antibody. Recent vaccine developments include expression of VP2 gene by a baculovirus and by a recombinant fowlpox virus.

Further reading

Nagarajan, M.M. and Kibenge, F.S.B. (1997). Infectious bursal disease virus: a review of molecular basis for variations in antigenicity and virulence. *Canadian Journal of Veterinary Research*, **61**, 81–88.

Chapter 64

Orthomyxoviridae

Key points

- Enveloped viruses with helical nucleocapsids and spherical or pleomorphic morphology
- Linear, negative-sense, single-stranded RNA
- Replication occurs in the nucleus
- Two important glycoproteins, one a haemagglutinin which binds to cell receptors and the other with neuraminidase activity, are present in the envelope
- Genome is segmented facilitating genetic reassortment
- Subtypes of Influenza A virus are important pathogens

The family *Orthomyxoviridae* (Greek *orthos*, proper and *myxa*, mucus) contains those viruses which cause influenza in humans and animals. Orthomyxoviruses are spherical or pleomorphic, enveloped viruses, 80 to 120 nm in diameter (Fig. 64.1). Long filamentous forms also occur. The envelope, which is derived from host cell membrane lipids, contains glycosylated and non-glycosylated viral proteins. Surface projections of glycoproteins form 'spikes' or peplomers which, in influenza A and B viruses, are of two types: a haemagglutinin (H) responsible for virus attachment and envelope fusion, and a neuraminidase (N) capable of cleaving viral receptors and promoting both entry of virus into cells and release of virions from infected cells.

Influenza viruses haemagglutinate erythrocytes from a wide range of species. Antibodies to the H glycoprotein are responsible for virus neutralization. The nucleocapsid has a helical symmetry. The genome, which is composed of six to eight segments, consists of linear, negative-sense, single-stranded RNA. Replication occurs in cell nuclei with release of virions by budding from plasma membranes. Virions are labile in the environment and are sensitive to heat, lipid solvents, detergents, irradiation and oxidizing agents.

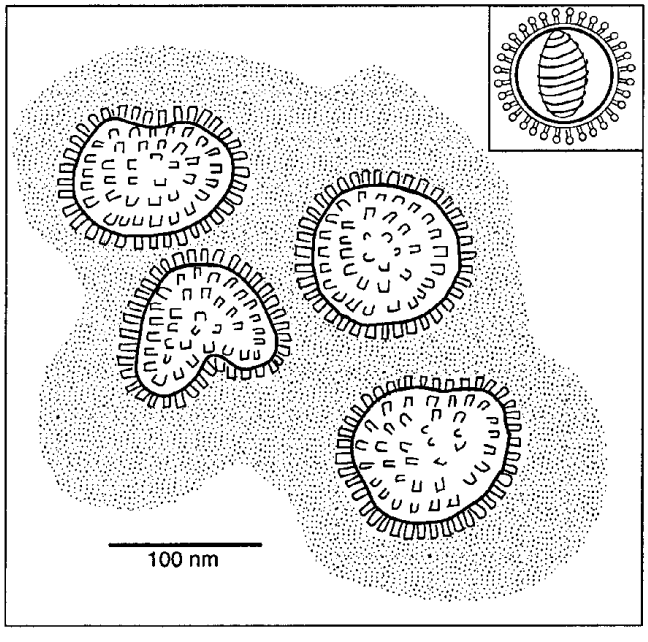


Figure 64.1 Influenza A virus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

The family contains four genera namely *Influenzavirus A*, *Influenzavirus B*, *Influenzavirus C* and *Thogotovirus*. Influenza B and C viruses are pathogens of humans; Thogotovirus and Dhori virus are tick-borne arboviruses isolated from camels, cattle and humans in parts of Africa, Europe and Asia. Influenza A virus, the most important member of the family, is a significant pathogen of animals and humans.

Isolates of influenza A virus are grouped into subtypes on the basis of their H and N antigens. Currently, 15 H antigens and nine N antigens are recognized. New subtypes of influenza A virus emerge periodically. Two mechanisms, point mutation and genetic reassortment, are responsible for the emergence of new subtypes. Point mutations give rise to antigenic drift in which variation occurs within a subtype. Genetic reassortment, a more complex process producing antigenic shift, results in the development of new subtypes. To assess the risk posed by the emergence of new variant viruses, a precise classification of isolates has been adopted by the World Health Organization. This system is based on the influenza virus type, host, geographical origin, strain number, year of isolation and subtype. An example of this classification system, influenza virus A/equine/Prague/1/56 (H7N7),

indicates that this virus was isolated from a horse in Prague during 1956. Antigenic subtypes of influenza A virus which cause disease in humans and animals are presented in Table 64.1.

Clinical infections

Influenza A viruses cause significant infections in humans, pigs, horses and birds. Antibodies to influenza A virus have been detected in cattle in association with respiratory disease, but their significance is unclear (Brown *et al.*, 1998). Aquatic birds, particularly ducks which are reservoirs of influenza A virus, provide a genetic pool for the generation of the new subtypes capable of infecting mammals. Migratory waterfowl disseminate the virus across international borders. Although isolates of influenza A virus are usually species specific, there are well documented instances of transfer between species. The viruses replicate in the intestinal tract of birds and transmission is by the faecal-oral route. Human influenza pandemics have been attributed to the combined effects of poor hygiene and the close association of concentrated human populations with domestic fowl and pigs. The frequency of genetic reassortment in these animal populations can lead to the emergence of virulent influenza virus subtypes which are capable of infecting humans, thereby initiating pandemics (Fig. 64.2). Avian influenza viruses usually replicate poorly in humans. However, both human and avian influenza subtypes replicate in pigs, in which species genetic reassortment readily occurs. Because the genome of influenza A virus is segmented, mixed infections frequently give rise to genetic reassortment with the emergence of new subtypes. Such novel subtypes are often implicated in major pandemics which occur at about 20 year intervals. As there is limited immunity to the new subtypes in the human population, spread from country to country tends to occur rapidly. It is estimated that the pandemic of 'Spanish flu' in 1918 was responsible for more than 20 million human deaths worldwide.

Sporadic, less serious outbreaks of human influenza, which are of relatively frequent occurrence, can be attributed to subtle antigenic changes arising from errors

Table 64.1 Antigenic subtypes of influenza A virus isolated from humans and animals.

Hosts	Antigenic subtypes	Comments
Humans	H2N8 (1890)[a] H3N8 (1900) H1N1 (1918) H2N2 (1957) H3N2 (1967)	Subtypes which have been found in pigs such as H1N1 have been implicated in human pandemics
Birds	Many antigenic subtypes represented by different combinations of haemagglutinin (H) and neuraminidase (N) peplomers have been recognized	Disease is usually associated with subtypes expressing H5 or H7. Wild birds, especially migrating ducks, act as carriers
Pigs	Predominantly H1N1 and H3N2	Severity of disease is determined by the antigenic subtype
Horses	Usually H7N7 or H3N8	Subtypes associated with disease, which are widely distributed geographically, are absent from Australia, New Zealand and Iceland

a year of recognition

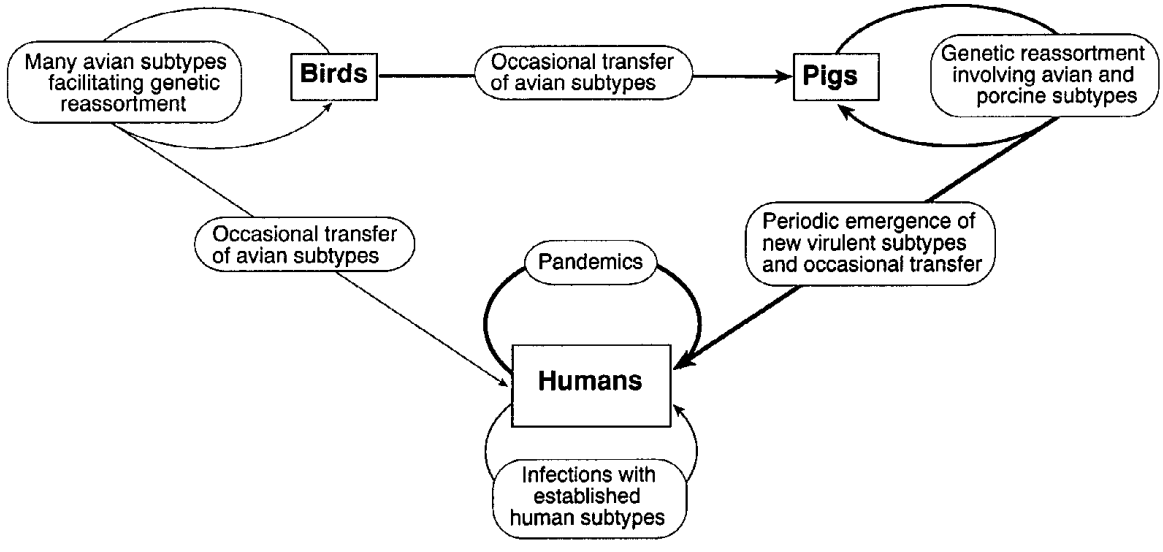


Figure 64.2 The circulation of subtypes of influenza A virus in bird and pig populations with the emergence of virulent subtypes which occasionally may be responsible for pandemics in the human population.

during replication of viral nucleic acid. If these subtypes exhibit antigenically distinct haemagglutinins, they are unaffected by existing neutralizing antibodies and a proportion of the susceptible population may become infected. Such outbreaks occur abruptly, typically in the winter months in temperate regions.

Subtypes of influenza A virus, which are well established as pathogens in animal populations, have also been implicated in human infections. In 1997, following a large epidemic of avian influenza in chickens, a H5N1 subtype was isolated from a fatal case in a young child in Hong Kong. This subtype had not previously been described outside of avian species. Human health fears prompted the destruction of 1.2 million birds. Fortunately human to human transmission did not occur to any significant extent, although other human cases did occur as a result of contact with infected poultry, By the end of 1997, 18 human cases had been confirmed resulting in 6 deaths.

Avian influenza (Fowl plague)

Many combinations of H and N antigens in influenza A virus are represented in isolates from avian species, particularly waterfowl. Influenza A virus subtypes are distributed worldwide and are frequently recovered from clinically normal birds. Outbreaks of severe clinical disease, usually caused by subtypes expressing H5 and H7 determinants, occur periodically in chickens and turkeys. In these species, acute infection is often referred to as fowl plague or highly pathogenic avian influenza and is categorised as a List A disease by the OIE.

Epidemiology

Infection is maintained in wild bird populations. Migrating waterfowl are considered to be responsible for spreading the virus to domestic birds. Although ducks become infected with influenza A virus, they rarely show signs of illness. A high level of subclinical infection occurs in susceptible young birds. Following replication in the intestinal tract, virus is shed in faeces. Live-bird markets may contribute to the spread of infection. Secondary spread can result from the movement of personnel and contaminated equipment between poultry farms.

Pathogenesis

Spread of influenza virus in tissues is dependent on the type of proteases present in a given tissue and the structure of the viral haemagglutinin molecule. The production of infectious virions requires cleavage of viral haemagglutinin. In the majority of influenza A virus subtypes, haemagglutinin cleavage takes place in the epithelial cells of the respiratory and digestive tracts. Because of the amino acid composition at their cleavage sites, haemagglutinins of virulent subtypes are susceptible to cleavage in many tissues, facilitating the development of generalized infection.

Clinical signs

The incubation period, which is variable, is up to seven days. Clinically, the disease may be inapparent, mild or, in some instances, severe with high mortality. Factors, such as overcrowding, poor ventilation and concurrent infections, may predispose to the development of severe disease. Highly virulent subtypes cause explosive outbreaks of disease with high mortality. Clinical signs are more apparent in birds which survive for a few days. Respiratory distress, diarrhoea, oedema in the cranial region, cyanosis, sinusitis and lacrimation are features of the clinical presentation. Infection of laying birds results in a dramatic drop in egg production.

Diagnosis

The severe form of the disease may be difficult to distinguish from velogenic viscerotropic Newcastle disease or from fowl cholera. Mild forms of the disease resemble other respiratory conditions in birds.

- Laboratory confirmation, which involves virus isolation and chacterization, is essential. Suitable specimens for laboratory examination include tracheal and cloacal swabs, faeces and pooled samples of organs.
- Tissue suspensions are inoculated into nine to 11 day old embryonated eggs. Allantoic fluid, harvested after incubation for four to seven days, is tested for haemagglutinating activity.
- The presence of influenza A virus can be confirmed by immunodiffusion using a suspension of chorioallantoic membrane from eggs inoculated with material from an outbreak and positive antiserum to the nucleocapsid or matrix antigens common to all influenza A viruses.
- Antisera with broad specificity may be used in haemagglutination inhibition (HI) or immunodiffusion tests to confirm that an isolate is influenza A virus. Definitive subtyping is carried out in reference laboratories using monospecific antisera prepared against the 15 haemagglutinins and nine neuraminidase determinants.
- All highly virulent avian influenza A subtypes possess either H5 or H7 antigens. However, numerous low virulence isolates expressing H5 and H7 determinants have been recorded. To assess pathogenicity, ten chickens should be inoculated intravenously at four to eight weeks of age. Isolates which cause more than 75% mortality within eight days are considered highly pathogenic.
- Genomic sequencing can be used to predict the amino acid composition at the cleavage site of the haemagglutinin molecule.
- Serological testing for antibodies to influenza virus can

be carried out using an agar gel immunodiffusion test or by competitive ELISA (Shafer *et al.*, 1998).

Control

Outbreaks of avian influenza in domestic species are notifiable to national regulatory authorities. In countries free of the disease, outbreaks are controlled by slaughter of affected flocks, imposition of movement restrictions and implementation of rigorous disinfection procedures. Imported birds are quarantined. In high risk areas along the migration routes of waterfowl, poultry should be housed in bird-proof buildings.

Vaccination is usually prohibited in those countries implementing a slaughter policy because of international trade restrictions and possible difficulties in establishing freedom from infection. Some countries accept the presence of mildly pathogenic subtypes because of the expense of implementing control measures. In such countries inactivated oil emulsion vaccines are available commercially and are used, particularly in turkeys, to protect against subtypes of low virulence. Recombinant haemagglutinin protein vaccines and recombinant fowlpox virus vector vaccines containing a haemagglutinin gene insert have been developed (Swayne *et al.*, 1997; Crawford *et al.*, 1999).

Vaccines, which are effective against a particular virulent influenza A subtype, may not be effective against new emerging subtypes. Due to the risk of reversion to virulence, live vaccines against influenza A virus are not used. However, clinical trials in humans with an attentuated cold-adapted, reassortant influenza virus vaccine have produced good results (Couch, 2000) and may lead to the development of similar vaccines for poultry.

Swine influenza

This highly contagious disease of pigs occurs worldwide. Swine influenza was first described in 1918, its occurrence coinciding with a major pandemic of human influenza. Two cocirculating subtypes, H1N1 and H3N2, are endemic in pig populations. In Europe during 1979, H1N1 isolates, clearly distinguishable from classical H1N1 subtypes and with haemagglutinins structurally similar to avian haemagglutinins, were identified. These H1N1 subtypes, which are more virulent than classical H1N1 isolates, now predominate in Europe. Acute respiratory disease of pigs in Japan (Ouchi *et al.*, 1996) and in the United Kingdom (Brown, 1998) has been attributed to the H1N2 subtype. There is convincing epidemiological evidence to support the view that transfer of virulent subtypes from pigs to humans is a major factor in the emergence of pandemics in the human population (Fig. 64.2).

Epidemiology

An outbreak of swine influenza is usually associated with the recent introduction of pigs into a herd. Virus, shed in high concentrations in the nasal secretions of infected pigs, spreads rapidly within a herd. The principal route of transmission is by direct contact. Airborne spread between farms may occur under suitable weather conditions in areas with high pig densities. Outbreaks of disease usually occur when environmental temperatures are low. Between outbreaks it is probable that virus circulates in herds without evidence of clinical disease and that some animals remain carriers for many months.

Pathogenesis and pathology

Infection is limited to the respiratory tract; the lungs are the major target organs. Following infection, virus multiplies in nasal, tracheal and bronchial epithelium. Spread of infection throughout the respiratory tract results in necrosis, extensive pneumonic change and consolidation. Lesions are often limited to the apical and cardiac lobes. The acute phase of the disease persists for more than 72 hours after which virus replication declines.

Clinical signs

Onset of the disease in a herd is often abrupt, many pigs becoming clinically ill simultaneously. The incubation period is up to three days. The severity of the illness ranges from subclinical to acute and is strongly influenced by the strain of the infecting virus. Secondary bacterial infections frequently complicate the course of the disease and delay recovery. Acute disease is characterized by huddling in groups, paroxysmal coughing, dyspnoea and fever. Some pigs may have a discharge from the eyes and nose. Most pigs recover within six days. Mortality is usually low except in very young pigs or when intercurrent infection is present. The economic impact of the disease is mainly attributable to loss of weight. In fully susceptible herds, abortion can occur in affected sows.

Diagnosis

- Samples suitable for virus isolation include nasal mucus and lung tissue from acute cases early in the disease. As the virus is labile, transport media should be used for rapid transfer of specimens to the laboratory. Isolation is usually carried out in embryonated eggs. After incubation for 72 hours, haemagglutinating activity is demonstrable in the allantoic fluid.
- Demonstration of a rise in antibody levels in paired serum samples using haemagglutination-inhibition test or ELISA procedures is indicative of infection.
- Viral antigen can be detected using immunofluorescence or ELISA.
- Viral nucleic acid can be detected with PCR.

Control

Good husbandry, including the elimination of stress

factors, may help to minimize losses from swine influenza. Measures should be implemented to prevent the introduction of infection. Inactivated vaccines are available commercially. Vaccination can be beneficial provided that the subtypes of virus incorporated in vaccines include those involved in the outbreaks.

Equine influenza

This economically important, acute respiratory disease of horses occurs worldwide except in Australia, New Zealand and Iceland. Two immunologically distinct subtypes of influenza A virus are described in horses. The virus, first isolated from horses in 1956, was designated A/equine/Prague/1/56 (H7N7) or influenza A/equine 1. In 1963, a second subtype was isolated in the USA and designated A/equine/Miami/2/63 (H3N8) or influenza A/equine 2. Infection or vaccination with one subtype does not induce protection against infection with the other subtype. Although the last outbreak of disease attributed to influenza A/equine 1 occurred in 1979, there is serological evidence that this subtype continues to circulate in the horse population.

Antigenic drift accounts for several variants of influenza A/equine 2 with two antigenically and genetically distinct lineages identified in Europe and the Americas (Oxburgh *et al.*, 1998). In contrast, the H3N8 subtype isolated from horses in China was more closely related to avian strains than to the H3N8 subtype circulating in horses elsewhere.

Epidemiology

Outbreaks are associated with movement and assembly of horses for shows, sales, racing or training. The initial source of infection is often a partially immune horse shedding virus without showing clinical signs. Equine influenza is highly contagious and spreads rapidly among susceptible horses. Large quantities of virus are shed in aerosols by the frequent coughing of affected animals. Infection can be acquired at distances up to 30 metres. Indirect transmission through contamination of clothing, equipment and vehicles can also occur.

Pathogenesis

The virus replicates in the epithelium of the respiratory tract resulting in destruction of ciliated epithelium and hypersecretion from submucosal glands.

Clinical signs

The incubation period is up to two days. Affected animals develop a high temperature, nasal discharge and dry cough. Anorexia and depression, although common, can vary in intensity. Ocular discharge, limb oedema and stiffness may also be present. Age and previous exposure or vaccination status may influence the severity of the clinical signs and the likelihood of secondary bacterial infection with the development of respiratory complications. Exercise exacerbates the clinical signs (Gross *et al.*, 1998). Animals with mild infections usually recover within three weeks. In severe cases, several months may be required for convalescence.

Diagnosis

Although clinical signs may be suggestive of equine influenza, laboratory confirmation is required.

- Nasopharyngeal swabs collected during the acute phase of the infection are suitable for isolation of the virus in embryonated eggs or in cell culture. New isolates should be closely monitored for antigenic drift.
- A commercial diagnostic kit, developed for the detection of the nucleoprotein of human influenza A virus, can be used for the diagnosis of equine influenza (Chambers *et al.*, 1994).
- Serological diagnosis of equine influenza is possible. Haemagglutination inhibition or single radial haemolysis tests on paired serum samples can be used for diagnosis. Serum used in the HI test must be pretreated in order to remove non-specific inhibitors.

Treatment and control

Supportive therapy and rest is indicated for affected horses. The antiviral drugs, amantidine and rimantidine, which have been shown to be effective for inhibiting replication of influenza A virus *in vitro*, are being evaluated for therapeutic use (Rees *et al.*, 1997). Several inactivated vaccines are commercially available. However, immunity is usually short-lived and booster injections are required in accordance with manufacturer's instructions. The incorporation of polymer adjuvants or Quil-A based immuno-stimulating complexes (ISCOMs) into vaccine preparations extends the duration of protective levels of immunity. Protective immunity generated by natural exposure is related both to a mucosal IgA immune response and to humoral IgGa and IgGb responses, a pattern of protective immunity not generated by conventional vaccines (Nelson *et al.*, 1998). Vaccinated horses generally exhibit milder clinical signs and shed virus for shorter periods than unvaccinated animals. Vaccine manufacturers must update vaccinal strains regularly. Vaccines should include antigenic material representative of the influenza A virus subtypes prevalent in the horse population. In addition to vaccination, control of equine influenza requires isolation of affected horses and cleaning, disinfection and isolation of infected premises. Animal movement should cease until contaminated premises have been cleaned and disinfected.

Chapter 65

Paramyxoviridae

Paramyxoviruses and orthomyxoviruses were formerly grouped together as the 'myxoviruses' (Greek *myxa*, mucus), a name which describes their affinity for mucous membranes. Paramyxoviruses are pleomorphic, 150 nm or more in diameter and enveloped (Fig. 65.1). They contain a single molecule of negative-sense, single-stranded RNA. Two types of glycoprotein 'spikes' or peplomers are present in the envelope: an attachment protein and a fusion protein (F). The attachment protein may either be a haemagglutinin-neuraminidase protein (HN) or a protein without neuraminidase activity (G). The attachment proteins allow the virus to bind to cell surface receptors and the fusion protein causes the virus envelope to fuse with the host cell membrane. Both types of peplomers can induce production of virus neutralizing antibodies. There is also an envelope associated non-glycosylated membrane protein (M). Paramyxoviruses may exhibit haemagglutinating, haemolytic and neuraminidase activities. The nucleocapsid, which has helical symmetry, is 13 to 18 nm in diameter and has a characteristic herringbone appearance. Replication occurs in the cell cytoplasm. Virions are released by budding from the plasma membrane at sites containing virus envelope proteins. The labile virions are sensitive to heat, desiccation, lipid solvents, non-ionic detergents and disinfectants.

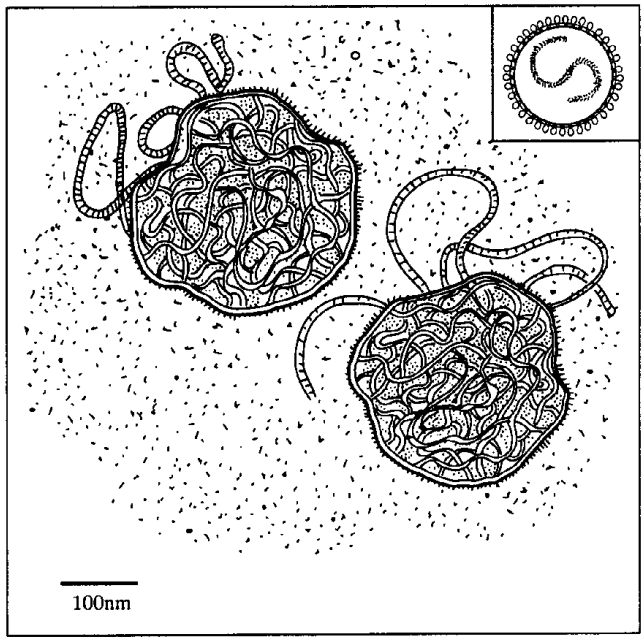


Figure 65.1 Paramyxovirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Key points

- Large pleomorphic enveloped viruses
- Negative sense single-stranded RNA
- Helically symmetrical nucleocapsid
- Replicate in the cytoplasm
- Subfamilies *Paramyxovirinae* and *Pneumovirinae* divided into five genera, each containing viruses of veterinary importance
- Cause rinderpest, peste des petits ruminants, canine distemper, Newcastle disease and a range of respiratory diseases in domestic animals

Recently the classification of the *Paramyxoviridae* has been changed to include a new genus, *Metapneumovirus*, and renaming of the genus *Paramyxovirus* as *Respirovirus* (Fig. 65.2). Although paramyxoviruses are genetically stable and do not exhibit recombination, some antigenic variation may occur through mutation.

Clinical infections

Paramyxoviruses, which have a narrow host range, infect mainly mammals and birds (Table 65.1). Following transmission through close contact or by aerosols, replication occurs primarily in the respiratory tract. Infection is generally cytolytic but persistent infections are described *in vitro*. Formation of syncytia and intracytoplasmic, acidophilic inclusions are features of infection with these viruses. Serious diseases caused by paramyxoviruses include rinderpest, peste des petits ruminants, canine distemper, Newcastle disease, measles and mumps.

Rinderpest

This acute disease, which occurs primarily in ruminants and is also referred to as cattle plague, has been recognized for centuries as a major cause of mortality in cattle and

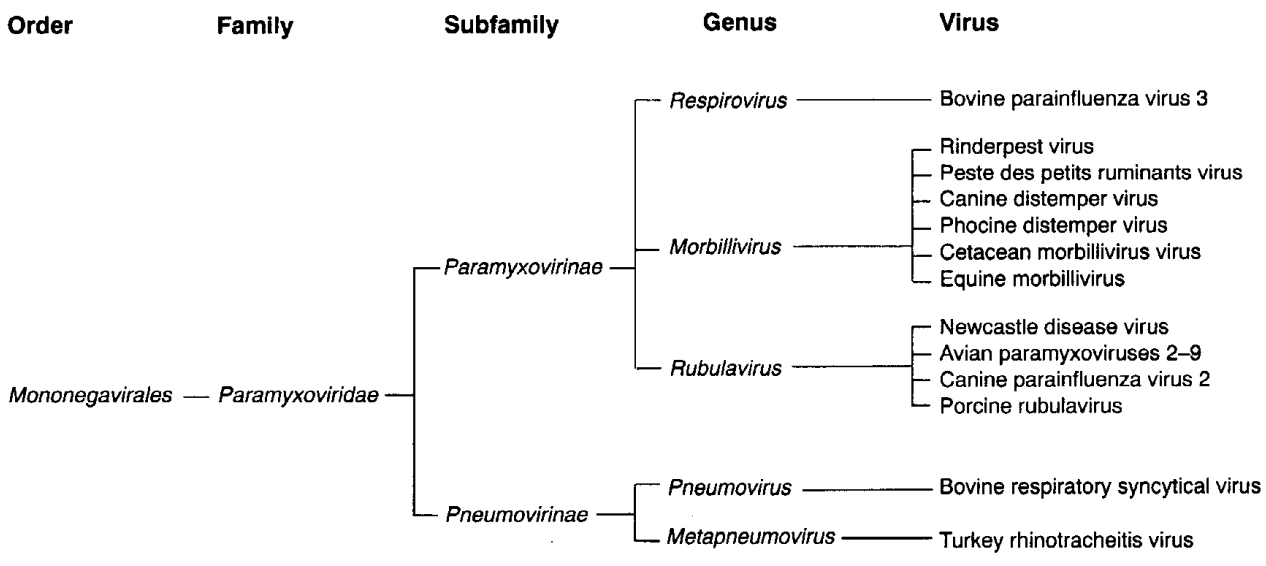


Figure 65.2 A classification of paramyxoviruses with emphasis on those of veterinary importance.

domestic buffalo. Rinderpest is endemic in parts of Africa, the Middle East and Asia. It is classed as a list A disease by the Office International des Epizooties (OIE).

Epidemiology

Although only one serotype of this morbillivirus is recognized, strains differ both in host range and in virulence. Individual host species exhibit differences in susceptibility to virus strains. Domestic cattle and buffalo and several wildlife species including giraffe, warthog, Cape buffalo and eland are highly susceptible to infection. Gazelles and small domestic ruminants are less susceptible. Asiatic breeds of domestic pig develop disease whereas infection in European breeds is subclinical. Transmission, which occurs through aerosols, usually requires close contact as the virus is labile and remains viable in the environment for short periods only. Virus shedding in all secretions and excretions begins a few days

Table 65.1 Paramyxoviruses of veterinary importance.

Genus	Virus	Comments
Morbillivirus	Rinderpest virus	Causes highly contagious disease in domestic and wild ruminants characterized by high morbidity and high mortality
	Peste des petits ruminants virus	Causes severe disease in small ruminants, particularly sheep and goats, resembling rinderpest with high morbidity and high mortality rates
	Canine distemper virus	Causes acute disease in dogs and wild carnivores characterized by multisystemic involvement and variable mortality
Rubulavirus	Newcastle disease virus (Avian paramyxovirus 1)	Causes Newcastle disease in domestic and wild birds. Isolates vary in virulence: velogenic, mesogenic and lentogenic strains. Generalized infection characterized by respiratory, intestinal and nervous signs
	Porcine rubulavirus	Causes blue eye disease; described only in Mexico
	Canine parainfluenza virus 2	Causes inapparent or mild respiratory disease in dogs; sometimes associated with kennel cough; related to or possibly a subtype of simian virus 5 (SV5)
Respirovirus	Bovine parainfluenza virus 3	Causes subclinical or mild respiratory disease in cattle and sheep. Sometimes associated with shipping fever in cattle. Predisposes to secondary bacterial infection particularly with *Mannheimia haemolytica*
Pneumovirus	Bovine respiratory syncytial virus	Common subclinical infection in adult cattle. Associated with respiratory disease outbreaks of varying severity in young cattle. Sheep and goats are also susceptible
Metapneumovirus	Turkey rhinotracheitis virus	Causes severe upper respiratory tract infection in turkeys with coryza and swollen sinuses. In chickens, the disease is referred to as swollen head syndrome

before clinical signs develop. In endemic areas, the disease is usually mild and is restricted to young cattle in which maternally-derived immunity has declined. As there is no carrier state, maintenance of infection requires continuous transmission to susceptible animals. Epidemics usually occur following movement of susceptible animals into an endemic area or the introduction of infected animals into susceptible populations. All ages of animals are affected in epidemics. Morbidity may reach 90% and mortality can approach 100%.

Pathogenesis

After inhalation of the virus, multiplication occurs in the pharyngeal and mandibular lymph nodes. Viraemia develops within three days resulting in spread to other lymphoid tissues and to the mucosae of the respiratory and digestive tracts. Leukopenia and immunosuppression follow necrosis in lymphoid tissues. Virus shedding, which continues throughout the acute phase of the disease, subsides a few days after body temperature returns to normal.

Clinical signs

After an incubation period of three to nine days, infected animals develop a fever and become anorexic and depressed. Mucosal erosions in the mouth and nasal passages become evident within five days . Profuse salivation is accompanied by an oculonasal discharge. About three days after the appearance of the mucosal ulcers, fever regresses and a profuse diarrhoea develops. The dark fluid faeces often contain mucus, necrotic debris and blood. Dehydration and wasting soon become evident. Severely affected animals may collapse and die within 12 days of the onset of clinical signs. In surviving animals, convalescence lasts several weeks. Secondary infections and activation of latent protozoal infections are frequent complications. Pregnant animals may abort during the convalescent period.

Diagnosis

Clinical and pathological findings may be sufficient for diagnosis in endemic areas. They may also be adequate in individual animals in outbreaks which have been confirmed by laboratory tests. In regions where rinderpest is uncommon or absent, laboratory confirmation is required to differentiate it from bovine viral diarrhoea, infectious bovine rhinotracheitis, malignant catarrhal fever and foot-and-mouth disease. When investigating an outbreak, specimens for laboratory examination should be collected from several febrile animals which have not developed diarrhoea.

- Specimens suitable for virus isolation include white cells from the buffy coat of heparinized blood samples, lymph node and spleen.
- The presence of rinderpest virus, which produces cytopathic effects in cell cultures, can be confirmed by immunofluorescence.
- Agar gel immunodiffusion or a counter immunoelectrophoresis test are used as rapid antigen detection tests. Specimens suitable for these procedures include ocular discharge and mesenteric lymph nodes.
- A reverse transcription polymerase chain reaction method, which can detect rinderpest virus and differentiate it from the virus of peste des petits ruminants, has been developed (Forsyth and Barrett, 1995).
- A competitive ELISA for detecting serum antibodies to rinderpest virus is the test recommended by OIE for international trade.
- Postmortem enteric lesions are characteristic but not pathognomonic. Congestion of the folds of the colonic mucosa often produces a zebra-stripe pattern.
- Syncytia may form in stratified squamous epithelium of the upper alimentary tract and in crypts of the small intestine.

Control

The Food and Agricultural Organization aims to achieve worldwide eradication of rinderpest by 2010. Factors which render this achievable include the availability of a vaccine that induces lasting immunity, reliable diagnostic tests and the absence of carrier animals and wild life reservoirs.

In countries free of rinderpest, control is based on restriction of animal movement, quarantine of imported animals and slaughter of infected animals. In endemic areas, control is achieved by vaccination of domestic cattle and buffaloes with a modified live tissue culture-based vaccine that induces immunity lasting at least five years. This stable freeze-dried vaccine is thermolabile following reconstitution. Recombinant vaccinia and capripox virus vaccines expressing either haemagglutinin protein or fusion protein of rinderpest virus have high heat stability and have been used to protect cattle (Inui *et al.*, 1995; Ngichabe *et al.*, 1997). Control of animal movement is the single most important measure in preventing disease transmission.

Peste des petits ruminants

This condition, also referred to as goat plague, is an acute contagious disease of ruminants, particularly goats. It is caused by the morbillivirus, peste des petits ruminants virus (PPRV), which is closely related to other members of the genus. Peste des petits ruminants (PPR), which occurs in sub-Saharan Africa, the Middle East, India and Pakistan, is a list A disease.

Epidemiology

Close contact is required for transmission of this labile virus which occurs by aerosols. The introduction of infection into a flock is invariably associated with

movement of animals. Although a carrier state is not known to occur, subclinical infection and the onset of viral shedding before overt clinical signs facilitate spread of infection. In West Africa, epidemics tend to occur during the rainy season, when flocks are gathered together in preparation for sales. Infection rates are similar in sheep and goats but the disease is generally more severe in goats.

Pathogenesis

The pathogenesis of PPR is similar to that of rinderpest. Mucosal erosions and profuse diarrhoea are features of the condition. During the acute phase of the disease, virus is shed in all secretions and excretions.

Clinical signs

The incubation period is about four days. The disease is particularly severe in young animals. Affected goats exhibit fever, dry muzzle and a serous nasal discharge which becomes mucopurulent. Erosions on the mucous membrane of the buccal cavity are accompanied by marked salivation. Ulcers develop in the mucosae of the alimentary, respiratory and urinary tracts. Conjunctivitis with ocular discharge is a feature of the disease. A profuse diarrhoea, which results in dehydration, develops within days of infection. Signs of tracheitis and pneumonia are common. There is a severe leukopenia which facilitates secondary bacterial infection. Pulmonary infections caused by *Pasteurella* species are common in the later stages of the disease. Pregnant animals may abort. Mortality rates in severe outbreaks often exceed 70% and acutely affected goats may die within ten days of exposure to the virus.

In sheep, infection with PPRV is subacute and characterized by fever, nasal catarrh, mucosal erosions and intermittent diarrhoea. Affected animals usually recover after ten to 14 days.

Diagnosis

Specimens for laboratory examination should be taken from animals in the acute phase of the disease. Suitable specimens include nasal and ocular swabs, unclotted blood and scrapings of buccal and rectal mucosae. Samples of lung, spleen and lymph node from animals slaughtered early in the course of the disease are also suitable. Laboratory confirmation is based primarily on virus isolation in tissue culture and on antigen detection. Rapid antigen detection methods include ELISA, counter immunoelectrophoresis and agar gel immunodiffusion. Specific primers for use in PCR are available. Antibodies can be detected by virus neutralization or by competitive ELISA.

Control

Slaughter policies apply in countries free from PPR. Quarantine and vaccination are used in regions where the disease is endemic. Modified live rinderpest vaccine, which provides adequate protection against PPRV, has been used in sheep and goats for many years. A modified live PPRV vaccine has recently been developed.

Canine distemper

This highly contagious disease of dogs and other carnivores has a worldwide distribution. Canine distemper virus (CDV), a pantropic morbillivirus, produces a generalized infection involving many organ systems.

Epidemiology

The wide host range of CDV includes members of the families *Canidae, Ailuridae, Hyaenidae, Mustelidae, Procyonidae, Ursidae, Viverridae* and *Felidae*. Outbreaks of disease have been documented in several wildlife species including foxes, skunks, racoons, black-footed ferrets and lions (Appel and Summers, 1995; Roelke-Parker *et al.*, 1996). The virus is relatively labile, requiring transmission by direct contact or by aerosols. In urban dog populations, the virus is maintained by infection in susceptible animals. Infection spreads rapidly among young dogs, usually between three and six months of age when maternally-derived immunity declines. The number of dogs in populations in rural areas is often too low to maintain continuous infection with the result that, irrespective of age, unvaccinated dogs are susceptible and significant outbreaks of the disease can occur.

Pathogenesis

The virus, which replicates in the upper respiratory tract, spreads to the tonsils and bronchial lymph nodes. A cell-associated viraemia follows with spread to other lymphoreticular tissues. Viral replication produces lymphocytolysis and leukopenia resulting in immunosuppression and allowing a secondary viraemia to develop. The extent of spread to tissues and organs is determined by the rapidity and effectiveness of the immune response. In the absence of a sufficiently vigorous response, dissemination and replication of CDV occurs in the respiratory, gastrointestinal, urinary and central nervous systems. Spread to the skin may also occur.

Virus infects both neurons and glial cells within the CNS and may persist there for very long periods. Old dog encephalitis is apparently associated with prolonged persistence of the virus in the brain, possibly as a result of non-cytolytic spread from cell to cell without budding from the cell membrane thus evading immune detection (Stettler *et al.*, 1997). This mechanism appears to be analogous to that causing subacute sclerosing panencephalitis of children which is associated with persistent infection with defective measles virus. The presence of viral antigen in these conditions stimulates a low grade prolonged inflammatory response eventually leading to the development of neurological signs.

Clinical signs

The incubation period is usually about one week but may extend to four weeks or more when nervous signs appear without prior evidence of infection. The severity and duration of illness are variable and are influenced by the virulence of the infecting virus, the age and immune status of the infected animal and the rapidity of its immune response to infection. The pyrexic response to infection is biphasic although the initial elevation of temperature may not be noticed. During the second period of pyrexia, oculonasal discharge, pharyngitis and tonsillar enlargement become evident. Coughing, vomiting and diarrhoea are often consequences of secondary infections. A skin rash and pustules may be present on the abdomen. Some affected dogs have hyperkeratosis of the nose and footpads, referred to as 'hardpad'. Acute disease, which may last for a few weeks, is followed either by recovery and life-long immunity or by the development of neurological signs and, eventually, death. Common neurological signs include paresis, myoclonus and epileptiform seizures. A grave prognosis is indicated in animals displaying neurological disturbance. Residual neurological deficits are common in dogs that survive. Old dog encephalitis, characterized by motor and behavioural deterioration, is invariably fatal.

Diagnosis

A febrile, catarrhal illness with neurological sequelae in young dogs is highly suggestive of canine distemper.

- Viral antigen may be demonstrated by immunofluorescence in conjunctival or vaginal impression smears or in smears of cells from the buffy coat.
- Cryostat sections of lymph nodes, urinary bladder and cerebellum are also suitable for the demonstration of viral antigen.
- Eosinophilic inclusions can be demonstrated in nervous and epithelial tissues.
- Serological demonstration either of IgM antibodies or of a fourfold rise in antibody titre between acute and convalescent sera may be determined by virus neutralization, ELISA or indirect immunofluorescence. Antibody may be detected in cerebrospinal fluid.
- Virus isolation may prove difficult. Urinary bladder and brain are suitable postmortem specimens for virus isolation. Cells from the buffy coat of heparinized blood are also suitable.

Control

Modified live vaccines, which are available commercially, provide adequate protection when administered to pups after maternally-derived antibody has declined to negligible levels, usually after 12 weeks of age. Most CDV vaccines are produced from egg-adapted or avian cell culture-adapted virus (Onderstepoort strain) or from canine cell culture-adapted virus (Rockborn strain). Because post-vaccinal encephalitis has been reported occasionally following the use of canine cell culture-adapted strains, the avian cell culture-adapted strain is considered to be safer. Heterotypic measles virus vaccines have been used in young pups to induce protection in the presence of moderate levels of maternally-derived antibody. Although many dogs remain immune for several years following vaccination, a proportion of vaccinated animals become susceptible after a year. Because of the labile nature of the virus, control can be achieved after an outbreak of disease in kennels by strict isolation and disinfection.

Infections caused by other morbilliviruses

During the late 1980s serious outbreaks of a disease, with clinical and pathological features similar to canine distemper, were recorded in the seal populations of the Baltic and the North Sea. The interest generated by these disease outbreaks of viral infections in marine mammals led to the recognition of several new morbilliviruses including phocine distemper virus, dolphin distemper virus and porpoise distemper virus. Serological evidence of morbillivirus infection has been recorded in several cetacean species.

Hendra virus, equine morbillivirus, was isolated during an outbreak of severe respiratory disease in Australia during 1994. Two humans in contact with infected horses were also affected. Fourteen horses and their trainer died. A related virus, Nipah virus, was isolated in Malaysia during 1999 following outbreaks of disease in pigs and humans working in affected pig units. The disease, which caused a febrile encephalitis, resulted in more than 100 deaths in humans. Although related to morbilliviruses, Hendra and Nipah viruses may be sufficiently different from them to warrant classification in a separate genus. It has been recently proposed that a new genus, *Henipavirus*, be created within the subfamily *Paramyxovirinae*.

Newcastle disease

A large number of avian paramyxovirus (APMV) isolates has been reported worldwide from a range of domestic and wild birds. Nine species of antigenically-distinct APMV are currently recognized in the genus *Rubulavirus*. New isolates are assigned to a species on the basis of antigenic relatedness in haemagglutination inhibition tests. Although infections with most avian paramyxovirses are associated with mild or inapparent disease, infections with APMV-2 and APMV-3 cause respiratory disease in turkeys.

The most important avian paramyxovirus is Newcastle disease virus (NDV), also designated avian paramyxovirus 1 (APMV-1), which causes Newcastle disease. This disease occurs in poultry worldwide. Newcastle disease was first described in 1926 when severe outbreaks were

reported in Newcastle, England and in Java. Other major outbreaks of the disease occurred in the Middle East during the late 1960s and in the 1970s when pigeons were the species primarily affected.

Epidemiology

A wide range of avian species including chickens, turkeys, pigeons, pheasants, ducks and geese are susceptible. Infection with NDV is probably endemic in wild birds especially waterfowl (Takakuwa *et al.*, 1998). Strains of NDV differ in their virulence. Isolates are categorized into four groups or pathotypes on the basis of virulence and tissue tropism in poultry:

- viscerotropic velogenic isolates causing severe fatal disease characterized by haemorrhagic intestinal lesions (Doyle's form).
- neurotropic velogenic isolates causing acute disease characterized by nervous and respiratory signs with high mortality (Beach's form).
- mesogenic isolates causing mild disease with mortality confined to young birds (Beaudette's form).
- lentogenic isolates causing mild or inapparent infection (Hitchner's form).

Virus is shed in all excretions and secretions. Transmission usually occurs by aerosols or by ingestion of contaminated feed or water. The relative stability of the virus permits mechanical transfer of infective material through the movement of personnel and equipment. Virus, which can survive in carcases for some weeks, is present in all organs of acutely affected birds and in eggs.

Captive and wild birds can contribute to the spread of infection. Pigeons are susceptible to all strains of NDV and may play a role in the transmission of Newcastle disease. Mesogenic isolates, which can be distinguished from other NDV isolates using monoclonal antibodies, were obtained from racing pigeons in Europe during the early 1980s. These isolates, often referred to as 'pigeon' paramyxovirus 1, are associated with clinical disease in pigeons resembling the neurotropic form of Newcastle disease. Outbreaks of Newcastle disease in poultry in the United Kingdom during 1984 were linked to feed contaminated by infected feral pigeons.

Pathogenesis

Viral replication, which occurs initially in the epithelia of the respiratory and intestinal tracts, is followed by haematogenous spread to the spleen and bone marrow. Secondary viraemia results in infection of other organs including lungs, intestine and CNS. The extent of spread within the body relates to strain virulence which is determined by the amino acid sequence of the F glycoprotein. The fusion (F) glycoprotein of NDV is synthesized in an infected cell as a precursor molecule (F_0) which is cleaved by host cell proteases to F_1 and F_2 subunits. If cleavage fails to occur, non-infectious particles are produced. The F_0 molecules of virulent strains of NDV possess basic amino acids at critical positions which facilitate cleavage by proteases in a wide range of host tissues. In contrast, the replication of lentogenic strains is confined to the respiratory and intestinal epithelia where suitable proteases are produced.

Clinical signs

The incubation period is usually about five days. Respiratory, gastrointestinal and nervous signs occur in chickens. The particular clinical presentation relates to the virulence of the virus strain, its tissue tropism and the age and immune status of the host. Highly virulent strains may produce sudden high mortality in a flock in the absence of premonitory clinical signs. The mortality rate in fully susceptible flocks may be close to 100%. When present, signs in these flocks include listlessness, weakness and a decrease in egg production. Viscerotropic strains tend to produce respiratory signs such as gasping and rales, oedema of the head and neck and greenish diarrhoea. Birds that survive the acute phase may develop neurological signs. Infection with neurotropic velogenic strains results in respiratory disease followed by nervous signs such as wing paralysis, leg paralysis, torticollis and muscle spasms. Mesogenic strains usually cause respiratory disease. Lentogenic strains do not produce disease in adult birds but may produce respiratory signs in young birds. Pathogenicity of NDV isolates relates not only to their virulence but also to host susceptibility. Infection in turkeys, which usually involves the respiratory and central nervous systems, is less severe than that in chickens. Pigeons infected with 'pigeon' paramyxovirus 1 present with neurological signs and diarrhoea, and mortality in affected birds may approach 10%.

Humans may develop a transitory conjunctivitis if exposed to high concentrations of NDV.

Diagnosis

A presumptive clinical diagnosis may be made when the characteristic signs and lesions associated with virulent strains are present. Laboratory confirmation by isolation and identification of the virus is necessary.

- Tracheal and cloacal swabs from live birds are suitable for virus isolation. Suitable postmortem specimens for laboratory examination include faeces, intestinal contents and portions of trachea, intestine, spleen, brain and lung. Samples may be stored at 4°C for up to four days.
- Virus isolation is carried out in embryonated eggs from specific pathogen free (SPF) flocks, usually by inoculation into the allantoic cavity. After incubation, allantoic fluid is tested for haemagglutination activity.
- Haemagglutination-inhibition test using specific antiserum confirms the presence of NDV.

- Virulence of NDV isolates is assessed using *in vivo* tests including the intracerebral pathogenicity index and the intravenous pathogenicity index in SPF chicks. The mean death time (MDT) using embryonated eggs has been employed to classify isolates as velogenic (embryonic death (ED) in less than 60 hours), mesogenic (ED between 60 and 90 hours) and lentogenic (ED in more than 90 hours).
- Demonstration of antibody to NDV is of diagnostic value only in unvaccinated flocks. The haemagglutination-inhibition test is the most widely used assay. Commercial ELISA kits are available.
- Demonstration of viral antigen in tracheal sections or impression smears using immunofluorescence is a less sensitive technique than virus isolation.

Control

General control measures include locating poultry farms several kilometers apart, bird-proofing of houses and food stores, controlled access to farms, thorough cleaning and disinfection of vehicles and equipment and restriction of movement between poultry farms. National control policies for Newcastle disease differ from country to country and range from compulsory vaccination to slaughter of infected flocks. A combination of vaccination and slaughter policies is frequently employed. Vaccination is particularly important for birds in breeder flocks. Lentogenic or mesogenic strains of NDV propagated in eggs or tissue culture are used in live vaccines. They are administered as a spray, in drinking water or by intranasal or intraconjunctival instillation. The presence of maternally-derived antibodies interferes with the efficacy of live vaccines. In order to avoid this undesirable effect, vaccination should be delayed until two to four weeks of age or live vaccine should be administered to day old chicks by conjunctival instillation or by a coarse spray. This method, which may result in respiratory disease, establishes active infection in some birds that persists until maternally-derived immunity has waned in the rest of the birds. Revaccination is then carried out when chicks are three or four weeks of age. A schedule of vaccination employing both live and inactivated vaccines gives good results. Vaccinated birds, although protected from clinical disease, can be infected with wild type virus and become shedders.

Blue eye disease in pigs

This condition, caused by porcine rubulavirus, was first observed in pigs during 1980 in Mexico. Blue eye disease is characterized by neurological signs, corneal opacity and reproductive failure. Morbidity and mortality is highest in young pigs. Diagnosis is based on clinical signs, histopathological changes and serological testing of paired serum samples. Suitable tests for the detection of antibodies include haemagglutination-inhibition, ELISA and virus neutralization. Methods aimed at preventing the introduction of the infection include strict isolation procedures combined with serological testing of replacement animals. Inactivated vaccines have been produced. The disease has not been reported outside Mexico.

Infection caused by bovine parainfluenza virus 3

Infection with bovine parainfluenza virus 3 (BPIV-3), occurs worldwide and is often subclinical. Transmission occurs by aerosols and direct contact. It is facilitated by overcrowding in poorly ventilated conditions. Although uncomplicated infections are frequently subclinical, mild respiratory disease may be seen. The virus is commonly isolated from animals during outbreaks of serious respiratory disease such as enzootic calf pneumonia and shipping fever, conditions in which other respiratory viruses and bacteria are often involved. Various stress factors such as transportation or adverse environmental conditions may contribute to the severity of the disease.

The virus infects ciliated epithelium of the respiratory tract, alveolar epithelium and macrophages. Infection causes destruction of ciliated epithelium resulting in interference with the mucociliary clearance mechanism. In addition, phagocytosis and intracellular destruction of bacteria by alveolar macrophages are depressed, predisposing to secondary bacterial infection in the lungs. Most uncomplicated infections with BPIV-3 are mild and are characterized by fever, nasal discharge and coughing. Most affected animals recover within a few days. The virus can be isolated in suitable bovine cell lines from nasal swabs or lung tissue. Samples should be taken from several animals in the early stages of the disease and transferred immediately to the laboratory in viral transport medium. Direct immunofluorescence for detecting viral antigen can be carried out on samples of nasal mucus or on cryostat sections of lung. Haemagglutination-inhibition tests, virus neutralization, ELISA and indirect immunofluorescence are commonly used to demonstrate a four-fold rise in antibody titre between acute and convalescent sera.

Both inactivated and modified live BPIV-3 vaccines are available, often combined with other respiratory viruses. Modified live vaccines are designed either for intranasal administration or for intramuscular injection. Immunity tends to be shortlived and reinfection may occur after some months.

Infection caused by bovine respiratory syncytial virus

Pulmonary disease, caused by bovine respiratory syncytial virus (BRSV), is recorded in beef and dairy calves worldwide. Infections occur in cattle, sheep and goats. The

virus is named for the characteristic syncytia which it induces in infected cells *in vivo* and *in vitro*.

Epidemiology

Infection in cattle is common. Moderate to severe respiratory signs often develop in infected calves. Infection in adult animals is usually mild or subclinical but severe disease may occasionally occur (Ellis *et al.*, 1996; Elvander, 1996). Persistent infection in individual animals is considered to be responsible for the maintenance of infection in herds. Transmission occurs through aerosols or through direct contact with infected animals. Most clinical cases are recorded during autumn and winter months. Transportation, overcrowding or adverse weather conditions can precipitate outbreaks of the disease. Concurrent infection with bovine viral diarrhoea virus results in more severe clinical signs than those encountered in infection with each individual virus (Pollreiz *et al.*, 1997).

Pathogenesis

The virus replicates in the epithelia of the respiratory system. Destruction of bronchiolar epithelium results in necrotizing bronchiolitis. Multinucleate cells are occasionally formed by fusion of infected type 2 pneumocytes. Bovine respiratory syncytial virus is considered to be immunosuppressive. This effect along with the accumulation of cellular debris and exudate in pulmonary airways facilitates bacterial proliferation.

Clinical signs

Affected animals are typically between three and nine months old. Clinical signs, which range from mild to severe, include fever, nasal and lacrimal discharge, coughing and polypnoea. As the disease progresses, open-mouth and abdominal breathing may be present. The course of the disease is usually up to two weeks. A biphasic pattern is commonly observed in outbreaks among beef calves. Mild respiratory disease is followed by apparent recovery and, within a few days, dyspnoea and pulmonary emphysema develop. Mortality in these outbreaks may reach 20 %.

Diagnosis

Clinical signs and pathological findings may permit a presumptive diagnosis. Laboratory confirmation is necessary for definitive diagnosis.

- Suitable specimens for laboratory examination include nasal swabs, bronchoalveolar lavage fluid, lung tissue and paired serum samples. Specimens should be collected from several animals in the affected group.
- As the virus is thermolabile, specimens must be transferred to the laboratory rapidly in a suitable transport medium.
- Virus isolation is not attempted routinely as it is difficult and requires several blind passages in cell culture.
- Commercial ELISA kits are available for the detection of viral antigen. Immunofluorescence is a rapid and useful technique. Viral antigen can be detected more readily in specimens from the lower respiratory tract than in nasal swabs.
- Polymerase chain reaction procedures have been described for the detection of specific viral nucleic acid (Valarcher *et al.*, 1999).
- Suitable serological tests for demonstrating a rising antibody titre include virus neutralization and ELISA. Serum samples should be taken early in the course of the disease as antibody levels tend to rise rapidly.

Control

Suitable control measures include reducing stress factors, maintaining good hygiene in calf pens, raising calves away from older age groups and implementing a closed herd policy. Modified live and inactivated vaccines, which are safe, are administered parenterally and tend to induce systemic antibody responses, mainly to the fusion surface glycoprotein. Although vaccination tends to reduce the likelihood of clinical disease in exposed animals, the duration of protection is short and frequent boosters may be required. An intranasal recombinant herpes-1 vaccine expressing the attachment (G) glycoprotein and a subunit fusion glycoprotein vaccine expressed in recombinant baculovirus-infected insect cells have recently been developed (Taylor *et al.*, 1998; Sharma *et al.*, 1996).

References

Appel, M.J.G. and Summers, B.A. (1995). Pathogenicity of morbilliviruses for terrestrial carnivores. *Veterinary Microbiology*, **44**, 187–191.

Ellis, J.A., Philibert, H., West, K. *et al.* (1996). Fatal pneumonia in adult dairy cattle associated with active infection with bovine respiratory syncytial virus. *Canadian Veterinary Journal*, **37**, 103–105.

Elvander, M. (1996). Severe respiratory disease in dairy cows caused by infection with bovine respiratory syncytial virus. *Veterinary Record*, **138**, 101–105.

Forsyth, M.A. and Barrett, T. (1995). Evaluation of polymerase chain reaction for the detection and characterization of rinderpest and peste des petits ruminants viruses for epidemiological studies. *Virus Research*, **39**, 151–163.

Inui, K., Barrett, T., Kitching, R.P. and Yamanouchi, K. (1995). Long term immunity in cattle vaccinated with a recombinant rinderpest vaccine. *Veterinary Record*, **137**, 669–670.

Ngichabe, C.K., Wamwayi, H.M., Barrett, T. *et al.* (1997). Trial of a capripoxvirus-rinderpest recombinant vaccine in African cattle. *Epidemiology and Infection*, **118**, 63–70.

Pollreiz, J.H., Kelling, C.L., Brodersen, B.W. *et al.* (1997). Potentiation of bovine respiratory syncytial virus infection in calves by bovine viral diarrhoea virus. *Bovine Practitioner*,

Chapter 66

Rhabdoviridae

Members of the family *Rhabdoviridae* (Greek *rhabdos*, rod) have characteristic rod shapes (Fig. 66.1). This family, along with the families *Paramyxoviridae*, *Bornaviridae* and *Filoviridae,* belongs to the order *Mononegavirales* (Fig. 66.2). Viruses in this order possess a linear, non-segmented RNA genome of negative polarity encased in a ribonucleoprotein complex. Rhabdoviruses of vertebrates are bullet- or cone-shaped while those infecting plants are generally bacilliform. It is a large family, containing viruses of vertebrates, invertebrates and plants. The family *Rhabdoviridae* comprises six genera; *Vesiculovirus*, *Lyssavirus*, *Ephemerovirus*, *Cytorhabdovirus*, *Novirhabdovirus* and *Nucleorhabdovirus*. Moreover, a large number of rhabdoviruses have yet to be assigned to a genus. *Vesiculovirus*, *Lyssavirus* and *Ephemerovirus* genera contain viruses which infect vertebrates. Infectious haematopoietic necrosis virus and related rhabdoviruses of fish are included in the genus *Novirhabdovirus*.

Rhabdoviruses usually contain five major proteins: a large RNA-dependent RNA polymerase (L), a surface glycoprotein (G), a nucleoprotein (N), a protein component of the viral polymerase (P) and a matrix protein (M). The G protein forms the surface peplomers which interact with host cell receptors, facilitating endocytosis of the virion. In addition, the G protein induces virus-neutralising antibodies and cell-mediated immunity. Replication occurs in the cytoplasm (with the exception of nucleorhabdoviruses). Newly synthesized nucleocapsids acquire envelopes from the plasma membrane as virions bud from the cell. Virions (100 to 430 nm x 45 to 100 nm) are stable in the pH range of 5 to 10. They are rapidly inactivated by heating at 56°C, by treatment with lipid solvents and by exposure to UV light.

Key points

- Enveloped RNA viruses with helical symmetry and rod-shaped morphology
- Rabies virus and related lyssaviruses
  - —present in saliva; transmitted by biting carnivores and bats
  - —cause encephalitis in mammals which is invariably fatal
- Vesicular stomatitis viruses
  - —transmitted by direct contact and environmental contamination or by arthropod vectors
  - —cause febrile disease with vesicular lesions especially in cattle, horses and pigs
- Bovine ephemeral fever virus
  - —transmitted by biting arthropods
  - —causes febrile transient illness with ill-defined clinical signs

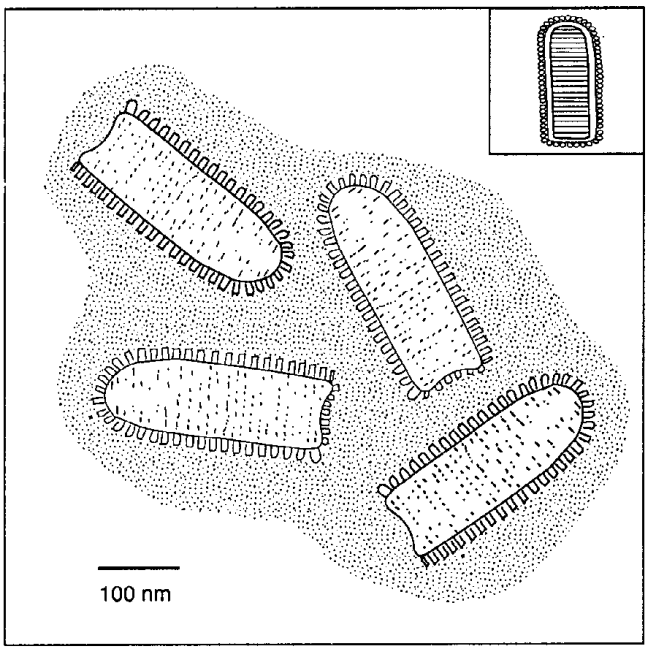


Figure 66.1 Rhabdovirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Clinical infections

Rhabdoviruses of veterinary importance are presented in Tables 66.1 and 66.2. They can be transmitted by bites of mammals, arthropod vectors or direct contact. Infection may also be acquired through environmental contamination.

The best known and most important member of the

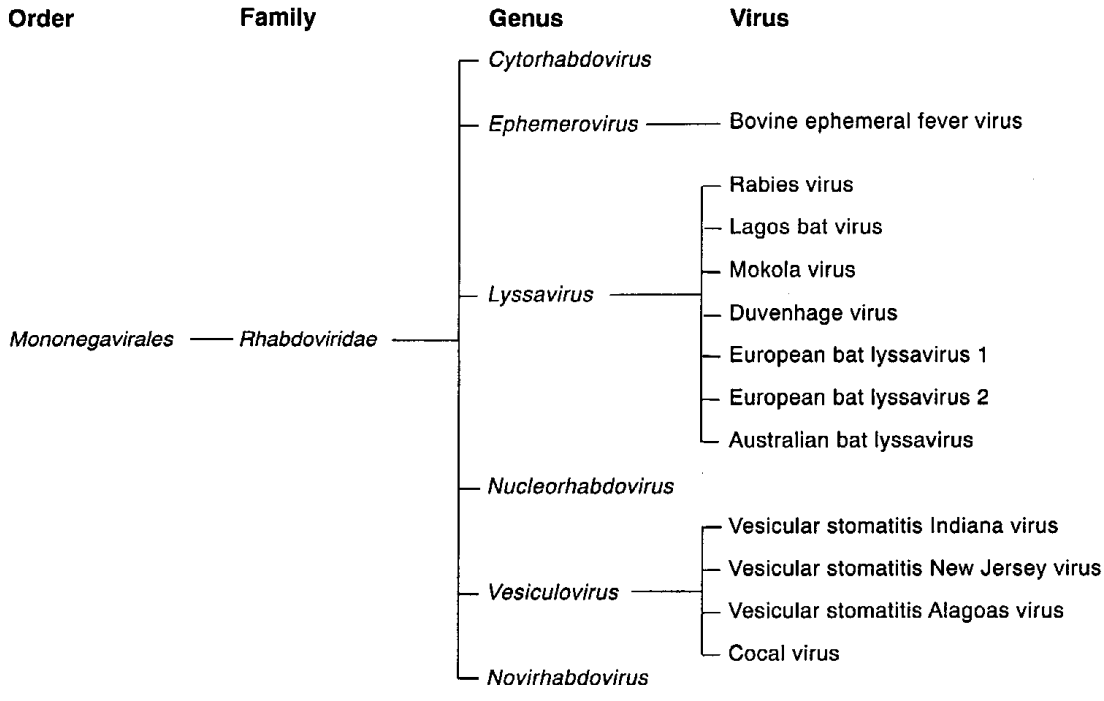


Figure 66.2 A classification of rhabdoviruses with emphasis on those of veterinary importance

Rhabdoviridae is rabies virus, a *Lyssavirus* (Greek *lyssa*, rage or fury). A number of distinct *Lyssavirus* genotypes produce clinical signs indistinguishable from rabies. More than 25 viruses isolated from animals have been classified in the genus *Vesiculovirus*. The most important vesiculoviruses which infect domestic animals are the vesicular stomatitis Indiana virus and the vesicular stomatitis New Jersey virus (Table 66.2). Bovine ephemeral fever virus, of significance in some countries, is the type species of the genus *Ephemerovirus*. Some fish diseases, such as infectious haematopoietic necrosis, viral haemorrhagic septicaemia and spring viraemia of carp are also caused by rhabdoviruses.

Rabies

This viral infection, which affects the central nervous system of most mammals including man, is invariably fatal. However, mammalian species vary widely in their susceptibility. Most clinical cases are due to infection with rabies virus (genotype 1). A number of other neurotropic lyssaviruses, closely related to the rabies virus, produce clinical signs indistinguishable from rabies (Table 66.1). Genetic sequencing and antigenic studies have been used to categorize lyssaviruses into seven genotypes and four serotypes (Smith, 1996; Gould *et al.*, 1998). Each genotype is assigned separate species status. Classical rabies caused by genotype 1 lyssavirus is endemic on continental land masses with the exception of Australia and Antarctica. Many island countries are also free of the disease.

Epidemiology

Several species-adapted strains of rabies virus have been described. Strains affecting a particular species are transmitted more readily to members of that species than to other animal species. In a given geographical region, rabies is usually maintained and transmitted by particular mammalian reservoir hosts. Two epidemiologically important infectious cycles are recognised, urban rabies in dogs and sylvatic rabies in wildlife. More than 95% of human cases are the result of bites from rabid dogs. Racoons, skunks, foxes and bats are important reservoirs of rabies virus in North America (Krebs *et al.*, 1998). In continental Europe, the principal reservoir is the red fox. The vampire bat is an important reservoir of the virus in Central and South America and in the Caribbean islands. In developed countries, the control of stray dogs and vaccination programmes have reduced the importance of urban rabies and have focused attention on wildlife reservoirs.

Species susceptibility to rabies virus is important epidemiologically. Domestic animals and man are considered to be moderately susceptible to the virus, whereas foxes, wolves, coyotes and jackals are considered to be highly susceptible. Although virus may be transmitted through scratching and licking, transmission usually occurs through bites. Infected animals may excrete virus in their saliva for some time before the onset of clinical signs.

Pathogenesis

Following introduction into the tissues, virus enters

Table 66.1 Lyssaviruses which cause rabies and rabies-like diseases.

Virus	Genotype	Serotype	Geographical distribution	Comments
Rabies virus	1	1	Apart from Australia and Antartica, rabies virus (genotype 1) occurs on all continents. Many island countries are free of the disease	Causes fatal encephalitis in many mammalian species. Transmitted by wildlife species including foxes, racoons and bats; domestic carnivores also involved in transmission. Rabies is a major zoonotic disease
Lagos bat virus	2	2	Africa	Isolated initially from fruit bats; also isolated from domestic animals with encephalitis
Mokola virus	3	3	Africa	Isolated initially from shrews; also isolated from domestic animals. Human infection reported
Duvenhage virus	4	4	Africa	Originally isolated from a human bitten by an insectivorous bat; additional cases reported in humans. Not reported in domestic animals
European bat lyssavirus 1	5	–	Europe	Identified with increasing frequency in insectivorous bats. Human infection reported
European bat lyssavirus 2	6	–	Europe	Isolated initially from a human with symptoms of rabies; present in insectivorous bats. Additional human cases reported; not reported in domestic animals
Australian bat lyssavirus	7	–	Australia	Identified in fruit bats and in insectivorous bats; human infection reported

peripheral nerve endings. There may be limited replication locally in myocytes or other tissue cells. The virus is transported to the central nervous system by retrograde axoplasmic flow and becomes widely disseminated in nervous tissue by intra-axonal spread. Clinical signs develop following neuronal damage caused by viral replication. Virus spreads centrifugally within nerve cell processes and is released at axon terminals where it infects many non-nervous tissues including the salivary glands. The presence of virus in saliva, especially in carnivores, is an important factor in rabies transmission.

Although rabies viral antigens are highly immunogenic, immune detection is delayed because intracellular transport prevents contact with the cells of the immune system in the early stages of infection.

Clinical signs

The incubation period, which is highly variable and can be as long as six months, is influenced by various factors including host species, virus strain, the amount of inoculum and the site of introduction of the virus. Large amounts of virus introduced into deep bite wounds in the head region are usually associated with short incubation periods. The clinical course in domestic carnivores, which usually lasts for days or for a few weeks, may encompass prodromal, furious (excitative) and dumb (paralytic) phases. In certain rabid animals, some of these phases may not be observed. In the prodromal phase, affected animals are often confused and disorientated; wild animals may lose their natural fear of humans. The furious phase is charaterized by an increase in aggressiveness and hyperexcitability, and there is a tendency to bite at inanimate objects and at other animals. Affected animals may roam over long distances. The furious form is observed more often in cats than in dogs. Foxes rarely exhibit this form of the disease. In dumb rabies, muscle weakness, difficulty in swallowing, profuse salivation and dropping of the jaw are the usual features. These clinical signs may be mistaken for those caused by a foreign body in the mouth or throat. The term hydrophobia, a synonym for rabies in humans, relates to the inability to swallow water because of pharyngeal paralysis.

Diagnosis

Ante-mortem diagnostic tests for rabies are not generally used. In endemic areas, suspect domestic carnivores which have bitten humans should be isolated and observed for up to 14 days. The brains of animals which develop clinical signs should be examined for the presence of virus. Rapid laboratory confirmation is essential for the implementation of appropriate treatment of human patients.

- Non-suppurative encephalitis characterized by perivascular lymphoid cuffing and intracytoplasmic inclusions (Negri bodies) may be demonstrable histologically.
- The direct fluorescent antibody test (FAT), which provides a rapid and specific diagnosis, may yield false-

negative results with autolysed brain specimens. The conjugated antisera usually used for diagnosis are specific for rabies virus (serotype 1).

- Rabies virus can be cultured in neuroblastoma cells or in baby hamster kidney cells. Culture of the virus is of value when results of the FAT are uncertain. Rabies virus, which is non-cytopathic, can be detected in tissue culture using conjugated antisera.
- Suckling mice, inoculated intracerebrally with brain tissue from suspect rabies cases, should be observed over several days for the development of disease. The FAT is used to confirm the presence of rabies virus in infected mice.
- Reverse transcriptase polymerase chain reaction (RT-PCR) has been used to detect viral RNA in brain samples. This test can distinguish rabies virus (genotype 1) from rabies-related lyssaviruses. The sensitivity of RT-PCR can be enhanced by combining the technique with ELISA which aids detection of amplified product (Whitby *et al.*, 1997).

Control

Most countries which are free of rabies rely on rigorous quarantine measures to prevent the introduction of disease. Movement of vaccinated domestic carnivores is permitted between some countries provided that strict identification and testing procedures are in place. In countries where rabies is endemic, control methods are aimed mainly at reservoir species. Urban rabies can be effectively controlled by vaccination and restriction of movement of dogs and cats and by the elimination of stray animals. Control of sylvatic rabies requires special measures. Regional depopulation of reservoir species, which has rarely been successful, is ecologically unacceptable. Vaccination of red foxes with live oral vaccines delivered in baits has eliminated sylvatic rabies from several regions of western Europe. Although attenuated virus vaccines were used initially, there was uncertainty about their ultimate safety. A vaccinia-rabies virus glycoprotein (VRG) vaccine was developed and has proved effective for vaccinating foxes (Pastoret and Brochier, 1999), coyotes (Fearneyhough *et al.*, 1998) and racoons (Hanlon *et al.*, 1998). The rapid increase in racoon rabies in the USA has proved difficult to control through vaccination (Smith, 1996). Laboratory based studies with recombinant rabies vaccines indicate that these vaccines are likely to be effective in the control of rabies in most wildlife species.

Commercial vaccines available for the immunization of domestic carnivores by parenteral inoculation contain inactivated virus (genotype 1) and are potent and safe. These inactivated rabies vaccines are considered to be effective against strains of genotype 1 virus. Their ability to induce protection against infection with other lyssavirus genotypes is variable and, in some instances, cross-protection may not occur.

Vesicular stomatitis

This febrile disease affects mainly horses, cattle and pigs. Other susceptible species include camels, several wildlife species and humans. Vesicular stomatitis is clinically similar to foot-and-mouth disease and is classified as a list A disease by the Office International des Epizooties. A number of closely related, antigenically distinct members of the genus *Vesiculovirus*, of which the type species is vesicular stomatitis Indiana virus, can cause the disease. Most outbreaks are associated with vesicular stomatitis Indiana virus or with vesicular stomatitis New Jersey virus, a more virulent virus. Cocal virus and vesicular stomatitis Alagoas virus, also referred to as subtypes 2 and 3 of vesicular stomatitis Indiana virus, have been isolated from outbreaks in horses and cattle in South America (Table 66.2).

Epidemiology

Infection is endemic in Central America and in regions of South America and the USA. Outbreaks of the disease occur every two to three years in tropical and subtropical regions, with clinical cases most common at the end of the rainy season and early in the dry season. Rapid spread from endemic areas to other regions may occur during some summer seasons. Disease outbreaks in temperate regions, which occur every five to ten years, usually cease abruptly with the onset of winter.

Table 66.2 Viruses of veterinary significance in the genera *Vesiculovirus* and *Ephemerovirus*.

Genus / Virus	Hosts	Comments
Vesiculovirus		
Vesicular stomatitis Indiana virus	Cattle, horses, pigs, humans	Causes febrile disease with vesicular lesions; resembles foot-and-mouth disease clinically. Occurs in North and South America
Vesicular stomatitis New Jersey virus	Cattle, horses, pigs, humans	Causes febrile disease with vesicular lesions; infection more severe than that caused by the Indiana virus. Occurs in North and South America
Vesicular stomatitis Alagoas virus (Brazil virus)	Horses, mules, cattle, humans	Originally isolated from mules in Brazil
Cocal virus (Argentina virus)	Horses	Isolated initially from mites in Trinidad; occurs in South America
Ephemerovirus		
Bovine ephemeral fever virus	Cattle	Causes febrile illness of short duration; occurs in Africa, Asia and Australia

Although the mode of transmission is incompletely understood, direct contact and insect vectors have been implicated. Virus is shed in saliva and can contaminate water and feed troughs. The involvement of insect vectors is inferred from the seasonal occurrence of cases and from the pattern of spread with clustering of cases along river valleys and in irrigated areas. Virus has been isolated from many insect species including blackflies, mosquitoes, sandflies and houseflies. Viral replication in blackflies has been demonstrated experimentally. It is unclear how biting insects acquire the virus from domestic animals as a viraemic phase has not been demonstrated in these species.

Pathogenesis

Virus probably enters the body through abrasions on the skin or mucous membranes or following an insect bite. Vesicles which develop at the site of infection may coalesce. Spread may occur locally by extension from primary lesions. Although secondary lesions at distant sites may develop, it is unclear how transfer of the virus occurs and if these lesions result from viraemia or following environmental contamination (Clarke *et al.*, 1996).

Clinical signs

The incubation period is up to five days. Subclinical infection is common. Affected animals, which are usually more than one year old, become febrile. Vesicles develop on the tongue and on oral mucous membranes, often accompanied by profuse salivation. Secondary lesions may occur on the coronary band and teats. Lameness is often a prominent feature of the disease in pigs. Mastitis may develop in cows with severe teat lesions. In the absence of secondary infection, lesions generally heal within two weeks.

The economic impact of the disease relates to production losses, culling and other disease control measures (Hayek *et al.*, 1998). Following infection, animals develop high levels of neutralising antibodies but the duration of protection is variable. Cross-protection between vesicular stomatitis Indiana virus and vesicular stomatitis New Jersey virus is limited.

Diagnosis

Prompt laboratory confirmation is required because of similarities between vesicular stomatitis, foot-and-mouth disease and swine vesicular disease. If horses present with vesicular lesions, infection with vesicular stomatitis virus should be considered.

- Suitable specimens for isolation of virus or the detection of viral antigen include epithelium from lesions and vesicular fluid.
- Viral antigen can be detected by CFT or ELISA.
- Virus may be isolated in suitable cell lines, in embryonated eggs or in suckling mice by intracerebral inoculation. The virus is cytopathic. The fluorescent antibody test, ELISA, CFT or the virus neutralization test are suitable procedures for identification of isolates.
- Electron microscopy can be used to identify virus in specimens or tissue culture.
- Antibody levels in recovered animals may be assayed by CFT, the virus neutralization test, competitive ELISA or IgM-specific capture ELISA. Because levels of complement-fixing and IgM antibodies persist for only short periods, assays based on procedures involving these antibodies can be used to confirm recent infections in endemic areas.

Treatment and control

- Specific treatment is not available. Measures aimed at minimising secondary infections may be beneficial.
- Suspected cases should be notified to the relevant authorities. Movement restrictions and a 30-day quarantine period following the last clinical case are recommended for infected premises. International trade restrictions are generally instituted following an outbreak.
- Insect-proof buildings and avoidance of habitats associated with insect vectors reduce the likelihood of infection.
- Although both inactivated and attenuated vaccines have been used, they are not commercially available.
- Vesicular stomatitis is a zoonotic disease.

Bovine ephemeral fever

This arthropod-borne viral disease of cattle and water buffalo occurs in tropical and subtropical regions of Africa, Asia and Australia. The virus causes subclinical infection in many other ruminant species including Cape buffalo, wildebeest, waterbuck and deer.

Epidemiology

Epidemiological evidence suggests that *Culicoides* species are involved in transmission of the virus. In tropical areas where bovine ephemeral fever is endemic, subclinical infections are common. Outbreaks often follow periods of rainfall. In more temperate regions, epidemics occur during summer months and tend to decline with the onset of winter. Transmission does not occur by direct contact or by fomites. There is no evidence that the virus persists in animals following recovery from acute illness. It is probable that virus persistence occurs in arthropod vectors.

Pathogenesis

Blood-sucking insects acquire the virus when feeding on animals during the brief viraemic stage of the disease. The virus, which multiplies in the insect vector, is shed in its saliva and is transmitted to a new host through wounds during feeding. Many of the changes observed in infected animals are attributed to host response rather than to direct viral damage.

Clinical signs

The incubation period is up to eight days. The severity of the clinical signs is influenced by the immune status of infected animals and the virulence of the virus strain. The disease tends to be more severe in well-fed animals and high-yielding dairy cows. A biphasic high fever is commonly observed. Affected animals become depressed, anorexic, lame and constipated. Milk production drops dramatically. Muscle stiffness and ruminal stasis may develop. Pregnant animals may abort. Recumbency may be accompanied by salivation and ocular and nasal discharge. Muscular fibrillation and paresis frequently occur, reflecting the accompanying hypocalcaemia. The disease is of short duration and affected animals usually recover after a few days. Most recovered animals develop a solid immunity.

Diagnosis

Diagnosis of bovine ephemeral fever is usually based on clinical signs. Virus neutralization tests or ELISA should be carried out on paired serum samples to detect a rise in virus-specific antibody. Other serological tests such as immunofluorescence are less useful as interpretation is complicated by cross-reacting antibodies induced by infections with related non-pathogenic ephemeroviruses such as Kimberley virus. Neutrophilia, increased plasma fibrinogen and decreased plasma calcium levels are commonly present. Isolation of bovine ephemeral fever virus is difficult.

Treatment

Affected animals should be rested. Anti-inflammatory drugs such as phenylbutazone, flunixin meglumine and ketoprofen have proved useful for treatment (Fenwick and Daniel, 1996). Intravenous or subcutaneous administration of calcium borogluconate is recommended. Oral drenching should be avoided during the acute phase of the illness because swallowing may be impaired.

Control

Vector control is usually impractical in endemic areas. Control is based on the use of vaccines, both inactivated and attenuated. Trials have been carried out using a subunit vaccine based on the envelope glycoprotein (Uren *et al.*, 1994). A recombinant virus-vectored glycoprotein vaccine has been developed.

References

Clarke, G.R., Stallknecht, D.E. and Howerth, E.W. (1996). Experimental infection of swine with a sandfly (*Lutzomyia shannoni*) isolate of vesicular stomatitis virus, New Jersey serotype. *Journal of Veterinary Diagnostic Investigation*, **8**, 105–108.

Fearneyhough, M.G., Wilson, P.J., Clark, K.A., *et al.* (1998). Results of an oral rabies vaccination program for coyotes. *Journal of the American Veterinary Medical Association*, **212**, 498–502.

Fenwick, D.C. and Daniel, R.C.W. (1996). Evaluation of the effect of ketoprofen on experimentally induced ephemeral fever in dairy heifers. *Australian Veterinary Journal*, **74**, 37–41.

Gould, A.R., Hyatt, A.D., Lunt, R. *et. al.* (1998). Characterization of a novel lyssavirus isolated from *Pteropid* bats in Australia. *Virus Research*, **54**, 165–187.

Hanlon, C.A., Niezgoda, M., Hamir, A.N., *et. al.* (1998). First North American field release of a vaccinia-rabies glycoprotein recombinant virus. *Journal of Wildlife Diseases*, **34**, 228–239.

Hayek, A.M., McCluskey, B.J., Chavez, G.T. and Salman, M.D. (1998). Financial impact of the 1995 outbreak of vesicular stomatitis on 16 beef ranches in Colorado. *Journal of the American Veterinary Medical Association*, **212**, 820–823.

Krebs, J.W., Smith, J.S., Rupprecht, C.E. and Childs, J.E. (1998). Rabies surveillance in the United States during 1997. *Journal of the American Veterinary Medical Association*, **213**, 1713–1728.

Pastoret, P.P. and Brochier, B. (1999). Epidemiology and control of fox rabies in Europe. *Vaccine*, **17**, 1750–1754.

Smith, J.S. (1996). New aspects of rabies with emphasis on epidemiology, diagnosis, and prevention of the disease in the United States. *Clinical Microbiology Reviews*, **9**, 166–176.

Uren, M.F., Walker, P.J., Zakrzewski, H., St. George, T.D. and Byrne, K.A. (1994). Effective vaccination of cattle using the virion G protein of bovine ephemeral fever as antigen. *Vaccine*, **12**, 845–850.

Whitby, J.E., Heaton, P.R., Whitby, H.E. *et al.* (1997). Rapid detection of rabies and rabies-related viruses by RT-PCR and enzyme-linked immunosorbent assay. *Journal of Virological Methods*, **69**, 63–72.

Further reading

Bridges, V.E., McCluskey, B.J., Salman, M.D. *et al.* Review of the 1995 vesicular stomatitis outbreak in the western United States. *Journal of the American Veterinary Medical Association*, **211**, 556–560.

Ministry of Agriculture, Fisheries and Food, UK (1998). *Quarantine and rabies, a reappraisal.* MAFF Publications, London.

Vanselow, B.A., Walthall, J.C. and Abetz, I. (1995). Field trials of ephemeral fever vaccines. *Veterinary Microbiology*, **46**, 117–130.

Chapter 67

Bornaviridae

The family *Bornaviridae*, which belongs to the order *Mononegavirales*, contains a single genus *Bornavirus*. The sole member of the genus is Borna disease virus (BDV). This enveloped virus, which has only recently been demonstrated by electron microscopy, is spherical with a diameter of about 90 nm (Fig. 67.1). The envelope surrounds an inner core, 50 to 60 nm in diameter. The genome consists of a single molecule of negative-sense, single-stranded RNA. Replication occurs in the nucleus of host cells with budding at the cell surface. This labile virus is sensitive to heat, lipid solvents and low pH values.

Key points

- Labile, enveloped virus with a particular affinity for nervous tissues
- Genome consists of negative-sense, single-stranded RNA
- Replicates in nuclei of neural cells
- The single genus *Bornavirus* contains one species
- Causes Borna disease, an infection of the CNS, fatal in many domestic species, particularly horses

Borna disease

First described over 200 years ago, this fatal neurological disease of horses is named after Borna, the town in Saxony, where it was first recorded. The disease occurs sporadically in Germany, Switzerland and other parts of Europe. Sero-epidemiological studies, however, indicate a wide geographical distribution. Antibodies reacting with BDV have been found in the sera of human patients with psychiatric disorders and a possible link between infection with the virus and human neurological disorders is being investigated (Boucher *et al.*, 1999).

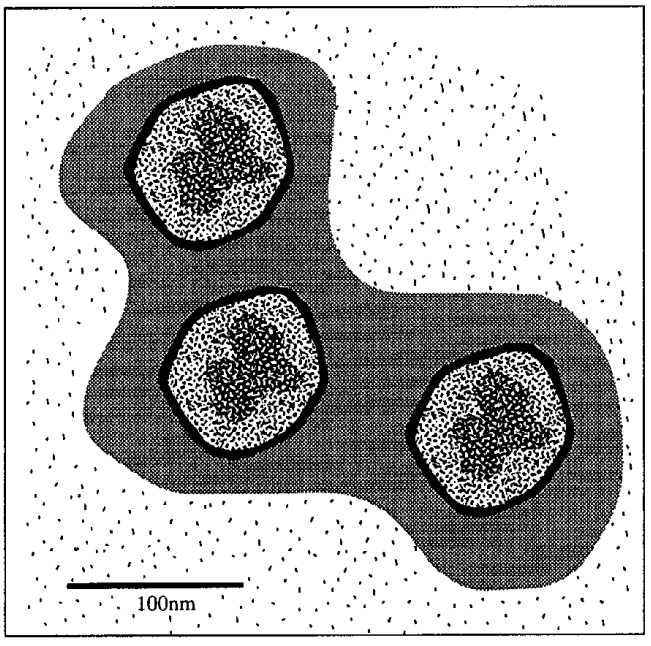


Figure 67.1 Bornavirus particles as they appear in an electron micrograph.

Epidemiology

Neurological disease attributed to BDV has been described in horses and cats. Serological evidence of infection has been recorded in other species including sheep, rabbits and ostriches. It is thought that virus may be transmitted through ingestion or inhalation. Most cases of Borna disease occur in spring and early summer; prevalence varies from year to year. It has been suggested that rodents may act as reservoir hosts. Persistent infections can be established experimentally in rats.

Pathogenesis and pathology

Following oronasal infection, the virus gains entry to the CNS by intra-axonal spread either through the olfactory nerve or through nerves supplying the oropharyngeal and intestinal regions. Spread within the CNS and into the peripheral nerves also occurs within axons. Viral antigens expressed on the surface of infected neurons induce a cell-mediated response involving cytotoxic $CD8^+$ T lymphocytes, resulting in destruction of the infected cells. A non-suppurative encephalitis with lymphocytic perivascular cuffing is largely confined to grey matter. Neuronal degeneration is prominent and intranuclear inclusions (Joest-Degen bodies) may be present particularly in the hippocampus. Antibody levels in

infected animals, which are relatively low and non-protective, are not linked to the pathogenesis of the disease. Despite persistence of infection, clinical signs do not develop in immunodeficient animals.

Clinical signs

Borna disease has been described mainly in young horses. The incubation period, which is highly variable, ranges from weeks to several months. Factors which may influence the severity of clinical signs include the age and immunological status of the infected animal and the strain of infecting virus. On farms where infection in horses is present, clinical disease is usually confined to individual animals. Clinical signs include fever, somnolence and evidence of neurological disturbance. Ataxia, pharyngeal paralysis and hyperaesthesia may be present. The course of the disease is up to three weeks and mortality rates may reach 100%. Surviving horses have permanent CNS damage and may exhibit recurrent episodes of neurological disturbance. 'Staggering disease' in cats has been associated with BDV infection (Lundgren *et al.*, 1995). In defined areas, some sheep flocks have large numbers of seropositive animals.

Diagnosis

Borna disease may vaguely resemble other neurological conditions in the horse. However, the distribution of lesions in the CNS differs from that in other equine encephalomyelitides and, if eosinophilic intranuclear inclusions (Joest-Degen bodies) are present, they may be confirmatory. Viral antigen can be demonstrated in the brain by immunohistochemical methods. Demonstration of antibodies in serum or in cerebrospinal fluid by immunofluorescence or ELISA may aid diagnosis. Reverse transcriptase-PCR for the demonstration of BDV-RNA is a valuable diagnostic tool.

Control

Although BDV does not appear to be readily transmitted by infected horses, seropositive animals should be isolated. Standard hygienic measures should be applied to suspect animals.

References

Boucher, J.M., Barbillon, E. and Cliquet, F. (1999). Borna disease: a possible emerging zoonosis. *Veterinary Research*, **30**, 549–557.

Lundgren, A.L., Zimmermann, W., Bode, L., *et al.* (1995). Staggering disease in cats: isolation and characterization of the feline Borna disease virus. *Journal of General Virology*, **75**, 2215–222.

Further reading

Huebner, J., Bode, L. and Ludwig, H. (2001). Borna disease virus infection in FIV-positive cats in Germany. *Veterinary Record*, **149**, 152.

Richt, J.A., Pfeuffer, I., Christ, M. *et al.* (1997). Borna disease virus infection in animals and humans. *Emerging Infectious Diseases*, **3**, 343–352.

Weissenböck, H., Suchy, A., Caplazi, P. *et al.* (1998). Borna disease in Austrian horses. *Veterinary Record*, **143**, 21–22.

Chapter 68

Bunyaviridae

The name of this family is derived from Bunyamwera, the place in Uganda where the type species Bunyamwera virus was first isolated. The family *Bunyaviridae* contains more than 300 viruses. Virions, (80 to 120 nm in diameter), are spherical and enveloped. Glycoprotein peplomers project from the surface of the envelope which encloses three circular, helical nucleocapsid segments (Fig. 68.1). The viruses are sensitive to heat, acid pH levels, lipid solvents, detergents and disinfectants. The genera in the family are *Bunyavirus*, *Phlebovirus*, *Nairovirus*, *Hantavirus* and *Tospovirus*. Based on antigenic relatedness, viruses within each genus are placed in serogroups. The genome consists of three single-stranded RNA segments designated small (S), medium (M) and large (L). Genetic reassortment occurs between closely-related viruses. Replication takes place in the cytoplasm of host cells. In the final stages of assembly, virions acquire envelopes by budding into the Golgi network. They are then transported through the cytoplasm in secretory vesicles and released by exocytosis at the cell surface.

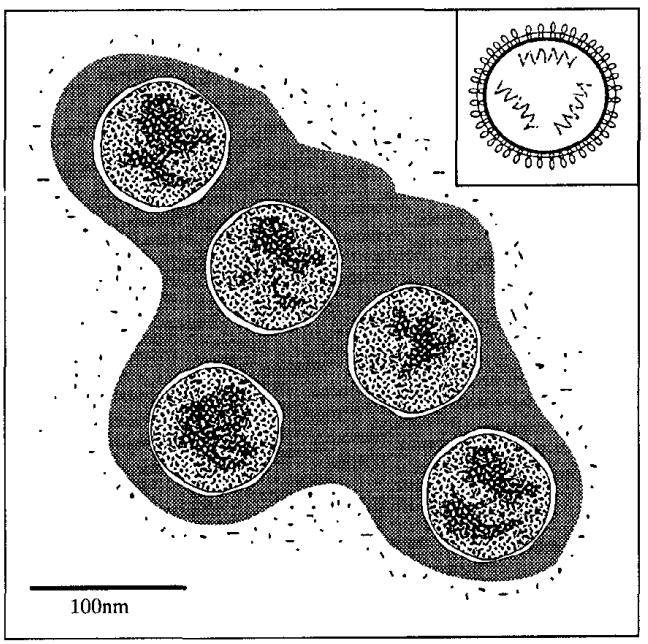


Figure 68.1 Bunyavirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Key points

- Medium-sized, enveloped, single-stranded RNA viruses
- Replicate in the cytoplasm
- Labile in the environment
- More than 300 viruses in the family, the majority are arthropod-borne
- The family is composed of five genera:
  - —*Bunyavirus* contains viruses which cause congenital defects in cattle and sheep
  - —*Phlebovirus* contains the virus which causes Rift Valley fever
  - —*Nairovirus* contains the virus which causes Nairobi sheep disease
  - —*Hantavirus* contains many viruses which cause haemorrhagic fever in humans; rodents act as reservoirs
  - —*Tospovirus* contains viruses of plants

Viruses in the genera *Bunyavirus, Phlebovirus, Nairovirus* and *Hantavirus* infect vertebrates; those in the genus *Tospovirus* infect plants.

Clinical infections

With the exception of viruses in the genus *Hantavirus*, bunyaviruses are arthropod-borne. These arboviruses are maintained in nature in complex life cycles involving replication in both arthropod vectors and vertebrate hosts. Infection of mammalian cells often results in cytolysis while infection of invertebrate cells is non-cytolytic and persistent. Mosquitoes are the most important vectors. Ticks, sandflies and midges may act as vectors for some bunyaviruses. Arthropod vectors acquire virus from vertebrate hosts during viraemic periods. Each bunyavirus species replicates in a limited number of vertebrate and invertebrate hosts.

Hantaviruses, which are primarily human pathogens, are maintained in nature by persistent infections in rodents

which shed virus in urine, faeces and saliva. Transmission between rodent hosts can occur by aerosols and biting. Individual hantaviruses are associated with particular rodent species. Many bunyaviruses infect humans and frequently cause serious diseases including California encephalitis, haemorrhagic fever with renal syndrome, hantavirus pulmonary syndrome and Crimean-Congo haemorrhagic fever. Such human infections are generally considered to be incidental and do not usually result in disease transmission.

Three important ruminant diseases, Rift Valley fever, Nairobi sheep disease and Akabane disease are caused by bunyaviruses (Table 68.1).

Rift Valley fever

This peracute or acute disease of domestic ruminants in Africa is characterized by abortion and also by high mortality rates in newborn animals. Although a wide range of ruminant species is susceptible to infection, Rift Valley fever occurs primarily in sheep, cattle and goats. Indigenous African ruminants are less susceptible to infection than imported species. Rift Valley fever is also an important zoonotic disease.

Epidemiology

Outbreaks of the disease tend to occur unpredictably in eastern and southern Africa at intervals of five or more years and are associated with abnormally heavy rains and a dramatic rise in vector populations. In exceptional years, large-scale epidemics have occurred in west Africa and in the Nile valley and delta. Transovarial transmission of Rift Valley fever virus (RVFV) occurs in *Aedes* species. During inter-epidemic periods, the virus is maintained in the eggs of floodwater species of mosquito laid in undrained shallow depressions at the edge of temporary pools. Eggs of these species must remain dry for a period before they hatch following re-immersion in water. During epidemics, RVFV replicates in both wild and domestic ruminants and can be transmitted by many species of mosquitoes. Humans are frequently infected during these epidemics. Infected ruminants develop marked viraemia for up to five days after infection. During this period, blood and tissues of affected animals are infectious. Direct and indirect transmission can occur through aerosols, contact with infected placentae or aborted foetuses, fomites or mechanical transfer on the mouthparts of flies. Abattoir workers and veterinarians are particularly at risk of acquiring infection.

Pathogenesis and pathology

Following infection and local replication, viraemia leads to invasion of the liver and other major organs. Cell necrosis is widespread, particularly in the liver. Extensive cytolysis results in foetal death.

Clinical signs

In mature sheep and goats, clinical signs include regurgitation of ingesta, foetid diarrhoea, blood-tinged mucopurulent nasal discharge and, occasionally, icterus. Pregnant ewes can abort. Abortion rates approaching 100% are not uncommon. Mortality rates in adult sheep may be up to 60%. In lambs the incubation period is up to 36 hours. Affected animals are pyrexic, listless and disinclined to move; they may show signs of abdominal pain. Affected lambs rarely survive more than 36 hours after the onset of clinical signs. The mortality rate in lambs less than one week old approaches 90%.

In cattle, mortality rates are usually below 10% and abortion rates range from 15 to 40%.

Human infections with RVFV are often inapparent or may present as a moderate to severe influenza-like illness. The haemorrhagic and encephalitic forms of the disease, which occur in a small number of patients, can be fatal.

Diagnosis

- The histopathological lesions, particularly in the livers of lambs, are considered pathognomonic.

Table 68.1 Bunyaviruses of veterinary importance.

Genus	Virus	Hosts	Comments
Phlebovirus	Rift Valley fever virus	Sheep, cattle, goats	Causes high mortality rates in neonatal animals and abortion in pregnant animals. Endemic in southern and eastern Africa, transmitted by mosquitoes. Important zoonotic disease
Nairovirus	Nairobi sheep disease virus	Sheep, goats	Causes severe, often fatal disease in susceptible animals. Present in central and eastern Africa. Transmitted by ticks
Bunyavirus	Akabane virus, Aino virus, Peaton virus	Cattle, sheep	Viruses belonging to the Simbu serogroup, transmitted by mosquitoes and midges. Widely distributed geographically in tropical and subtropical regions of the Old World. Associated with congenital defects and abortion
	Cache Valley virus	Sheep	Belongs to the Bunyamwera serogroup; transmitted by mosquitoes. Occasionally associated with congenital defects in sheep flocks in North America

Demonstration of viral antigen in fixed tissues by immunohistochemical methods is confirmatory.

- Virus can be isolated in suitable cell cultures, susceptible laboratory animals or embryonated eggs. Specimens for laboratory examinations include blood from viraemic animals, foetal organs and postmortem specimens of liver, spleen and brain. Because the virus can infect laboratory personnel, it should be handled only in properly equipped laboratories.
- Rapid confirmatory tests include detection of viral antigen in serum by ELISA or in impression smears by immunofluorescence.
- Demonstration of antibodies by ELISA in a serum sample or of seroconversion in paired serum samples by virus neutralization, ELISA or hamagglutination inhibition may be used for confirmation of the disease.

Control

Although vector control and environmental management can assist in limiting the spread of Rift Valley fever, such measures are often not feasible. Modified live vaccine containing Smithburn's attenuated strain of RVFV is widely used in endemic regions and during outbreaks. This vaccine is unsafe in pregnant animals because it may cause congenital defects or abortion. A mutagen-attenuated vaccine, developed for use in humans, is effective and appears to be safe for use in pregnant animals (Morrill *et al.*, 1997). Inactivated vaccines prepared from highly immunogenic virulent strains of RVFV are suitable for use in pregnant animals and may be used in Rift Valley fever-free countries bordering endemic regions.

Nairobi sheep disease

This severe, tick-borne viral infection of sheep and goats occurs in central and eastern Africa. The causal agent, Nairobi sheep disease virus (NSDV) is closely related to Ganjam virus of sheep and goats in India. Although humans are susceptible to NSDV, infection appears to be uncommon.

Epidemiology

The brown ear tick (*Rhipicephalus appendiculatus*) is the principal vector of the virus. Transovarial and transstadial infection occurs with transmission by all stages of the tick. In endemic areas, lambs and kids exposed to infection while protected by maternally-derived antibody develop active immunity. Outbreaks of disease arise from movement of susceptible animals into endemic areas or from the introduction of infected ticks into Nairobi sheep disease-free areas.

Pathogenesis

Following inoculation by an infected tick, virus replicates in the endothelium, in liver, spleen, lungs and other organs.

Clinical signs

The incubation period is up to six days. There is marked pyrexia and depression followed within 48 hours by foetid dysentery. A mucopurulent nasal discharge and conjunctivitis may be observed. Pregnant animals often abort. The mortality rate ranges from 30% to 90%; death may occur up to 11 days after the onset of clinical signs. The disease is more severe in native breeds of sheep than in Merinos. Clinical signs are milder in goats than in sheep.

Diagnosis

A history of a high mortality rate in a flock recently introduced into an endemic area may point to Nairobi sheep disease. Specimens suitable for virus isolation in cell culture include blood, mesenteric lymph nodes and spleen. Direct immunofluorescence is useful for identification of the virus in tissue culture cells. Viral antigen can be detected directly in tissue specimens by AGID. The indirect immunofluorescent test is recommended for the detection of antibodies to NSDV.

Control

Dipping is used to control the tick vector and animals at risk should be vaccinated. Modified live and inactivated vaccines have been used experimentally but the limited demand has not justified commercial production.

Akabane disease

Bunyaviruses, which belong to the Simbu serogroup and include Akabane, Aino and Peaton viruses, cause congenital defects, such as arthrogryposis and hydranencephaly, and abortion in cattle and sheep. One of the most important and virulent viruses in this group is Akabane virus. Serological studies indicate a widespread distribution in tropical and subtropical regions in the Middle East, Asia, Australia and Africa.

Sporadic epidemics, associated with developmental defects, have been described in Japan, Australia, Israel and parts of Africa. Viruses in the group are transmitted by midges and mosquitoes. Disease outbreaks appear to coincide with movements of the vectors or following the introduction of susceptible animals into endemic areas. Encephalomyelitis and polymyositis develop in foetuses infected with Akabane virus. The extent and degree of the pathological change relate to the stage of gestation at which infection occurred. The most severe damage, involving neurological defects, is evident in calves born to cows infected at about 28 weeks of gestation. Clinical signs are not observed in the dam. Diagnosis is based on gross pathological findings in the foetal CNS and the detection of specific neutralizing antibody in the sera of aborted calves or of newborn animals prior to suckling.

Vector control and vaccination are the methods used to prevent outbreaks of the disease. An inactivated vaccine is available in Japan and Australia. In order to minimize the occurrence of congenital defects in cattle and sheep, breeding stock should be introduced into endemic areas in advance of the breeding season.

Reference

Morrill, J.C., Mebus, A. and Peters, C.J. (1997). Safety and efficacy of a mutagen-attenuated Rift Valley fever virus vaccine in cattle. *American Journal of Veterinary Research*, **58**, 1104–1109.

Chapter 69

Picornaviridae

Picornaviruses (Spanish *pico*, very small), which are icosahedral and non-enveloped, contain a molecule of single-stranded RNA. Virions are 30 nm in diameter (Fig. 69.1). The capsid is composed of 60 identical subunits, each containing four major proteins VP1, VP2, VP3 and VP4. The VP4 protein is located on the inner surface of the capsid. Viral replication occurs in the cytoplasm in membrane-associated complexes and infection is usually cytolytic. Picornaviruses are resistant to ether, chloroform and non-ionic detergents. Individual genera differ in their thermal lability and pH stability. The family comprises six genera: *Enterovirus*, *Rhinovirus*, *Cardiovirus*, *Aphthovirus*, *Hepatovirus* and *Parechovirus*. It has recently been proposed by the ICTV to create three new genera: *Erbovirus*, *Kobuvirus* and *Teschovirus* (Pringle, 1999).

Viruses of veterinary importance in the family *Picornaviridae* are presented in Fig. 69.2. Apthoviruses are unstable at pH values below 6.5 and rhinoviruses are unstable below pH 5.0. Viruses in the other genera are stable at acid pH values. Some viruses in the genera

Key points

- Non-enveloped, positive-sense, single-stranded RNA viruses with icosahedral symmetry
- Replicate in cytoplasm
- Resistant to many organic solvents; individual members differ in their susceptibility to pH change
- Four genera, *Aphthovirus*, *Enterovirus*, *Cardiovirus* and *Hepatovirus* contain viruses of veterinary significance
- Aphthoviruses cause foot-and-mouth disease
- Enteroviruses cause swine vesicular disease, Teschen/Talfan disease, reproductive problems and enteritis in pigs
- Cardioviruses cause encephalomyocarditis in young pigs
- Hepatoviruses cause encephalomyelitis in chickens

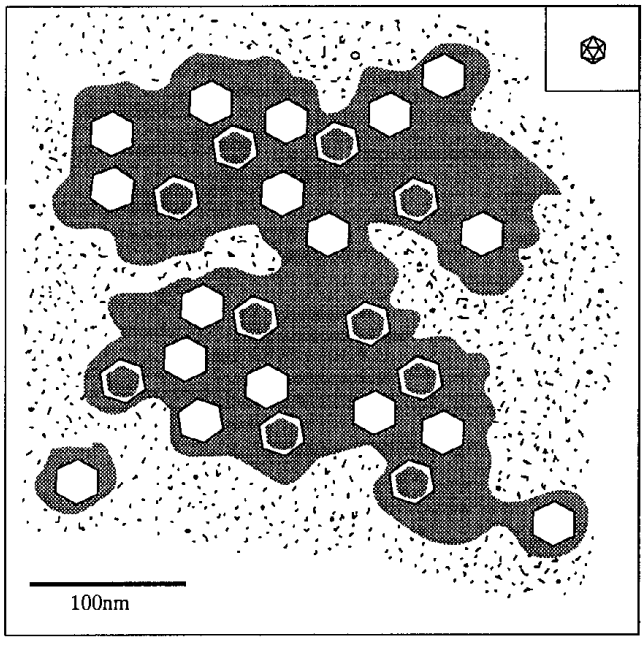


Figure 69.1 Picornavirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Hepatovirus and *Parechovirus*, including hepatitis A virus, are important human pathogens. Poliomyelitis virus which causes serious neurological disease in humans is an enterovirus.

Clinical infections

With the exception of foot-and-mouth disease virus and encephalomyocarditis virus, picornaviruses typically infect a single, or a limited number of, host species. Transmission usually occurs by the faecal-oral route but may also occur by fomites or by aerosols. Some picornaviruses, notably foot-and-mouth disease virus (Bergmann *et al.*, 1996; Mezencio *et al.*, 1999) and swine vesicular disease virus (Lin *et al.*, 1998), can produce persistent infections. Antigenic variation, which may contribute to the development of persistent infection (Woodbury, 1995), has been attributed to a number of molecular mechanisms including genetic recombination. Mixed infections with different serotypes of foot-and-mouth disease virus are known to occur in individual

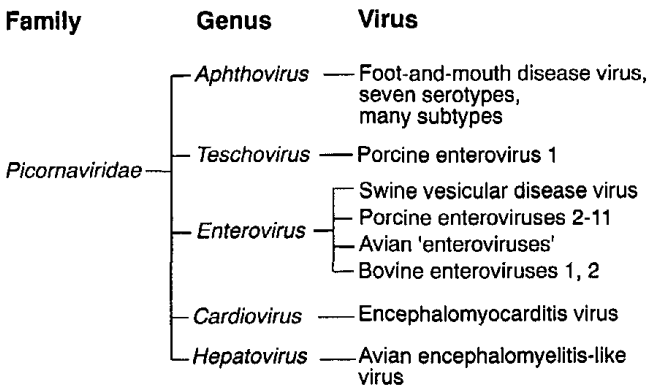


Figure 69.2 Viruses of veterinary importance in five genera of the family *Picornaviridae*.

animals, particularly in African Cape buffaloes. Although infections with enteroviruses are common in many vertebrate species, significant disease occurs only in pigs, poultry and humans (Table 69.1). Rhinoviruses, which are associated with the common cold in humans, are considered to be minor pathogens in cattle.

Swine vesicular disease

This mild vesicular disease of pigs occurs sporadically in parts of Europe and Asia. Because it is indistinguishable clinically from foot-and-mouth disease, accurate laboratory diagnosis of swine vesicular disease (SVD) is essential. Swine vesicular disease virus (SVDV), an enterovirus, is closely related to human coxsackievirus B5. The pig is the natural host for the virus. Laboratory workers handling contaminated material can become infected.

Epidemiology

In the presence of organic matter, SVDV is stable for long periods in the environment and transmission can occur directly or indirectly. Spread of disease from farm to farm is dependent on the movement of infected pigs or contaminated vehicles or fomites. Tissues of infected pigs, which contain large quantities of virus, remain infective despite the low pH values associated with rigor mortis. The virus exhibits prolonged survival in refrigerated pork.

Pathogenesis

Following entry through damaged skin or by ingestion, the virus replicates locally and subsequently spreads via the lymphatics to the blood stream. During a viraemia of short duration, many organs and tissues become infected. Virus shedding, which is greatest during the first week following infection, begins before clinical signs become evident. The faeces and tissue material from infected pigs may contain viable virus for months (Lin *et al.*, 1998).

Clinical signs

The incubation period is up to seven days. Infection is usually characterized by a mild febrile illness. Subclinical disease is common. Transient fever is followed by the development of vesicular lesions on the feet, particularly on the coronary bands. Less commonly, vesicles appear on the lips, tongue and snout. Lameness, dullness and inappetence may be present. Lameness is exacerbated by penning on concrete floors. Affected animals remain in good condition and the lesions heal within a few weeks.

Diagnosis

Laboratory tests are essential for differentiating SVD from other important vesicular diseases of pigs.

- An ELISA is available for the rapid detection of viral antigen in vesicular fluid or in epithelial tissues.
- Samples should be inoculated onto monolayers of susceptible cells. The virus produces cytopathic effects.
- Several serological procedures are suitable for herd screening for antibodies to SVDV. Virus neutralization and ELISA are the tests most frequently used. Virus neutralization, although the standard test, requires tissue culture procedures and takes longer to complete than ELISA. A limitation of ELISA is that it may occasionally yield false-positive results.

Control

Effective vaccines against SVD have been prepared. Commercial vaccines are not available because vaccination is not considered suitable as a control measure in countries free of the important vesicular diseases. In most countries, SVD is a notifiable disease and eradication

Table 69.1 Enteroviruses of veterinary importance.

Virus	Comments
Swine vesicular disease virus	Produces mild vesicular disease, indistinguishable from foot-and-mouth disease
Porcine enterovirus 1 (Porcine teschovirus)	Virulent strains which occur in eastern Europe and the Malagasy Republic cause Teschen disease; widely distributed mild strains cause Talfan disease
Porcine enteroviruses 2 to 11	Infection usually asymptomatic; occasionally cause mild posterior paresis of hind limbs, reproductive problems, diarrhoea or pneumonia
Avian 'enteroviruses'	Virulent strains cause avian encephalomyelitis, nephritis in chickens and hepatitis in ducks and turkeys
Bovine enteroviruses 1 and 2	Isolated from both normal cattle and animals with enteric, respiratory and reproductive disease

policies are enforced with restrictions on the importation of pigs and pork. Control measures, applied following an outbreak of SVD, include thorough cleaning and disinfection of premises, control of pig movement and boiling of waste food fed to pigs.

Porcine enteroviral encephalomyelitis

This condition, also referred to as Teschen/Talfan disease, was first described in Teschen, Czechoslovakia during 1929 and, subsequently, caused significant losses in several European countries. The clinical presentation varies in accordance with the virulence of the infecting strain of porcine enterovirus (PEV). Severe clinical disease is now rare and is largely confined to eastern Europe and the Malagasy Republic.

Epidemiology

Thirteen serotypes of porcine enteroviruses are recognized. Most important neurotropic strains belong to serotype PEV1, which includes both the highly virulent isolates associated with Teschen disease and the less virulent but more widely-distributed strains which cause endemic posterior paresis (Talfan disease). It has recently been proposed to rename PEV1 as porcine teschovirus, type species in the newly created genus *Teschovirus*. Other enterovirus serotypes associated with encephalomyelitis include PEV2, PEV3, PEV4, PEV5, PEV6, PEV8, PEV12 and PEV13. In common with swine vesicular disease, transmission occurs by the faecal-oral route either directly or indirectly. Clinical disease is most severe in young pigs in herds which have not been previously exposed to infection. In herds which are endemically infected, sporadic clinical disease tends to occur after mixing of weaned pigs when maternally-derived immunity has declined.

Pathogenesis

Following ingestion, virus replicates in the tonsils, intestines and associated lymph nodes. Viraemia and invasion of the central nervous system may follow, particularly when virulent strains are involved. Faecal excretion of virus may continue for several weeks.

Clinical signs

Fever, depression and listlessness may be followed about one week after infection by neurological signs. Weakness and incoordination progress to paraplegia and paralysis. Pigs with posterior paralysis may adopt a dog-sitting posture. Severely affected animals exhibit nystagmus, opisthotonos, convulsions and coma. Mortality rate in these animals is high. Mildly affected pigs usually recover.

Diagnosis

- A mild to severe non-suppurative encephalomyelitis is demonstrable histologically.
- Virus can be isolated in porcine kidney cell lines from specimens of brain and spinal cord. The virus produces a cytopathic effect.
- Virus neutralization and ELISA are the most frequently used serological methods for demonstrating antibodies to PEV1. Because antibodies to PEV1 strains are common in pig populations, it is necessary to demonstrate a four-fold rise in titre between acute and convalescent sera for disease confirmation.

Control

Both inactivated and modified live vaccines are effective. Inactivated vaccines are available commercially. Teschen disease is notifiable in many countries. Outbreaks can be controlled by slaughter, strict sanitary measures and ring vaccination.

Reproductive disorders caused by porcine enteroviruses

Several enterovirus serotypes including PEV1, PEV3, PEV6 and PEV8 are associated with the SMEDI syndrome (stillbirths, mummification, embryonic death and infertility) in pigs. Although enteroviruses capable of causing the SMEDI syndrome are widely distributed in commercial pig herds, they are pathogenic only for embryos and foetuses. Clinical disease follows infection of naive, pregnant animals. Cross-protection between serotypes does not occur. Transmission is by the faecal-oral route. Infection of the alimentary tract is followed by viraemia and transplacental spread to developing foetuses. The clinical effects of infection depend on the stage of gestation. Infection during early to mid-gestation results in embryonic death and mummification, whereas infection during the later stages of pregnancy may result in stillbirths or the birth of live piglets. A susceptible sow, therefore, may give birth to mummified, stillborn and live piglets, reflecting the stage of foetal development at the time of infection. The clinical presentation is indistinguishable from porcine parvovirus infection, a more common cause of the SMEDI syndrome. Laboratory confirmation requires either isolation of the virus from lung tissues of stillborn piglets or demonstration of antibody in the serum of stillborn or newborn piglets prior to ingestion of colostrum. Commercial vaccines are not available. Gilts should be exposed to PEV prior to breeding by contact with older sows, faeces or mummified foetuses.

Avian encephalomyelitis

This viral disease of young birds has been recorded in domestic fowl, pheasants, quail and turkeys. Avian encephalomyelitis (AE) is of considerable economic importance in chickens. Although avian encephalomyelitis-like virus (AEV) was formerly considered to be

an enterovirus, it has been recently shown to be most closely related to hepatitis A virus and tentatively assigned to the genus *Hepatovirus* (Todd *et al.*, 1999). Horizontal and vertical transmission occurs. The virus produces enteric infection and is shed in the faeces. A proportion of the eggs of infected hens are infected. Chicks infected *in ovo* hatch normally but shed virus and infect other chicks in the incubator shortly after hatching. Infection of the intestinal tract is followed by viraemia, and a competent immune response is required to prevent infection of the central nervous system. Clinical signs, which usually become evident within two weeks, include ataxia and fine tremors of the head and neck. Progressive paralysis leads to death due to inanition or following trampling. Non-suppurative encephalomyelitis and lymphocytic accumulations in viscera, particularly the pancreas, are characteristic. Demonstration of viral antigen in tissues by immunofluorescence or by virus isolation from brain or pancreas is confirmatory. Serological testing of paired sera may be of diagnostic value. Control is achieved by vaccination of breeding flocks with a modified live vaccine to ensure the presence of maternally-derived antibodies in chicks.

Foot-and-mouth disease

This highly contagious disease of even-toed ungulates is characterized by fever and the formation of vesicles on epithelial surfaces. Foot-and-mouth disease (FMD) is a List A disease of major importance internationally on account of its rapid spread and the dramatic economic losses which it causes in susceptible animals. Isolates of foot-and-mouth disease virus (FMDV) are grouped in seven serotypes, recognised as separate species, with differing geographical distributions (Table 69.2). Infection with one serotype does not confer immunity against the other serotypes. A large number of subtypes is recognized within each serotype. Nucleotide sequencing of the VP1 gene, which encodes for a capsid protein, is used for comparing an isolate from an outbreak with other isolates of the same serotype in order to determine possible sources of infection.

Table 69.2 The geographical distribution of foot-and-mouth disease virus serotypes. Australia, New Zealand, North and Central America, the Carribean countries and the countries of western Europe are free from the disease[a].

Foot-and-mouth disease virus serotypes	Geographical distribution
O, A and C	South America
O, A and C	Eastern European countries
O, A, C, SAT1, SAT2 and SAT3	Africa
O, A, C and Asia1	Asia

a large outbreak in UK during 2001

Epidemiology

Cattle, sheep, goats, pigs and domesticated buffalo are susceptible to FMD. Several wildlife species including African buffalo, elephants, hedgehogs, deer and antelopes are also susceptible. Large numbers of virus particles are shed in the secretions and excretions of infected animals. Virus shedding begins during the incubation period, about 24 hours before the appearance of clinical signs. Transmission can occur by direct contact, by aerosols, by mechanical carriage by humans or vehicles, on fomites and through animal products such as meat, offal, milk, semen or embryos. Infected groups of animals, particularly pigs, shed large quantities of virus in aerosols (Donaldson *et al.*, 2001). Under favourable conditions of low temperature, high humidity and moderate winds, virus in aerosols may spread up to 10 km over land. Turbulence is generally less marked over water than over land. In 1981, virus was carried a distance of more than 200 km from France to the south coast of England.

The virus, which is moderately resistant to environmental factors, is sensitive to acid and alkaline conditions outside the range pH of 6.0 to 9.0. Virus can remain infective on soil for three days in summer and for up to 28 days in the winter. Following death, lactic acid production in muscle inactivates the virus but it may survive in offal and in bone marrow. Foot-and-mouth virus can persist in the pharyngeal region of carrier animals which have recovered from FMD. It can also persist in vaccinated animals infected with a subtype different from the vaccinal subtype. Infection may persist for up to three years in cattle, for many months in sheep and for up to five years in the African Cape buffalo. Transmission of infection from persistently-infected African Cape buffalo to domestic cattle has been recorded. Because of their large respiratory volume and the low dose of virus required to establish infection, cattle are highly suseptible. It is unclear if virus can persist in pigs (Bergmann *et al.*, 1996; Mezencio *et al.*, 1999).

Pathogenesis

Although infection usually occurs through inhalation, virus can also gain entry to tissues through ingestion, insemination and inoculation, and through contact with abraded skin. Primary viral replication, after inhalation, takes place in the mucosal and lymphatic tissues of the pharynx. Viraemia follows primary multiplication with further viral replication in lymph nodes, mammary glands and other organs as well as the epithelial cells of the mouth, muzzle, teats, interdigital space and coronary band. In these areas of stratified squamous epithelium, vesicle formation results from swelling and rupture of keratinocytes in the stratum spinosum.

Clinical signs

The incubation period ranges from 2 to 14 days, but is generally shorter than a week. Infected cattle develop fever, inappetence and a drop in milk production. Profuse salivation, with characteristic drooling and smacking of lips, accompanies the formation of oral vesicles which rupture, leaving raw, painful ulcers. Ruptured vesicles in the interdigital cleft and on the coronary band lead to lameness. Vesicles may also appear on the skin of the teats and udders of lactating cows. Although the ulcers tend to heal rapidly, there may be secondary bacterial infection which exacerbates and prolongs the inflammatory process. Infected animals lose condition. Mature animals seldom die. Calves may die from acute myocarditis. Although the virus does not cross the placenta, abortion probably relates to the pyrexial response.

In pigs, foot lesions are severe and the hooves may slough. Marked lameness is the most prominent sign in this species. The disease in sheep, goats and wild ruminants is generally mild, presenting as fever accompanied by lameness which spreads rapidly through groups of animals.

Human infection, usually mild, has been described on rare occasions in laboratory personnel working with the virus and in individuals handling infected animals.

Diagnosis

Foot-and-mouth disease clinically resembles other vesicular diseases of domestic animals including vesicular stomatitis in cattle and pigs, swine vesicular disease and vesicular exanthema in pigs (Table 69.3). Consequently, FMD requires laboratory confirmation. Laboratory procedures involving FMDV must be carried out in purpose-built laboratories.

- Diagnosis is based on the demonstration of FMDV antigen in samples of tissue or in vesicular fluid.
- Epithelium collected from an unruptured or recently ruptured vesicle is suitable for antigen demonstration.
- In convalescent animals and in persistent or subclinical infections, samples of oesophageal/pharyngeal fluid can be obtained with a probang (sputum) cup.
- Appropriate serological tests include ELISA and CFT. The preferred test is ELISA which is sensitive and specific, and is available in kit form.
- The polymerase chain reaction has been adapted for the amplification of genome fragments of FMDV.
- Virus isolation is carried out in special cell lines such as primary bovine thyroid or kidney cells.
- Demonstration of specific antibody by virus neutralization or ELISA can be used to confirm a diagnosis in unvaccinated animals. In endemic areas, interpretation of antibody titres may prove difficult.

Control

In countries which are free from FMD, it is notifiable and affected and in-contact animals are slaughtered. Following an outbreak, movement restrictions are applied and infected premises must be thoroughly cleaned and disinfected. Mild acids, such as citric acid and acetic acid, and alkalis such as sodium carbonate are effective disinfectants. Reserves of inactivated virus are maintained in several countries to provide an adequate supply of vaccine at short notice in the event of a major outbreak of the disease. Although ring vaccination around an affected premises may help to limit the spread of the disease, it may also allow the development of the carrier state in animals subsequently exposed to the virus.

In countries where FMD is endemic, efforts are generally directed at protecting high-yielding dairy cattle by a combination of vaccination and control of animal movement. Vaccines for FMD, incorporating adjuvant, are derived from tissue culture-propagated virus which has been chemically inactivated. They are usually multivalent, containing three or more virus strains. Protection against antigenically similar strains of virus is satisfactory and lasts for up to six months. Research is continuing into the

Table 69.3 Susceptibility of farm animals to viruses which cause vesicular diseases.

Virus	Species			
	Cattle	Sheep / goats	Pigs	Horses
Foot-and-mouth disease virus	Susceptible	Susceptible	Susceptible	Resistant
Swine vesicular disease virus	Resistant	Resistant	Susceptible	Resistant
Vesicular exanthema of swine virus	Resistant	Resistant	Susceptible	Resistant
Vesicular stomatitis virus	Susceptible	Resistant	Susceptible	Susceptible

development of improved vaccines based principally on peptide synthesis or recombinant DNA technology (Doel, 1996).

Infections caused by equine rhinoviruses

Genomic and other studies have shown that equine rhinovirus 1 is closely related to foot-and-mouth disease virus. It has been renamed equine rhinitis A virus and has been placed in the genus *Aphthovirus*. Infection with rhinoviruses appears to be widespread and most horses are exposed early in life. Although equine rhinitis A virus and equine rhinitis B virus (equine rhinovirus 2) have been associated with acute respiratory disease (Carman *et al.*, 1997; Klaey *et al.*, 1998), they are generally considered to be minor respiratory pathogens. They may contribute to the development of disease following surgery or strenuous exercise or when present in mixed infections with bacteria or other viruses. Viraemia and prolonged shedding of virus in urine occurs in infection with equine rhinitis A virus.

Infection with encephalomyocarditis virus

Rodents are considered to be the natural hosts of encephalomyocarditis virus (EMCV). However, this cardiovirus has a wide host range including humans, monkeys and pigs. Infection in pigs is usually subclinical but sporadic deaths and minor outbreaks have been described. Rats and mice, the principal reservoirs, excrete virus in faeces and urine. The virus is stable in the environment. Pigs acquire infection by ingesting contaminated feed. Pig-to-pig transmission may also occur (Koenen *et al.*, 1999). Following ingestion, viraemia develops within a few days. Subsequently, high titres of virus can be demonstrated in the myocardium, spleen and mesenteric lymph nodes. Transplacental infection may occur. Virus isolates associated with myocardial disease are apparently distinct from those responsible for reproductive disease (Koenen *et al.*, 1999).

Outbreaks of disease are generally restricted to one particular age group. The severity of the disease relates to the strain of virus and the age of the infected pigs. Piglets may die suddenly as a result of heart failure. Hydrothorax, hydropericardium and ascites may be demonstrated at postmortem examination. Areas of myocardial necrosis with associated lymphoid infiltration may be demonstrable histologically. Changes in the CNS are minimal. Reproductive failure in sows is characterized by mummified foetuses and stillbirths. Laboratory confirmation relies on virus isolation and identification. Virus neutralization and haemagglutination tests can be used to detect specific antibodies. Control of rodents is important in reducing the likelihood of infection. An inactivated vaccine is available commercially in the USA.

References

Bergmann, I.E., Malirat, V., de Mello, P.A. and Gomes I. (1996). Detection of foot-and-mouth viral sequences in various fluids and tissues during persistence of the virus in cattle. *American Journal of Veterinary Research*, **57**, 134–137.

Carman, S., Rosendal, S., Huber, L. *et al.* (1997). Infectious agents in acute respiratory disease in horses in Ontario. *Journal of Veterinary Diagnostic Investigation*, **9**, 17–23.

Doel, T.R. (1996). Natural and vaccine-induced immunity to foot-and-mouth disease: the prospects for improved vaccines. *Revue scientifique et technique Office International des Epizooties*, **1**, 883–911.

Donaldson, A.I., Alexandersen, S., Sorensen, J.H. and Mikkelsen, T. (2001). Relative risks of the uncontrollable (airborne) spread of FMD by different species. *Veterinary Record*, **148**, 602–604.

Klaey, M., Sanchez-Higgins, M., Leadon, D.P. *et al.* (1998). Field case study of equine rhinovirus 1 infection: clinical signs and clinicopathology. *Equine Veterinary Journal*, **30**, 267–269.

Koenen, F., Vanderhallen, H., Castryck, F. and Miry, C. (1999). Epidemiologic, pathogenic and molecular analysis of recent encephalomyocarditis outbreaks in Belgium. *Journal of Veterinary Medicine, Series B*, **46**, 217–231.

Lin, F., MacKay, D.K.J. and Knowles N.J. (1998). The persistence of swine vesicular disease virus infection in pigs. *Epidemiology and Infection*, **121**, 459-472.

Mezencio, J.M.S., Babcock, G.D., Kramer, E. and Brown, F. (1999). Evidence for the persistence of foot-and-mouth disease virus in pigs. *Veterinary Journal*, **157**, 213–217.

Pringle, C.R. (1999). Virus taxonomy — 1999. *Archives of Virology*, **144**, 421–429.

Todd, D., Weston, J.H., Mawhinny, K.A. and Laird, C. (1999). Characterization of the genome of avian encephalomyelitis virus with cloned cDNA fragments. *Avian Diseases*, **43**, 219–226.

Woodbury, E.L. (1995). A review of the possible mechanisms for the persistence of foot-and-mouth disease virus. *Epidemiology and Infection*, **114**, 1–13.

Chapter 70

Caliciviridae

Caliciviruses (Latin *calix*, cup) have cup-shaped depressions, demonstrable by electron microscopy, on the surface of virions. The virions, 27 to 40 nm in diameter, are icosahedral and non-enveloped (Fig. 70.1). The genome consists of a single molecule of linear, positive-sense, single-stranded RNA. Replication takes place in the cytoplasm of infected cells and virions are released by cell lysis. Many caliciviruses have not yet been cultured. The virions are resistant to ether, chloroform and mild detergents. They are relatively resistant to heat but are sensitive to acid pH values.

Caliciviruses, which are closely related to picornaviruses, were formerly grouped within the *Picornaviridae*. Currently, the family *Caliciviridae* is divided into four genera: *Vesivirus*, *Lagovirus* and two unnamed genera of human caliciviruses referred to as 'Norwalk-like viruses' and 'Sapporo-like viruses'. The genus *Vesivirus* contains vesicular exanthema of swine virus, the type virus of the family, San Miguel sea lion virus and feline calicivirus. The *Lagovirus* genus contains two viruses of lagomorphs, rabbit haemorrhagic disease virus and European brown hare syndrome virus. The human caliciviruses cause gastroenteritis. 'Norwalk-like viruses' are also referred to as small, round, structured viruses groups 1 and 2 because of the absence of surface detail and their fuzzy appearance when viewed by electron microscopy. Hepatitis E virus of humans, formerly classified in the *Caliciviridae*, has been placed in a newly-created genus 'Hepatitis E-like viruses' which has yet to be assigned to a new family.

Key points

- Small, non-enveloped, single-stranded RNA viruses with icosahedral symmetry
- Replicate in the cytoplasm
- Stable in the environment
- Four genera:
  - —*Vesivirus* contains viruses which cause vesicular exanthema of swine and feline calicivirus infection
  - —*Lagovirus* contains viruses which cause rabbit haemorrhagic disease and European brown hare syndrome
  - —Two genera contain human caliciviruses which cause gastroenteritis

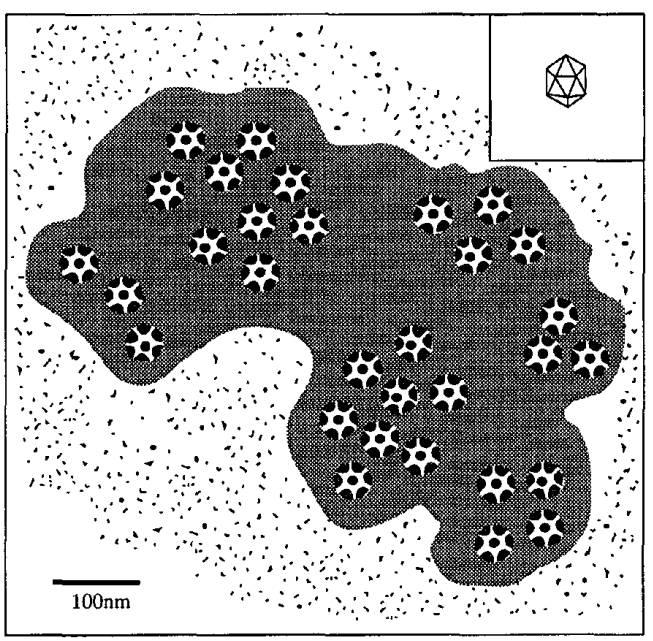


Figure 70.1 Calicivirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Clinical infections

Caliciviruses have been recovered from many species including humans, cats, pigs, marine mammals, rabbits, hares, cattle, dogs, reptiles, amphibians and insects. They are associated with a wide range of conditions including respiratory disease, vesicular lesions, necrotizing hepatitis and gastroenteritis (Table 70.1). Infections with caliciviruses, which are frequently persistent, may be inapparent, mild or acute. Transmission occurs directly or indirectly without vector involvement.

Vesicular exanthema of swine

First reported in southern California in 1932, vesicular exanthema of swine (VES), an acute, highly contagious disease, became widespread throughout the USA during

Table 70.1 Caliciviruses of veterinary importance.

Virus	Hosts	Comments
Vesicular exanthema of swine virus (13 serotypes)	Pigs	Acute, contagious vesicular disease, clinically similar to foot-and-mouth disease. Occurred in the U.S.A. before 1956. May have arisen from feeding sea lion and seal meat contaminated with San Miguel sealion virus
San Miguel sea lion virus (17 serotypes)	Marine mammals, Opal eye fish	Associated with cutaneous vesicles and premature parturition in pinnipeds; when inoculated into pigs, causes vesicular exanthema
Feline calicivirus	Domestic and wild cats	Important cause of upper respiratory tract infection in cats worldwide
Rabbit haemorrhagic disease virus	European rabbits	Acute fatal disease in European rabbits over two months of age
European brown hare syndrome virus	European brown hare	Related to rabbit haemorrhagic disease virus. Causes hepatic necrosis and widespread haemorrhages with high mortality
Canine calicivirus	Dogs	Occasionally associated with diarrhoea

the 1950s. An eradication campaign included rigorous implementation of laws relating to the cooking of garbage. The last outbreak of VES was recorded in 1956 and the country was declared free of the disease in 1959. A reservoir of the virus exists in marine mammals. In 1972, the San Miguel sea lion virus (SMSV) was isolated from California sea lions, which had developed vesicles on their flippers. Premature parturition occurred in infected animals. Subsequently, SMSV has been isolated from a number of other marine mammals and from the opal eye fish. Strains of SMSV produce VES when inoculated into pigs and it is thought that the original outbreak of VES arose through the feeding of uncooked swill containing meat from infected marine mammals. These vesicular viruses show antigenic heterogeneity. There are 13 serotypes of vesicular exanthema of swine virus and 17 serotypes of SMSV.

The incubation period of VES is up to 72 hours and the course of the disease is approximately two weeks. Vesicles appear on the tongue, lips, snout, interdigital spaces and coronary bands. Affected pigs are febrile and acutely lame. Although morbidity is high, mortality is low. The disease is clinically indistinguishable from foot-and-mouth disease, vesicular stomatitis and swine vesicular disease. Because of its similarity to foot-and-mouth disease, vesicular exanthema of swine is an important disease. Weight loss in fattening pigs and mortality in neonatal pigs are economically important consequences of the infection.

Vesicular fluid and the overlying flap of epithelium are rich in virus. Isolates can be identified by ELISA, CFT, immunoelectron microscopy and virus isolation in pig kidney cell lines.

Feline calicivirus infection

Infections caused by feline calicivirus (FCV) account for about 40% of upper respiratory tract inflammatory disease in cats worldwide. All species of *Felidae* are considered to be susceptible but natural disease tends to be confined to domestic cats and to cheetahs in captivity. There is a high degree of antigenic heterogeneity among FCV isolates. Sequence analysis studies have shown that individual isolates of FCV exist as quasispecies which evolve and exhibit antigenic drift. Significant alterations in the antigenic profiles of sequential virus isolates from carrier cats are thought to be influenced by immune selection and may play an important part in viral persistence (Radford *et al.*, 1998).

Epidemiology

Although cats of all ages are susceptible to infection with FCV, acute disease occurs most commonly in kittens as maternally-derived antibody wanes between two and six months of age. Infected cats excrete large amounts of virus in oronasal secretions. Many cats remain persistently infected after recovery from acute infection or following subclinical infection while protected by maternally-derived antibody or by vaccination. Infection is maintained in the cat population by these carriers which shed virus continuously from the oropharynx for months and, occasionally, for years.

Pathogenesis

Virus replication occurs primarily in the oropharynx with rapid spread throughout the upper respiratory tract and to the conjunctivae. A transient viraemia occurs. Infections range from subclinical to severe, reflecting differences in strain virulence. Virulent strains of FCV can cause interstitial pneumonia in young kittens. The virus has been recovered from the joints of lame cats.

Clinical signs

The incubation period is up to five days. Clinical signs, which are usually confined to the upper respiratory tract and the conjunctivae, are often less severe than those caused by feline herpesvirus 1 infection. Fever, oculonasal discharge and conjunctivitis are accompanied by the development of characteristic vesicles on the tongue and oral mucosa. These vesicles rupture leaving shallow ulcers. Morbidity may be high but mortality is usually low. Stiffness and shifting lameness, which usually

resolve within a few days, are sometimes seen during the acute phase of FCV infection or following inoculation with FCV vaccine. An association between infection with FCV and chronic gingivitis and stomatitis, when infection with feline immunodeficiency virus is also present, has been suggested.

Diagnosis

- Upper respiratory tract signs along with ulcers on the oral mucosa are suggestive of infection with FCV. Differentiation from feline herpesvirus 1 infection requires laboratory testing.
- Feline calicivirus can be isolated in feline cell lines from oropharyngeal swabs or from lung tissue. Isolation of FCV may not be aetiologically significant because of the large numbers of carrier animals in cat populations.
- Demonstration of a rising antibody titre in paired serum samples is required for laboratory confirmation.

Control

Vaccination and management practices aimed at reducing exposure to the virus are the main methods of control. Inactivated vaccines for parenteral administration and modified live vaccines for either parenteral or intranasal administration are available. Although vaccination protects effectively against clinical disease, it does not prevent subclinical infection or the development of a carrier state. Vaccines are based on a limited number of FCV isolates which cross-react with a broad spectrum of field isolates. An increasing number of recent field isolates are not neutralized by vaccine-induced antisera *in vitro* (Lauritzen *et al.*, 1997). Live vaccines for administration by injection may cause clinical signs if given by other routes.

Rabbit haemorrhagic disease

This is a highly contagious, acute and often fatal disease of European rabbits (*Oryctolagus cuniculus*). Rabbits under two months of age are not susceptible. Rabbit haemorrhagic disease (RHD) was first reported in China during 1984 and has since been encountered in many parts of the world. This virus (RHDV) is considered to be a mutant form of a non-pathogenic virus, termed rabbit calicivirus, which has been endemic in commercial and wild rabbits in Europe for many years. Rabbit haemorrhagic disease virus has been used for biological control of rabbits in Australia and New Zealand.

Epidemiology

Virus is shed in all excretions and secretions. Among rabbits in close contact, transmission is mainly by the faecal-oral route. Infection may also occur by inhalation or through the conjunctiva. Mechanical transmission by a variety of insects including mosquitoes and fleas has been demonstrated. The virus survives in the environment and indirect transmission through contaminated foodstuff or fomites may occur. Spread of virus between units and between countries may result from uncontrolled movement of infected rabbits or from contact with infected rabbit meat, insects or fomites. The virus was inadvertently released from a research facility in Australia during 1995 and, subsequently, was illegally introduced into New Zealand in 1997. In Europe, outbreaks of RHD have been variable in severity and this has been attributed to the presence of rabbit calicivirus infection.

Pathogenesis and pathology

Cells of the mononuclear phagocyte lineage are considered to be the major targets of the virus (Ramiro-Ibanez *et al.*, 1999). Rabbits under two months of age do not develop clinical signs. The reason for this resistance is unclear but it may have a physiological basis.

Severe hepatic necrosis is the most obvious lesion in affected rabbits. In addition, there may be evidence of disseminated intravascular coagulation.

Clinical signs

The incubation period is up to three days. The disease is characterized by high morbidity and high mortality. The course is short with death occurring within 36 hours of the onset of clinical signs. Acutely affected animals are pyrexic and depressed and have an increased respiratory rate. A serosanguineous nasal discharge, haematuria and neurological signs including convulsions may be present. Rabbits may be found dead or die in convulsions. A few rabbits may present with milder, subacute signs during the later stages of an epizootic. Some animals may survive for a few weeks with jaundice, weight loss and lethargy.

Diagnosis

High mortality in rabbits along with characteristic gross lesions including necrotic hepatitis and congestion of spleen and lungs are suggestive of RHD. Culture of RHDV has been unsuccessful. High concentrations of virus are present in affected livers. Confirmation is based on detection of virus by electron microscopy or of viral antigen by ELISA, immunofluorescence or haemagglutination using human erythrocytes. Reverse transcriptase PCR has been developed for the detection of RHDV nucleic acid. Suitable serological tests for the detection of specific antibodies to the virus include haemagglutination-inhibition and ELISA.

Control

In countries where RHD is endemic, control is achieved by vaccination. Inactivated and adjuvanted vaccines prepared from clarified liver suspensions of experimentally-infected rabbits are usually administered at about 10 weeks of age. Novel vaccines, based on recombinant myxoma virus expressing RHDV capsid protein or on virus-like particles

from capsid protein produced in baculovirus expression systems, are being developed.

References

Lauritzen, A., Jarrett, O. and Sabara, M. (1997). Serological analysis of feline calicivirus isolates from the United States and United Kingdom. *Veterinary Microbiology*, **56**, 55–63.

Radford, A.D., Turner, P.C., Bennett, M. *et al.* (1998). Quasispecies evolution of a hypervariable region of the feline calicivirus capsid gene in cell culture and in persistently infected cats. *Journal of General Virology*, **79**, 1–10.

Ramiro-Ibanez, F., Martin-Alonso, J.M., Garcia-Palencia, P. *et al.* (1999). Macrophage tropism of rabbit haemorrhagic disease virus is associated with vascular pathology. *Virus Research*, **60**, 21–28.

Chapter 71

Astroviridae

The family *Astroviridae* (Greek *aster*, star) contains viruses with a surface structure which imparts a star-like appearance. Astroviruses, 28 to 30 nm in diameter, are non-enveloped with icosahedral symmetry (Fig. 71.1). The genome consists of a single molecule of positive-sense, linear, single-stranded RNA. These viruses are resistant to low pH values, various detergents and heating at 60°C for 5 minutes. Replication occurs in the cytoplasm of host cells and virions are released by cell lysis. Trypsin is required for cultivation of these viruses. *Astrovirus* is the only genus in the family.

Key points

- Small, single-stranded RNA viruses with icosahedral symmetry
- Replicate in the cytoplasm
- Single genus *Astrovirus* contains viruses which produce mild gastroenteritis in most domestic species; in ducks serious disease may occur

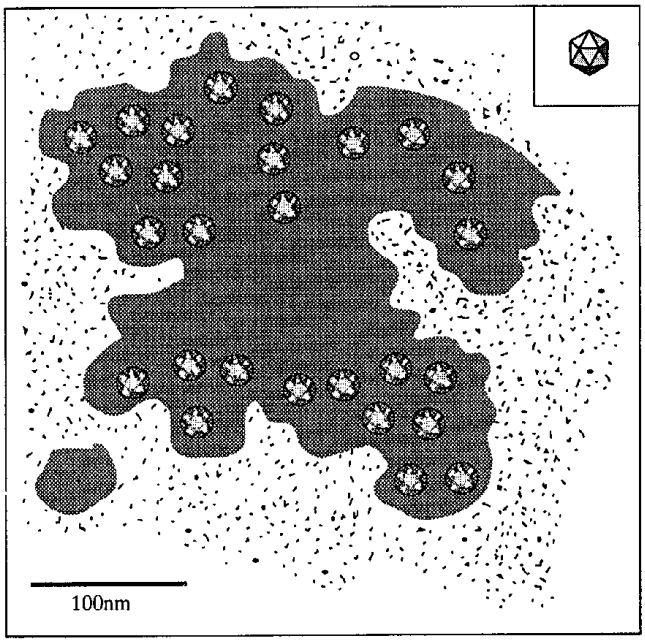


Figure 71.1 Astrovirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Clinical infections

Astroviruses, which are distributed worldwide, are associated with self-limiting gastroenteritis in animals and humans. These viruses have been detected in the faeces of humans, cattle, pigs, sheep, dogs, cats, deer, mice, ducks and turkeys. Isolates from different host species are antigenically distinct and species-specific. Two serotypes of bovine astrovirus are recognised. Transmission occurs by the faecal-oral route. Following an incubation period of up to four days, diarrhoea may develop. Infections are mild in most species; in ducklings a severe hepatitis may develop.

Diagnosis is based on the detection of astroviruses in faeces using electron microscopy or ELISA. Detection of viral RNA using reverse transcriptase PCR and virus isolation in primary cell lines or embryonated eggs are also possible. Because astrovirus infections are generally mild, vaccines have not been developed except for duck astrovirus. Control is based on husbandry practices appropriate for the prevention of neonatal enteritis.

Chapter 72

Coronaviridae

Members of the family *Coronaviridae* (Latin *corona*, crown) are large, pleomorphic, enveloped viruses. They contain a single molecule of linear, positive-sense, single-stranded RNA. Club-shaped glycoprotein peplomers projecting from the envelope impart a crown-like appearance to the virus (Fig. 72.1). Each peplomer is composed of a large viral glycoprotein (spike or S protein) which is responsible for attachment to cells. The S protein is the main antigenic component which induces the production of neutralizing antibodies during natural infection. Hypervariable domains in the S protein facilitate the production of virus escape mutants, capable of evading the host immune response. Along with the family *Arteriviridae*, the family *Coronaviridae* belongs to the order *Nidovirales*. There are two genera in the family, *Coronavirus* and *Torovirus*. Coronaviruses, which are almost spherical with a diameter of 120 to 160 nm, have helical nucleocapsids. Toroviruses, which have a tubular nucleocapsid, may be disc-shaped, kidney-shaped or rod-shaped and are 120 to 140 nm in diameter.

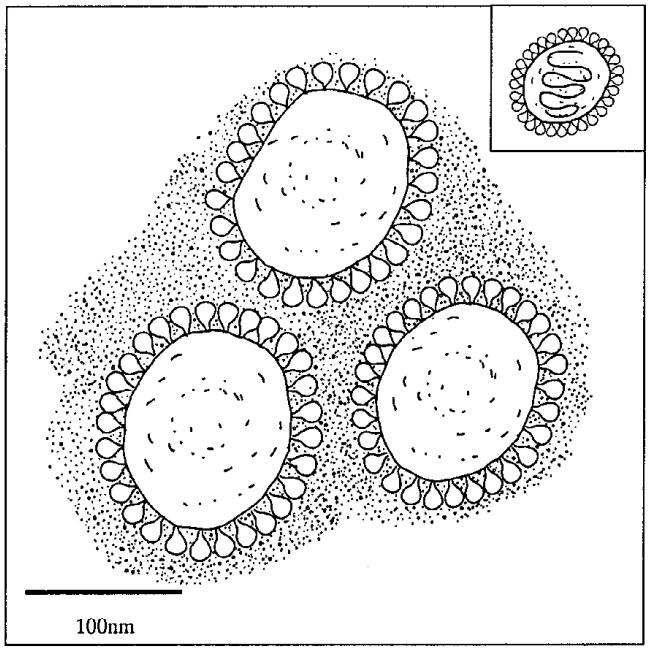


Figure 72.1 Coronavirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Key points

- Enveloped, pleomorphic, single-stranded RNA viruses
- Replicate in the cytoplasm
- Labile in the environment
- Two genera:
  - —*Coronavirus*, helical nucleocapsid
  - —*Torovirus*, tubular nucleocapsid
- Coronaviruses:
  - —Systemic disease in cats
  - —Enteric and systemic disease in pigs
  - —Respiratory disease in poultry
  - —Enteric disease in cattle

Coronaviruses replicate in the cytoplasm of cells. Newly synthesized virions acquire their envelopes from the membranes of the endoplasmic reticulum and the Golgi complex. They are incorporated into vesicles and transported to the cell surface where the virions are released following fusion of the vesicles with the plasma membrane. Genetic recombination can occur at high frequency between related coronaviruses.

With the exception of infectious bronchitis virus, coronaviruses are usually difficult to grow in cell culture. The virions are sensitive to heat, lipid solvents, formaldehyde, oxidizing agents and non-ionic detergents. The stability of coronaviruses at low pH values is variable; some are stable at values as low as pH 3.0.

Clinical infections

Coronaviruses can infect a number of mammalian and avian species and many display tropisms for respiratory and intestinal epithelium. The coronaviruses of veterinary importance and the clinical consequences of infection are indicated in Table 72.1. Infections, which are usually mild or inapparent in mature animals, may be severe in young animals. Coronaviruses are aetiologically important in humans as a cause of the common cold.

Although evidence of torovirus infection has been

Table 72.1 Coronaviruses of veterinary significance.

Virus	Consequences of infection
Feline coronavirus (FCoV)	Replicates in enterocytes; subclinical infection common. May produce mild gastroenteritis in young kittens; also referred to as feline enteric coronavirus (FECV). Feline infectious peritonitis virus (FIPV) is considered to have derived from strains of FCoV which initially replicated in enterocytes and subsequently in macrophages; causes sporadic fatal disease of young cats often presenting clinically as an effusive peritonitis
Transmissible gastroenteritis virus (TGEV)	Highly contagious infection with vomiting and diarrhoea in piglets; high mortality in newborn piglets. A deletion-mutant of TGEV, porcine respiratory coronavirus, induces partial immunity to TGEV
Porcine epidemic diarrhoea virus	Causes enteric infection similar to that caused by TGEV but with lower neonatal mortality
Porcine haemagglutinating encephalomyelitis virus	Nervous disease or vomiting and emaciation (vomiting and wasting disease) in young pigs. Infection is widespread but clinical disease is uncommon
Infectious bronchitis virus	Acute, highly contagious respiratory infection in young birds; causes a drop in egg production in layers
Turkey coronavirus	Infectious enteritis (bluecomb disease)
Bovine coronavirus	Diarrhoea in calves; associated with winter dysentery in adult cattle
Canine coronavirus	Asymptomatic infection or diarrhoea in dogs

found in pigs, sheep, goats and cats (Muir *et al.*, 1990), the clinical significance of these infections is questionable. Two toroviruses have been implicated in enteric diseases of domestic animals (Table 72.2).

Feline infectious peritonitis

This disease, feline infectious peritonitis (FIP), caused by certain strains of feline coronavirus, is a worldwide and invariably fatal, sporadic disease of domestic cats and other *Felidae*. Strains of feline coronavirus vary in pathogenicity. The term feline enteric coronavirus (FECV) has been used to describe strains that cause mild or inapparent enteritis, while the term feline infectious peritonitis virus (FIPV) was applied to those strains aetiologically implicated in FIP. It is believed that FIPV arises as a mutant of the widely distributed FECV resulting in an alteration in tropism from enteric epithelial cells to macrophages (Pedersen and Floyd, 1985; Poland *et al.*, 1996). Current thinking envisages a single virus termed feline coronavirus (FCoV) which encompasses strains of varying virulence. Although genomic studies have so far failed to demonstrate consistent genetic changes distinguishing enteric from peritonitis strains, they have shown a high degree of genetic relatedness between isolates of the two strains from the same locations (Vennema *et al.*, 1998). Two serotypes of FCoV are described. Serotype 1 accounts for most field infections whereas serotype 2 is thought to have arisen due to a recombinational event between FCoV and canine coronavirus (Herrewegh *et al.*, 1998).

Epidemiology

Feline infectious peritonitis occurs sporadically in catteries or multicat households. The incidence is reported to be higher in pedigree cats (Sparkes *et al.*, 1992). Although cats of any age may be affected, those less than one year of age appear to be most susceptible. A second peak of disease in cats over ten years of age has been noted (Barr, 1998). Infected cats shed virus in faeces and oronasal secretions. Transmission is mainly by ingestion or inhalation. Infection is acquired by young kittens from their mothers or from other adult cats (Addie and Jarrett, 1992). In infected households about 15% of cats are persistently infected carriers, responsible for maintaining the infection. The majority of cats are transiently infected but reinfection can occur.

Pathogenesis

The pathogenesis of FIP is outlined in Fig. 72.2. Infection with FIPV does not always result in clinical disease. Factors which may influence the development of the disease include the age, immune status and genetic characteristics of the host and the emergence of virulent virus strains (Addie *et al.*, 1995). In some instances, probably due to mutational changes in the virus, the emergence of a virulent FIPV strain results in systemic invasion with replication in macrophages. In most infected kittens, the development of effective cell-mediated immunity (CMI) restricts viral replication and ultimately eliminates infection. Some individual animals with less effective CMI may shed virus intermittently while remaining clinically normal. When CMI is severely impaired or defective, virus replication continues leading to B cell

Table 72.2 Toroviruses of veterinary significance.

Virus	Hosts	Comments
Equine torovirus (Berne virus)	Horses	Isolated from rectal swab of a horse with diarrhoea in Berne, Switzerland. Clinical disease appears to be uncommon
Bovine torovirus (Breda virus)	Calves	Diarrhoea in newborn calves, particularly if deprived of colostrum

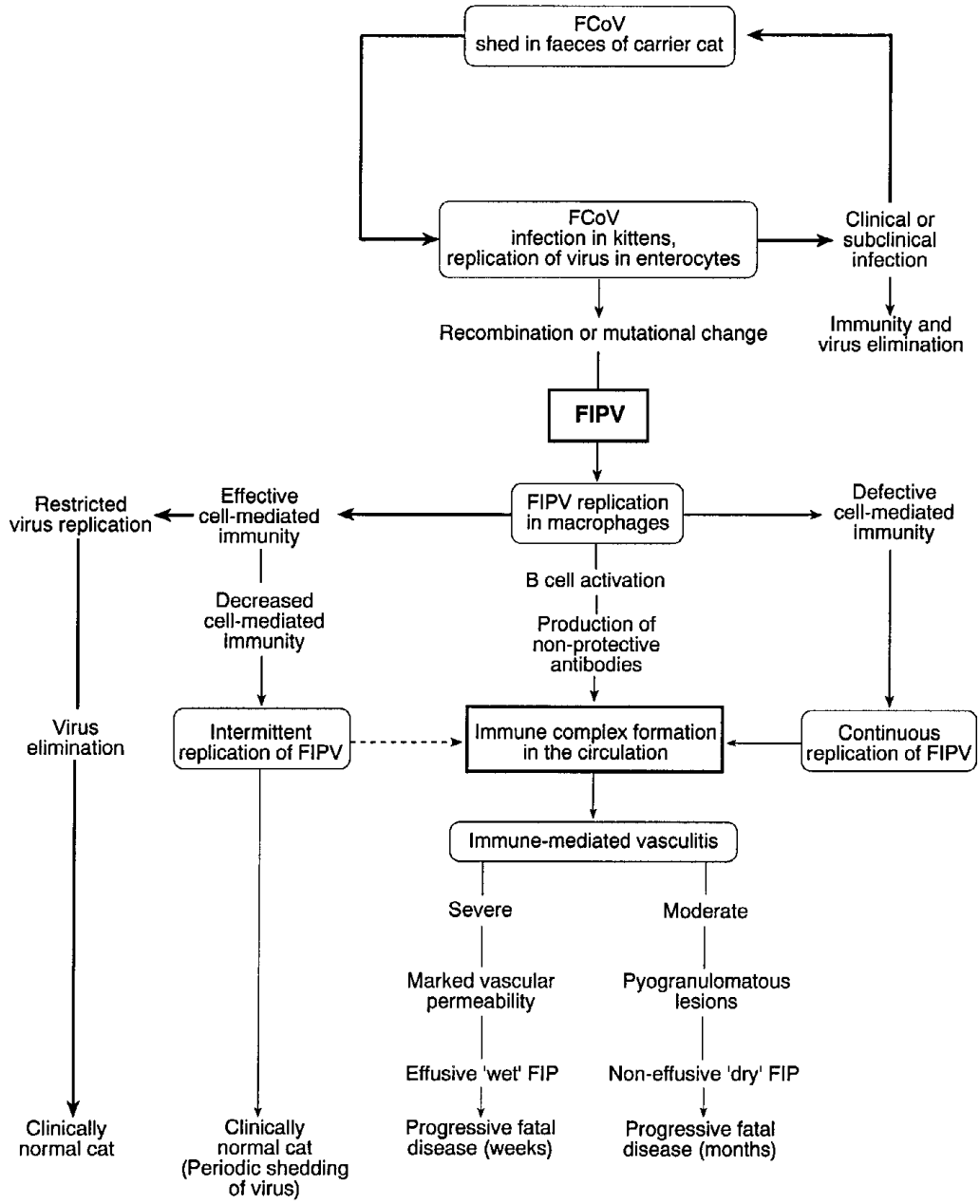


Figure 72.2 Proposed relationship between infection with feline enteric coronavirus and the emergence of feline infectious peritonitis virus leading to the development of feline infectious peritonitis.

FCoV, feline coronavirus; FIPV, feline infectious peritonitis virus; FIP, feline infectious peritonitis

activation and the production of non-protective antibodies. The immune complexes, formed from these antibodies and FIPV, activate complement leading to immune-mediated vasculitis. The severity of this vasculitis influences the clinical presentation and the rate of progression of the disease. In addition to type III hypersensitivity, there is evidence of type IV hypersensitivity in FIPV-induced lesions (Paltrinieri *et al.*, 1998a; Paltrinieri *et al.*, 1998b).

Clinical signs

The incubation period ranges from weeks to months. The onset of clinical signs may be either sudden or slow and insidious. Early signs, which are generally non-specific, include anorexia, weight loss, listlessness and dehydration. Affected cats often present with icterus.

Cats with the effusive form of the disease have fibrin-rich exudates in the abdominal or thoracic cavities. If the pleural effusion is marked, dyspnoea develops. The effusive form of the disease usually leads to death within eight weeks.

In the non-effusive form of FIP, clinical findings are less characteristic. Signs referable to lesions in organs or

tissues in the abdominal cavity are present in about 50% of affected cats. Anterior uveitis, chorioretinitis and neurological signs may be evident in up to 30% of cases. The course of the disease is usually protracted with animals surviving for weeks or months. Infection with feline leukaemia virus or with feline immunodeficiency virus may increase susceptibility to FIPV and contribute to the severity of the clinical signs.

Diagnosis

- Currently, histological examination of affected tissues is the only procedure available for the definitive diagnosis of FIP.
- Pleural or peritoneal fluid, which may contain fibrin strands, clots on standing. It has a high protein content. A gamma globulin content exceeding 32% of total protein is suggestive of FIP (Weiss, 1991).
- Typical haematological changes include neutrophilia, lymphopenia and, in chronic cases, a normocytic, normochromic, non-regenerative anaemia.
- A serum hyperproteinaemia is frequently present due to a hypergammaglobulinaemia. Serum liver enzymes and total bilirubin may also be raised.
- Diagnostic serological tests, including IFA and ELISA do not distinguish between cats infected with FCoV and FIPV. In indirect immunofluorescence, antibody titres may be very high in some FIP cases. In other cases antibody titres are negligible (Sparkes *et al.*, 1991).
- RT-PCR can be used to detect virus shed in the faeces and for the identification of carriers (Addie and Jarrett, 2001).

Treatment and control

There is no specific treatment for FIP. Supportive therapy and broad spectrum antibiotics may be useful for treating affected cats in good physical condition (Weiss, 1994).

An intranasal vaccine employing a temperature-sensitive mutant strain of FIPV has recently been developed. Although some efficacy and safety test results have been favourable (Postorino Reeves *et al.*, 1992; Hoskins *et al.*, 1994; Fehr *et al.*, 1997), other studies have failed to demonstrate significant protective immunity.

The creation and maintenance of coronavirus-negative catteries is an effective method of control but is extremely difficult to achieve. Measures aimed at reducing the incidence of the disease include breeding from bloodlines free of FIP, rearing litters of kittens in isolation (Addie and Jarrett, 1990) and reducing stress in the catteries.

Canine coronavirus infection

The importance of canine coronavirus (CCV) as a cause of disease is uncertain as the agent can be isolated from normal dogs and those with diarrhoea (Tennant *et al.*, 1993). This virus is related antigenically to the feline coronaviruses.

Epidemiology

Serological studies indicate that infection is common (Tennant *et al.*, 1991). Infections with CCV may spread rapidly among susceptible dogs at shows and in kennels. Seroprevalence may approach 100% in kennels and range from 6% to 75% in pet dog populations.

Infection is acquired from the faeces of infected animals. Infected dogs usually shed the virus for up to 9 days and intermittent shedding may continue for months. The virus is not particularly resistant in the environment and carrier dogs are required for its maintenance. Mucosal immunity appears to be more important than circulating antibody for protection of dogs from reinfection. In the absence of frequent re-exposure to the virus, the duration of immunity may be relatively short.

Pathogenesis

Canine coronaviruses withstand the acid environment of the stomach and infect enterocytes in the duodenum. Infection spreads rapidly to involve other parts of the small intestine. Diarrhoea may follow loss of digestive and absorptive capacity in the small intestine as a result of damage to mature enterocytes at the tips of villi. Recovery is rapid in uncomplicated cases.

Clinical signs

Although clinical disease has been recorded in dogs, foxes and coyotes, infection with CCV is often asymptomatic. Dogs of all ages can become infected; serious illness is most likely to occur in pups. The incubation period is up to three days. Clinical signs, which are variable and non-specific, include anorexia, depression, vomiting and diarrhoea. Most animals recover in seven to ten days. Occasionally, illness may be protracted due to secondary bacterial, parasitic or other viral infections. The mortality rate is low.

Diagnosis

- Virus may be detected in faeces by electron microscopy.
- Virus can be isolated in a number of cell lines but the procedure is slow and unreliable.
- Serum neutralization or indirect immunofluorescence tests can be used to demonstrate an increasing antibody titre.

Treatment and control

- Supportive treatment, including fluid replacement therapy and antibiotic administration, should be instituted when required.
- Although inactivated vaccines are available and can be used in pregnant bitches to boost colostral immunity, the degree of protection induced by these vaccines is

uncertain.
- Contact with infected animals and faeces should be minimized.
- Effective disinfection of premises and utensils can be achieved with 3% sodium hypochlorite or 2% formalin.

Transmissible gastroenteritis

Transmissible gastroenteritis (TGE) is a highly contagious, coronaviral disease of young pigs which occurs worldwide. There is one serotype of transmissible gastroenteritis virus (TGEV) which is closely related antigenically to feline coronavirus and canine coronavirus. A relatively non-pathogenic respiratory variant of TGEV, referred to as porcine respiratory coronavirus (PRCV), was first recognized in 1984. This virus spread to pig populations in many European countries and has now been identified in the USA and in some Asian countries. Infection with PRCV is usually subclinical.

Epidemiology

Transmission of TGEV is usually by the faecal-oral route. The virus is moderately stable in the presence of proteolytic enzymes and at pH 3.0, ensuring survival in the stomach and small intestine. Viral shedding in faeces can persist for up to two weeks. Outbreaks of TGE tend to occur in winter. In fully susceptible herds, the virus spreads rapidly infecting animals of all ages. The disease, however, is most severe in newborn piglets. Outbreaks usually terminate in a few weeks if no new susceptible animals are introduced into the herd.

Pathogenesis

Following ingestion, the virus replicates mainly in mature enterocytes at the tips of the villi in the small intestine. Viral replication results in villous atrophy throughout the length of the small intestine. Digestion and cellular transport of nutrients and electrolytes are severely disrupted resulting in the accumulation of fluid in the intestinal lumen and diarrhoea. Young piglets are particularly susceptible to the ensuing dehydration and metabolic acidosis.

Clinical signs

The incubation period is up to three days. Vomiting and watery diarrhoea may be evident in affected piglets less than seven days old. Rapid dehydration and weight loss follow. Mortality may approach 100% in newborn piglets and is usually confined to animals under three weeks of age. Inappetence and transient diarrhoea may be observed in older pigs. Subclinical infections also occur. Sows quickly become immune and maternally-derived immunity reduces the severity of clinical signs in piglets. Outbreaks usually last a few weeks. However, TGEV infection may become endemic in a herd if consecutive litters become infected as maternally-derived immunity wanes. Clinically, such infections are usually mild.

Diagnosis

The sudden onset and rapid spread of diarrhoea among newborn pigs along with almost 100% mortality is highly suggestive of TGE. Postmortem examination of washed small intestine discloses paper-thin walls, due to villous atrophy. The walls of the jejunum and ileum are affected while those of the duodenum are usually normal.

- Viral antigen can be detected in mucosal smears or cryostat sections of the small intestine by immunofluorescence. It may be necessary to euthanize some piglets in the early stages of the disease in order to obtain suitable specimens for laboratory examination. Viral antigens can be demonstrated in faeces by ELISA.
- Virus can be isolated from faeces in a swine testis cell line.
- Serological testing for antibodies can be carried out using virus neutralization. However, virus neutralization does not distinguish antibodies to TGEV from those induced by PRCV infection. A blocking ELISA, which is based on the use of monoclonal antibody directed against a glycoprotein epitope present in TGEV but absent from PRCV, can be used to distinguish infections caused by these two viruses.

Treatment and control

- There is no specific treatment but fluid replacement therapy may be beneficial. Maintaining the farrowing house at an optimal temperature may enhance survival.
- In acute outbreaks of TGE, deliberate exposure of pregnant sows to the virus may reduce neonatal mortality. After exposure, sows due to farrow should be moved to clean premises. Newborn piglets born to exposed sows will usually receive passive antibody protection following suckling.
- Modified live and inactivated vaccines are available. Modified live vaccines are administered orally to sows five to seven weeks before farrowing and a booster inoculation is administered parenterally one week before parturition. Vaccination reduces mortality but does not eliminate infection.
- Serious outbreaks of TGE have become rare in European pig populations endemically infected with PRCV. Porcine respiratory coronavirus is spread in pig herds by aerosols. Sows infected with PRCV usually transfer substantial colostral protection to their litters (Wesley and Woods, 1993).

Porcine epidemic diarrhoea

This porcine disease, which is clinically similar to TGE, is described in Europe and Asia but not in America. There is only one serotype of porcine epidemic diarrhoea

virus (PEDV), a coronavirus serologically unrelated to TGEV.

Epidemiology
The virus is transmitted by the faecal-oral route. Spread of the virus to susceptible herds occurs directly through infected pigs and indirectly through contaminated fomites or vehicles. The rate of spread of infection within a farm is slower than that of TGEV.

Pathogenesis
Virus replication, which occurs in enterocytes in the small intestine and colon, results in shortening of villi. The rate and severity of cell destruction is less marked than with TGEV.

Clinical signs
The incubation period is up to four days. The age of animals affected and the associated morbidity and mortality are variable. On some farms, animals of all ages become sick and the mortality rate in piglets under one week may approach 50%. Watery diarrhoea, which may be preceded by vomiting, is the main clinical presentation. Occasionally a few animals die suddenly, with back muscle necrosis evident at postmortem. The virus may persist on large breeding farms by infecting consecutive litters of pigs. Most affected pigs recover after about one week and mortality rates are usually low.

Diagnosis
- Direct immunofluorescence, using cryostat sections of small intestine from pigs euthanized during the phase of acute diarrhoea, is sensitive and reliable, particularly in specimens from newborn piglets.
- Viral antigen may be detected by ELISA in faecal material or intestinal contents collected during the acute phase of the disease.
- Antibodies can be detected in paired serum samples using a blocking ELISA or by indirect immunofluorescence on PEDV-positive cryostat sections of intestine.

Treatment and control
- There is no specific treatment and vaccines are not available.
- Appropriate hygienic measures and the control of animal and human movement onto a farm are necessary for disease prevention.
- During an outbreak of the disease on a breeding farm, good hygiene slows the spread of infection. Deliberate spread of the virus to pregnant sows using infected faecal material stimulates colostral immunity and shortens the course of the disease outbreak.

Porcine haemagglutinating encephalomyelitis virus infection

This coronavirus disease of young pigs, also known as vomiting and wasting disease, is caused by haemagglutinating encephalomyelitis virus of which there is only one serotype. The virus agglutinates red cells of several animal species.

Epidemiology
Infection is common and probably worldwide. The virus is shed in nasal secretions and readily transmitted by aerosols. Infection persists on breeding farms as a subclinical respiratory condition. In herds where infection is endemic, immune sows transfer protective antibodies to their offspring and piglets are protected until they have developed an age-related resistance. Pigs which become subclinically infected develop an active immunity at eight to 16 weeks of age.

Pathogenesis
The virus replicates locally in the upper respiratory tract and tonsils before spreading via the peripheral nervous system to the medulla oblongata. It then spreads to various other parts of the central nervous system. Viral damage to the vagal sensory ganglion and to the intramural plexus of the stomach are considered to be responsible for vomiting and delayed gastric emptying.

Clinical signs
Clinical signs develop in pigs less than three weeks of age after an incubation period of up to seven days. Signs of acute encephalomyelitis, vomiting and wasting are the predominant clinical features. Newborn piglets become severely dehydrated and may die. The mortality rate is often 100% in young pigs. Older pigs continue to vomit and become emaciated. Survivors may be permanently stunted.

Diagnosis
- For the isolation of virus or demonstration of viral antigen in cryostat sections by immunofluorescence, samples of brain stem must be collected within two days of the onset of clinical signs. Porcine thyroid cells are suitable for isolation.
- Lesions of non-suppurative encephalomyelitis may be evident.
- A significant rise in antibody titre may be demonstrable in paired serum samples by virus neutralization or haemagglutination inhibition tests.

Treatment and control
- No specific treatment is available.
- Due to the sporadic nature of the disease, vaccination is unwarranted.
- Appropriate measures should be taken to prevent the

introduction of infection into breeding units. If infection is introduced, it is important to ensure that infection of sows results in an adequate antibody response for the protection of litters.

Infectious bronchitis

Infectious bronchitis, caused by infectious bronchitis virus (IBV), is a highly contagious, economically important, worldwide disease of poultry which affects the respiratory, reproductive and renal systems. Many serotypes, often with different virulence and tissue tropisms probably as a result of mutation or recombination, are recognised.

Epidemiology

The chicken is the main host although IBV has been isolated from pigeons and pheasants. The most important route of transmission is by aerosols and spread of infection occurs rapidly among susceptible birds. Morbidity may approach 100%. Virus, shed from the respiratory tract for a few weeks after infection, may be recovered over a period of weeks from the faeces and from eggs of infected birds. Infection may persist in the digestive tract of individual birds.

Pathogenesis

The respiratory system is the primary site of virus replication. Viraemia follows within one to two days of exposure. The virus becomes widely distributed throughout the body, particularly in the oviducts, kidneys and bursa of Fabricius. The distribution and severity of lesions in these tissues are influenced by the virulence of the infecting strain.

Clinical signs

The incubation period is up to 48 hours. Age, immune status and strain of virus strongly influence the nature and severity of disease observed in a flock. In general, disease is most severe in young birds, particularly when secondary infections are present. In chickens less than three weeks of age there is gasping and nasal exudate. Infection may result in stunting and some birds may die suddenly from occluded bronchi. In older birds, rales and gasping are usually observed. Mortality rates are generally low in the absence of secondary infections. The course of the disease is up to seven days in individual birds and outbreaks last about ten to 14 days in flocks. Layers show signs of rales followed by a marked reduction in egg production which slowly returns to normal. Egg quality, with soft-shelled and mis-shaped eggs, may be poor for several weeks. Infection with nephrotropic strains of IBV are associated with interstitial nephritis and mild respiratory signs with moderate to high levels of mortality.

Diagnosis

- Virus isolation is often feasible in the acute stage of the disease. Although specimens from the respiratory tract are those most suitable for virus isolation, samples from kidney, oviduct and faeces can also be used. Material is usually inoculated into the allantoic sac of nine to ten day old embryonated eggs. A number of passages may be required to produce the characteristic stunting and curling of the embryo. Tracheal explants from day-old specific-pathogen-free chicks may also be used for virus isolation.
- Serological tests, including virus neutralization, agar gel immunodiffusion, haemagglutination inhibition and ELISA, can be used to demonstrate a rise in antibody titre between acute and convalescent serum samples.

Treatment and control

- There is no specific treatment. The administration of antibiotics may reduce mortality due to secondary bacterial infections.
- Both live and killed vaccines with adjuvant are available. Live vaccines are usually administered in the drinking water or by aerosol to chicks up to 14 days of age and again at about 4 weeks of age. A high-passage vaccinal virus, which is less pathogenic, is used for primary immunization. A more virulent strain of virus is used for booster vaccination. Following primary immunization with live vaccines, killed vaccines are useful in layer and breeder flocks to prevent losses in egg production and to ensure a high level of yolk sac-derived immunity in chicks.

Bovine coronavirus infection

Bovine coronavirus (BCV) is one of the causes of calf diarrhoea and is also associated with winter dysentery in adult housed cattle. There is evidence of its involvement in the bovine respiratory disease complex (Kapil and Goyal, 1995). The virus, which exists as a single serotype, haemagglutinates red cells of mice, rats and hamsters.

Epidemiology and pathogenesis

Virus is mainly transmitted by the faecal-oral route. However, coronaviruses have also been recovered from the respiratory tract of calves (McNulty *et al.*, 1984) and infected calves often harbour BCV in both the enteric and respiratory tracts. Infection is usually endemic on farms, maintained by clinically-affected calves and persistently-infected, clinically normal, calves and cows. The virus replicates and destroys mature enterocytes in the small intestine and colon, resulting in a malabsorptive diarrhoea. The severity of disease is influenced by the age of the animal at the time of infection and type of management. Risk factors, which include changes in diet, cold temperatures, close confinement and the presence of other microorganisms such as *Campylobacter jejuni*, appear to

be particuiarly important in the development and pathogenesis of winter dysentery.

Clinical signs

In calves, the incubation period is up to two days and clinical signs are usually observed between three to 21 days of age. There is profuse diarrhoea which may result in dehydration, acidosis and death. If milk is withdrawn and the calves are fed oral electrolytes, the diarrhoea usually ceases in a few days. Respiratory tract infections are generally mild but may predispose to more severe secondary infections.

In adult animals the incubation period of winter dysentery is three to seven days. There is a sudden onset of diarrhoea accompanied by a dramatic drop in milk yield. The faeces of some animals may contain blood or blood clots. A nasolacrimal discharge and cough may accompany the diarrhoea. Herd outbreaks can last for two weeks.

Diagnosis

- Faeces or intestinal contents for laboratory examination should be collected early in the course of the disease.
- Typical coronavirus particles can be demonstrated in faecal samples by direct electron microscopy (EM). Immune EM is preferable as it is more sensitive and specific. Alternative diagnostic methods of detection include ELISA and reverse passive haemagglutination.
- Immunofluorescence can be used to detect viral antigen in cryostat sections of distal small intestine or colon.
- Isolation of virus in tissue culture is difficult.

Treatment and control

Treatment is supportive but non-specific. Control of the disease in calves is based on vaccination and good management practices. Both live and inactivated vaccines have been developed and can be used orally in calves to stimulate active immunity. They may also be administered parenterally in cows to increase the level of antibody in colostrum and milk. However, no vaccine is available for the prevention of winter dysentery.

References

Addie, D.D. and Jarrett, O. (1990). Control of feline coronavirus infection in kittens. *Veterinary Record*, **126**, 164.

Addie, D.D. and Jarrett, O. (1992). A study of naturally occurring feline coronavirus infections in kittens. *Veterinary Record*, **130**, 133–137.

Addie, D. and Jarrett, O. (2001). Use of reverse-transsciptase polymerase chain reaction for monitoring the shedding of feline coronavirus by healthy cats. *Veterinary Record*, **148**, 649–653.

Addie, D.D., Toth, S., Murray, G.D. and Jarrett, O. (1995). Risk of feline infectious peritonitis in cats naturally infected with feline coronavirus. *American Journal of Veterinary Research*, **56**, 429–434.

Barr, F. (1998). Feline infectious peritonitis. *Journal of Small Animal Practice*, **39**, 501–504.

Fehr, D., Holznagel, E., Bolla, S. *et al.* (1997). Placebo-controlled evaluation of a modified live virus vaccine against feline infectious peritonitis: Safety and efficacy under field conditions. *Vaccine*, **15**, 1101–1109.

Herrewegh, A.A.P.M., Smeenk, I., Horzinek, M.C. *et al.* (1998). Feline coronavirus type II strains 79–1683 and 79–1146 originate from a double recombination between feline coronavirus type 1 and canine coronavirus. *Journal of Virology*, **72**, 4508–4514.

Hoskins, J.D., Taylor, H.W. and Lomax, T.L. (1994). Challenge trial of an intranasal feline infectious peritonitis vaccine. *Feline Practice*, **22**, 9–13.

Kapil, S. and Goyal, S.M. (1995). Bovine coronavirus-associated respiratory disease. *Compendium on Continuing Education for the Practicing Veterinarian*, **17**, 1179–1181.

McNulty, M.S., Bryson, D.G., Allan, G.M. and Logan, E.F. (1984). Coronavirus infection of the bovine respiratory tract. *Veterinary Microbiology*, **9**, 425–434.

Muir, P., Harbour, D.A., Gruffydd-Jones, T.J. *et al.* (1990). A clinical and microbiological study of cats with protruding nictitating membranes and diarrhoea: isolation of a novel agent. *Veterinary Record*, **127**, 324–330.

Paltrinieri, S., Cammarata, M.P., Cammarata, G. and Mambretti, M. (1998a). Type IV hypersensitivity in the pathogenesis of FIPV-induced lesions. *Journal of Veterinary Medicine, Series B*, **45**, 151–159.

Paltrinieri, S., Cammarata, M.P., Cammarata, G. and Comazzi, S. (1998b). Some aspects of humoral and cellular immunity in naturally occurring feline infectious peritonitis. *Veterinary Immunology and Immunopathology*, **65**, 205–220.

Pedersen, N.C. and Floyd, K. (1985). Experimental studies with three new strains of feline infectious peritonitis virus: FIPV-UCD2, FIPV-UCD3 and FIPV-UCD4. *Compendium on Continuing Education for the Practicing Veterinarian*, **7**, 1001–1011.

Poland, A., Vennema, H., Foley, J.E. and Pedersen, N.C. (1996). Two related strains of feline infectious peritonitis virus isolated from immunocompromised cats infected with a feline enteric coronavirus. *Journal of Clinical Microbiology*, **34**, 3180–3184.

Postorino Reeves, N.C., Pollock, R.V.H. and Thurber, E.T. (1992). Long-term follow-up study of cats vaccinated with a temperature-sensitive feline infectious peritonitis vaccine. *Cornell Veterinarian*, **82**, 117–123.

Sparkes, A.H., Gruffydd-Jones, T.J. and Harbour, D.A. (1991). Feline infectious peritonitis: a review of clinico-pathological changes in 65 cases, and a critical assessment of their diagnostic value. *Veterinary Record*, **129**, 209–212.

Sparkes, A.H., Gruffydd-Jones, T.J., Howard, P.E. and Harbour, D.A. (1992). Coronavirus serology in healthy pedigree cats. *Veterinary Record*, **131**, 35–36.

Tennant, B.J., Gaskell, R.M., Jones, R.C. and Gaskell, C.J. (1991). Prevalence of antibodies to four major canine viral

diseases in dogs in a Liverpool hospital population. *Journal of Small Animal Practice*, **32**, 175–179.

Tennant, B.J., Gaskell, R.M., Jones, R.C. and Gaskell, C.J. (1993). Studies on the epizootiology of canine coronavirus. *Veterinary Record*, **132**, 7–11.

Vennema, H., Poland, A., Foley, J. and Pedersen, N.C. (1998). Feline infectious peritonitis viruses arise by mutation from endemic feline enteric coronaviruses. *Virology*, **243**, 150–157.

Weiss, R.C. (1991). The diagnosis and clinical management of feline infectious peritonitis. *Veterinary Medicine*, **86**, 308–319.

Weiss, R.C. (1994). Feline infectious peritonitis virus: advances in therapy and control. In *Consultations in Feline Internal Medicine 2*. Ed. J.R. August. W.B. Saunders, Philadelphia. pp. 3–12.

Wesley, D. and Woods, R.D. (1993). Immunization of pregnant gilts with PRCV induces lactogenic immunity for protection of nursing piglets from challenge with TGEV. *Veterinary Microbiology*, **38**, 31–40.

Further reading

Clark, M.A. (1993). Bovine coronavirus. *British Veterinary Journal*, **149**, 51–70.

Gamble, D.A., Lobbiani, A., Gramegna, M. *et al.* (1997). Development of a nested PCR assay for detection of feline infectious peritonitis virus in clinical specimens. *Journal of Clinical Microbiology*, **35**, 673–675.

Olsen, C.W. (1993). A review of feline infectious peritonitis virus: molecular biology, immunopathogenesis, clinical aspects, and vaccination. *Veterinary Microbiology*, **36**, 1–37.

Siddell, S.G. (1995). *The Coronaviridae*. Plenum Press, New York.

Chapter 73

Arteriviridae

Arteriviruses, formerly classified as members of the family *Togaviridae*, have recently been assigned to the family *Arteriviridae*. Their genome organization and mode of replication are similar to those of members of the *Coronaviridae* and these two families make up the order *Nidovirales*. In the family *Arteriviridae* there is a single genus, *Arterivirus*. The name of the genus derives from the disease, equine arteritis, which is caused by the type species. Arteriviruses are spherical, 40 to 60 nm in diameter and possess a lipid-containing envelope which has ring-like surface structures (Fig. 73.1). The icosahedral nucleocapsid contains a molecule of linear single-stranded RNA. Replication takes place in the cytoplasm of infected cells. Arteriviruses, which are relatively labile, are sensitive to heat, low pH, lipid solvents, detergent treatment, UV irradiation and many disinfectants.

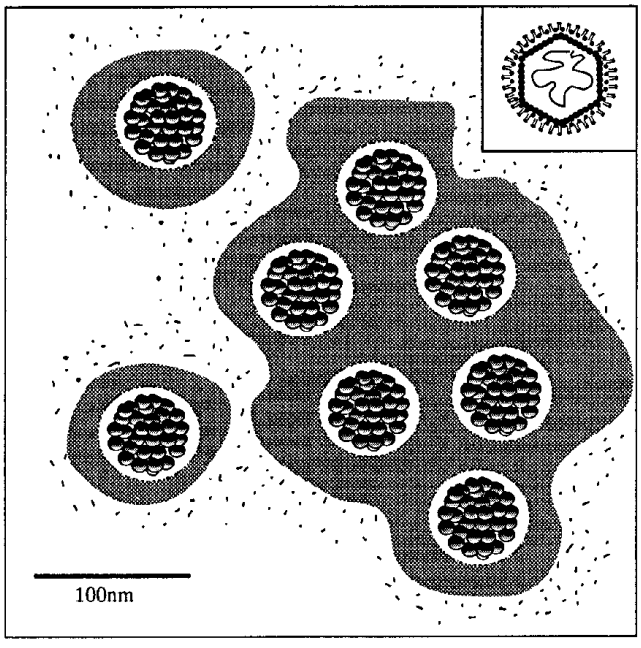


Figure 73.1 Arterivirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Key points

- Medium-sized, enveloped, single-stranded RNA viruses
- Icosahedral symmetry
- Replicate in cytoplasm of macrophages and endothelial cells
- Cause equine viral arteritis and porcine respiratory and reproductive syndrome

Clinical infections

Members of the genus are host-specific and antigenically unrelated. Infections have been described in horses, pigs, mice and monkeys. The primary target cells are macrophages. Infection is spread horizontally by aerosol, by biting or by venereal transmission. Infections are frequently persistent.

Equine viral arteritis

Although infection with equine arteritis virus (EAV) occurs worldwide, outbreaks of clinical disease are comparatively rare. Upper respiratory tract infection, ventral oedema and abortion are prominent clinical features. Biological and genomic differences have been demonstrated in isolates of EAV but antigenic variation is limited. Only one serotype has been recognized.

Epidemiology

Horses, donkeys and mules are susceptible to infection. The percentage of seropositive animals is higher in Standardbreds than in Thoroughbreds. Whether this reflects differences in susceptibility or in degree of exposure as a result of management practices is unclear. Although infection is prevalent in some horse populations, outbreaks of disease are sporadic. The frequency of confirmed outbreaks of equine viral arteritis has increased in recent years. Factors which may have contributed to this increase include increased international movement of horses, more extensive use of artificial insemination and greater awareness of the disease.

During the acute phase of infection, virus is spread primarily by aerosols from the respiratory tract. Virus is also shed in faeces, urine and vaginal secretions. Close contact facilitates spread of infection. Virus is usually eliminated from mares and geldings within one to two months but may persist in about 35% of infected stallions. Carrier stallions are asymptomatic and shed virus continuously in semen. More than 80% of mares covered by carrier stallions may become infected. Persistent infection, which does not impair the fertility of stallions, appears to be testosterone-dependent (McCollum *et al.*, 1994). Mares infected venereally may spread virus horizontally to in-contact susceptible animals. Abortion or infection of the foal may result when pregnant mares are infected.

Pathogenesis and pathology

Following aerosol transmission, replication occurs in pulmonary macrophages. There is subsequent spread to the bronchial lymph nodes and viraemia. Pathological changes, arising from infection of endothelial cells and widespread necrotic arteritis, include oedema, congestion and haemorrhage in many tissues. Aborted foetuses, which often exhibit autolysis, rarely display characteristic lesions.

Clinical signs

The incubation period ranges from three to 14 days. Many infections are subclinical. The disease tends to be more severe in very young and aged animals and in those subjected to stress. Affected animals present with fever, anorexia, depression, conjunctivitis, rhinitis and stiff gait. Oedema, which is usually prominent, may involve the eyelids, ventral abdomen and limbs, particularly the hind limbs. Urticarial-type lesions commonly affect the skin of the neck. Acute, often fatal, respiratory disease has been recorded in foals (Del Piero *et al.*, 1997). Clinical signs in donkeys are similar to those in horses but generally milder. Animals which recover are usually immune for several years.

Diagnosis

Because equine viral arteritis resembles a number of other equine diseases in its clinical presentation, definitive diagnosis requires laboratory confirmation. Internationally accepted testing procedures have been published (Timoney, 1996).

- Virus isolation is carried out in appropriate cell lines such as rabbit or equine kidney cells. Samples suitable for isolation procedures include nasopharyngeal and conjunctival swabs, placental tissue and foetal tissue and fluids.
- Viral RNA can be detected in semen and other specimens using the reverse transcriptase polymerase chain reaction.
- Acute and convalescent blood samples can be submitted for serology. Several serological tests including virus neutralization, complement fixation, indirect fluorescent antibody, agar gel immunodiffusion and ELISA have been used. The virus neutralization test, considered to be sensitive and highly specific, is the most widely used test.
- Carrier stallions can be identified by serological testing. If stallions are seropositive, virus isolation from semen should be attempted. The sperm-rich fraction of semen is suitable for virus isolation. Alternatively, carrier animals can be mated to seronegative mares which are monitored for seroconversion.

Treatment

Supportive therapy is indicated in severe cases. The carrier state in stallions cannot be eliminated by chemotherapy.

Control

Persistently-infected stallions should be identified and their breeding activities confined to seropositive or vaccinated mares. In order to reduce the risk of colt foals becoming carriers, vaccination at six to 12 months of age is recommended (Timoney and McCollum, 1996). Two types of vaccine are available:

- —A modified live tissue culture-adapted vaccine induces good protection against clinical disease but not against infection. Use of this vaccine is contraindicated in pregnant mares and in foals under six weeks of age.
- —An inactivated whole-virus vaccine with adjuvant is reported to be safe for pregnant mares but requires booster injections at six to 12 month intervals (Fukunaga, 1994).

Porcine respiratory and reproductive syndrome

This economically important condition is characterized by reproductive failure in sows and pneumonia in young pigs. The syndrome was first described in the USA in 1987. Despite attempts at controlling spread, the disease is now endemic in many countries. The aetiological agent, originally called Lelystad virus, was first isolated in the Netherlands (Wensvoort *et al.*, 1991). It was characterized as an arterivirus and renamed porcine respiratory and reproductive syndrome virus (PRRSV). Significant antigenic and genomic differences between American and European isolates of the virus are evident.

Epidemiology

Natural infection occurs in pigs and wild boars. Virus, which is shed in saliva, urine, semen and faeces, is highly infectious. Nose-to-nose contact is considered to be the most likely route of transmission. Airborne transmission

between farms was important during the early acute outbreaks of the disease when large quantities of virus were excreted. Now it appears to be important only when pig population densities are high and when weather conditions are suitable. Virus survival, which is prolonged in winter months when low temperature and high humidity prevail, facilitates transmission. Infection is generally introduced onto farms by infected pigs or by infected semen. On endemically infected farms, virus is transmitted either continuously or in waves. Maintenance of infection on farms is multifactorial (Albina, 1997). Maternally-derived immunity is of such short duration that piglets become susceptible to infection at four to 10 weeks of age. Susceptible replacement pigs maintain infection in endemic herds. Infection may spread in a slow and unpredictable manner with the result that some animals in infected herds remain susceptible. Immunocompetent pigs, which display a progressive decline in antibody levels over a period of several months, may become susceptible to reinfection. In experimentally infected pigs, infection persisted for up to 157 days (Wills *et al.*, 1997).

Pathogenesis and pathology

Infection occurs most frequently by the respiratory route. The virus has an affinity for pulmonary alveolar macrophages and the lungs are probably the target organs (Van Reeth, 1997). Early antibody responses are not effective in clearing virus infection. Antibody-dependent enhancement of infection of pulmonary alveolar macrophages has been described in the disease. After transportation to regional lymph nodes, virus spreads to tissue macrophages throughout the body. Transplacental infection of foetuses occurs. For reasons which are unclear, reproductive failure is more difficult to induce experimentally in early gestation than in late gestation (Kranker *et al.*, 1998). Foetal and placental abnormalities are not consistently present and the mechanism of foetal death and reproductive failure is uncertain. Although the virus does not appear to have a systemic immunosuppressive effect, it predisposes to infection with other microorganisms such as *Streptococcus suis*, porcine respiratory coronavirus and *Haemophilus parasuis* (Albina *et al.*, 1998).

Clinical signs

Introduction of PRRSV to a breeding herd is usually followed by reproductive failure which may take the form of abortions, early farrowing, increased numbers of stillborn and mummified foetuses, weak neonatal pigs and delayed return to service in affected sows. A 'rolling inappetence', progressively affecting animals in an infected herd, has been described. In some cases, cyanosis of the ears and vulva along with erythematous plaques on the skin ('blue-eared disease') have been described. Respiratory distress and increased preweaning mortality are important features of the disease in neonatal pigs. Subclinical infection is common. Factors which may exacerbate clinical disease include concentrated numbers of pigs, the virulence of the strain of PRRSV and the presence of slatted floors. Although sporadic respiratory and reproductive problems are the main clinical manifestations in most affected herds, in a few endemic herds, chronic disease problems predominate (Zimmermann *et al.*, 1997).

Diagnosis

- Laboratory confirmation is usually required because the clinical presentation, particularly in endemic herds, is variable.
- Serology is the most widely used diagnostic method. Several serological tests are available including ELISA, virus neutralization, indirect fluorescent antibody and immunoperoxidase monolayer assay. However, these tests do not distinguish carrier from vaccinated animals.
- The presence of PRRSV may be demonstrated by virus isolation, direct FA staining, *in situ* hybridization or reverse transcriptase polymerase chain reaction. Suitable samples for submission include serum, foetal fluids and lung tissue.

Treatment

There is no specific treatment. Supportive therapy and antibiotic administration to suppress secondary infections may be beneficial.

Control

Vaccination and effective hygiene and health management are important for controlling infection.

- A commercial modified live vaccine is available for use in pigs 3 to 18 weeks of age and is suitable for use in non-pregnant sows before breeding. It is not suitable for use in boars and pregnant sows or in herds free from PRRSV infection. An inactivated vaccine with adjuvant is also available (Plana-Duran *et al.*, 1997). Vaccination provides reasonable protection from the clinical effects of infection.
- Stabilization of the sow herd is required to avoid subpopulations of non-immune sows and to break the cycle of reinfection. Replacement sows should be introduced to the herd only when effective isolation and acclimatization procedures are instituted. Other control measures, relating to the weaning and rearing of piglets on infected premises and to strategies for the elimination of infection from herds, have been proposed (Dee and Joo, 1997; Dee and Molitor, 1998).

References

Albina, E. (1997). Epidemiology of porcine reproductive and respiratory syndrome (PRRS): an overview. *Veterinary Microbiology*, **55**, 309–316.

Albina, E., Piriou, L., Hutet, E. *et al.* (1998). Immune response

Chapter 74

Flaviviridae

The family name of the *Flaviviridae* (Latin *flavus*, yellow) is derived from yellow fever, a disease of humans caused by a flavivirus, with jaundice as a major clinical feature. Members of the family are 40 to 60 nm in diameter with icosahedral capsids and tightly adherent envelopes which contain either two or three virus-encoded proteins, depending on the genus (Fig. 74.1). The genome is composed of positive-sense single-stranded RNA. Replication of virus occurs in the cytoplasm with maturation in cytoplasmic vesicles and release by exocytosis. The mature virions are generally labile, being sensitive to heat, detergents and organic solvents.

The family comprises three genera namely *Flavivirus*, *Pestivirus* and *Hepacivirus* (Fig. 74.2). Two genera, *Flavivirus* and *Pestivirus* contain viruses of veterinary importance. The genus *Flavivirus* contains approximately 70 members assigned to several serologically defined groups. Most members of the genus are arboviruses, which require either mosquitoes or ticks as vectors.

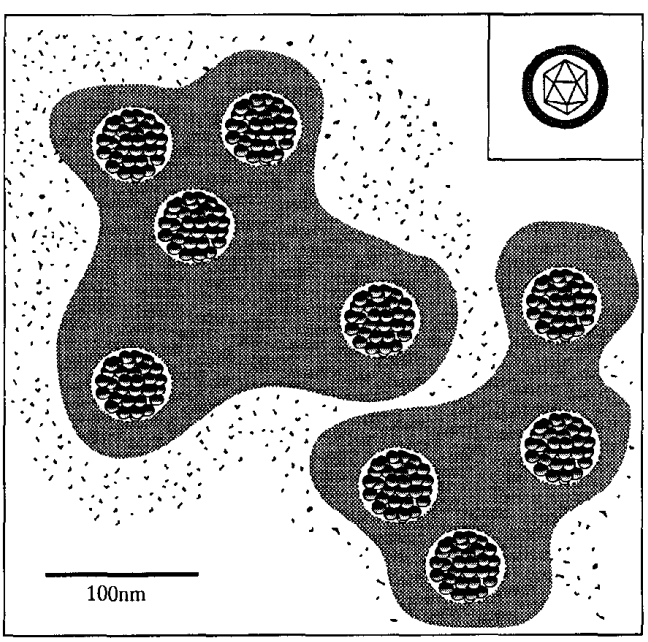


Figure 74.1 Flavivirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Key points

- Enveloped labile viruses
- Positive-sense, single-stranded RNA
- Replicate in the cytoplasm
- Two genera of veterinary importance, *Flavivirus* and *Pestivirus*
- Most viruses in the genus *Flavivirus* are transmitted by arthropods and cause encephalitis
- Pestiviruses, which are transmitted directly or indirectly, cause bovine viral diarrhoea, border disease and classical swine fever

Viruses in the genus agglutinate goose red cells. The genus *Pestivirus* contains three viruses of veterinary importance namely bovine viral diarrhoea virus, border disease virus and classical swine fever virus. Pestiviruses possess four structural proteins: a capsid protein and three envelope glycoproteins designated E^{rns} (soluble ribonuclease), E1 and E2. The immunodominant major glycoprotein, E2 (gp53) induces neutralizing antibodies.

Clinical infections

In the genera *Flavivirus* and *Pestivirus* there are several viruses of particular veterinary importance (Table 74.1). Three members of the genus *Flavivirus,* louping ill virus, Japanese encephalitis virus and Wesselsbron virus, cause disease in domestic animals. In addition, infection with West Nile virus, an important human pathogen, causes fatal disease in horses. Other members of the genus which are important human pathogens include yellow fever virus, dengue virus, Japanese encephalitis virus, tick-borne encephalitis virus and St. Louis encephalitis virus. The sole member of the *Hepacivirus* genus, hepatitis C virus, is an important cause of hepatitis in humans.

The four recognised members of the *Pestivirus* genus which infect domestic species are closely related antigenically. Bovine viral diarrhoea virus can infect both cattle and sheep, as well as other ruminants and pigs. Six distinct genotypes have recently been defined within the

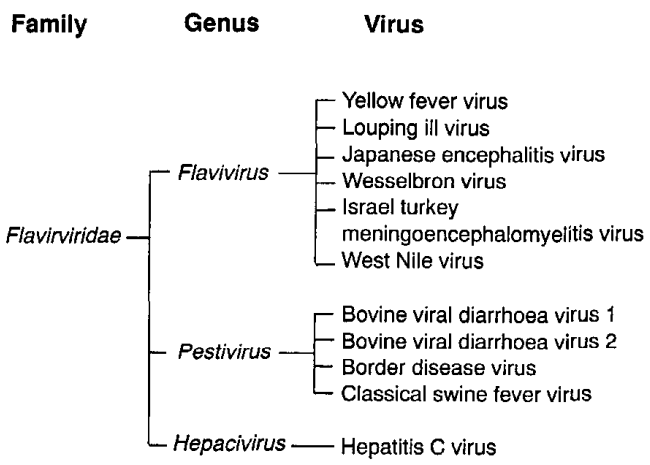


Figure 74.2 Viruses in the family *Flaviviridae*, with emphasis on those of veterinary importance.

genus on the basis of sequence differences in the gene coding for the envelope glycoprotein E2 (gp53) of various isolates (van Rijn *et al.*, 1997): classical swine fever virus, border disease virus, classical bovine viral diarrhoea virus (isolates predominantly from cattle), atypical bovine viral diarrhoea virus (isolates from cattle, sheep and pigs), deer pestivirus and giraffe pestivirus. Pestivirus infections may be inapparent, acute or persistent and are economically important worldwide.

Louping ill

The name 'louping ill' derives from the Scottish vernacular for 'leaping' or 'bounding', an allusion to the abnormal gait of some affected animals. Louping ill is a viral disease primarily of sheep. Although the virus is pathogenic for humans, infection is rare. The disease, which is largely confined to Britain and Ireland, has also been described in Norway, Spain, Bulgaria and Turkey. Isolates from Spain and Turkey are distinct from each other and also from isolates in Britain, Ireland and Norway (Marin *et al.*, 1995). Louping ill virus belongs to a group of serologically related viruses, the tick-borne encephalitis group or complex. The members of this group are distributed in northern temperate latitudes and are mainly human pathogens.

Epidemiology

Louping ill virus is transmitted by the tick *Ixodes ricinus* and the seasonal incidence and regional distribution of the disease reflect periods of tick activity in a suitable habitat such as upland grazing. Two main periods of tick activity occur, the first in spring and a second in late summer or early autumn. The host range of *I. ricinus* is wide and infection with louping ill virus can occur in many vertebrate species including sheep, cattle, horses, deer, red grouse and humans. Red grouse are particularly susceptible to infection with mortality reaching 80% in experimental infections. In areas where louping ill is

Table 74.1 Viruses of veterinary importance in the genera *Flavivirus* and *Pestivirus*.

Genus	Virus	Hosts	Comments
Flavivirus	Louping ill virus	Sheep, cattle, horses, red grouse and humans	Present in defined regions of Europe. Transmitted by the tick *Ixodes ricinus* and produces encephalitis in sheep and other species
	Japanese encephalitis virus	Waterfowl, pigs, horses and humans	Widely distributed in Asia. Transmitted by mosquitoes. Waterfowl are reservoir hosts. Infection in pigs results in abortion and neonatal mortality
	Wesselsbron virus	Sheep	Occurs in parts of sub-Saharan Africa. Transmitted by mosquitoes. Produces generalized infection, hepatitis and abortion
	Israel turkey meningo-encephalomyelitis virus	Turkeys	Reported in Israel and South Africa. Transmitted by mosquitoes. Progressive paresis and paralysis
	West Nile virus	Birds, humans, horses	Birds are the natural hosts. Transmitted by mosquitoes. Serious nervous disease reported sporadically in humans and horses
Pestivirus	Bovine viral diarrhoea virus types 1 and 2	Cattle (sheep, pigs)	Occurs worldwide. Causes inapparent infection, bovine viral diarrhoea and mucosal disease. Congenital infection may result in abortion, congenital defects and persistent infection due to immunotolerance
	Border disease virus	Sheep	Occurs worldwide. Infection of pregnant ewes may result in abortion and congenital abnormalities
	Classical swine fever (hog cholera) virus	Pigs	Highly contagious, economically important disease with high mortality. Generalized infection with nervous signs and abortion; congenital tremors in piglets

endemic, infection can result in a dramatic reduction in the population of red grouse. They are, therefore, not considered to be important maintenance hosts. Louping ill virus is maintained in endemic areas through a sheep-tick cycle. Trans-stadial but not transovarial transmission of the virus occurs in the tick. Rarely, contaminated instruments may be responsible for virus transmission. On farms where infection is endemic, losses occur mainly in sheep under two years of age. Most sheep acquire life-long immunity. Young lambs are protected by colostral antibody.

Pathogenesis

Viral replication occurs initially in lymph nodes draining sites of inoculation. Viraemia follows with dissemination to other lymphatic organs and sometimes to the brain and spinal cord. The speed and onset of the immune response are important in preventing spread of virus to the central nervous system. Immunosuppression caused by infection with *Ehrlichia phagocytophila*, the agent of tickborne fever, is considered to be responsible for increased mortality in sheep with louping ill.

Clinical signs

Following infection, sheep develop a febrile response which may go unnoticed. The temperature then returns to normal and, in a proportion of animals, rises again as neurological signs develop. These signs include hyperexcitability, fine muscular tremor, incoordination and exaggerated limb movements. Most affected animals develop convulsions prior to coma and death. Some sheep recover but exhibit mild, residual neurological signs. In cattle, the course of louping ill is more protracted. Affected animals may become recumbent but usually remain bright and many eventually recover. In humans, influenza-like clinical signs are followed, in most cases, by mild neurological disturbance.

Diagnosis

- A history of neurological signs or unexplained deaths in sheep in endemic areas during periods of tick activity may indicate louping ill. Laboratory confirmation is usually required.
- A non-suppurative encephalomyelitis is usually detectable histologically. Lesions are most pronounced in the brain stem and spinal cord. A specific diagnosis may be possible using an immunoperoxidase technique to detect viral antigen.
- Specimens of brain, collected aseptically into 50% glycerol saline, may be inoculated into tissue culture or intracerebrally into baby mice in order to isolate the virus.
- Antibody to the virus can be detected using complement fixation, gel diffusion, haemagglutination inhibition or IFA tests. The virus agglutinates goose red cells. The detection of IgM antibodies is indicative of acute infection.

Treatment

No specific treatment is available. Careful nursing and sedation may aid recovery.

Control

- Inactivated vaccines are protective. In the past, formalized vaccines produced from infected sheep brains were responsible, in some instances, for causing scrapie in vaccinated animals. Tissue culture-derived virus is now used for inactivated vaccines.
- Animals to be retained for breeding are vaccinated at six to 12 months of age. Colostrum from vaccinated ewes protects lambs during the first year of life. A booster injection may be advisable for ewes in their second pregnancy to enhance colostral antibody levels.
- Land improvement may help to reduce tick populations. Dipping of sheep may also reduce risk of infection.

Japanese encephalitis

This disease, which mainly affects humans, has a wide geographical distribution in Asia. Infection can occur in a number of animal species including horses and pigs. The virus is transmitted by mosquitoes and is maintained by a mosquito-aquatic bird cycle. The disease is of reduced importance in horses because of their declining numbers in endemic areas and the use of effective vaccines. The pig is an important amplifying host because of its close association with human populations in parts of Asia. Infection can cause reproductive failure in sows. Litters of infected sows may contain mummified and stillborn foetuses, weak piglets with neurological signs and clinically normal piglets. Confirmation is based either on virus isolation or on the demonstration of specific antibody. Both inactivated and live attenuated vaccines have been used to control the disease.

Wesselsbron disease

This disease, caused by a flavivirus, has a wide host range including domestic species, wild mammals and humans. However, clinical disease is usually encountered in sheep; infections in other species tend to be mild or subclinical. Infection in humans may result in febrile influenza-like symptoms. The virus is transmitted by mosquitoes. Infection is widespread in sub-Saharan Africa. The disease in sheep is similar to Rift Valley fever but is clinically less severe. It is characterized by abortion, neonatal mortality and congenital abnormalities such as hydranencephaly and arthrogryposis. The disease is most severe in newborn lambs which present with fever, depression, general weakness and polypnoea. Confirmation is based on virus isolation, intracerebral inoculation of baby mice and the demonstration of specific antibody. An attenuated vaccine which provides life-long

immunity is available. Pregnant animals should not be vaccinated because of the risk of abortion.

Bovine viral diarrhoea and mucosal disease

Infection with bovine viral diarrhoea virus (BVDV), also known as bovine virus diarrhoea virus, is common in cattle populations throughout the world. The virus can cause both acute disease, bovine viral diarrhoea (BVD), and a protracted form of illness, mucosal disease, arising from persistent infection. Using cell culture, cytopathic and non-cytopathic biotypes are recognized. The biotype most often isolated from cattle populations is non-cytopathic. Cytopathic isolates can arise from non-cytopathic BVDV as a result of recombination events including incorporation of host RNA and duplication of viral RNA sequences (Meyers *et al.*, 1996). Two genotypes now considered separate species, BVDV 1 (classical BVDV isolates) and BVDV 2 (atypical BVDV isolates), are recognized on the basis of differences in the 5' untranslated region of the viral genome. Both genotypes contain cytopathic and non-cytopathic isolates, and produce similar clinical syndromes in cattle. However, only type 2 isolates have been associated with thrombocytopenia and a haemorrhagic syndrome, first described in North America (Rebuhn *et al.*, 1989). Isolates of BVDV used in vaccines and for diagnostic tests generally belong to the type 1 genotype.

Epidemiology

When cattle are infected initially with BVDV, they shed virus for a short period and may transmit virus to other animals. Persistently-infected animals, which shed virus in secretions and excretions, are particularly important sources of infection. Persistent infection develops when infection of the foetus with a non-cytopathic strain occurs before day 120 of gestation. About 1% of animals in an infected population are persistently-infected and viraemic. Although persistently-infected cows may breed successfully, they can transmit virus transplacentally to calves during successive gestations. This form of disease transmission is relatively common. The presence of cattle with persistent infection in a herd results in constant exposure of the other cattle to virus, producing a high level of herd immunity. In such herds, more than 80% of animals are serologically positive.

Virus is excreted in the semen of both persistently-infected and transiently-infected bulls. Infection may be transmitted through natural service or by artificial insemination. Embryo transfer from animals with persistent or transient infection can lead to infection in recipient cows. If pregnant animals are inoculated with live vaccines, their calves may develop persistent infection. Due to the instability of the virus, indirect transmission rarely occurs through farm workers, equipment and biting flies. Although cattle are the primary hosts, the virus can infect most even-toed ungulates. Interspecies spread of bovine and ovine pestiviruses has been demonstrated under natural conditions but its epidemiological significance is uncertain.

Pathogenesis

The virus is usually acquired by the oronasal route and initial replication occurs in the oronasal mucosa. In the subsequent viraemia, virus spreads throughout the body either free in the serum or in association with leukocytes. Both B and T lymphocyte numbers decrease. As the virus has an immunosuppressive effect, infection may predispose calves to respiratory and enteric disease. The outcome of transplacental spread depends on the age of the foetus at the time of infection. During the first 30 days of gestation, infection may result in embryonic death with return of the dam to oestrus. The effects of foetal infection between 30 and 90 days of gestation include abortion, mummification and congenital abnormalities of the CNS, often cerebellar hypoplasia. Foetuses, which become infected after day 120 of gestation, can mount an active immune response and are usually normal at birth. If virus invades the foetus before the development of immune competence, immunotolerance to the agent develops with persistent infection for the lifetime of the animal. The virus involved in this persistent infection is non-cytopathic. Later, usually between six months and two years of age, a cytopathic biotype emerges as a consequence of mutation of the non-cytopathic virus or of recombination with nucleic acid of the host cell or other non-cytopathic biotypes. These may lead to the development of mucosal disease in some animals.

Cytopathic isolates differ from their non-cytopathic counterparts by producing an 80 kDa non-structural protein (NS3). The role of NS3 in the pathogenesis of mucosal disease is unclear. Cytopathic isolates have a particular tropism for gut-associated lymphoid tissues.

Clinical signs

Most BVDV infections are subclinical. Outbreaks of BVD are usually associated with high morbidity and low mortality. When present, clinical signs include inappetence, depression, fever and diarrhoea. Significant mortality has been described in some outbreaks of BVD (David *et al.*, 1994). Peracute BVD is characterized by high fever, severe diarrhoea and dehydration. Ulceration of the oral mucous membrane and the epithelia of the interdigital cleft and coronary band may also be present. In some cases thrombocytopenia results in bloody diarrhoea, epistaxis and petechiae in the mouth, conjunctiva and sclera.

Although a significant proportion of persistently-infected animals are clinically normal, some are born undersized and demonstrate retarded growth rate and poor viability. Increased susceptibility to enteritis and pneumonia has been reported. Mucosal disease is usually sporadic in occurrence. The condition affects persistently-infected

animals, usually between six months and two years of age. Clinical signs include depression, fever, profuse watery diarrhoea, nasal discharge, salivation and lameness. Ulcerative lesions are present in the mouth and interdigital clefts. Case fatality rate is 100%; death usually occurs within weeks of the onset of clinical signs. A few animals may survive for several months before dying from severe debilitation.

Diagnosis

A tentative diagnosis may be possible on the basis of clinical signs and pathological findings. Laboratory confirmation requires demonstration of antibody, viral antigen or viral RNA. Seroconversion and the presence of viraemic animals are necessary for confirmation of established infection in a herd.

- Specimens suitable for laboratory examinations include the buffy coat from whole blood, spleen, lymph node and lesions from the gastrointestinal tract.
- Virus can be isolated in cell cultures. Sequential samples taken three weeks apart should be used to confirm persistent infection. Before foetal calf serum is used in cell culture medium, it should be screened for the presence of virus and antibody.
- Viral antigen can be detected in frozen sections or buffy coat smears by immunofluorescence. An ELISA or immunoperoxidase technique may also be used.
- Dot blot, *in situ* hybridization and PCR techniques for the detection of viral RNA have been described.
- Virus neutralization and ELISA are the most commonly used methods for the detection of antibodies to BVDV. Demonstration of a four-fold increase in antibody titre of paired serum samples is necessary to confirm recent infection.

Treatment and control

Supportive therapy may be of benefit in outbreaks of bovine viral diarrhoea. Treatment of animals with mucosal disease is of no benefit. Most losses arising from BVDV infections in herds result from the effects of prenatal infections and mucosal disease. Control strategies are directed at preventing infections which can lead to the birth of persistently-infected animals.

- Killed, attenuated live and temperature-sensitive mutant virus vaccines have been developed. Live vaccines may cause foetal infections and immunosuppression. In addition, they may precipitate mucosal disease in some persistently-infected animals. Killed vaccines may be used in pregnant animals but require regular boosters to maintain protection. Vaccines produced from a single strain or genotype of virus may not be fully protective due to antigenic variation, a feature of BVDV isolates. Vaccines have generally been evaluated for their ability to prevent acute disease and manufacturers do not claim that vaccination prevents foetal infection (van Campen and Woodard, 1997). Exposure of replacement stock before breeding commences to a persistently-infected animal may help to maintain herd immunity.
- The elimination of BVDV from a herd requires the identification and removal of persistently-infected animals. The dam, sire and progeny of persistently-infected animals should be tested since the virus can be passed from parent to offspring.
- Herd immunity wanes following the removal of persistently-infected animals. Therefore, all newly-acquired cattle should be tested before being introduced to the herd.
- Systematic testing of bulk milk or pooled blood samples for antibodies is important in national eradication programmes.

Border disease

This congenital disorder of lambs, also known as hairy shaker disease, occurs worldwide. Border disease, which was first reported from the Welsh-English border, is caused by infection of the foetus with a non-cytopathic pestivirus. Border disease virus (BDV) is closely related to bovine viral diarrhoea virus and it has been suggested that they may be a single species. Pestivirus isolates from sheep can infect other domestic ruminants and pigs. Moreover, pestivirus isolates from a number of domestic species can infect pregnant sheep causing border disease in their offspring.

Epidemiology

Persistently-infected animals shed virus continuously in excretions and secretions. These animals tend to have a low survival rate under field conditions, although some may survive for several years without developing clinical signs. Persistently-infected ewes may give birth to persistently-infected lambs. Acute infections in susceptible sheep are transient and result in immunity to challenge with homologous strains of BDV. Infected rams shed virus in semen and may infect susceptible ewes. In addition to sheep-to-sheep contact, transmission can occur through contaminated needles during flock vaccination. Other ruminant species shedding pestivirus are potential sources of infection for sheep.

Pathogenesis

Virus is probably acquired by the oronasal route. In susceptible pregnant ewes, infection results in placentitis and invasion of the foetus. The immune response of the ewe does not protect the developing foetus. The age of the foetus at the time of infection ultimately determines the outcome. The foetus develops immune competence between 60 and 80 days of gestation. Foetal death may follow infection prior to the development of immune competence, the outcome being resorption, abortion or mummification. Foetuses which survive become

immunotolerant and remain persistently infected. These animals may be clinically normal at birth or may display tremors and hairy birthcoat, consequences of viral interference with organogenesis. Congenital defects in affected lambs include skeletal growth retardation, hypomyelinogenesis and enlarged primary hair follicles with reduced numbers of secondary follicles. Infection after day 80 of gestation induces an immune response with elimination of the virus and the birth of a healthy lamb. Foetal infection during mid-gestation when the immune system is developing may result in lesions in the central nervous system including cerebral cavitation and cerebellar dysplasia. Immune-mediated reactions have been suggested as the possible explanation for these severe lesions. Some persistently-infected sheep may develop a condition similar to mucosal disease of cattle. Cytopathic isolates of BDV have been recovered from the intestines of such affected animals.

Clinical signs

In flocks infected with BDV, there may be an increase in the number of abortions and weak neonatal lambs. Characteristic signs of infection in newborn lambs include altered body conformation, changes in fleece quality and tremors. Hairs projecting above the wool, particularly along the neck and back, impart a halo effect that is most noticeable in fine coated breeds. Affected lambs are often small and their survival rate is poor. The survival rate is influenced not only by the severity of the neurological dysfunction but also by the standard of animal care. In well-nursed lambs, the neurological signs gradually abate and such animals may eventually become clinically normal.

Diagnosis

- The characteristic clinical signs are diagnostic.
- Dysmyelination may be demonstrable histologically in the central nervous system. Immunocytochemical staining can be used to demonstrate virus in brain tissue.
- Virus isolation is possible in susceptible bovine or ovine cell lines. Immunocytochemical staining is used to demonstrate the presence of non-cytopathic virus.
- Samples suitable for virus isolation include whole blood and tissues from affected lambs. Pre-colostral blood from lambs is preferable because antibody acquired from colostrum may interfere with virus isolation.
- Viral antigen can be detected by immunofluorescence staining of frozen sections or by immunoperoxidase staining of fixed sections. An ELISA is used for detecting viral antigen in blood.
- Serological testing, employing methods such as serum neutralization and ELISA, can be used to determine the extent of infection in a flock.

Control

Control should be based on identification and removal of persistently-infected animals and precautions to avoid introduction of infected animals into a flock. Where such a policy is not feasible, breeding stock should be deliberately mixed with persistently-infected animals at least two months before mating. A commercial inactivated, adjuvanted vaccine is available, containing BDV and BVDV-1 (Nettleton *et al.*, 1998).

Classical swine fever (Hog cholera)

This highly contagious, potentially fatal disease of pigs, although still present in many countries, has been eradicated from North America, Australia and most European countries. It is classed as a List A disease by the Office International des Epizooties. In recent years, sporadic outbreaks have occurred in the United Kingdom, Italy, Belgium, the Netherlands and Germany. Isolates of classical swine fever virus (CSFV), the causal agent of the disease, can be placed in two major groups on the basis of nucleotide sequence data. Recent European isolates, placed in Group 2, are distinct from those which caused swine fever outbreaks during the 1940s and 1950s (Lowings *et al.*, 1996). Isolates, although conforming to a single major antigenic type and being mostly non-cytopathic, differ substantially in virulence.

Epidemiology

Pigs, both domestic and feral, are the natural hosts of CSFV and direct contact between infected and susceptible animals is the main means of transmission. In endemic areas, the disease is spread principally by movement of infected pigs. Shedding of virus may begin before clinical signs become evident. Virulent virus is shed in all excretions and secretions until the time of death at about 20 days post infection. Virus strains of moderate virulence may result in chronic infection with continuous or intermittent shedding by infected pigs. In addition, congenital infections with strains of low virulence may result in the birth of persistently-infected piglets. Although infected animals have been found in the wild boar populations in Europe, their significance as reservoirs of infection is unclear. Spread between premises can occur indirectly, particularly in regions with a high density of pig farms. The virus can be transmitted mechanically by personnel, vehicles and biting arthropods. The virus, which is relatively fragile and does not persist in the environment, is not spread over long distances by air movement. Despite its lability, CSFV can survive for long periods in protein-rich biological materials such as meat or body fluids, particularly if chilled or frozen. Although legislation is in place in most European countries prohibiting the feeding of uncooked swill, recent outbreaks of classical swine fever can still be traced to waste food fed to pigs.

Pathogenesis and pathology

Pigs are usually infected by the oronasal route. The tonsil is the primary site of viral multiplication. Virus spreads to regional lymph nodes and viraemia develops after further viral multiplication. Virus, which has an affinity for vascular endothelium and reticuloendothelial cells, can be isolated from all major organs and tissues. In acute swine fever, vascular damage in conjunction with severe thrombocytopenia results in widespread petechial haemorrhages. A non-suppurative encephalitis with prominent perivascular cuffing is present in most CSFV-infected pigs. Virus strains of reduced virulence can cause a mild form of the disease. In pregnant sows, virus may be transmitted to foetuses. The outcome of transplacental infection is determined by the age of the foetus and the virulence of the invading strain of virus. Infection early in gestation results in foetal death with resorption or abortion. *In utero* infection may also result in stillbirths, weak newborn piglets with congenital tremors and, occasionally, clinically normal piglets. Piglets with immune tolerance to the virus remain persistently-infected and excrete virus continuously. Animals, which are clinically normal when born, may subsequently develop late-onset disease. The factors which precipitate late-onset disease are unclear.

Clinical signs

Following an incubation period of up to ten days, affected animals develop high fever and become inappetent and depressed. Sick pigs are inclined to huddle together. Vomiting and constipation are followed by diarrhoea. Some animals may die soon after developing convulsions. A swaying gait usually precedes posterior paresis. Most cases of acute classical swine fever succumb within 20 days after infection.

Signs of disease are milder in infections caused by strains of low virulence. Partial recovery from an initial phase of acute illness may be followed by relapse and death. Some pigs may survive for several months but exhibit marked growth retardation. Abortion, mummification, malformations and stillbirths may be encountered in breeding herds. Live-born infected piglets, often exhibiting tremors, may die soon after birth. Some affected piglets may present with haemorrhages in the skin. Congenital malformations include deformities of the head and limbs and cerebellar hypoplasia.

Diagnosis

Although clinical signs and history may provide evidence for a tentative diagnosis, laboratory confirmation is essential, particularly with infections caused by strains of reduced virulence.

- In acute disease, haemorrhages are present in many internal organs and on serosal surfaces. Petechiae are often present on kidney surfaces and in lymph nodes. Other gross pathological features of diagnostic significance are splenic infarction and 'button' ulcers in the mucosa of the terminal ileum near the ileocaecal valve.
- Rapid confirmation is possible using direct immunofluorescence on frozen sections of tonsillar tissue, kidney, spleen, distal ileum and lymph nodes. As pigs can be infected with BVDV, monoclonal antibodies specific for CSFV may be required to reach a definitive diagnosis.
- Virus isolation can be carried out in porcine cell lines using homogenates of spleen and tonsil. As most isolates are non-cytopathic, immunostaining is necessary for the demonstration of viral antigen.
- Serological testing is useful on farms infected with strains of low virulence or for serological surveys. Virus neutralization and ELISA are the tests most widely used. A blocking ELISA has been developed for distinguishing CSFV from BVDV (Wensvoort *et al.*, 1988).

Control

- The disease is notifiable in many countries which have adopted slaughter policies and banned vaccination. Pigs and pig products should not be imported from countries where infection with CSFV is present. Swill must be boiled before being fed to pigs.
- In countries where the disease is endemic or during the early stages of an eradication programme, vaccination may be used. Live vaccines attenuated either by serial passage in rabbits (Chinese strain) or in tissue culture (Japanese guinea pig strain or French Thiverval strain) are currently used. These vaccines are safe and effective. Vaccinated animals cannot be distinguished serologically from naturally-infected animals. The recently-developed recombinant E2 marker vaccine used in conjunction with a specific ELISA capable of detecting antibodies to the other main envelope glycoprotein, E^{rns}, may offer a way of distinguishing vaccinated from naturally infected pigs (Baars *et al.*, 1998).

References

Baars, J., Bonde Larsen, A. and Martens, M. (1998). Porcilis pestis: the missing link in the failing non-vaccination policy for classical swine fever. *The Pig Journal*, **41**, 26–38.

David, G.P., Crawshaw, T.R., Gunning, R.F. *et al.* (1994). Severe disease in adult dairy cattle in three UK dairy herds associated with BVD virus infection. *Veterinary Record*, **134**, 468–472.

Lowings, P., Ibata, G., Needham, J. and Paton, D. (1996). Classical swine fever virus diversity and evolution. *Journal of General Virology*, **77**, 1311–1321.

Marin, M.S., McKenzie, J., Gao, G.F. *et al.* (1995). The virus causing encephalomyelitis in sheep in Spain: a new member of the tick-borne encephalitis group. *Research in Veterinary*

Chapter 75

Togaviridae

Viruses in the family *Togaviridae* (Latin *toga*, cloak) are enveloped RNA viruses, approximately 70 nm in diameter, with icosahedral symmetry. The envelope, which contains glycoprotein spikes, is closely bound to an icosahedral capsid (Fig. 75.1). Togaviruses agglutinate goose and chick erythrocytes. There are two genera, *Alphavirus* and *Rubivirus*, in the family. The sole member of the genus *Rubivirus* is rubella virus, which causes German measles in children and young adults.

The genus *Alphavirus* includes more than 25 species, a number of which are important animal pathogens. Alphaviruses are divided, on the basis of genomic composition, into a number of groups including Venezuelan equine encephalitis virus (VEEV) complex, eastern equine encephalitis virus (EEEV) complex, Semliki forest virus complex and western equine encephalitis virus (WEEV) complex. Western equine encephalitis virus has been shown to have arisen by recombination between EEEV and Sindbis-like viruses, probably between 1,300 and 1,900 years ago (Weaver *et al.*, 1997).

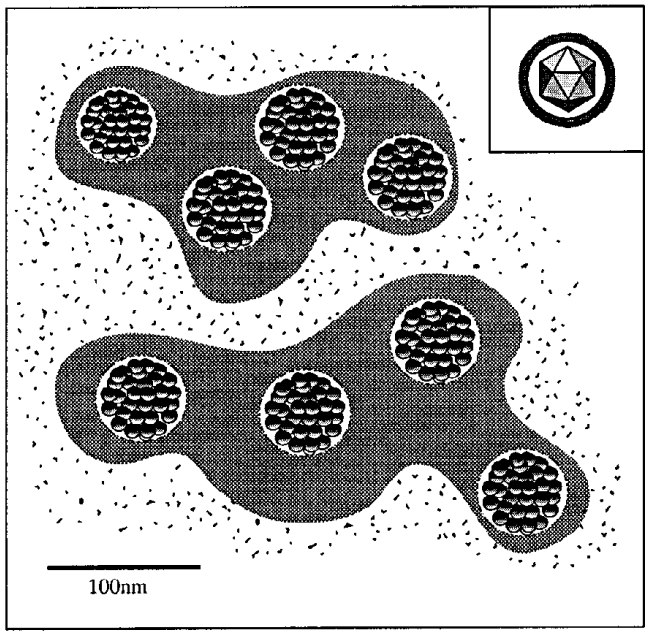


Figure 75.1 Togavirus particles as they appear in an electron micrograph and a diagrammatic representation (inset).

Key points

- Enveloped, RNA viruses with icosahedral symmetry
- Replicate in cell cytoplasm
- Labile in the environment
- Genus *Alphavirus*
  - —arthropod-borne
  - —cause eastern equine encephalitis, western equine encephalitis and Venezuelan equine encephalitis
  - —major cause of zoonotic infections

Replication of alphaviruses, which contain positive-sense single-stranded RNA, occurs in the cytoplasm and nucleocapsids are assembled in the cytosol. In vertebrates, alphavirus infection results in cytolysis. The viral envelope is acquired as the nucleocapsid buds into the plasma membrane which contains virus-derived glycoprotein spikes. Viral infection of invertebrate cells is usually non-cytolytic and is persistent. In this instance virus assembly occurs in association with intracellular membranes rather than through the plasma membrane.

Mature virions of alphaviruses are sensitive to pH changes, heat, detergents and disinfectants, and are not stable in the environment. Alphaviruses, in common with certain members of the *Flaviviridae*, *Reoviridae*, *Rhabdoviridae* and *Bunyaviridae*, are termed arboviruses indicating that they are **ar**thropod-**bo**rne. This term, however, has no taxonomic significance.

Clinical infections

Domestic animals and humans are usually considered to be 'dead-end' hosts of alphaviruses because they do not develop a sufficiently high titre of circulating virus to act as reservoir hosts. A number of important equine diseases are caused by infection with members of the genus *Alphavirus* (Table 75.1). The three equine encephalitis viruses (Venezuelan, eastern and western), which are confined to the western hemisphere, are transmitted by mosquitoes. Getah virus occurs mainly in south-east Asia and in Australia. A number of outbreaks of disease caused by this virus have been recorded in Japan.

Table 75.1 Alphaviruses of veterinary significance.

Virus	Vector	Comments
Eastern equine encephalitis virus	Mosquito (*Culiseta melanura*, *Aedes* species)	Infection endemic in passerine birds which frequent freshwater swamps of eastern North America, Caribbean Islands and parts of South America. Causes disease in horses, humans and pheasants
Venezuelan equine encephalitis virus	Mosquito (*Culex* species)	Infection endemic in small mammals in Central and South America. Causes outbreaks of disease in horses, donkeys and humans in endemic regions, occasionally spreading to southern USA
Western equine encephalitis virus	Mosquito (*Culex tarsalis* and other *Culex* species, *Aedes* species)	Infection of passerine birds widespread in the Americas. Causes mild disease in horses and humans
Getah virus	Mosquito	Causes sporadic disease in horses in south-east Asia and Australia characterized by fever, urticaria and oedema of the limbs. Subclinical infection occurs in pigs

Equine encephalitides

The viruses which cause Venezuelan, eastern and western encephalitis are important in the Americas. These three viruses produce similar clinical signs although infections caused by the virus of western equine encephalitis tend to be milder.

Epidemiology

The equine encephalitides share some common epidemiological features. The peak periods of these diseases coincide with times of maximum vector numbers usually in late summer following heavy rainfall. The regional distribution of the viruses is related to that of mosquito vectors. Numbers of clinical cases drop dramatically when vector numbers decrease due to cold or drought.

Eastern equine encephalitis virus (EEEV) occurs principally in Atlantic coast areas of North America. However, EEEV has also been isolated in Michigan, the Caribbean islands and South America. The virus is maintained in cycles of infection involving passerine birds and the irrigation ditch mosquito (*Culiseta melanura*), which inhabits freshwater swamps (Fig. 75.2). Following infection, a high titre of virus develops in many wild birds without evidence of disease. High mortality rates have, however, been recorded in pheasants, emus and whooping cranes. Virus can be transmitted between pheasants by pecking and cannibalism. Periodic epidemic outbreaks of infection in wild birds, which may lead to infection of humans and horses, involve additional mosquito species such as *Aedes sollicitans* and *Coquillettidia perturbans* which feed both on birds and on mammals. Infection usually results in sporadic disease in man, horses and pheasants. Epizootics, which tend to occur in the autumn, disappear with the arrival of the first frosts. Overwintering mechanisms for virus maintenance are unclear, although wild birds are considered to be possible reservoirs. Transovarial transmission in mosquitoes has not been demonstrated.

Isolates of Venezuelan equine encephalitis virus (VEEV) comprise a complex of six subtypes. Epidemic forms of Venezuelan equine encephalitis (VEE) are caused by two highly virulent subtype I serotypes of the virus (I-AB and I-C). Other subtypes are considered to be either non-pathogenic or of low pathogenicity for horses. The viruses are maintained in sylvatic cycles involving rodents and mosquitoes (*Culex* species) in swampy habitats. Phylogenetic studies suggest that the viruses implicated in epizootics are derived from mutation of viruses involved in enzootic cycles (Weaver *et al.*, 1992). Epizootics of VEE occurred regularly between 1962 and 1972 in the northern part of South America and in Central America, extending at one stage as far as Texas. Following a quiescent period of 20 years, a limited outbreak of disease occurred in Venezuela in 1992 followed by an extensive epizootic in Venezuela and Columbia in 1995 (Weaver *et al.*, 1996). Horses inoculated with a virulent subtype of VEEV develop a viraemia of sufficiently high titre to allow transmission by feeding mosquitoes.

Although western equine encephalitis (WEE) has traditionally occurred in the USA west of the Mississippi, it is also present in many other parts of the American continent. Infections tend to recur in certain areas. The cycle of infection involves mosquitoes, usually *Culex tarsalis*, and indigenous wild birds in which infection is inapparent. Horses are infected incidentally and are 'dead-end' hosts because levels of virus in the blood remain low. Epizootics are rare. Overwintering mechanisms of the virus are unclear but may involve birds, reptiles or mosquitoes.

Pathogenesis

Following inoculation by a feeding mosquito, viral replication occurs near the site of entry and in the regional lymph nodes. Viraemia, ranging from barely detectable to high levels, is accompanied by fever. When disease is severe, the virus invades the central nervous system resulting in neuronal necrosis and perivascular lymphoid cuffing.

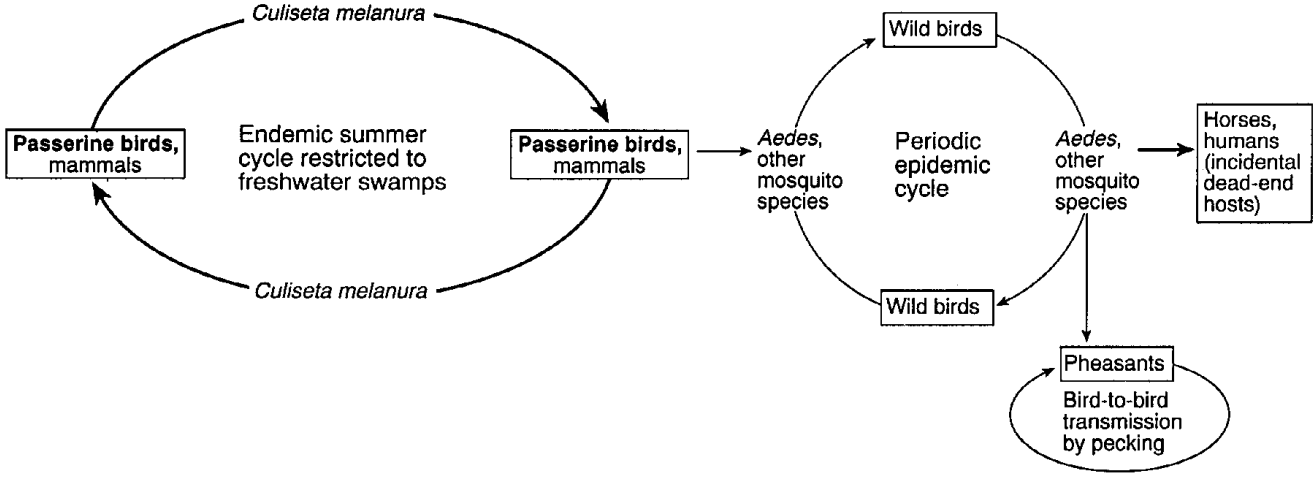


Figure 75.2 Endemic and epidemic transmission cycles of eastern equine encephalitis virus in North America. The endemic cycle which occurs in summer, is restricted to swampy regions. The epidemic cycle, which occurs periodically, involves mosquito species and wild birds not usually associated with swamps. Infection in horses and humans, which are 'dead-end' hosts, is a 'spill over' from the epidemic cycle. In farmed pheasants, bird-to-bird transmission may occur through pecking.

Clinical signs

Diseases caused by the three equine encephalitis viruses are clinically similar. The incubation period may be up to nine days. Clinical signs, which usually last from four to nine days, range from mild fever and depression to fatal febrile encephalomyelitis. Neurological signs include photophobia, blindness, head pressing, circling, ataxia and inability to swallow. Affected horses exhibit severe depression with low carriage of the head and a wide-based stance. Terminally, animals become recumbent and semi-comatose with convulsions prior to death. The case fatality rate is 90% for EEE, 50%-80% for VEE and 20%-40% for WEE.

Diagnosis

Clinical signs along with a history of previous cases of equine encephalitis in the same region may be suggestive of the disease. However, laboratory confirmation is usually required. Due to the possibility of human infection, care must be taken during specimen collection.

- Virus isolation provides a definitive diagnosis. Isolation is carried out in cell culture or in suckling mice. Whole blood or serum, collected during the pyrexic phase of the disease, is suitable for virus isolation. Brain or cerebrospinal fluid can be collected at postmortem. When VEE is suspected, isolates should be typed in order to distinguish virulent from non-virulent subtypes.
- An immunohistochemical staining technique for the detection of EEEV antigen in fixed brain sections has been described (Patterson *et al.*, 1996).
- Diagnosis of WEE or EEE is usually based on serology. Paired serum samples should be collected to demonstrate a rising titre. Suitable testing methods include ELISA, plaque reduction neutralization assay, haemagglutination inhibition and complement fixation. An IgM capture ELISA has been used to provide evidence of infection in single serum samples. The vaccination status of an animal must be considered in interpreting the results of serological tests. The interpretation of serological results for VEEV is complicated by the presence of antibodies produced in response to inapparent infections with non-virulent subtypes.

Treatment and control

Although supportive palliative treatments may be beneficial, the prognosis is generally poor. Control is based on vaccination of horses and implementation of measures aimed at reducing mosquito populations.

- Monovalent, bivalent and trivalent vaccines are available. Vaccines for EEE and WEE are inactivated. A live attenuated TC-83 VEEV vaccine provides effective protection and has been used successfully to prevent epizootics of VEE.
- Vector control measures include spraying of vector habitats, destruction of mosquito breeding areas, use of insect repellants and stabling of horses at night in insect-proof buildings.

References

Patterson, J.S., Maes, R.K., Mullaney, T.P. and Benson, C.L. (1996). Immunohistochemical diagnosis of eastern equine encephalomyelitis. *Journal of Veterinary Diagnostic Investigation*, **8**, 156–160.

Weaver, S.C., Bellew, L.A. and Rico-Hesse, R. (1992).

Chapter 76

Prions: unconventional infectious agents

At present, conventional infectious agents have not been implicated aetiologically in the transmissible spongiform encephalopathies (TSEs), a unique group of neurodegenerative diseases. Intensive research efforts to elucidate the cause of TSEs have been inconclusive. It has been proposed that these diseases are caused by unconventional infectious agents termed prions (Prusiner, 1982). These infectious agents are 'unconventional' because they appear to be devoid of nucleic acid, unlike viruses and other microbial agents. In addition, they are non-immunogenic and are extremely resistant to inactivation by heating, exposure to chemicals and irradiation. The 'prion theory' proposes that they are derived from a native glycoprotein (Prusiner *et al.*, 1999). This native glycoprotein, PrP^{C}, (cellular prion protein) is associated with the plasma membrane of many cell types particularly neurons and lymphocytes. Structurally, PrP^{C} is composed of more α-helices than β-sheets. Following exposure to abnormal prion protein (PrP^{Sc}, scrapie prion protein), PrP^{C} is altered post-translationally to a structure similar to that of the PrP^{Sc} in which the β-sheet content predominates. As more PrP^{C} is converted to PrP^{Sc}, this protease-resistant molecule gradually accumulates especially in the long-lived cells of the CNS (Fig. 76.1). Recent studies suggest that PrP^{Sc} is formed from PrP^{C} on cell membranes in caveolae-like structures before these fuse with endosomes. During normal metabolic turnover in cells, most membrane glycoproteins are transported in endosomes to lysosomes for degradation. However, because it is protease-resistant, PrP^{Sc} accumulates in cytoplasmic vesicles particularly lysosomes (Prusiner *et al.*, 1999).

The mechanism by which PrP^{Sc} induces the structural alteration in native PrP^{C} has not yet been defined. However, the newly formed PrP^{Sc} closely resembles in its three dimensional structure the 'infecting' PrP^{Sc}, implying that the latter has a central role in the initiation of the chain reaction which results in the intracellular accumulation of large amounts of PrP^{Sc}.

Experimentally, mild acidification and reduction procedures induce structural rearrangement in PrP^{C} producing a highly soluble monomeric form of PrP (β-PrP) which is rich in β-sheets (Jackson *et al.*, 1999). This β-PrP can revert to an α-configuration or, alternatively, it can act as a stable 'seed' inducing polymerization with the *in vitro* formation of insoluble fibrillar structures similar to the prion fibrils which can be recovered from detergent-treated extracts of the brains of animals with TSEs. These experiments demonstrate that binding of several β-PrP molecules together can lead to the formation of an irreversible β-sheet configuration, the probable basis of PrP^{Sc} accumulation in TSEs.

Key points

- Prions are proteinaceous particles apparently devoid of nucleic acid
- Aetiologically implicated in the transmissible spongiform encephalopathies, fatal neurodegenerative diseases with long incubation periods
- Neuropathological changes, which include vacuolation of both neurons and neuropil without evidence of an inflammatory response, are associated with the accumulation of abnormally-folded host-derived prion protein
- Transmissible spongiform encephalopathies include:
  - —Scrapie in sheep
  - —Bovine spongiform encephalopathy
  - —Feline spongiform encephalopathy
  - —Transmissible mink encephalopathy
  - —Kuru and Creutzfeldt-Jakob disease in humans

The formation of PrP^{Sc} from PrP^{C} in TSEs may be initiated following exposure to an external source of PrP^{Sc}, usually by ingestion (Fig. 76.2). Rarely, random spontaneous conversion of native PrP^{C} to PrP^{Sc} may initiate the process in an individual. A third mechanism which predisposes to configurational change in PrP^{C} relates to mutation in the *PrP* gene as occurs in the Gerstmann-Sträussler-Scheinker syndrome in humans.

The *PrP* gene of an infected animal determines the primary amino acid sequence of the prion protein in that animal. The resistance of some species to infection by prions derived from another species is termed the 'species barrier'. This barrier is attributed to differences between the amino acid sequences of the prion proteins in the two species. On initial transfer of PrP^{Sc} between species, the

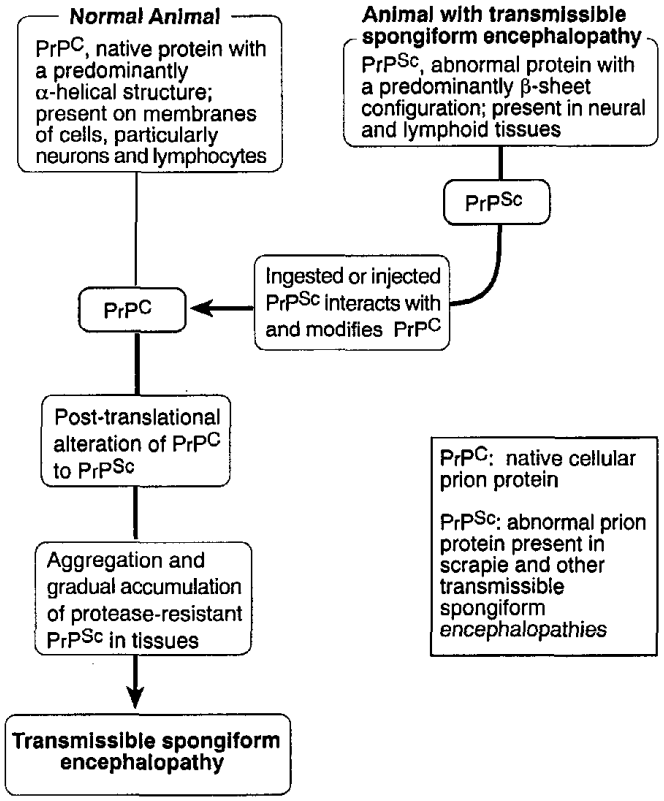


Figure 76.1 Outline of the proposed mechanisms involved in the pathogenesis of transmissible spongiform encephalopathies

incubation period tends to be relatively long. Subsequent transfer between members of the recipient species leads to shorter incubation periods. The presence of a 'species barrier' may explain the resistance of humans to infection with PrP^{Sc} derived from sheep with scrapie. Prion strains from infected sheep have been described on the basis of bioassays in mice. Strain differentiation is based on incubation periods, mortality patterns, lesion profiles and titratable infectivity in the brains of mice of known genotypes. Prion diversity is thought to be determined by the conformation and glycosylation patterns of PrP^{Sc} and may account for the observed differences in prion strains. The 'protein only' theory of prion composition is contested (Chesebro, 1998). The existence of strains has been cited as evidence which supports a requirement for nucleic acid in prions.

Prions are stable at a wide pH range and are remarkably resistant to most methods of biological inactivation. Early evidence of the resistance of these agents to chemical inactivation was provided by retrospective studies on 18,000 sheep, which had been inadvertently exposed to infection when inoculated with a formalized louping ill vaccine. The vaccine, prepared from brains, spinal cords and spleens of sheep, contained the scrapie agent which was not inactivated by treatment with formaldehyde (Greig, 1950). Approximately 10% of the vaccinated animals developed scrapie. Treatment of prions with alcohols and aldehydes that fix proteins may help to stabilize rather than inactivate these agents. Physical methods for inactivating the agents of bovine spongiform encephalopathy (BSE) and scrapie have been intensively investigated because safe disposal of infected carcases is mandatory. Although autoclaving at temperatures of 132°C has been recommended, it does not ensure prion inactivation. Paradoxically, autoclaving at temperatures above 138°C may increase the resistance of prions to inactivation. It has recently been shown that high concentrations of sodium hypochlorite or hot solutions of sodium hydroxide inactivate a thermostable strain of scrapie agent (Taylor, 2000). The inclusion of a formic acid step in formaldehyde fixation of brain tissue reduces the infectivity of the agents of scrapie, BSE and Creutzfeldt-Jacob disease without adversely affecting the quality of histological sections.

Clinical infections

Diseases attributed to prions occur sporadically and are significantly influenced by the genome of the affected animal. These slowly progressive neurodegenerative diseases, which are characterized by long incubation periods and spongiform changes in the brain, have been described in many animal species and in humans. Transmissible spongiform encephalopathies have been recognized in both ruminants and carnivores (Table 76.1).

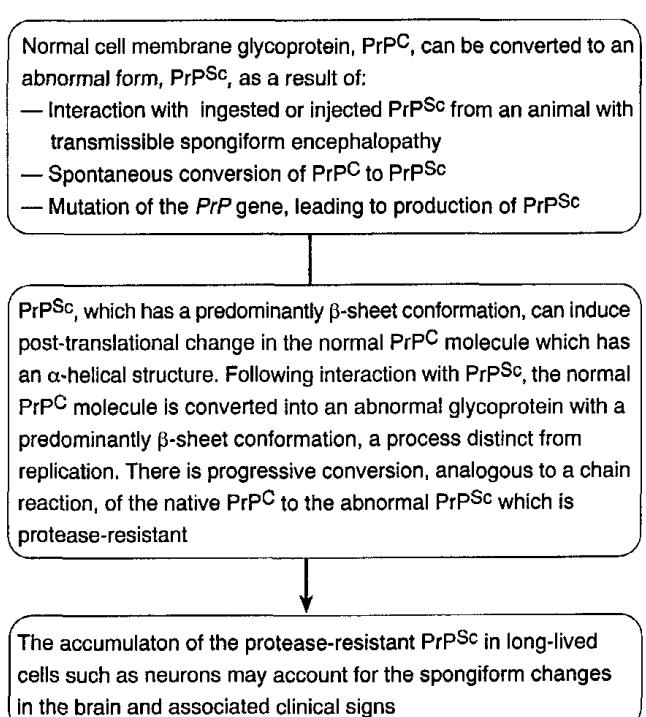


Figure 76.2 An outline of the probable mechanisms which lead to the accumulation of PrP^{Sc} in neurons.

Table 76.1 Transmissible spongiform encephalopathies of animals.

Disease	Comments
Scrapie	Recognized in sheep in parts of Europe for 300 years; apart from Australia and New Zealand, now occurs worldwide. Occurs also in goats
Bovine spongiform encephalopathy	First reported in England in 1986; developed into a major epidemic over a ten year period. Prevalence declined with the implementation of effective control measures. Occurs at lower frequency in many other European countries
Feline spongiform encephalopathy	First recorded during the bovine spongiform encephalopathy epidemic in the early 1990s. Most cases reported in the UK
Transmissible mink encephalopathy	First recognized in caged mink in Wisconsin in 1947; attributed to the feeding of scrapie-infected sheep meat
Spongiform encephalopathy in captive ruminants	First recorded during the bovine spongiform encephalopathy epidemic in 1986. Reported in greater kudu, nyala, oryx and some other captive ruminants in zoological collections
Chronic wasting disease	First recognized in captive mule deer in Colorado in 1980. Occurs in deer and elk populations in the wild in North America

Similarities in the neuropathological features of scrapie in sheep and kuru in humans suggested that the two diseases had a similar aetiology (Hadlow, 1959). Subsequently, it was established that kuru, like scrapie, was caused by an unconventional transmissible agent. A number of other similar neurodegenerative diseases of humans, some of which are genetically determined, are classified as TSEs. The TSEs which have been described in humans are presented in Table 76.2. In humans, these TSEs may occur as infectious, genetic or sporadic diseases. In scrapie, there is convincing evidence for the importance of the genetic constitution of certain breeds of sheep in determining susceptibility to the disease.

Scrapie

This insidious, fatal, neurological disease of adult sheep and goats occurs worldwide except in Australia and New Zealand.

Epidemiology

The mode of transmission of scrapie is not clearly understood. Potential modes for natural infection include ingestion, entry through superficial abrasions and transmission from ewe to lamb. There is evidence that transmission tends to occur during the perinatal period and that exposure to placental material from affected ewes may be important. Pastures grazed by affected animals appear to remain contaminated for years.

Particular polymorphisms of the *PrP* gene are associated with an increased incidence of scrapie in certain breeds of sheep. Some research workers have inferred from breeding experiments that scrapie is an exclusively genetic disease. However, Australia and New Zealand are free of scrapie despite the presence of animals with scrapie-associated *PrP* alleles (Hunter *et al.*, 1997). The *PrP* gene-coding sequence in sheep is highly polymorphic. Polymorphisms at codons 136, 154 and 171 of the *PrP* gene, producing amino acid substitutions in PrP, are important in influencing susceptibility to scrapie. In many breeds, the valine 136 glutamine 171 arginine 154 (denoted VRQ) allele is strongly associated with susceptibility to scrapie (Laplanche *et al.*, 1999).

Pathogenesis and pathology

Following natural infection, PrP^{Sc} is usually first detected in tissues of the lymphoreticular system including the spleen, the palatine tonsil and the retropharyngeal and mesenteric lymph nodes. In lymph nodes, replication apparently occurs in follicular dendritic cells. After oral exposure it is thought that the portal of entry to neural tissues is in the duodenum and ileum. The agent then spreads through fibres of the autonomic nervous system to the spinal cord and the medulla oblongata (van Keulen *et al.*, 1999). Neuronal and neuropil vacuolation and astrogliosis are associated with accumulation of PrP^{Sc} in the CNS.

Clinical signs

The disease has a long incubation period. Neurological signs develop predominantly in sheep of breeding age with a peak incidence between three and four years of age. Initially, affected animals may present with restlessness or nervousness, particularly after sudden noise or movement. Fine tremors of the head and neck and incoordination with a tendency to exhibit jerky movements are characteristic signs. Pruritis may result in loss of wool. In some affected sheep, a nibbling reflex can be elicited by scratching the back. Progression of the disease leads to emaciation. Death usually occurs within six months from the onset of clinical signs.

Diagnosis

Clinical signs and histopathological examination of the CNS form the basis for diagnosis. Characteristic microscopic changes include neuronal vacuolation and degeneration, vacuolar change in the neuropil and astrocytosis, particularly in the medulla. No obvious inflammatory response is evident. Confirmatory methods include immunohistochemical staining for PrP^{Sc}, immunoblotting to detect proteinase-K-resistant PrP^{Sc} and electron microscopy to detect scrapie-associated fibrils in detergent-treated extracts of brain. Antemortem detection

Table 76.2 Transmissible spongiform encephalopathies of humans.

Disease	Comments
Kuru	Described in members of the Fore population in Papua New Guinea. Acquired through ritualistic cannibalism; brain tissue was the primary source of infection.
Creutzfeldt-Jakob disease (CJD)	
Sporadic CJD	Attributed to somatic mutation in *PrP* gene or to random conversion of PrP^{C} into PrP^{Sc}
Iatrogenic CJD	Transmitted by medical or surgical procedures which utilized contaminated human tissues
Variant CJD	Considered to be a consequence of exposure to PrP^{Sc} from cattle with bovine spongiform encephalopathy
Familial CJD	Germline mutations in *PrP* gene
Gerstmann-Sträussler-Scheinker syndrome	Germline mutations in *PrP* gene
Fatal familial insomnia	Germline mutation in *PrP* gene

methods, based on demonstration of PrP^{Sc} in lymphoid tissues of the palatine tonsil and nictitating membrane by histochemical methods, are being developed.

Control

In the European Union, scrapie has been designated a notifiable disease. Countries free of the disease impose strict quarantine procedures. Slaughter policies have been enforced with different degrees of success in several countries. In Australia and New Zealand an eradication policy, implemented soon after the introduction of the disease, was successful. Eradication was abandoned in the United States because of the cost and difficulties in implementation. A control policy involving flock certification and movement restrictions is now in place. Breeding scrapie-resistant sheep may be a realistic method for reducing the frequency of the disease (Parry, 1983).

Bovine spongiform encephalopathy

This condition is a progressive, neurodegenerative disease of adult cattle, first recognized in England in 1986 (Wells *et al.*, 1987). More than 170,000 cases of the disease were subsequently confirmed and an estimated one million animals were infected. This common source epidemic peaked in 1993 when more than 300 new cases were identified each week. Since then, there has been a steady decline in numbers of confirmed cases. The disease has been reported in several countries in animals imported from Great Britain. In addition, indigenous cattle in a number of European countries, including Switzerland, Ireland, France and Portugal, have developed the disease.

The prion strain causing bovine spongiform encephalopathy (BSE) is not considered to be species-specific; infection has been reported in exotic ungulates in zoological collections following ingestion of feed derived from contaminated bovine tissues. In addition, feline spongiform encephalopathy was first recorded in the early 1990s in association with the BSE epidemic. In 1996, a novel form of human prion disease termed variant Creutzfeldt-Jakob disease (vCJD) was recognized in Great Britain. Molecular strain-typing studies and experimental transmission in transgenic and conventional mice indicated that vCJD and BSE are caused by indistinguishable prion strains. The extent of exposure of the human population to the agent cannot be accurately estimated because of uncertainties about risk factors and about the length of the incubation period in vCJD (Collinge, 1999).

Epidemiology

The BSE epidemic, which started simultaneously at several geographical locations in Great Britain, was attributed to contaminated meat-and-bone meal (MBM) prepared from slaughterhouse offal and fed as a protein dietary supplement to cattle. It is postulated that the scrapie agent crossed the species barrier into cattle in the early 1980s following changes in the rendering process which allowed survival of increased amounts of scrapie PrP (PrP^{Sc}) in the MBM. Recycling of infected tissues from animals with BSE, prior to the recognition of the disease and the imposition of control measures, resulted in extension of the epidemic. Because of the high ratio of sheep to cattle, the frequency of endemic scrapie and the heavy reliance on MBM as a supplement for dairy cattle in Great Britain, the epidemic has been largely confined to that country (Nathanson *et al.*, 1999). As a result of the banning of ruminant-derived MBM in 1988, there was a marked decline in the prevalence of BSE in Great Britain after 1993. Despite this decline, animals born after imposition of the ban have developed BSE. This was ascribed to the continued use of ruminant-derived MBM and also to cross-contamination in feed mills by rations specified for pigs and poultry. Additional stringent regulations, introduced in 1996, banned the inclusion of mammalian-derived MBM from farm animal feed.

Horizontal transmission of BSE does not appear to occur and, although maternal transmission may occur at a low rate, it is considered to be of minimal importance in the spread of disease. Susceptibility of cattle to BSE appears to be independent of sex, breed and genotype.

Pathogenesis and pathology

The pathogenesis of BSE is poorly defined. The agent of

BSE has been found in the distal ileum following experimental oral exposure. In naturally occuring cases of the disease, the agent has not been demonstrated in lymphoreticular tissues or peripheral nodes. Accumulation of PrP^{Sc} in the CNS is associated with vacuolation and glial proliferation.

Clinical signs

The mean incubation period is about five years. Neurological signs, which are highly variable, include changes in behaviour and deficits in posture and movement. Loss of weight and decreased milk production also occur. Other clinical signs include tremors, hyperaesthesia, apprehension, bruxism, exaggerated menace reflex and head shyness. Ataxia, hypermetria and a tendency to fall become increasingly evident in the later stages of the disease. The clinical course may extend over many days or months.

Diagnosis

Bovine spongiform encephalopathy can be confirmed by histopathological examination of brain tissue and appropriate additional methods. Examination of brain tissue may be confined to a coronal section of the medulla at the obex. Characteristic neuropathological changes including neuropil vacuolation and astrocytosis are consistently present at this site. Additional confirmatory methods include immunohistochemical staining for PrP^{Sc}, immunoblotting to demonstrate proteinase K-resistant PrP^{Sc} and electron microscopy to detect prion fibrils in detergent-treated extracts of brain.

Control

Bovine spongiform encephalopathy is a notifiable disease in countries of the European Union. In some countries, the entire herd, in which an affected animal is detected, is slaughtered. In other countries, clinically affected animals only are slaughtered. Ruminant-derived protein should be excluded from ruminant rations. Carcases of infected animals should be incinerated at high temperatures to ensure destruction of the thermostable agent. Buildings and equipment can be decontaminated by the application of high concentrations of sodium hypochlorite or of heated strong solutions of sodium hydroxide (Taylor, 2000).

References

Chesebro, B. (1998). BSE and prions: uncertainties about the agent. *Science*, **279**, 42–43.

Collinge, J. (1999). Variant Creutzfeldt-Jakob disease. *Lancet*, **354**, 317–323.

Greig, J.R. (1950). Scrapie in sheep. *Journal of Comparative Pathology*, **60**, 263–266.

Hadlow, W.J. (1959). Scrapie and kuru. *Lancet*, **ii**, 289–290.

Hunter, N., Cairns, D., Foster, J.D. *et al.* (1997). Is scrapie solely a genetic disease? *Nature*, **386**, 137.

Jackson, G.S., Hosszu, L.L.P., Power, A. *et al.* (1999). Reversible conversion of monomeric human prion protein between native and fibrillogenic conformations. *Science*, **283**, 1935–1937.

Laplanche, J.L., Hunter, N., Shinagawa, M. and Williams, E. (1999). Scrapie, chronic wasting disease and transmissible mink encephalopathy. In *Prion Biology and Diseases*. Ed. S.P. Prusiner. Cold Spring Harbor Laboratory Press, New York. pp. 393–429.

Nathanson, N., Wilesmith, J., Wells, G.A. and Griot, C. (1999). Bovine spongiform encephalopathy and related diseases. In *Prion Biology and Diseases*. Ed. S.B. Prusiner. Cold Spring Harbor Laboratory Press, New York. pp. 431–463.

Parry, H.B. (1983). *Scrapie disease in sheep*. Academic Press, New York.

Prusiner, S.B. (1982). Novel proteinaceous infectious particles cause scrapie. *Science*, **216**, 136–144.

Prusiner, S.B., Peters, P., Kaneko, K. *et al.* (1999). Cell biology of prions. In *Prion Biology and Diseases*. Ed. S.B. Prusiner. Cold Spring Harbor Laboratory Press, New York. pp. 349–391.

Taylor, D.M. (2000). Inactivation of transmissible degenerative encephalopathy agents: a review. *Veterinary Journal*, **159**, 1–7.

van Keulen, L.J.M., Schreuder, B.E.C., Vromans, M.E.W. *et al.* (1999). Pathogenesis of natural scrapie in sheep. In *Proceedings of Characterization and Diagnosis of Prion Diseases in Animals and Man*, Tubingen.

Wells, G.A.H., Scott, A.C., Johnson, C.T. *et al.* (1987). A novel progressive spongiform encephalopathy in cattle. *Veterinary Record*, **121**, 419–420.

Further reading

Chin, J. (2000). *Control of Communicable Diseases Manual*. Seventeenth Edition. American Public Health Association, Washington, D.C. pp. 183–186.

Prusiner, S.B. (1999). *Prion Biology and Diseases*. Ed. S.B. Prusiner. Cold Spring Harbor Laboratory Press, New York.

Prusiner, S.B. (1997). Prion diseases and the BSE crisis. *Science*, **278**, 245–251.

Prusiner, S.B. (2001). Shatuck lecture – neurodegenerative diseases and prions. *New England Journal of Medicine*, **344**, 1516–1526.

Raeber, A.J. and Aguzzi, A. (2000). Engulfment of prions in the germinal centre. *Immunology Today*, 21, 66–67.

Wilesmith, J.W. (1993). Epidemiology of bovine spongiform encephalopathy and related diseases. *Archives of Virology*, **7**, 245–254.